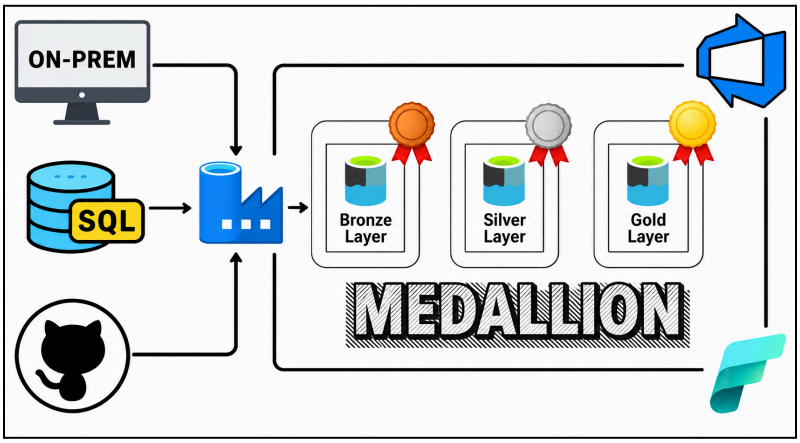
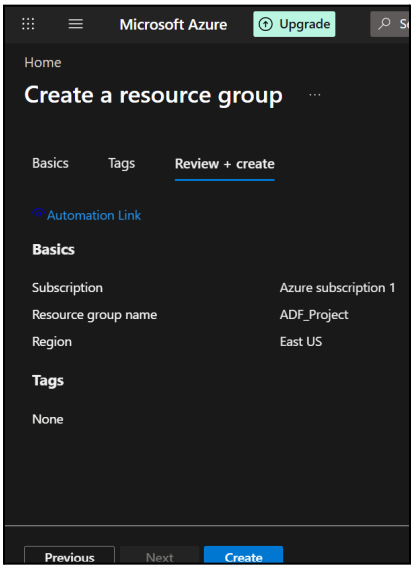
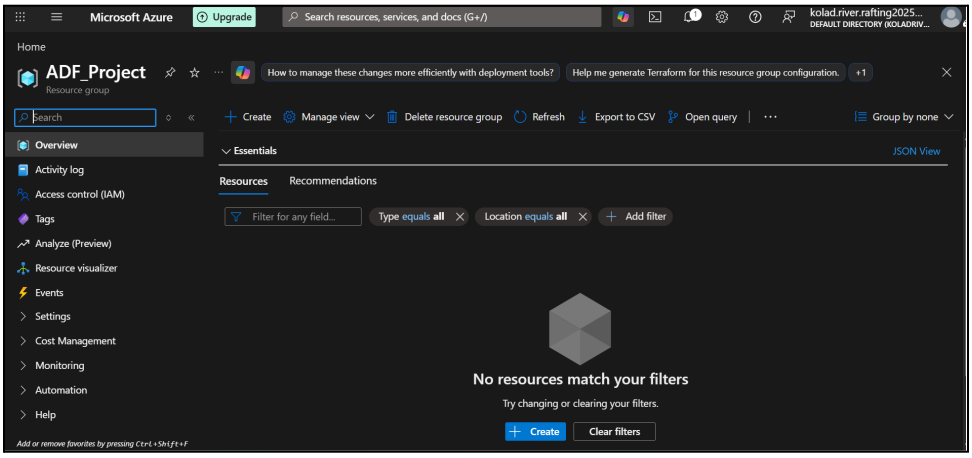
Architecture :



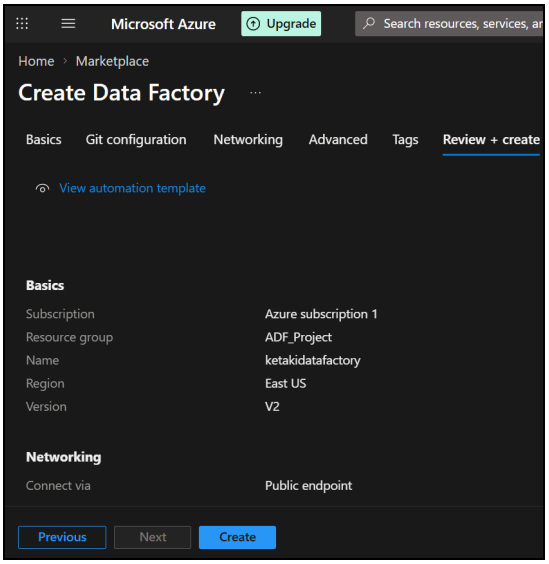
Creating a resource group :



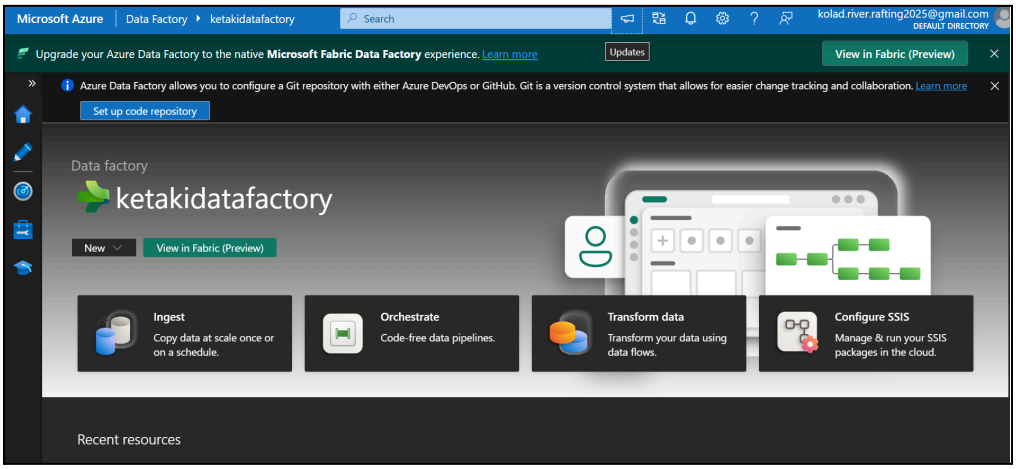
Click on create to go to marketplace :



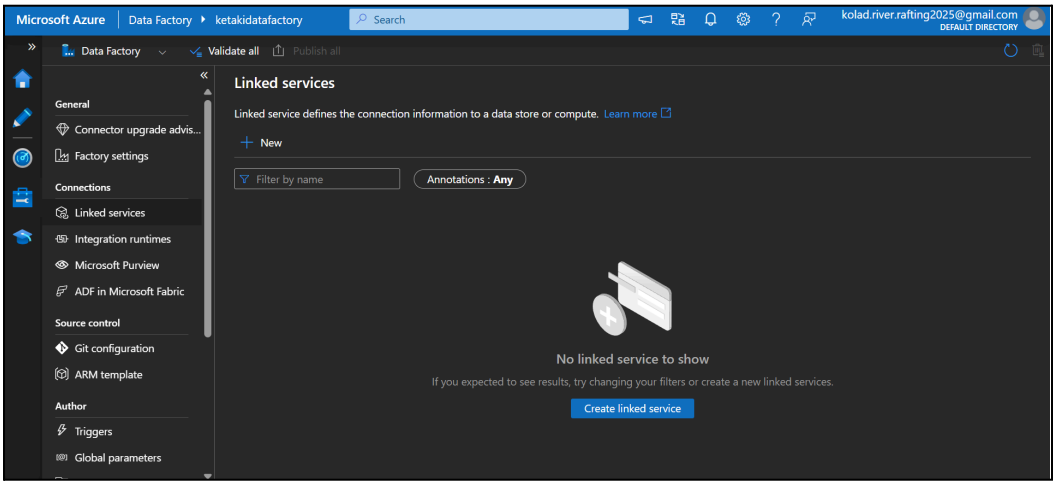
Type azure data factory and configure :



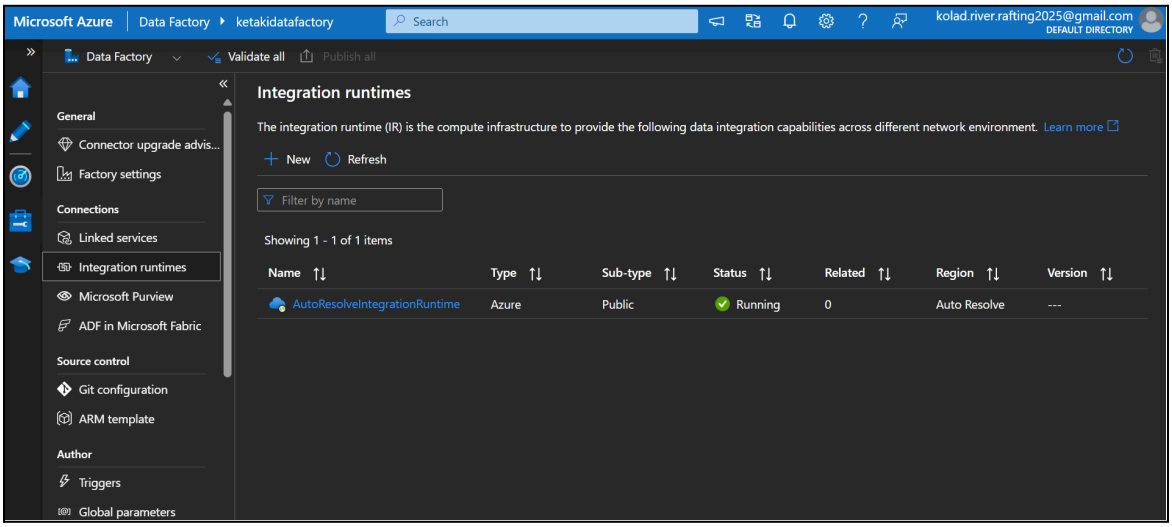
Launch azure data factory :



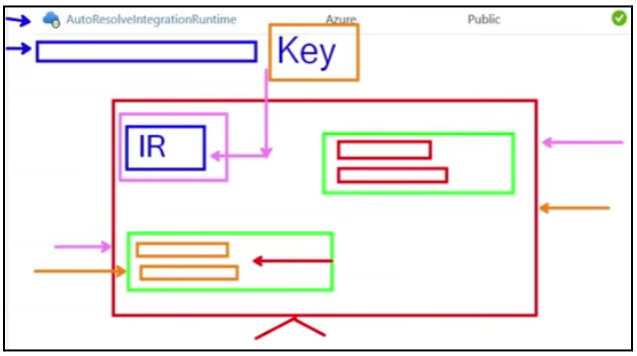
Linked service -> connection



Default by azure :

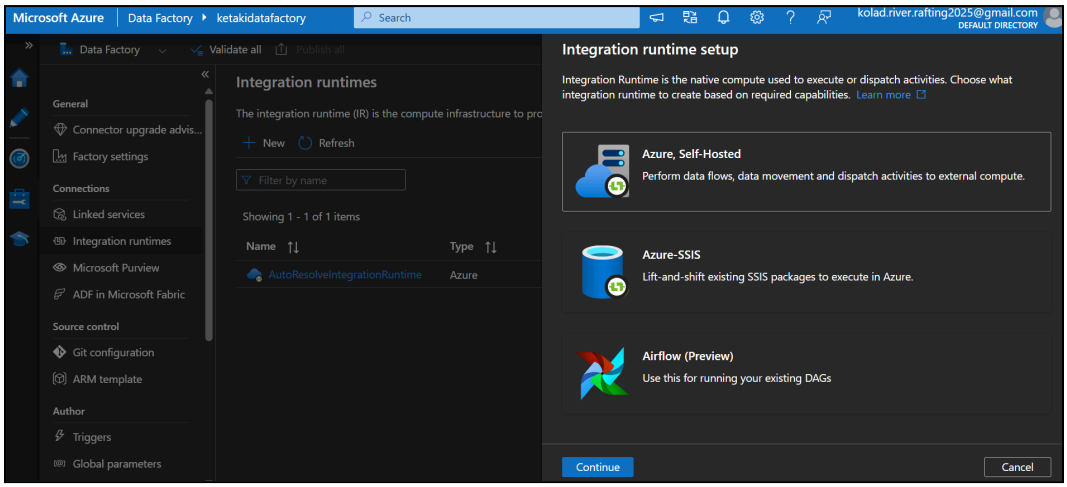


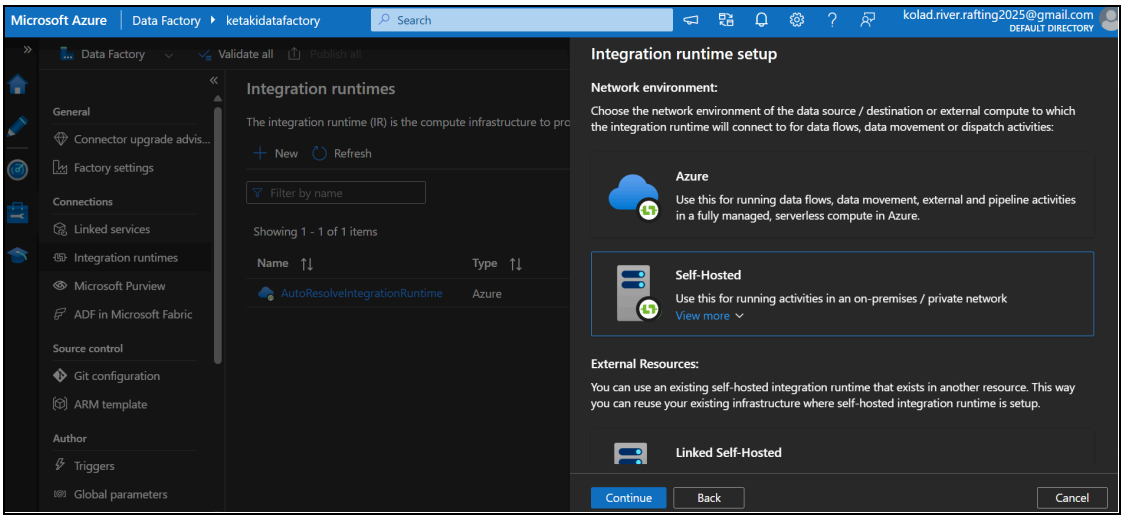
Creating integration runtime -> getting a key -> download integration runtime application -> registering this application with the key

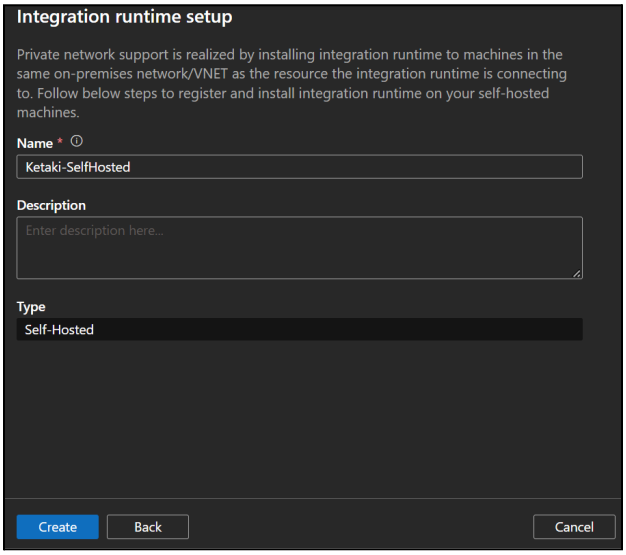


38:03 On-Prem To Azure Migration

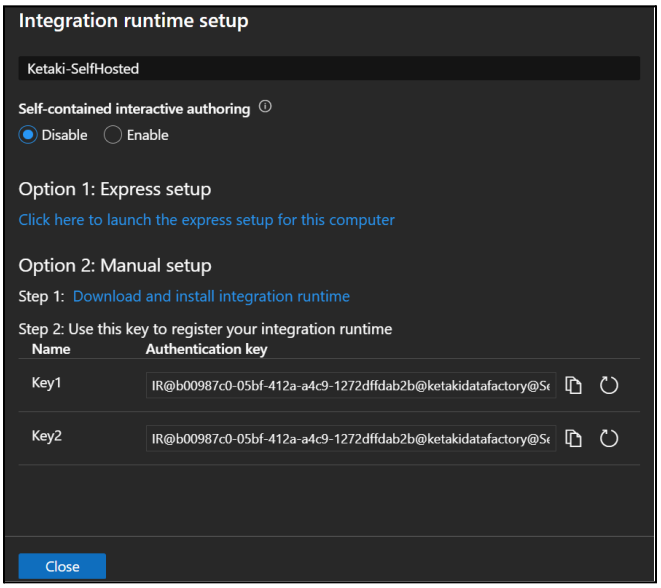
Choosing azure self hosted :

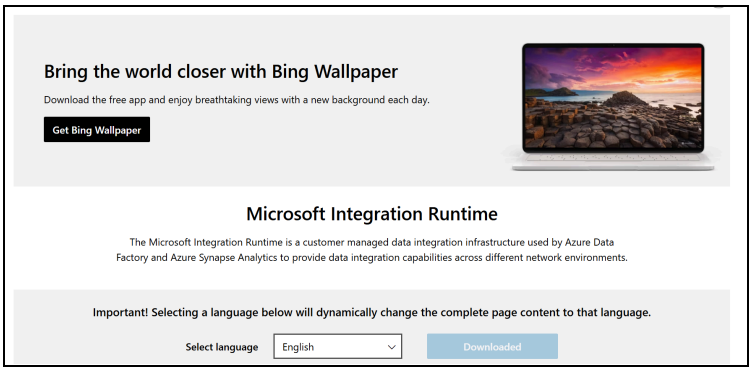


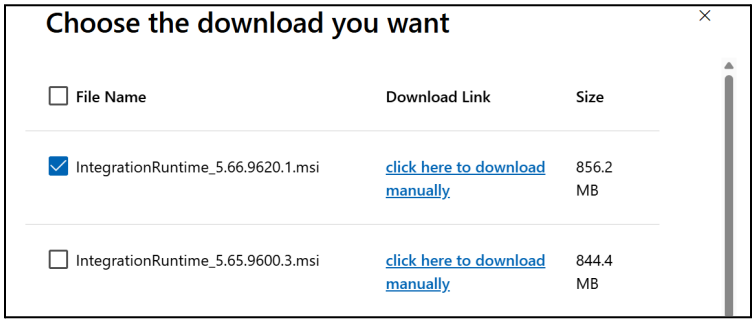


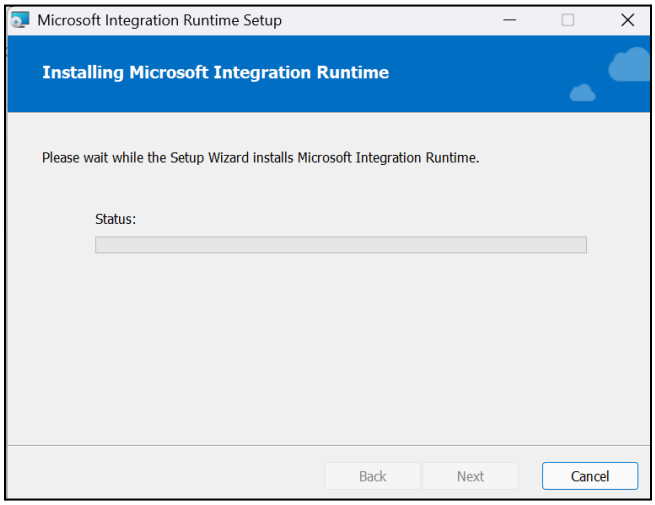


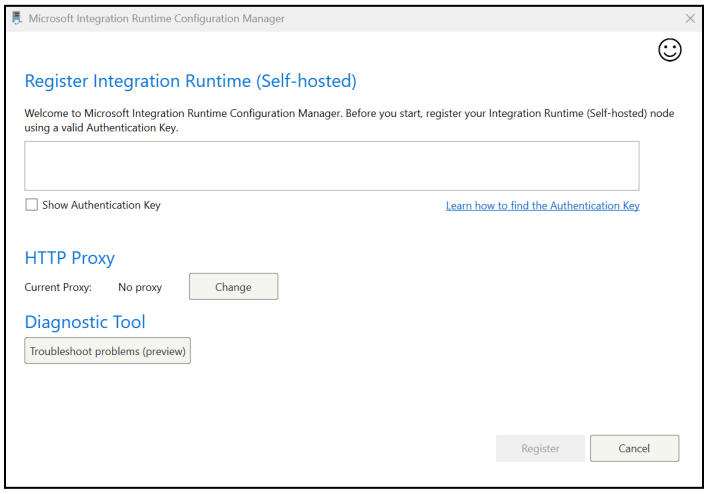
Click on manual setup :

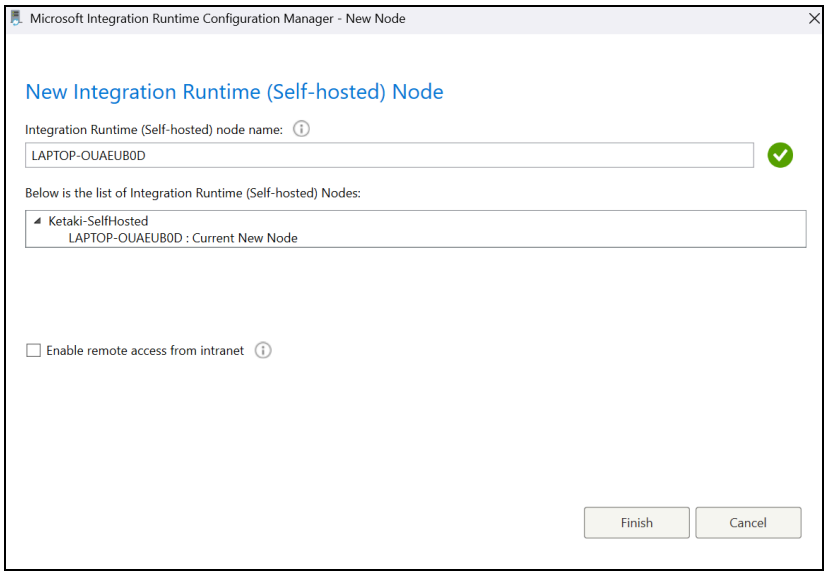


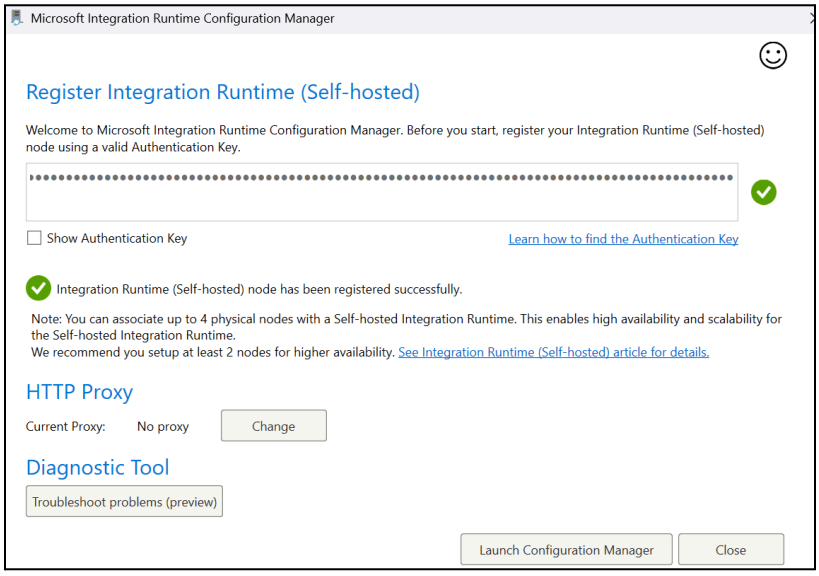

Bring the world closer with Bing Wallpaper
Download the free app and enjoy breathtaking views with a new background each day.
Get Bing Wallpaper
Microsoft Integration Runtime
The Microsoft Integration Runtime is a customer managed data integration infrastructure used by Azure Data Factory and Azure Synapse Analytics to provide data integration capabilities across different network environments.
Important! Selecting a language below will dynamically change the complete page content to that language.
Select language
English
Downloaded

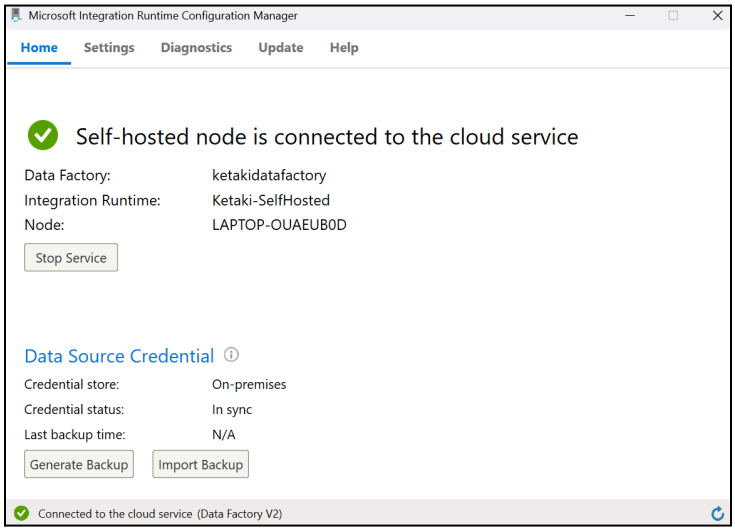

Choose the download you want
File Name
Download Link
Size
IntegrationRuntime_5.66.9620.1.msi
click here to download manually
856.2 MB
IntegrationRuntime_5.65.9600.3.msi
click here to download manually
844.4 MB

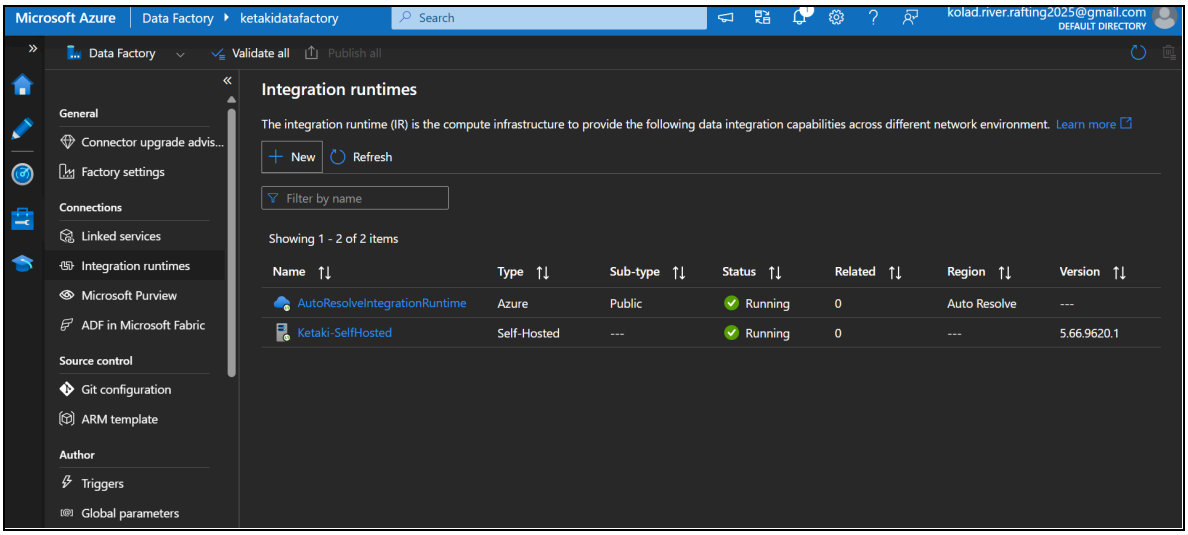

Microsoft Integration Runtime Setup
Installing Microsoft Integration Runtime
Please wait while the Setup Wizard installs Microsoft Integration Runtime.
Status:
Back
Next
Cancel


Microsoft Integration Runtime Configuration Manager
Register Integration Runtime (Self-hosted)
Welcome to Microsoft Integration Runtime Configuration Manager. Before you start, register your Integration Runtime (Self-hosted) node using a valid Authentication Key.
Show Authentication Key
Learn how to find the Authentication Key
HTTP Proxy
Current Proxy:
No proxy
Change
Diagnostic Tool
Troubleshoot problems (preview)
Register
Cancel


Microsoft Integration Runtime Configuration Manager - New Node
New Integration Runtime (Self-hosted) Node
Integration Runtime (Self-hosted) node name:
LAPTOP-OUAEUB0D
Below is the list of Integration Runtime (Self-hosted) Nodes:
Ketaki-SelfHosted
LAPTOP-OUAEUB0D : Current New Node
Enable remote access from intranet
Finish
Cancel

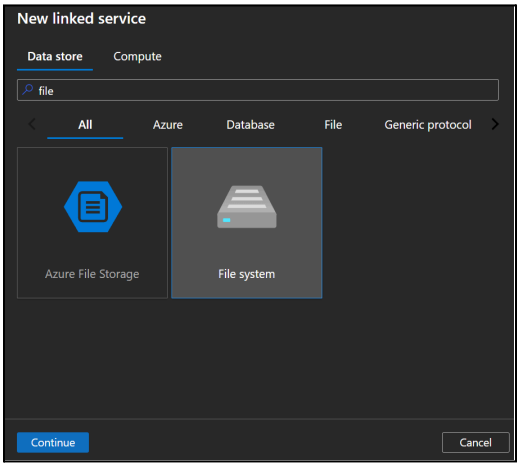

Microsoft Integration Runtime Configuration Manager
Register Integration Runtime (Self-hosted)
Welcome to Microsoft Integration Runtime Configuration Manager. Before you start, register your Integration Runtime (Self-hosted) node using a valid Authentication Key.
Show Authentication Key
Learn how to find the Authentication Key
Integration Runtime (Self-hosted) node has been registered successfully.
Note: You can associate up to 4 physical nodes with a Self-hosted Integration Runtime. This enables high availability and scalability for the Self-hosted Integration Runtime.
We recommend you setup at least 2 nodes for higher availability. See Integration Runtime (Self-hosted) article for details.
HTTP Proxy
Current Proxy: No proxy
Change
Diagnostic Tool
Troubleshoot problems (preview)
Launch Configuration Manager
Close


Microsoft Integration Runtime Configuration Manager
Home
Settings
Diagnostics
Update
Help
Self-hosted node is connected to the cloud service
Data Factory: ketakidatafactory
Integration Runtime: Ketaki-SelfHosted
Node: LAPTOP-OUAEUB0D
Stop Service
Data Source Credential
Credential store: On-premises
Credential status: In sync
Last backup time: N/A
Generate Backup
Import Backup
Connected to the cloud service (Data Factory V2)

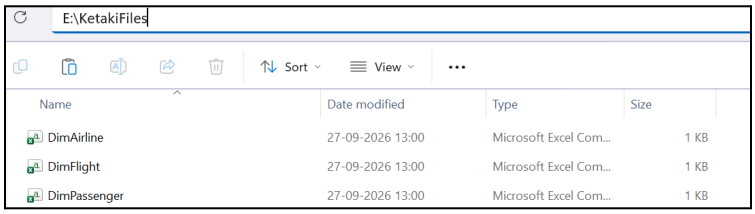


Data in private network , data in local drive , data in hosted drives -> self hosted integration runtime

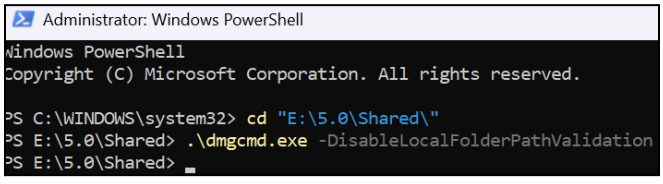
Creating new linked service :



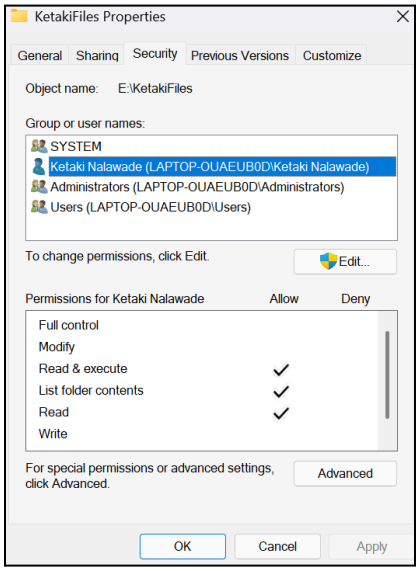
Use self hosted integration runtime for onprem and private networks
Download the files in E :

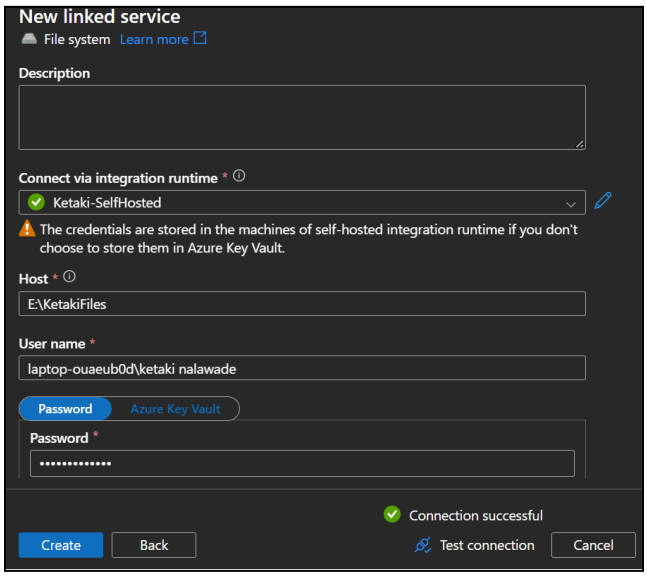


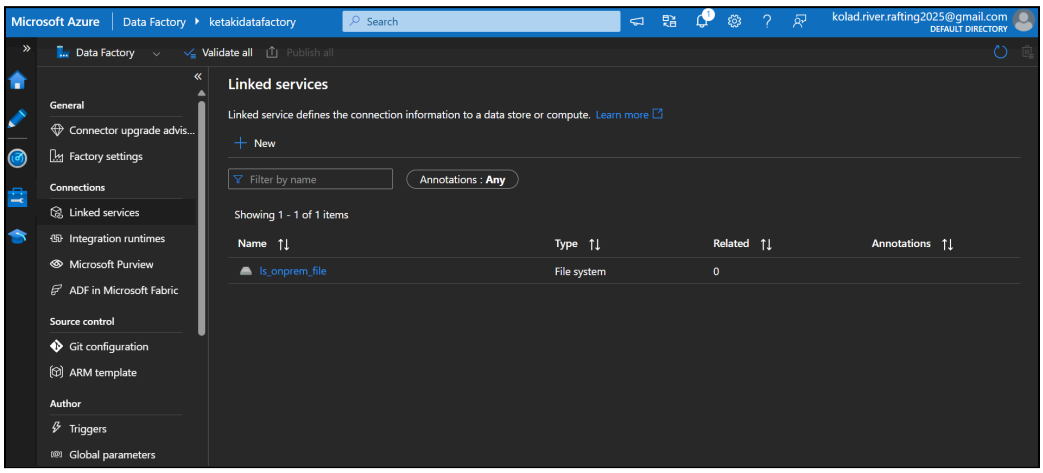
Run powershell as administrator :



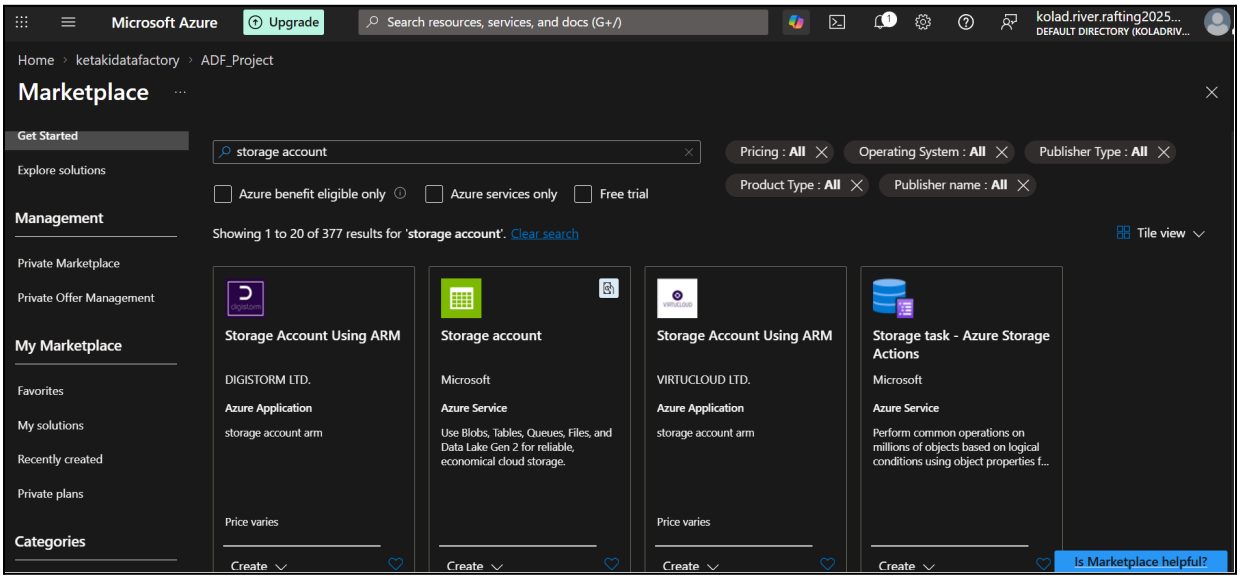
I tried using a Microsoft account but it did not access the E drive so I used the local user and its password.

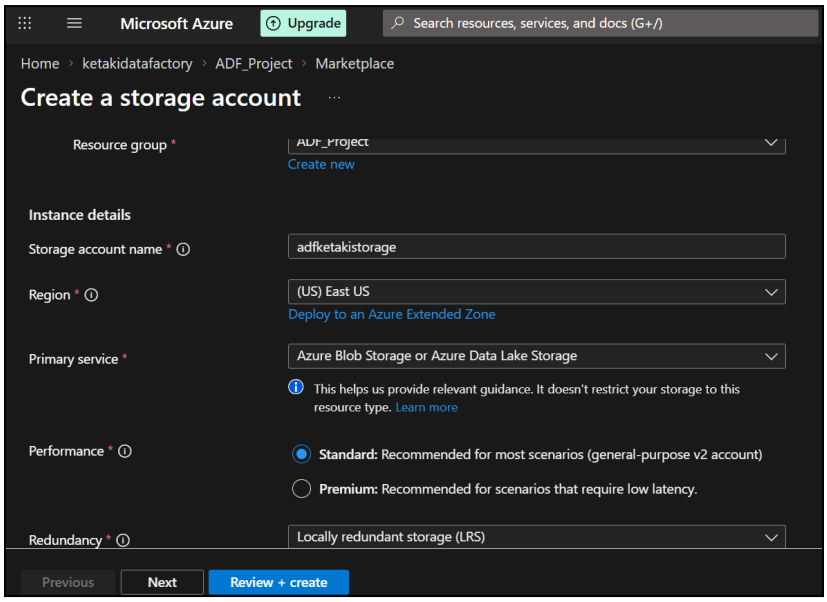




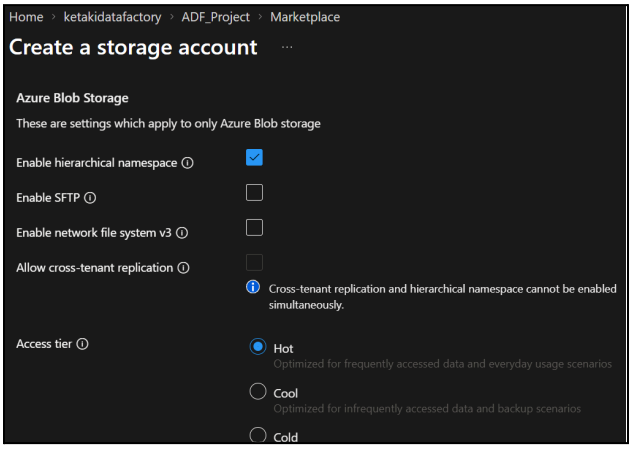


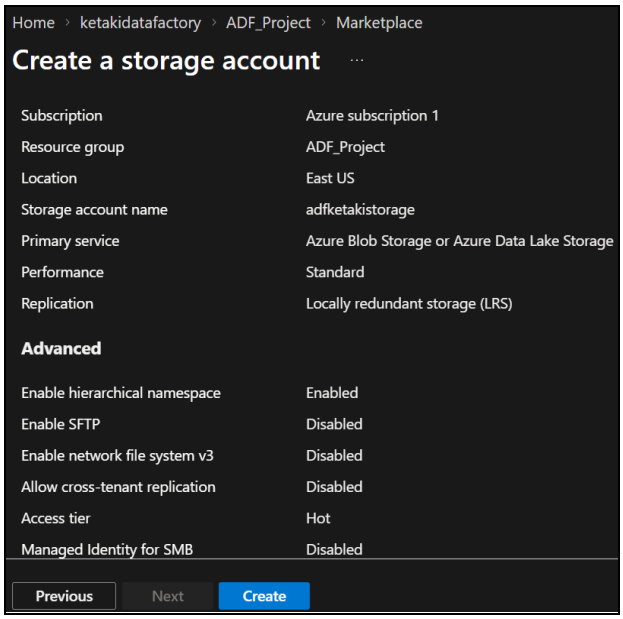
Creating storage account :



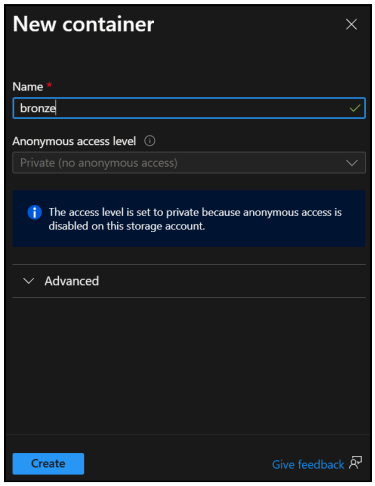


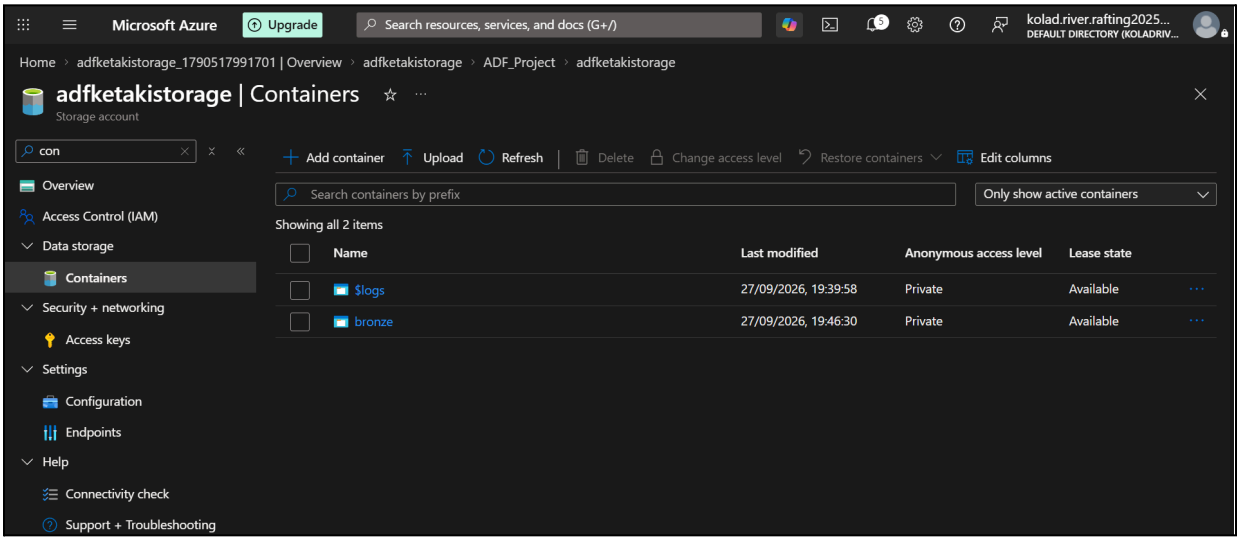
Tick hierarchical namespace :



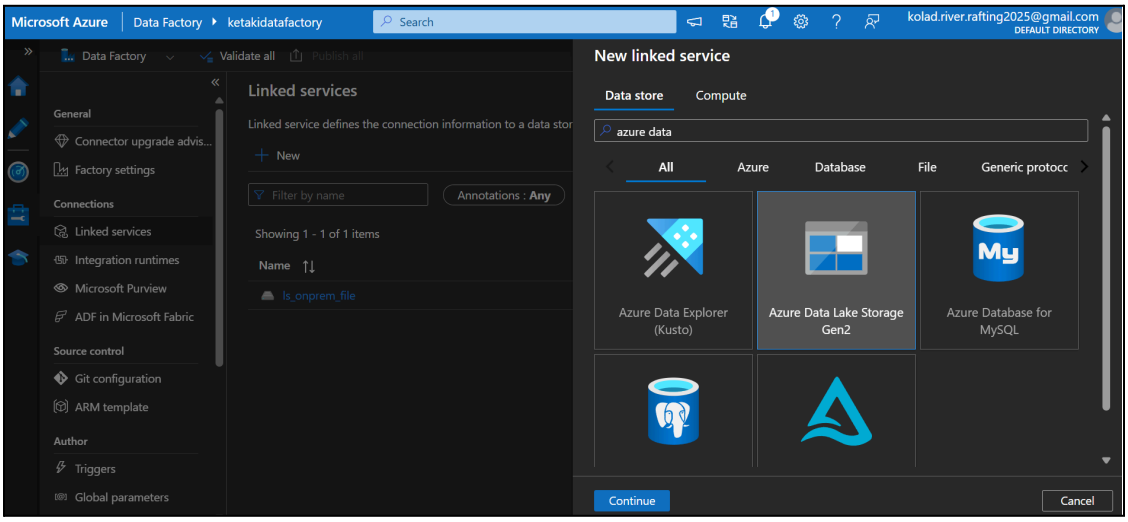


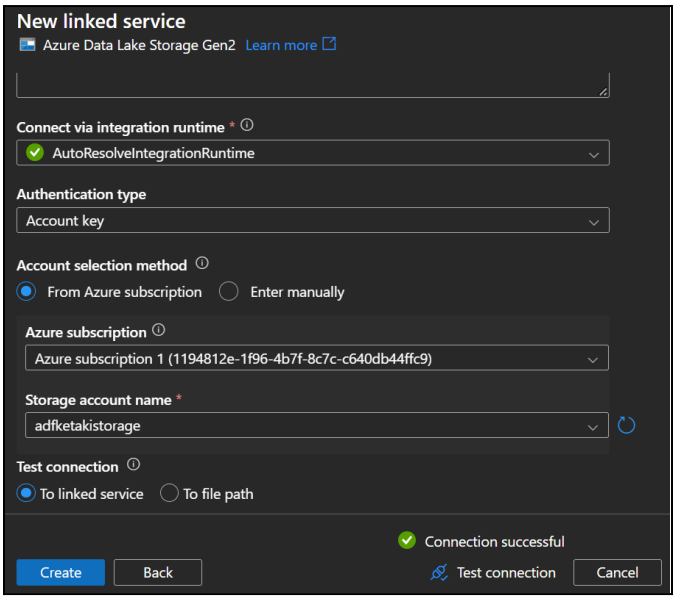
Creating containers :



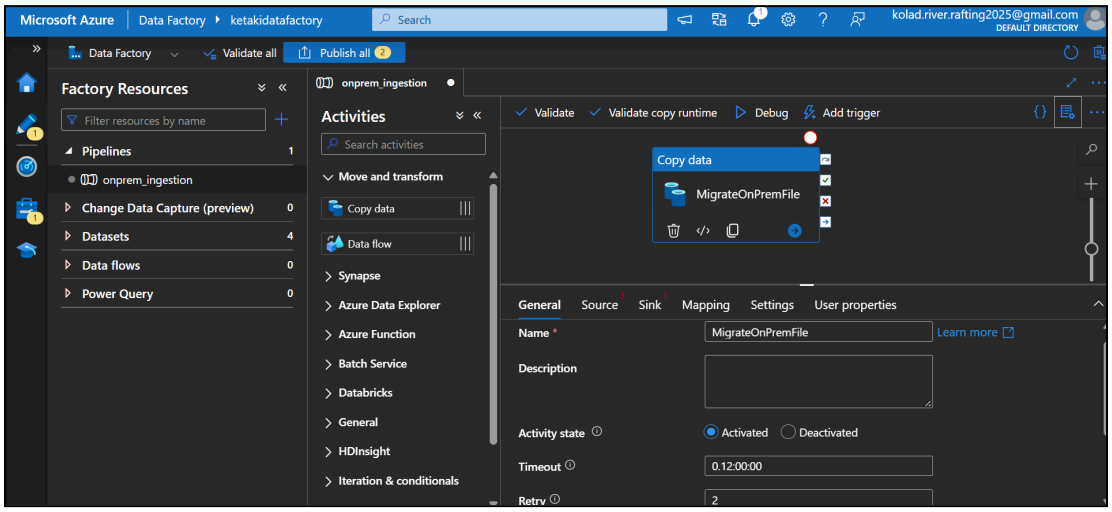


Creating new link service :

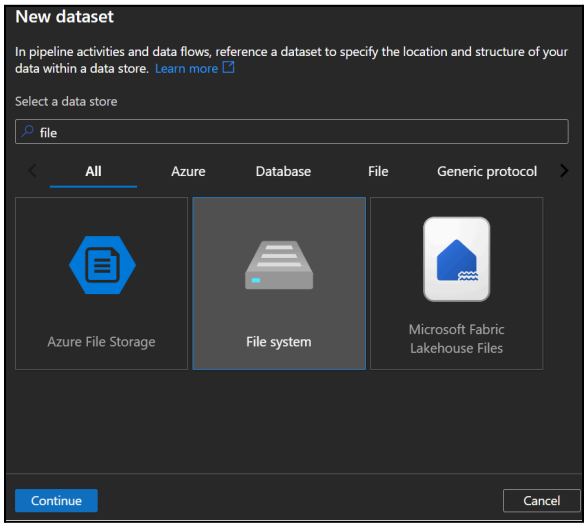


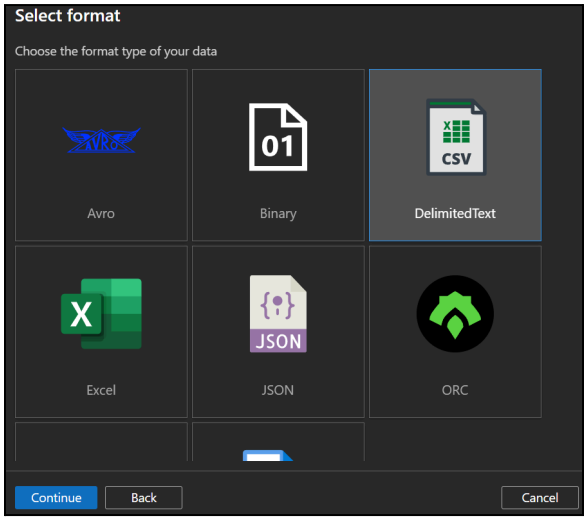


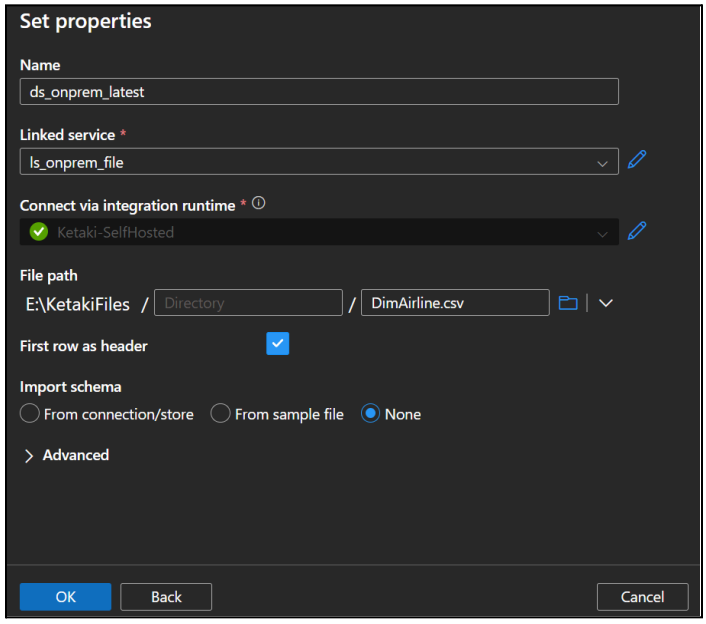
Creating new pipeline :

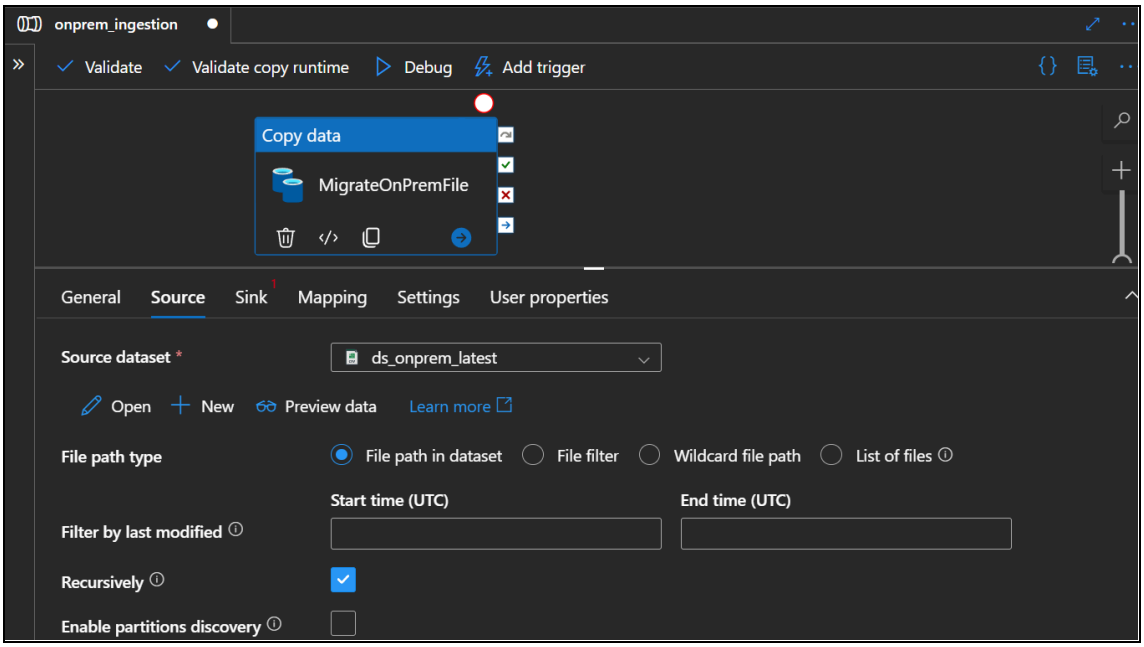


For source :

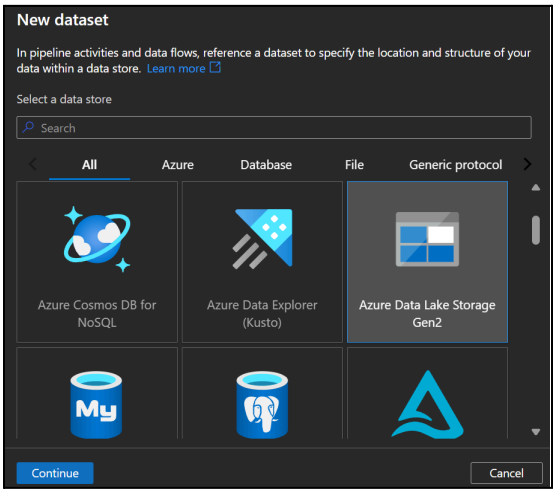

New dataset
In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. Learn more
Select a data store
file
All
Azure
Database
File
Generic protocol
Azure File Storage
File system
Microsoft Fabric Lakehouse Files
Continue
Cancel

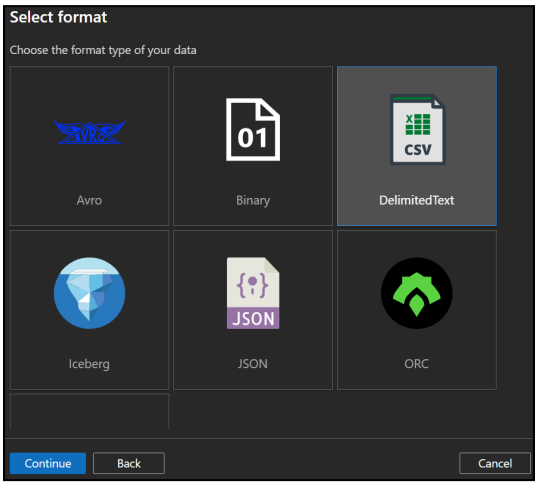

Select format
Choose the format type of your data
Avro
Binary
DelimitedText
Excel
JSON
ORC
Continue
Back
Cancel

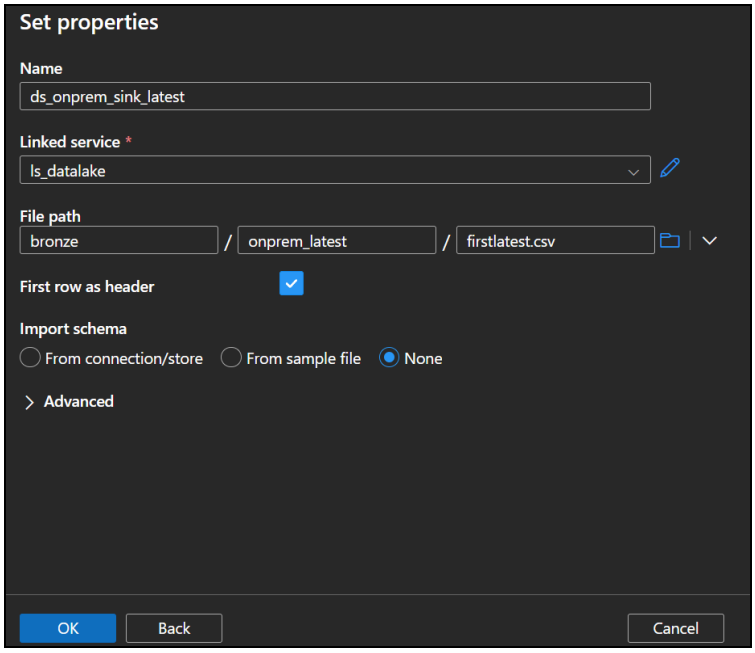

Set properties
Name
ds_onprem_latest
Linked service *
ls_onprem_file
Connect via integration runtime *
Ketaki-SelfHosted
File path
E:\KetakiFiles /
Directory
/
DimAirline.csv
First row as header
Import schema
From connection/store
From sample file
None
Advanced
OK
Back
Cancel

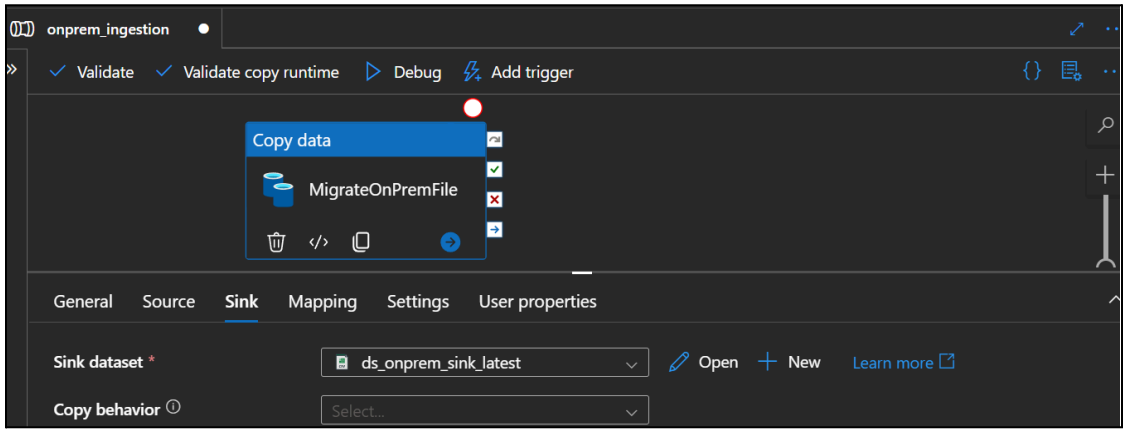

onprem_ingestion
Validate
Validate copy runtime
Debug
Add trigger
Copy data
MigrateOnPremFile
General
Source
Sink
Mapping
Settings
User properties
Source dataset *
ds_onprem_latest
Open
New
Preview data
Learn more
File path type
File path in dataset
File filter
Wildcard file path
List of files
Start time (UTC)
End time (UTC)
Filter by last modified
Recursively
Enable partitions discovery

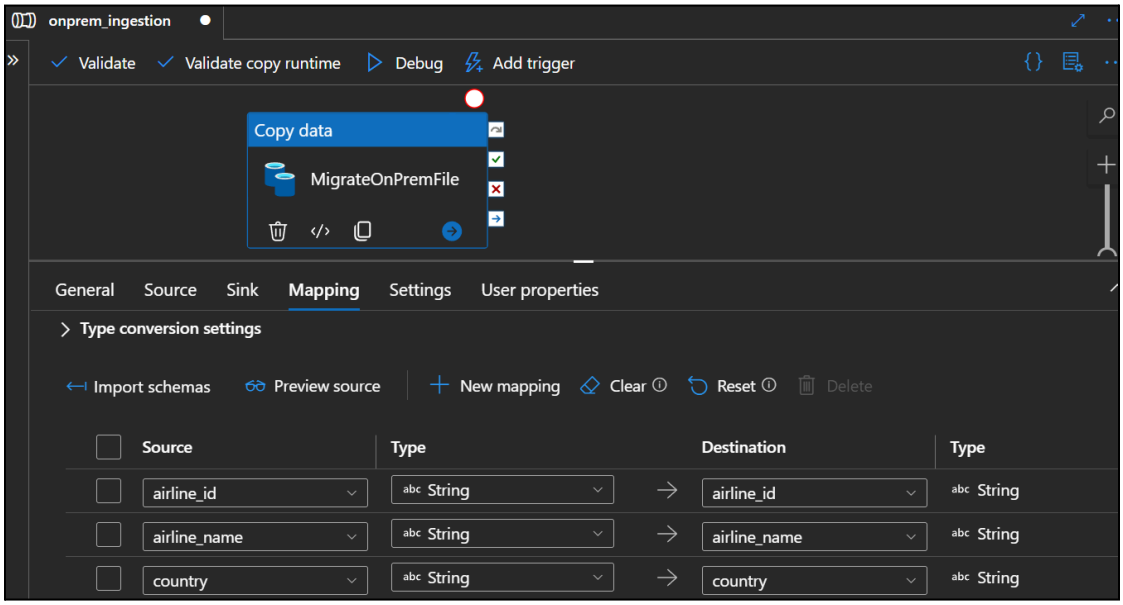
For sink :

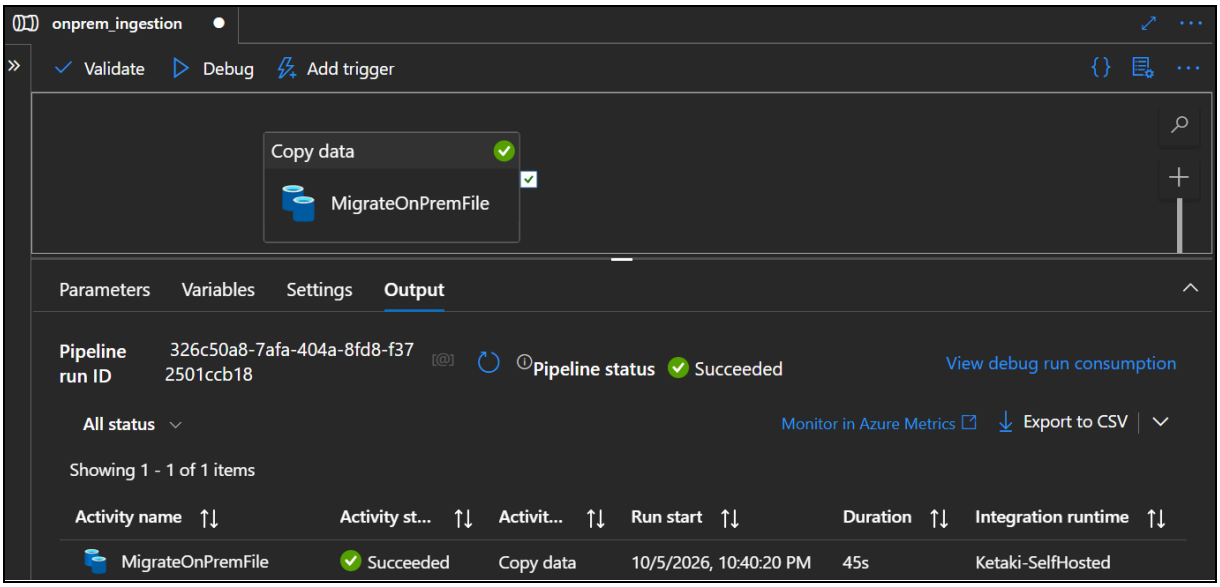

New dataset
In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. Learn more
Select a data store
Search
All
Azure
Database
File
Generic protocol
Azure Cosmos DB for NoSQL
Azure Data Explorer (Kusto)
Azure Data Lake Storage Gen2
Continue
Cancel


Select format
Choose the format type of your data
Avro
Binary
DelimitedText
Iceberg
JSON
ORC
Continue
Back
Cancel

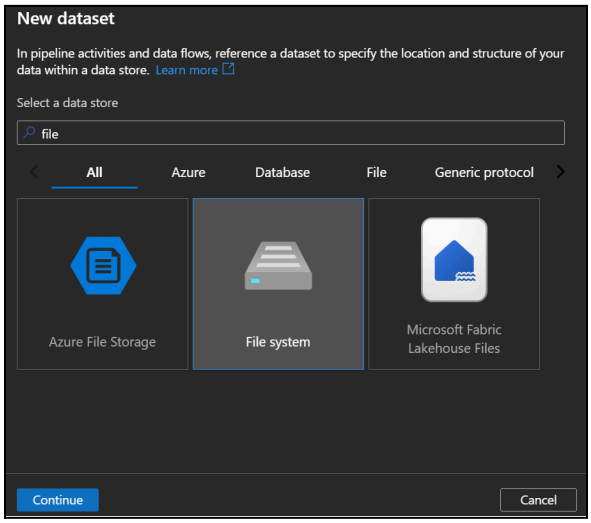

Set properties
Name
ds_onprem_sink_latest
Linked service *
ls_datalake
File path
bronze
onprem_latest
firstlatest.csv
First row as header
Import schema
From connection/store
From sample file
None
Advanced
OK
Back
Cancel

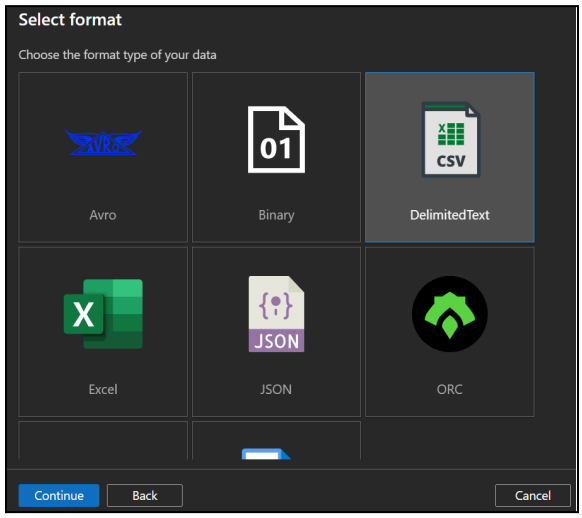

onprem_ingestion
Validate
Validate copy runtime
Debug
Add trigger
Copy data
MigrateOnPremFile
General
Source
Sink
Mapping
Settings
User properties
Sink dataset *
ds_onprem_sink_latest
Open
New
Learn more
Copy behavior
Select...

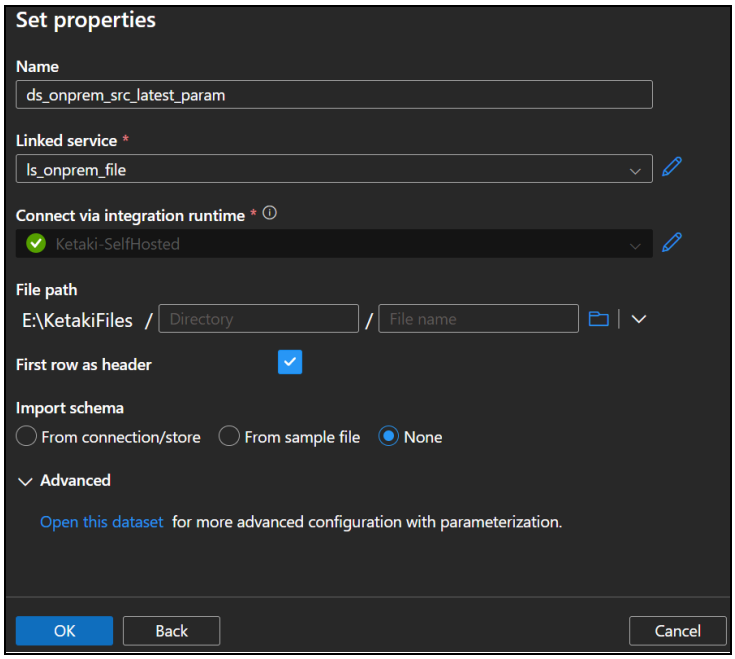

onprem_ingestion
Validate
Validate copy runtime
Debug
Add trigger
Copy data
MigrateOnPremFile
General
Source
Sink
Mapping
Settings
User properties
Type conversion settings
Import schemas
Preview source
New mapping
Clear
Reset
Delete
Source
Type
Destination
Type
airline_id
String
airline_id
String
airline_name
String
airline_name
String
country
String
country
String

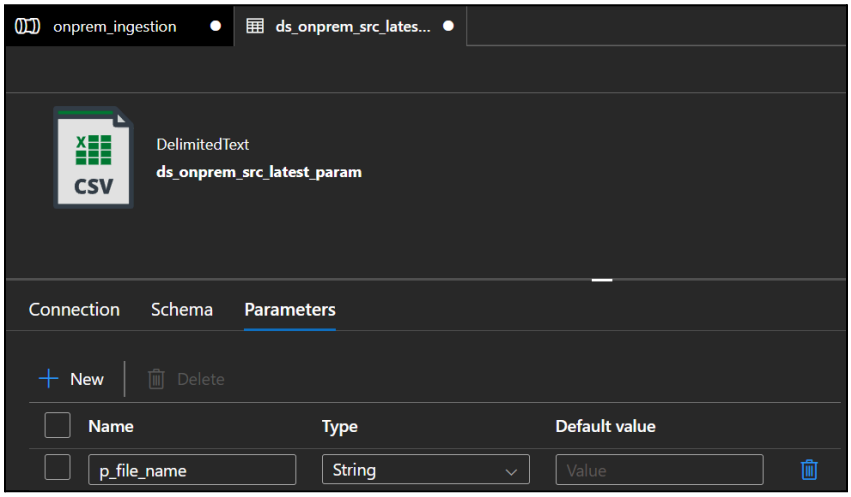


Param Source :

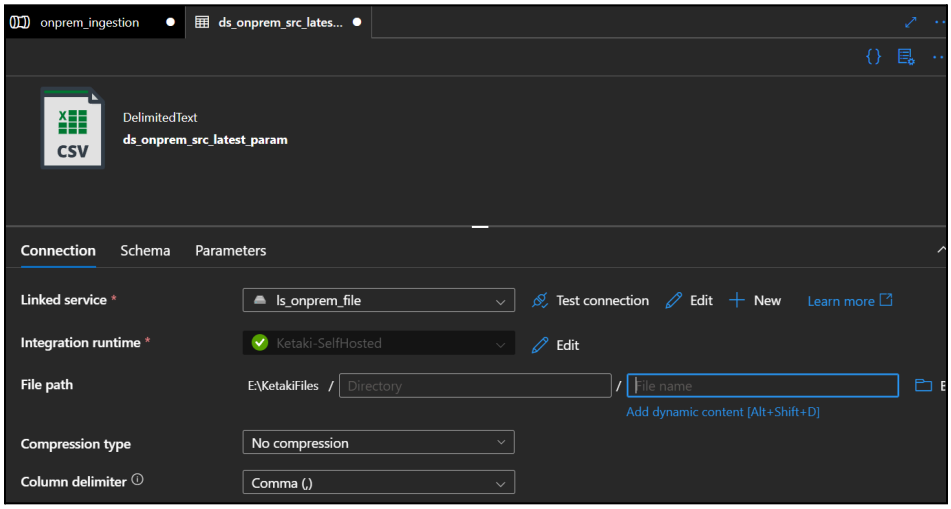


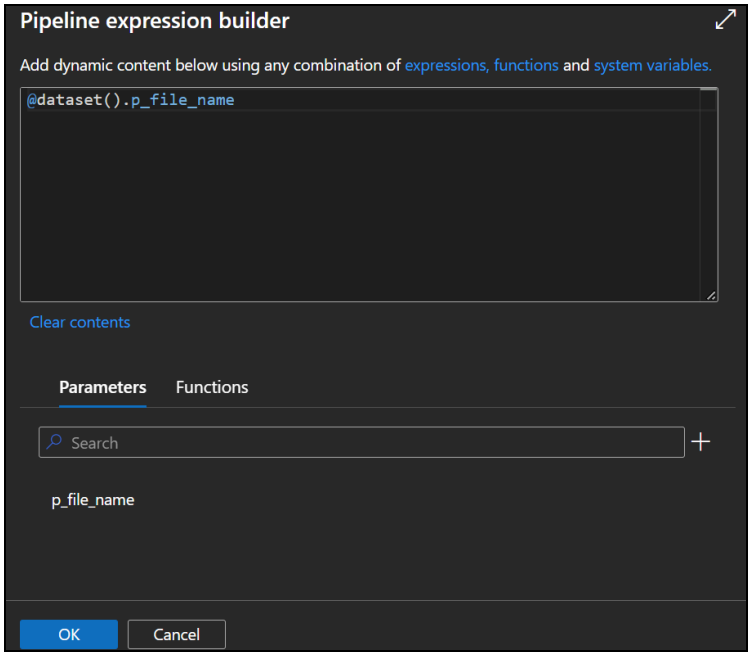


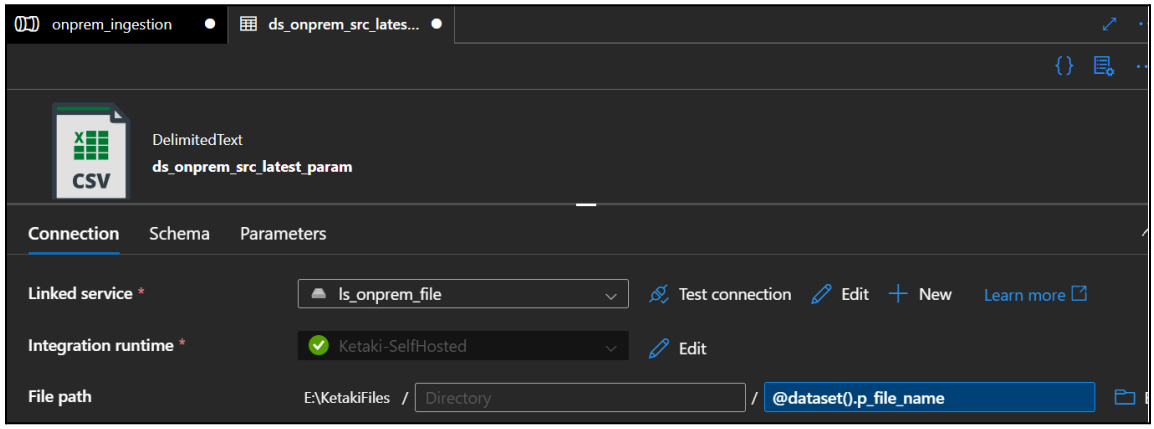


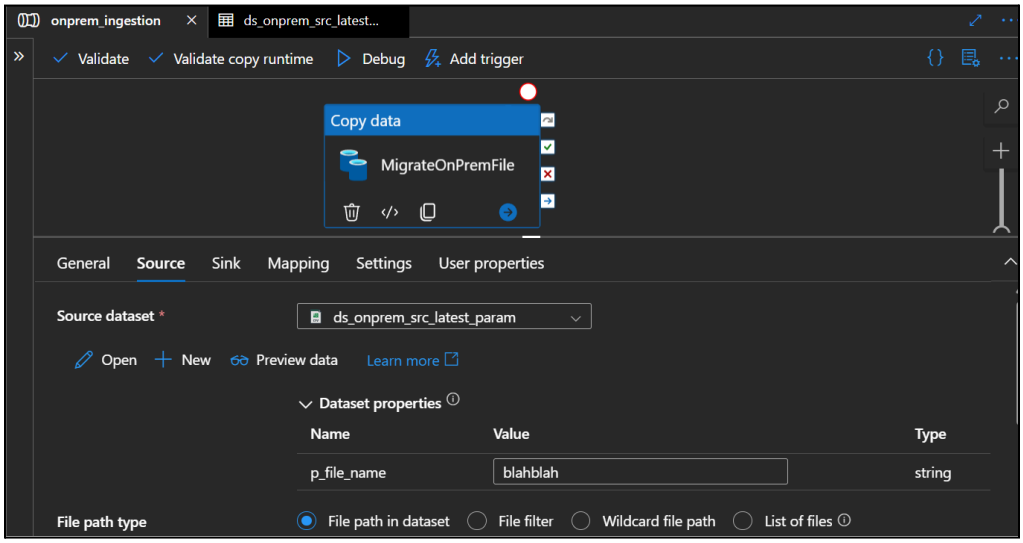


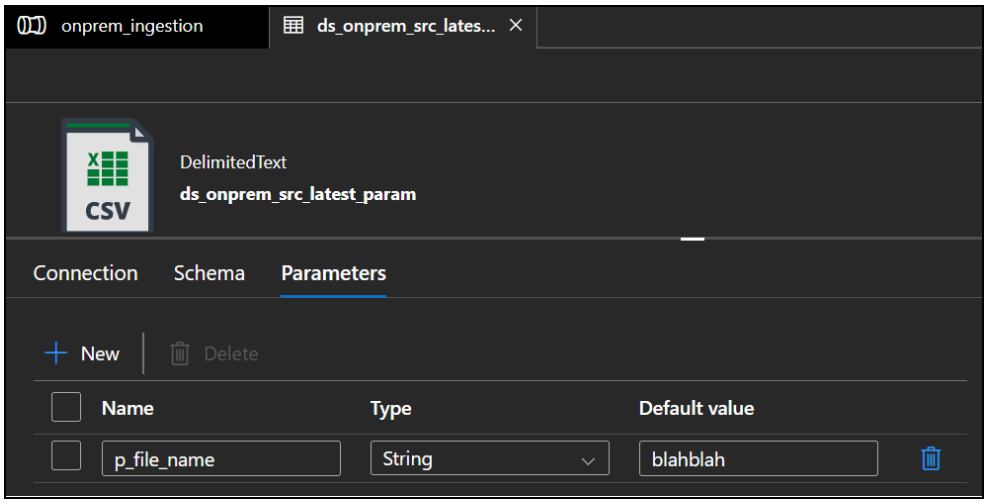
Click on add dynamic content :



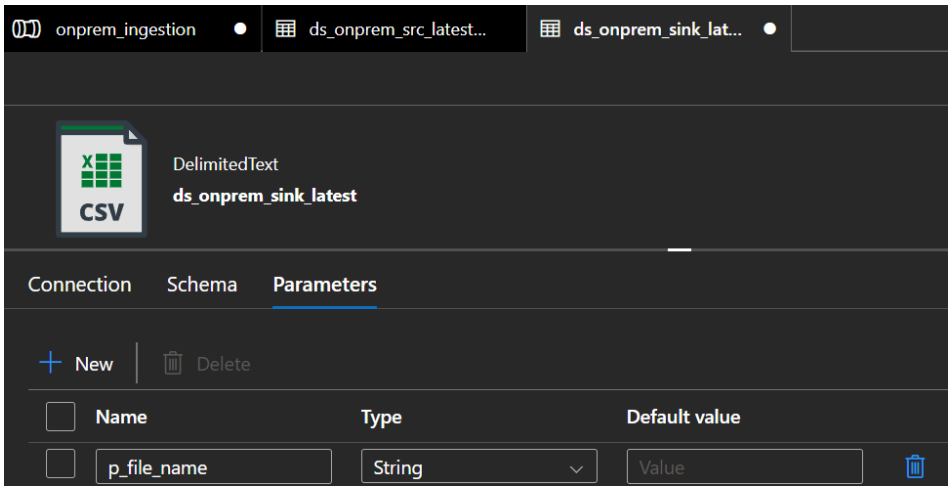

Pipeline expression builder
Add dynamic content below using any combination of expressions, functions and system variables.
@dataset().p_file_name
Clear contents
Parameters
Functions
Search
p_file_name
OK
Cancel

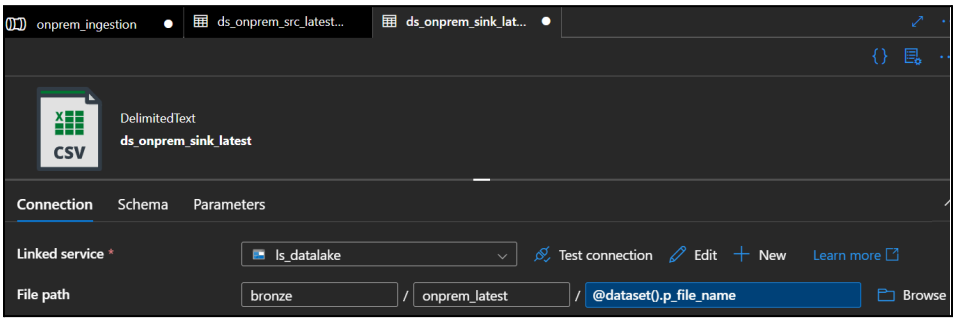

onprem_ingestion
ds_onprem_src_lates...
DelimitedText
ds_onprem_src_latest_param
Connection
Schema
Parameters
Linked service *
ls_onprem_file
Test connection
Edit
New
Learn more
Integration runtime *
Ketaki-SelfHosted
Edit
File path
E:\KetakiFiles
/
Directory
/
@dataset().p_file_name

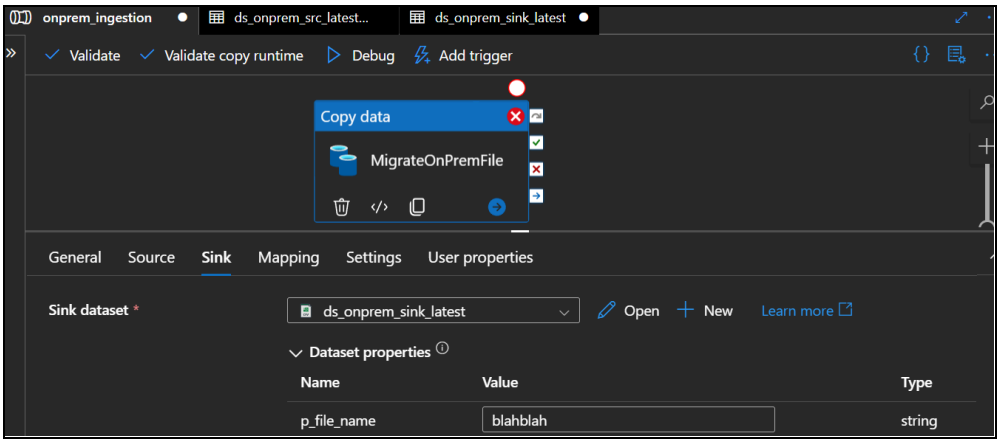

onprem_ingestion
ds_onprem_src_latest...
Validate
Validate copy runtime
Debug
Add trigger
Copy data
MigrateOnPremFile
General
Source
Sink
Mapping
Settings
User properties
Source dataset *
ds_onprem_src_latest_param
Open
New
Preview data
Learn more
Dataset properties
Name
Value
Type
p_file_name
blahblah
string
File path type
File path in dataset
File filter
Wildcard file path
List of files


onprem_ingestion
ds_onprem_src_lates...
DelimitedText
ds_onprem_src_latest_param
Connection
Schema
Parameters
New
Delete
Name
Type
Default value
p_file_name
String
blahblah

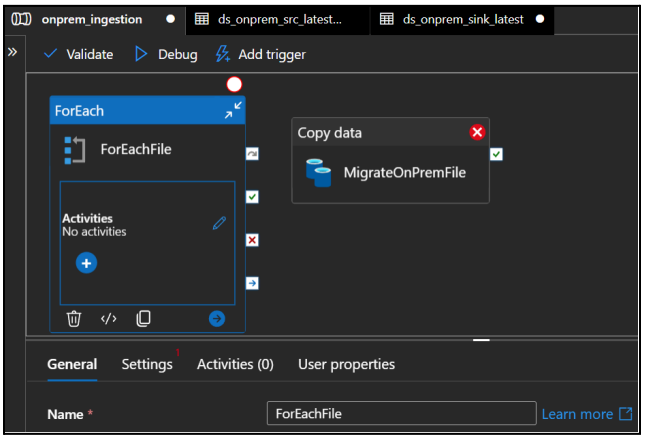
Onprem sink :


onprem_ingestion
ds_onprem_src_latest...
ds_onprem_sink_lat...
DelimitedText
ds_onprem_sink_latest
Connection
Schema
Parameters
New
Delete
Name
Type
Default value
p_file_name
String
Value


onprem_ingestion
ds_onprem_src_latest...
ds_onprem_sink_lat...
DelimitedText
ds_onprem_sink_latest
Connection
Schema
Parameters
Linked service *
ls_datalake
Test connection
Edit
New
Learn more
File path
bronze
onprem_latest
@dataset().p_file_name
Browse

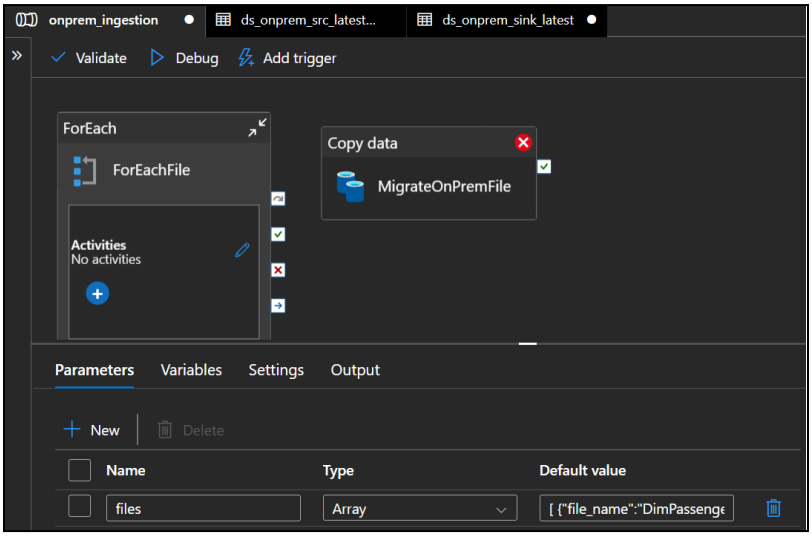


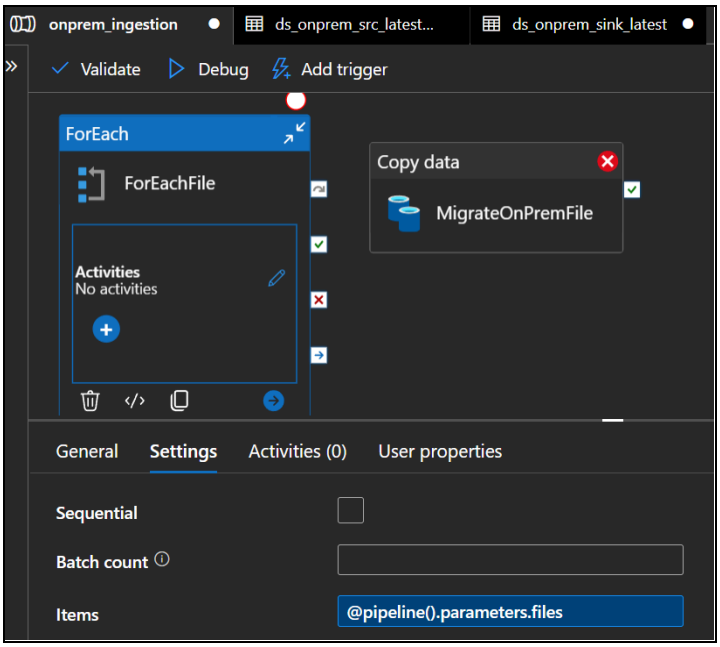
1:25:20 Dynamic ADF Pipelines with Dynamic Mapping



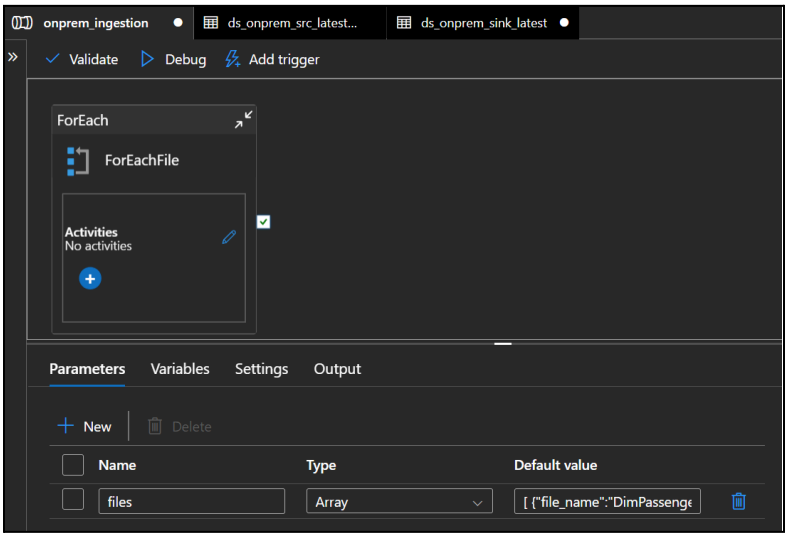
Paste this in Default value :

[{"file_name":"DimPassenger.csv"},
{"file_name":"DimAirline.csv"},
{"file_name":"DimFlight.csv"}]

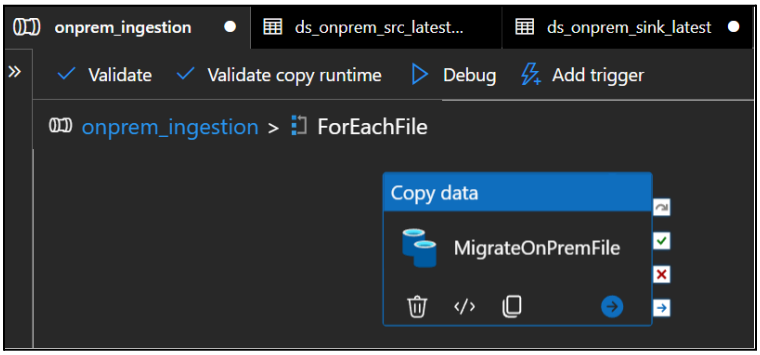




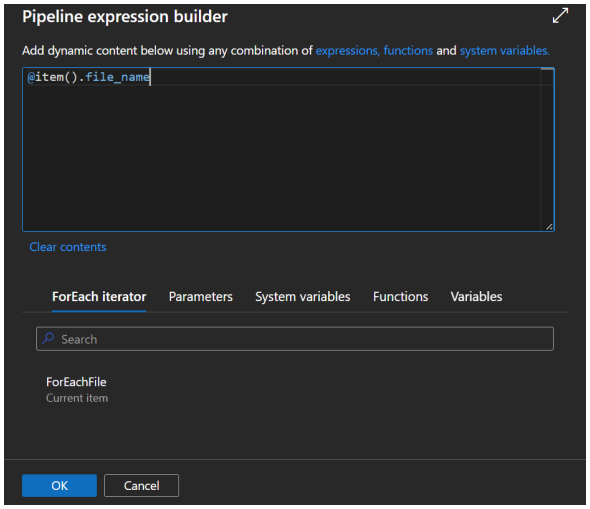
Click on copy data and control + x :

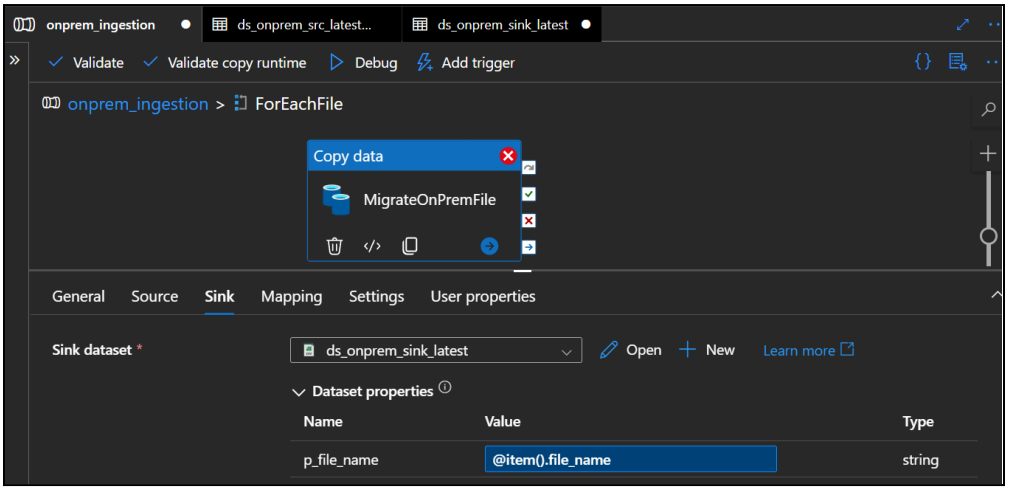


Click on edit (pencil button in foreach) and control+v:

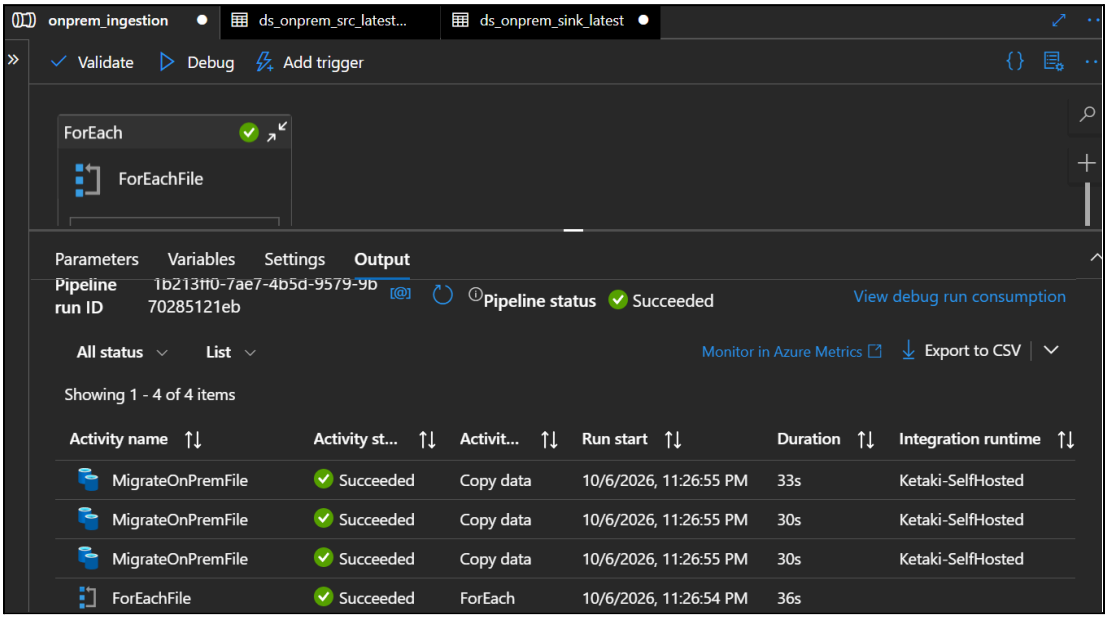


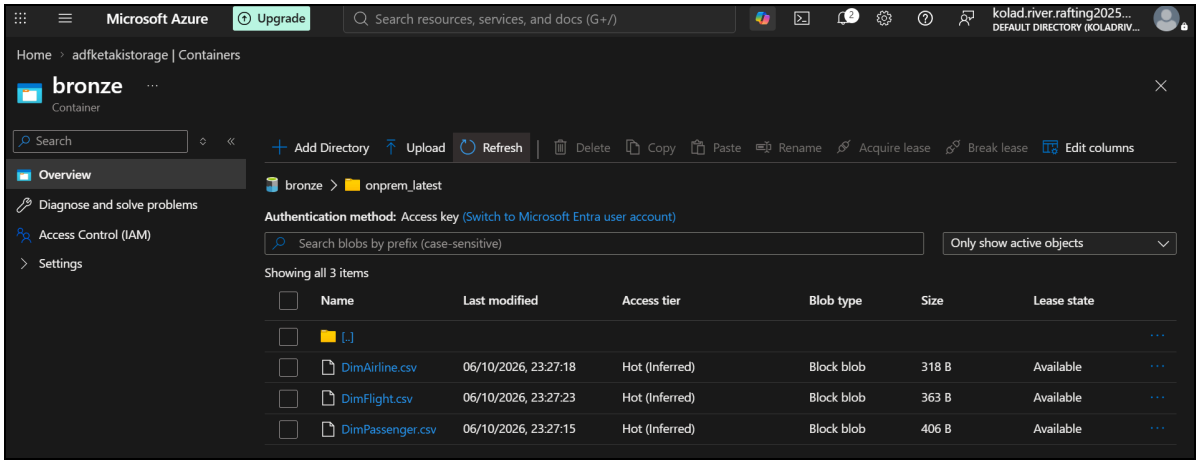
In both src and sink :



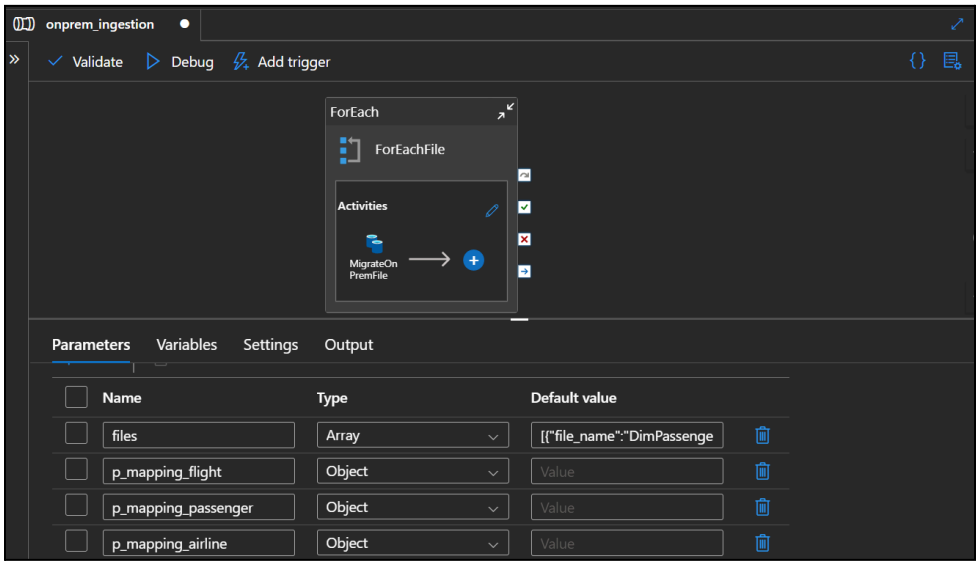


Remove mapping otherwise it fails

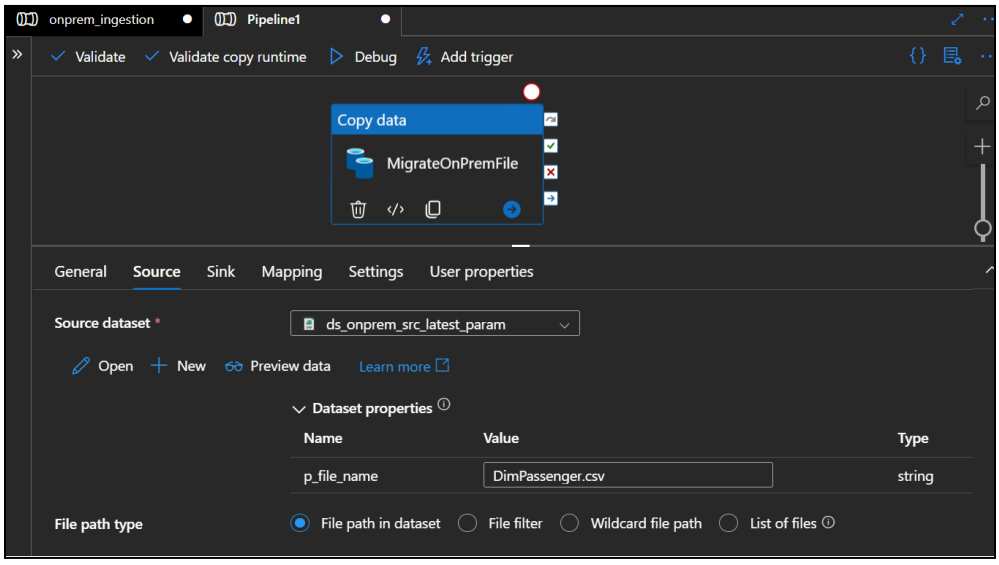


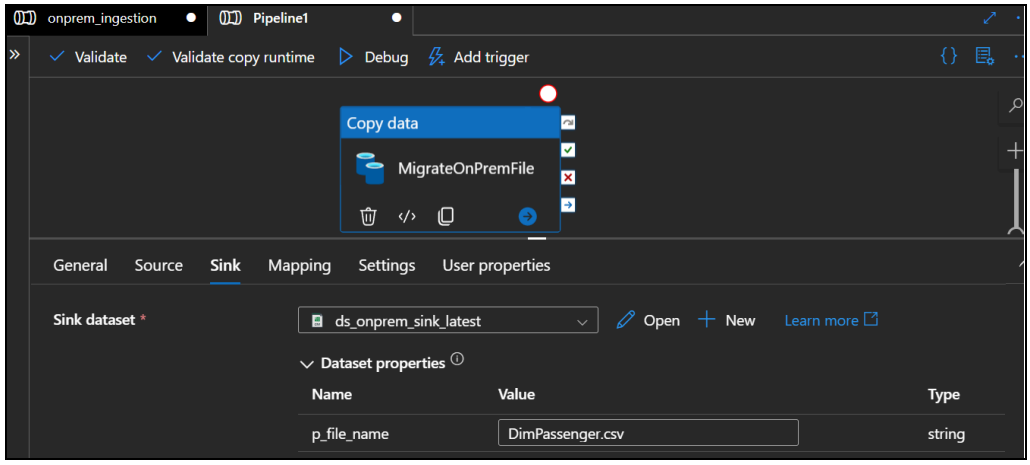


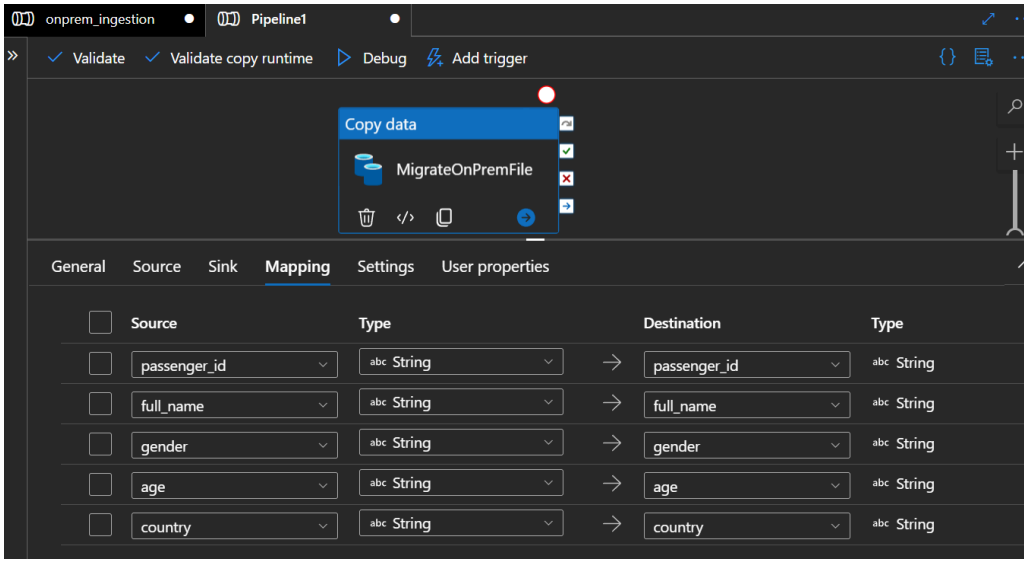
Object type means json :



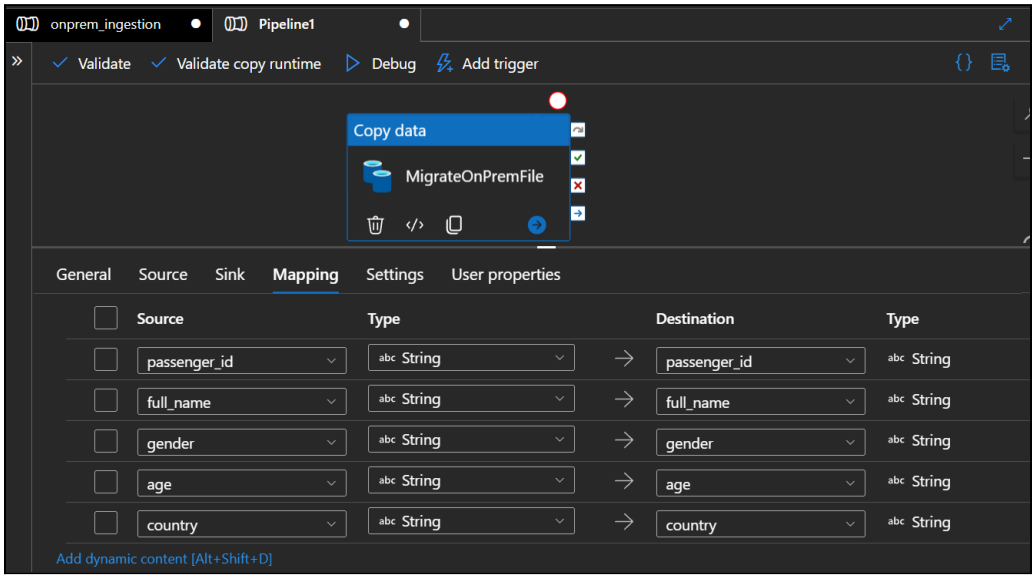
Create new Pipeline :

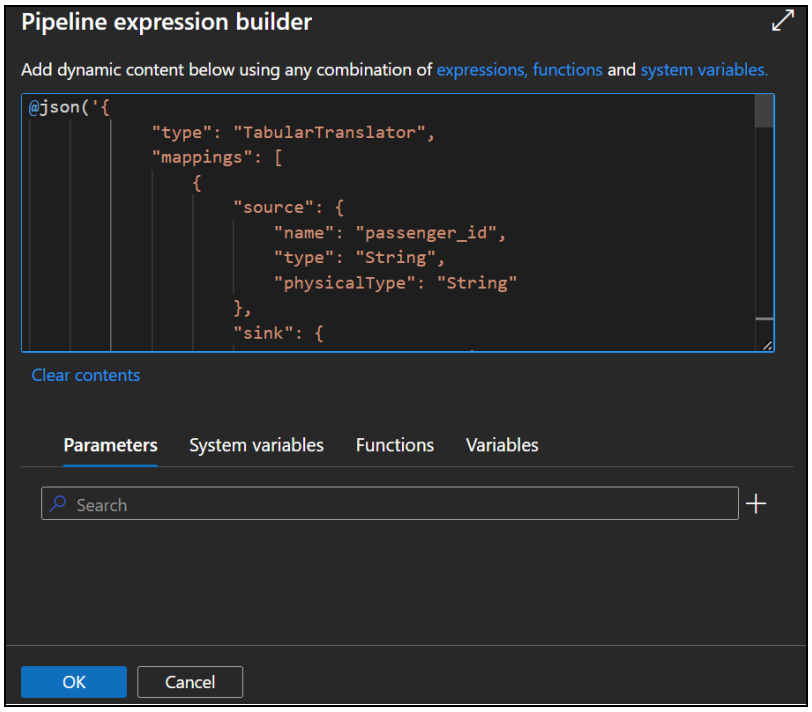


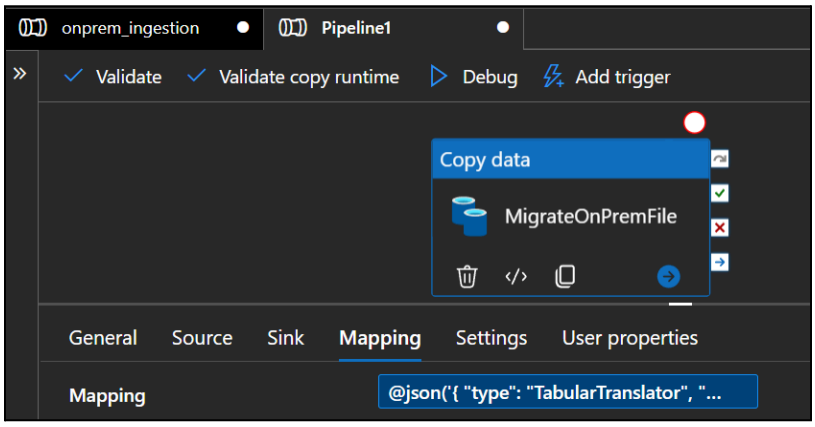


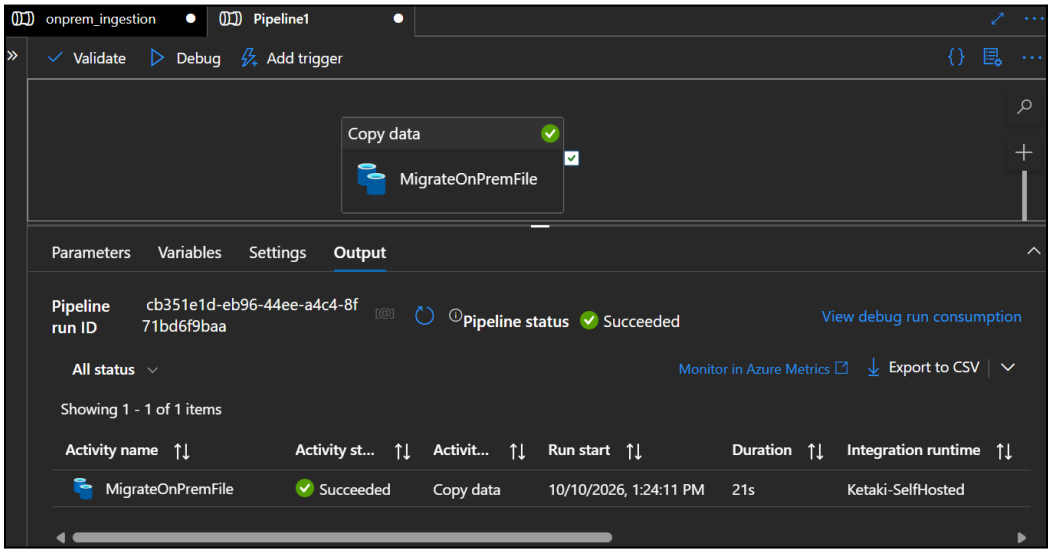


Click on code and copy from translator and then click on add dynamic content :

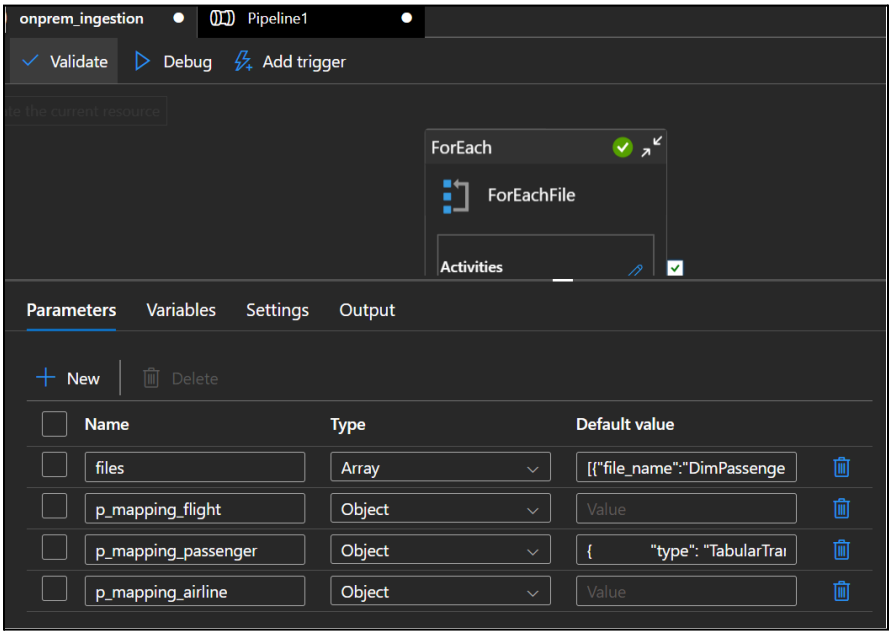



Pipeline expression builder
Add dynamic content below using any combination of expressions, functions and system variables.
@json('{
"type": "TabularTranslator",
"mappings": [
{
"source": {
"name": "passenger_id",
"type": "String",
"physicalType": "String"
},
"sink": {
Clear contents
Parameters
System variables
Functions
Variables
Search
OK
Cancel

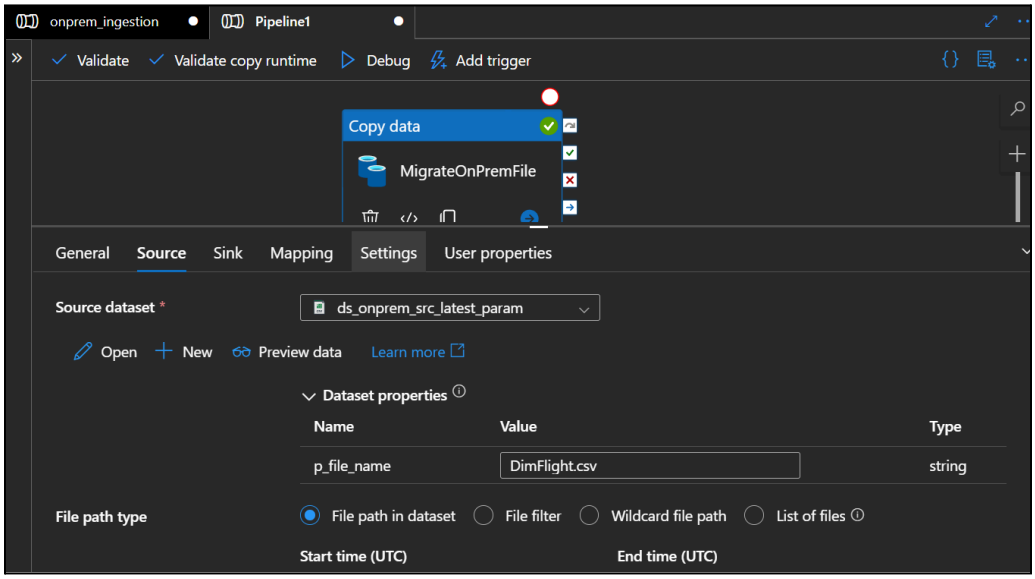

onprem_ingestion
Pipeline1
Validate
Validate copy runtime
Debug
Add trigger
Copy data
MigrateOnPremFile
General
Source
Sink
Mapping
Settings
User properties
Mapping
@json('{ "type": "TabularTranslator", "...

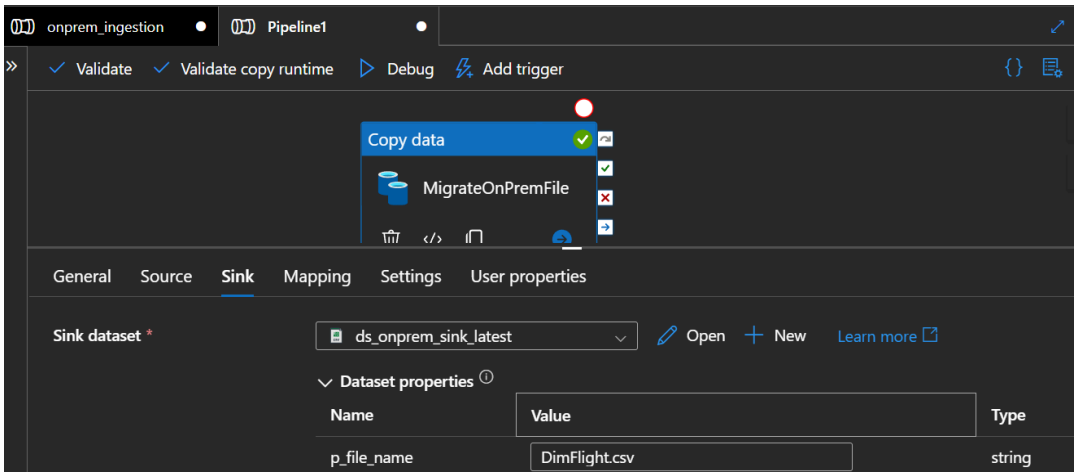

onprem_ingestion
Pipeline1
Validate
Debug
Add trigger
Copy data
MigrateOnPremFile
Parameters
Variables
Settings
Output
Pipeline run ID
cb351e1d-eb96-44ee-a4c4-8f71bd6f9baa
Pipeline status
Succeeded
View debug run consumption
All status
Monitor in Azure Metrics
Export to CSV
Showing 1 - 1 of 1 items
Activity name
Activity st...
Activit...
Run start
Duration
Integration runtime
MigrateOnPremFile
Succeeded
Copy data
10/10/2026, 1:24:11 PM
21s
Ketaki-SelfHosted

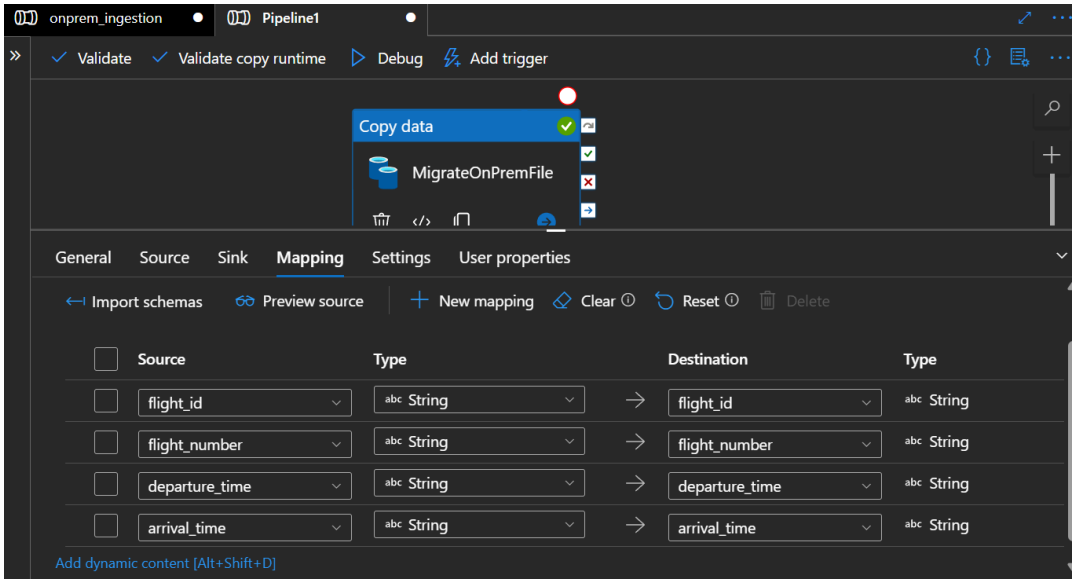
String : use json in expression builder
Object : no need to use json function

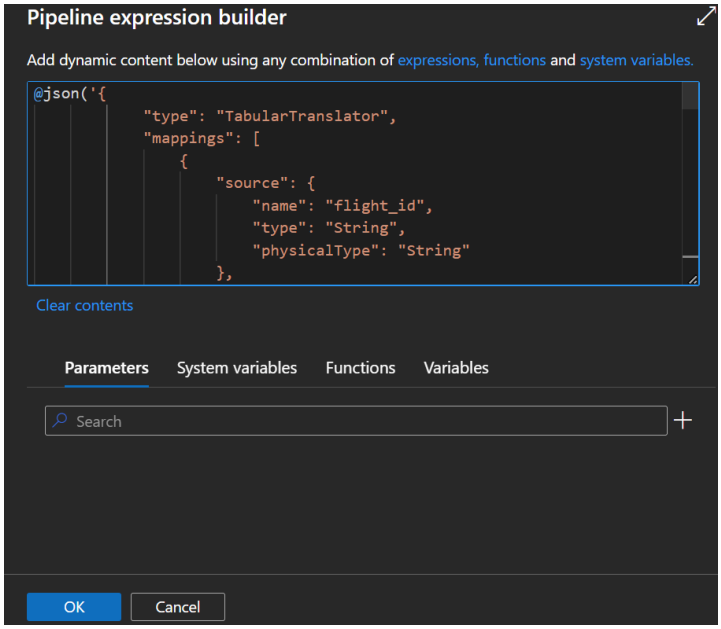


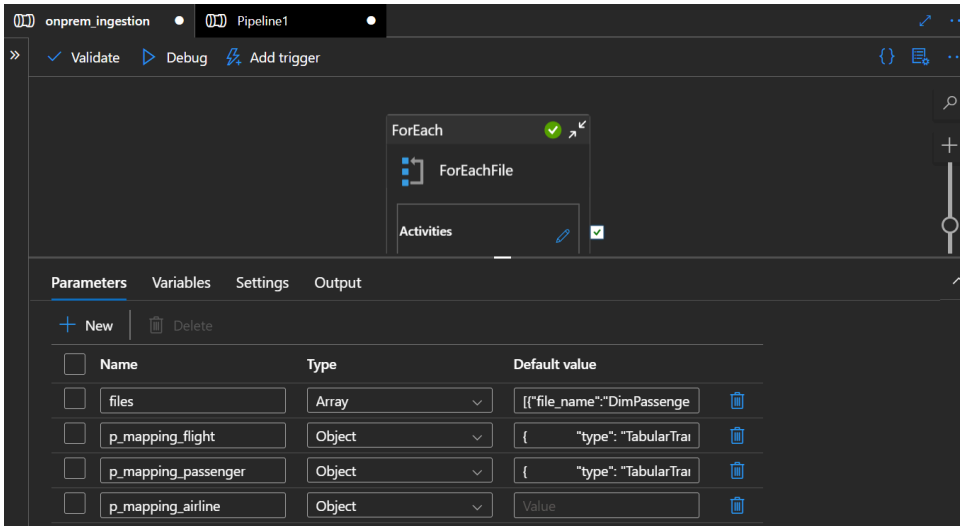
For other files search for code in the pipeline1:



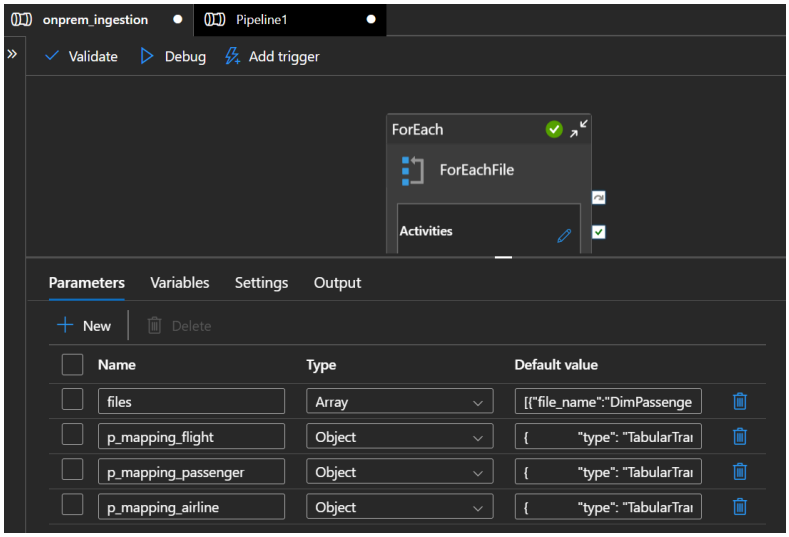

onprem_ingestion
Pipeline1
Validate
Validate copy runtime
Debug
Add trigger
Copy data
MigrateOnPremFile
General
Source
Sink
Mapping
Settings
User properties
Sink dataset *
ds_onprem_sink_latest
Open
New
Learn more
Dataset properties
Name
Value
Type
p_file_name
DimFlight.csv
string

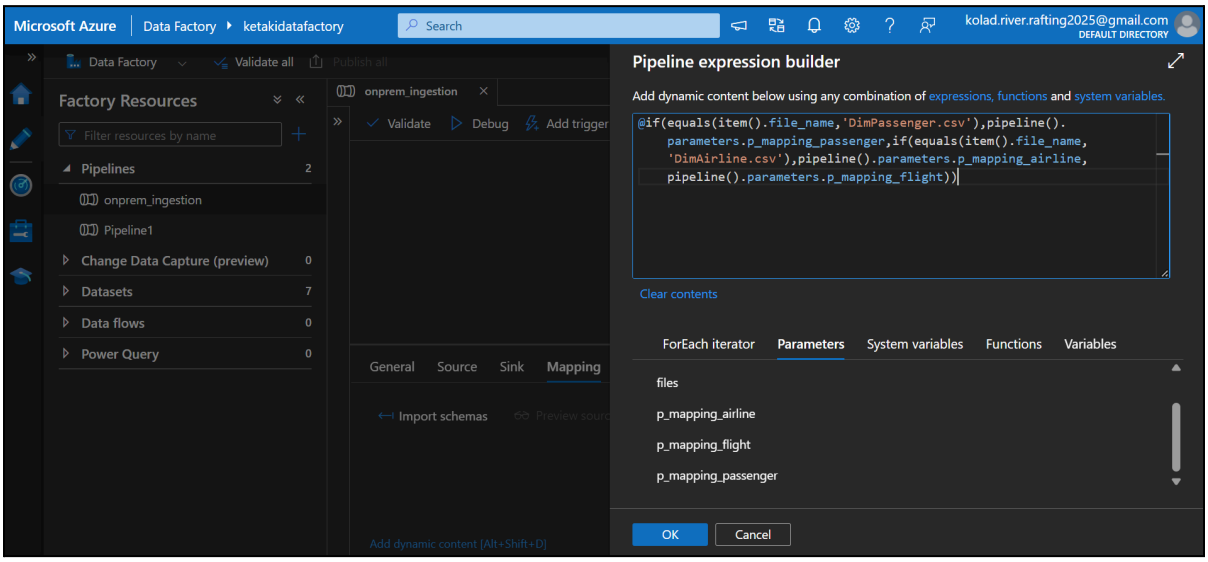

onprem_ingestion
Pipeline1
Validate
Validate copy runtime
Debug
Add trigger
Copy data
MigrateOnPremFile
General
Source
Sink
Mapping
Settings
User properties
Import schemas
Preview source
New mapping
Clear
Reset
Delete
Source
Type
Destination
Type
flight_id
String
flight_id
String
flight_number
String
flight_number
String
departure_time
String
departure_time
String
arrival_time
String
arrival_time
String
Add dynamic content [Alt+Shift+D]


Pipeline expression builder
Add dynamic content below using any combination of expressions, functions and system variables.
@json('{
"type": "TabularTranslator",
"mappings": [
{
"source": {
"name": "flight_id",
"type": "String",
"physicalType": "String"
},
Clear contents
Parameters
System variables
Functions
Variables
Search
OK
Cancel



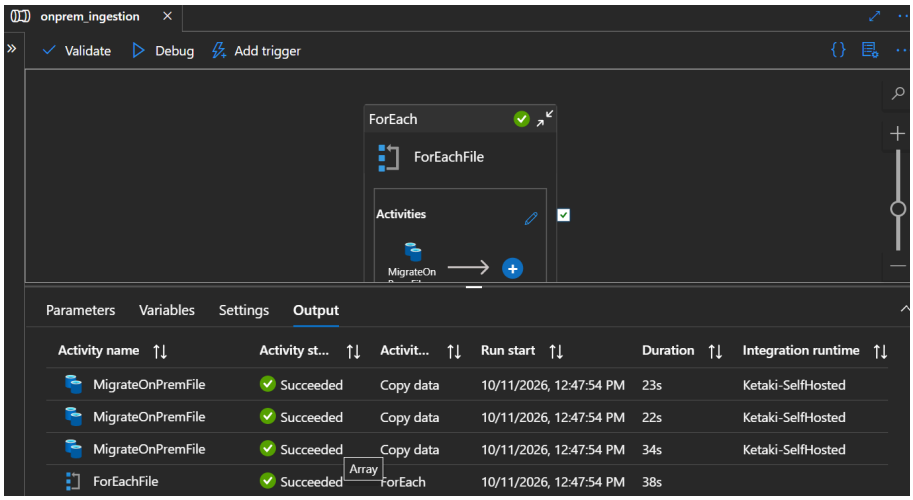
Similarly for all 3 files :

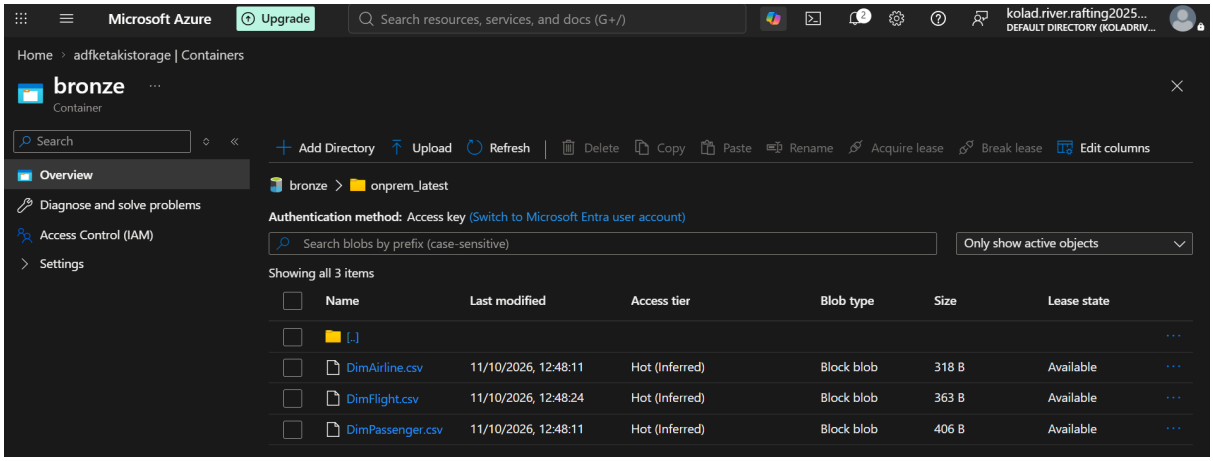




```
@if(equals(item().file_name,'DimPassenger.csv'),pipeline().parameters.p_mapping_passenger,if(equals(item().file_name,'DimAirline.csv'),pipeline().parameters.p_mapping_airline,pipeline().parameters.p_mapping_flight))
```

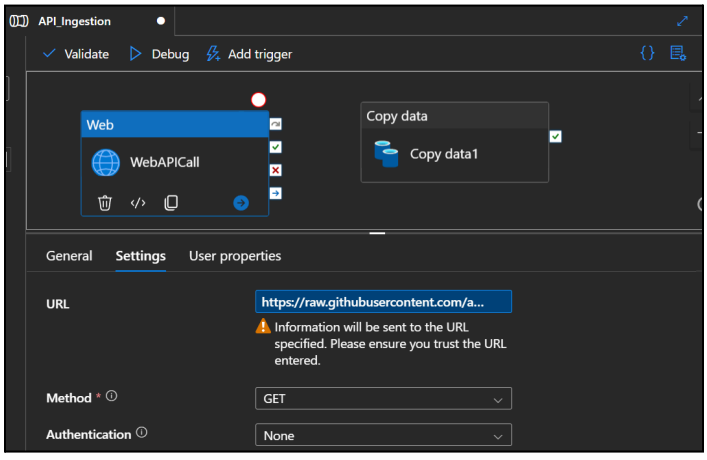
After running the pipeline :



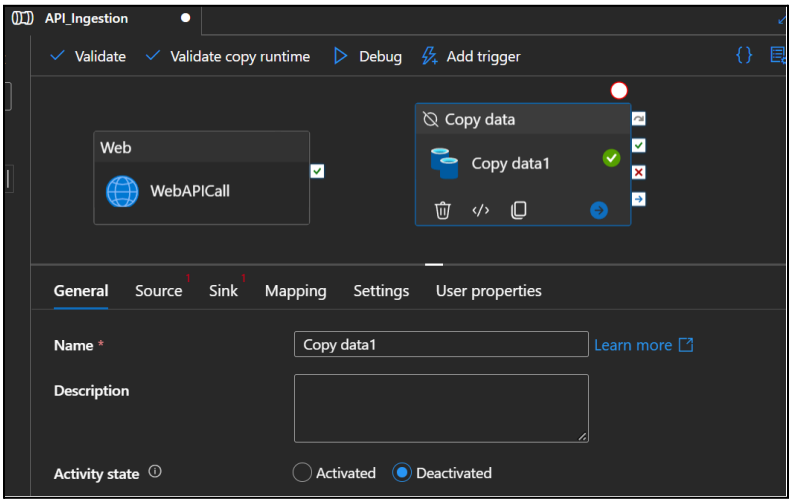


API To Azure Migration

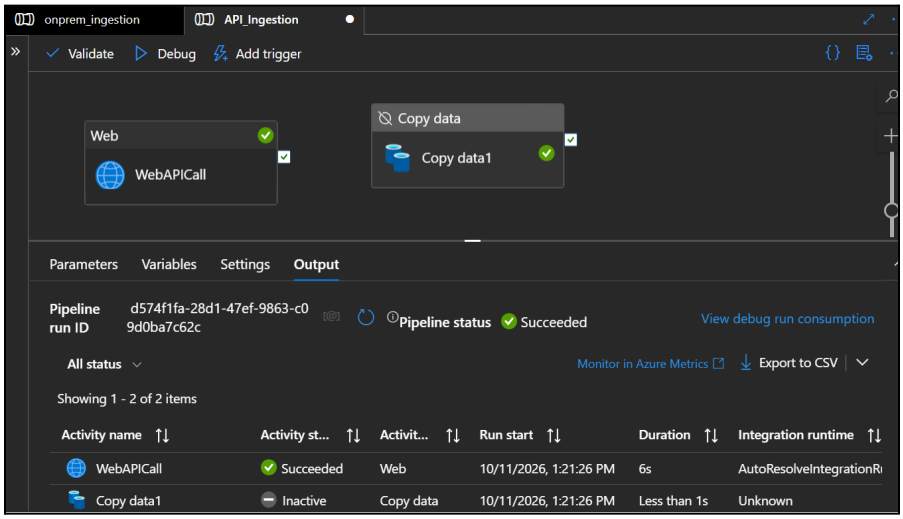
Create API_Ingestion pipeline , Copy the url of the raw .json file from github



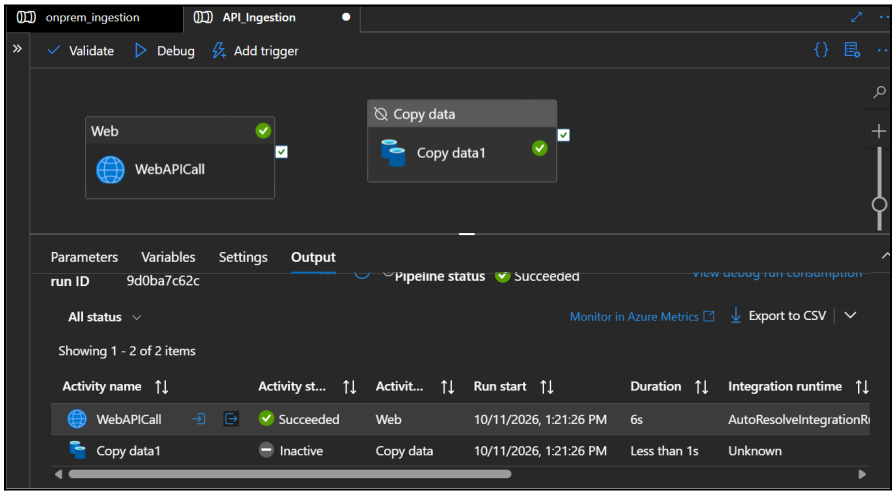
Click on deactivated in copy data



Debugging :



Click on that output symbol in webapicall :



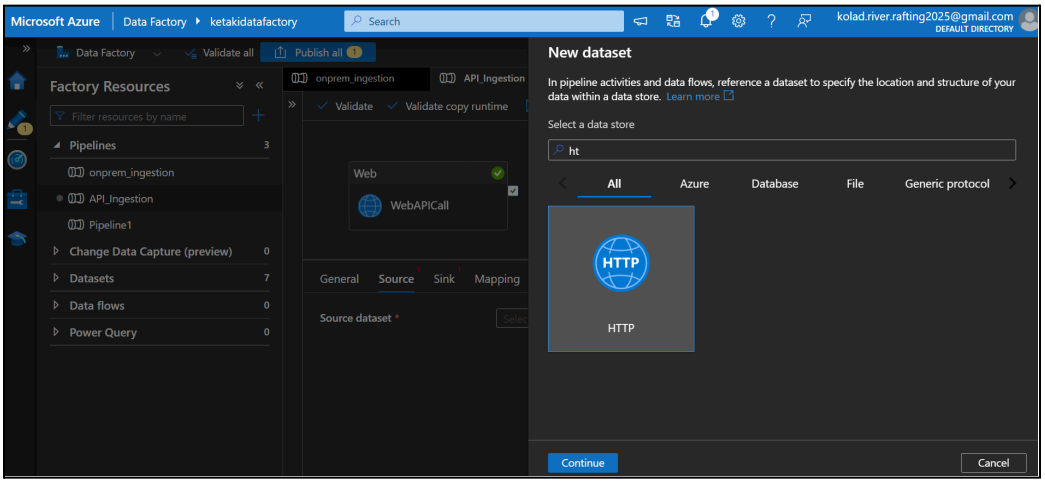
It looks like this :

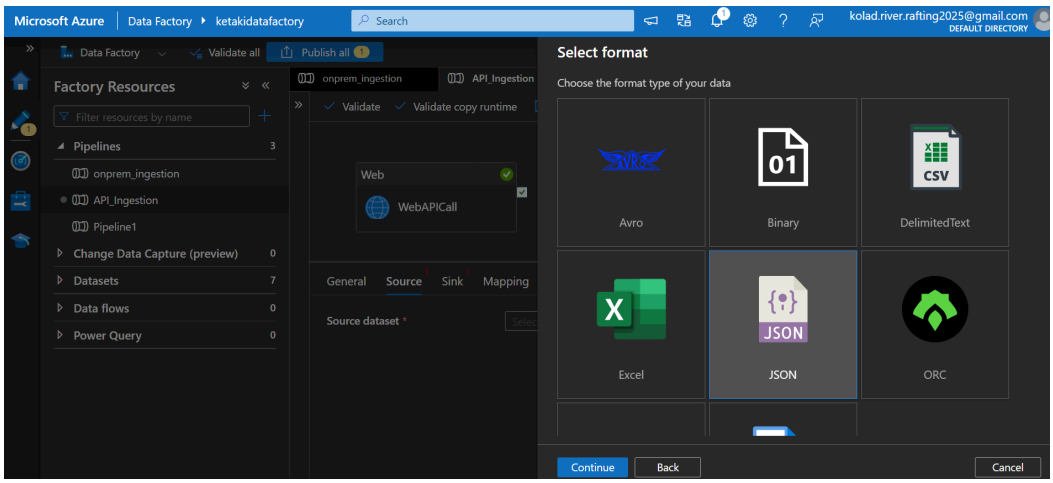
Output

Copy to clipboard

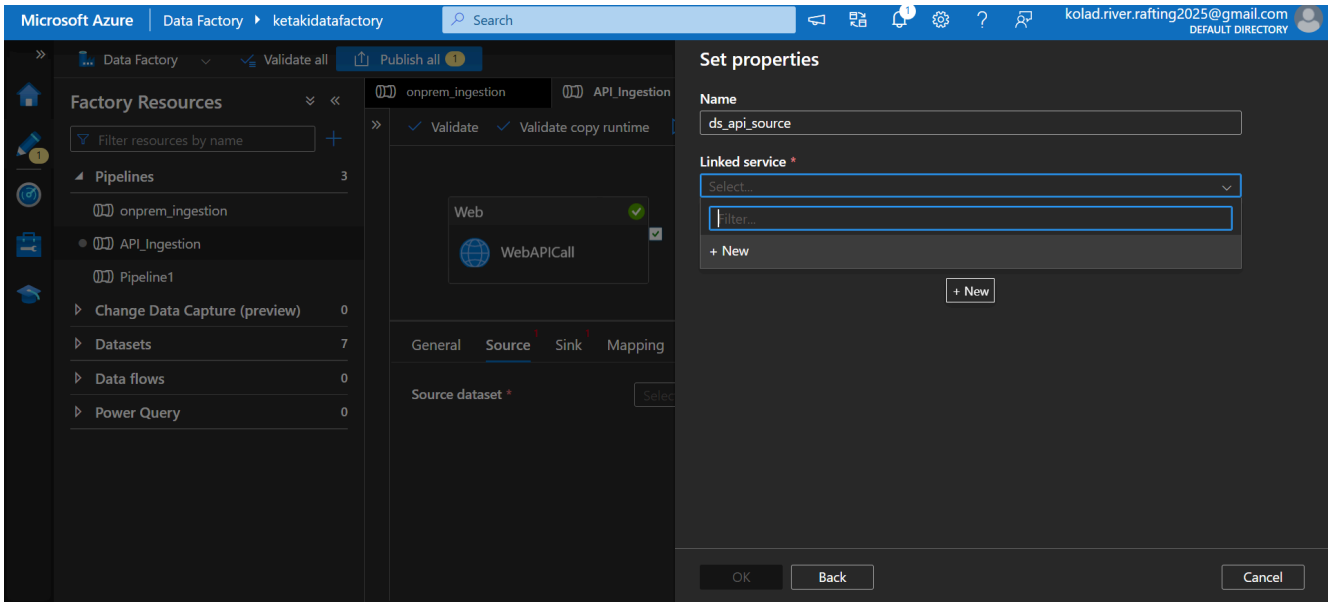

```
{
    "Response": "[\n  {\n    \"airport_id\": 1,\n    \"airport_name\": \"JFK International\",\n    \"city\": \"New York\",\n    \"country\": \"USA\"\n  },\n  {\n    \"airport_id\": 2,\n
\"airport_name\": \"Heathrow\",\n    \"city\": \"London\",\n    \"country\": \"UK\"\n  },\n  {\n    \"airport_id\": 3,\n    \"airport_name\": \"Chhatrapati Shivaji\",\n    \"city\": \"Mumbai\",\n
\"country\": \"India\"\n  },\n  {\n    \"airport_id\": 4,\n    \"airport_name\": \"Beijing Capital\",\n    \"city\": \"Beijing\",\n    \"country\": \"China\"\n  },\n  {\n    \"airport_id\": 5,\n
\"airport_name\": \"Pearson Intl\",\n    \"city\": \"Toronto\",\n    \"country\": \"Canada\"\n  },\n  {\n    \"airport_id\": 6,\n    \"airport_name\": \"Charles de Gaulle\",\n    \"city\": \"Paris\",\n
\"country\": \"France\"\n  },\n  {\n    \"airport_id\": 7,\n    \"airport_name\": \"Haneda\",\n    \"city\": \"Tokyo\",\n    \"country\": \"Japan\"\n  },\n  {\n    \"airport_id\": 8,\n
\"airport_name\": \"Dubai Intl\",\n    \"city\": \"Dubai\",\n    \"country\": \"UAE\"\n  },\n  {\n    \"airport_id\": 9,\n    \"airport_name\": \"Frankfurt\",\n    \"city\": \"Frankfurt\",\n    \"country\":
\"Germany\"\n  },\n  {\n    \"airport_id\": 10,\n    \"airport_name\": \"Sydney Airport\",\n    \"city\": \"Sydney\",\n    \"country\": \"Australia\"\n  }\n]",
    "ADFHttpStatusCodeInResponse": "200",
    "ADFWebActivityResponseHeaders": {
        "Connection": "keep-alive",
        "Content-Security-Policy": "default-src 'none'; style-src 'unsafe-inline'; sandbox",
        "Strict-Transport-Security": "max-age=31536000",
        "X-Content-Type-Options": "nosniff",
        "X-Frame-Options": "deny",
        "X-XSS-Protection": "1; mode=block",
        "X-GitHub-Request-Id": "F526:CB645:F5F85:17EA44:6ACB4000",
        "x-github-edge-region": "iad",
        "X-Served-By": "cache-iad-kcgs7200122-IAD",
        "X-Cache": "MISS",
        "X-Cache-Hits": "0",
        "X-Timer": "S1791705088.477160,VS0,VE50",
        "Vary": "Authorization;Accept-Encoding",
        "Access-Control-Allow-Origin": "*",
        "Cross-Origin-Resource-Policy": "cross-origin",
```


Click on copy data and do the following for source :

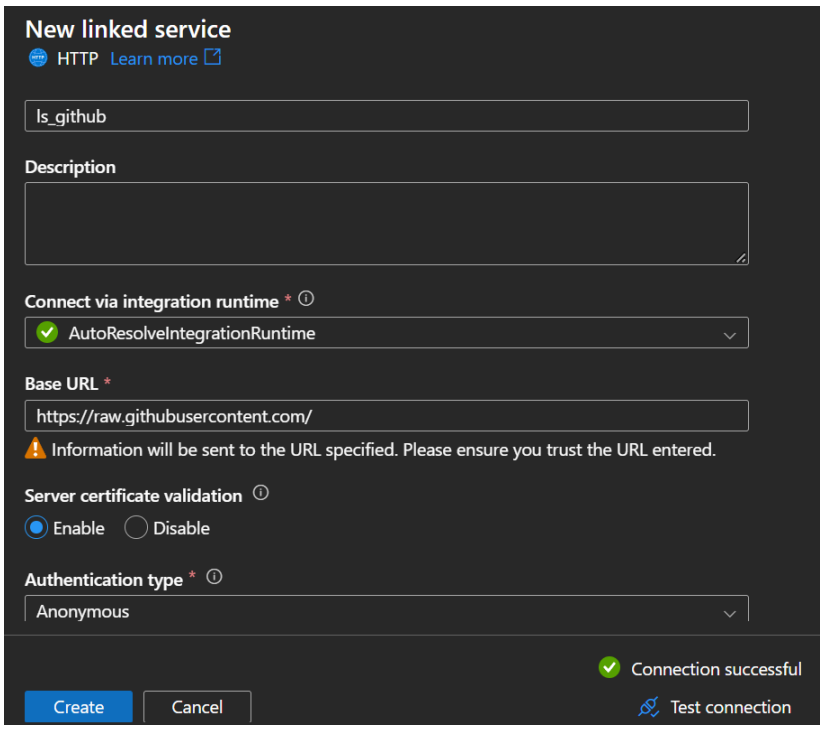


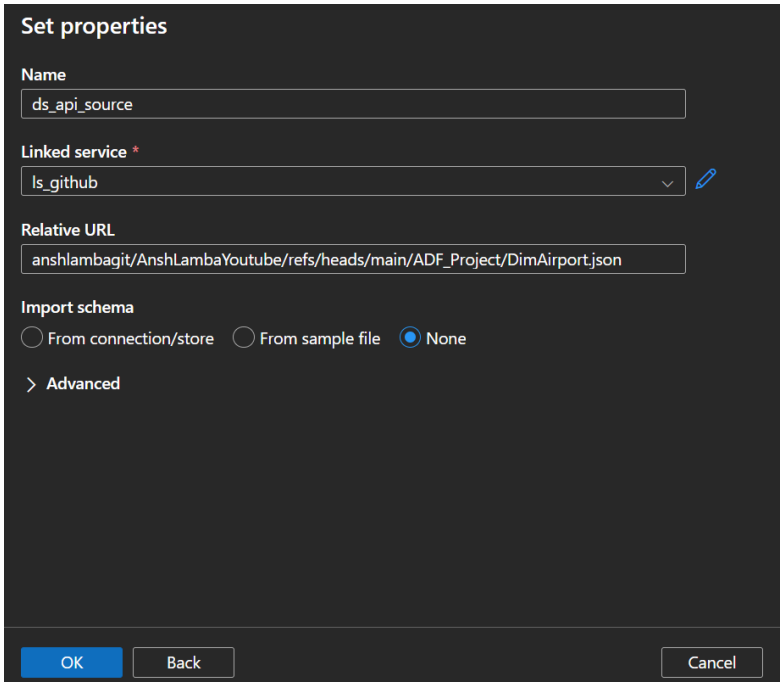


Creating a link service :

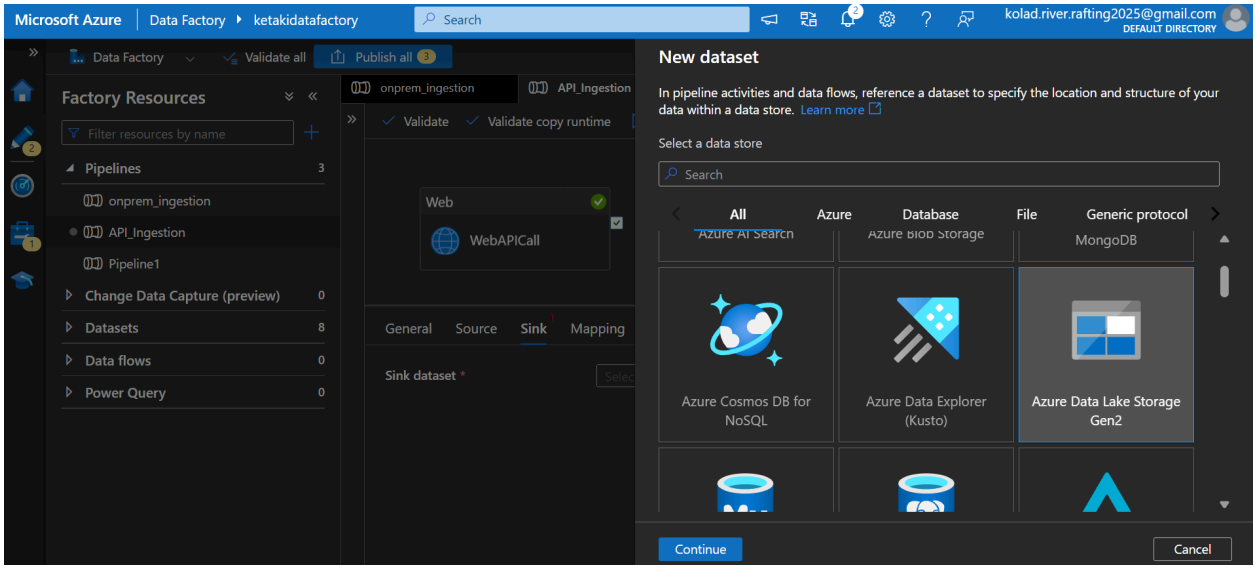


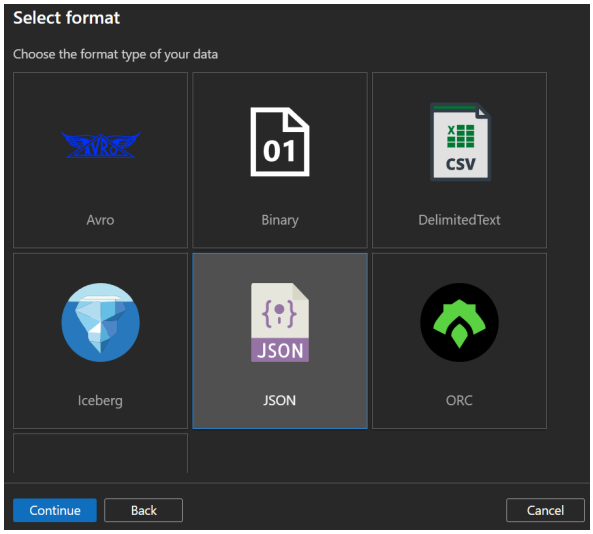
https://raw.githubusercontent.com/anshlambagit/AnshLambaYoutube/refs/heads/main/ADF_Project/DimAirport.json

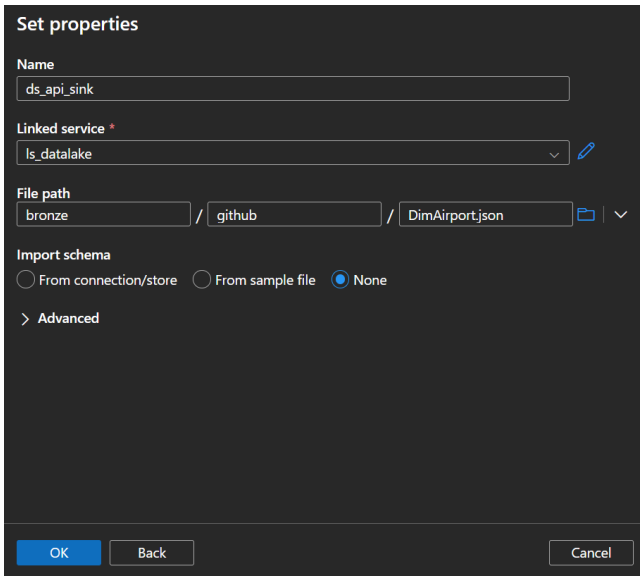


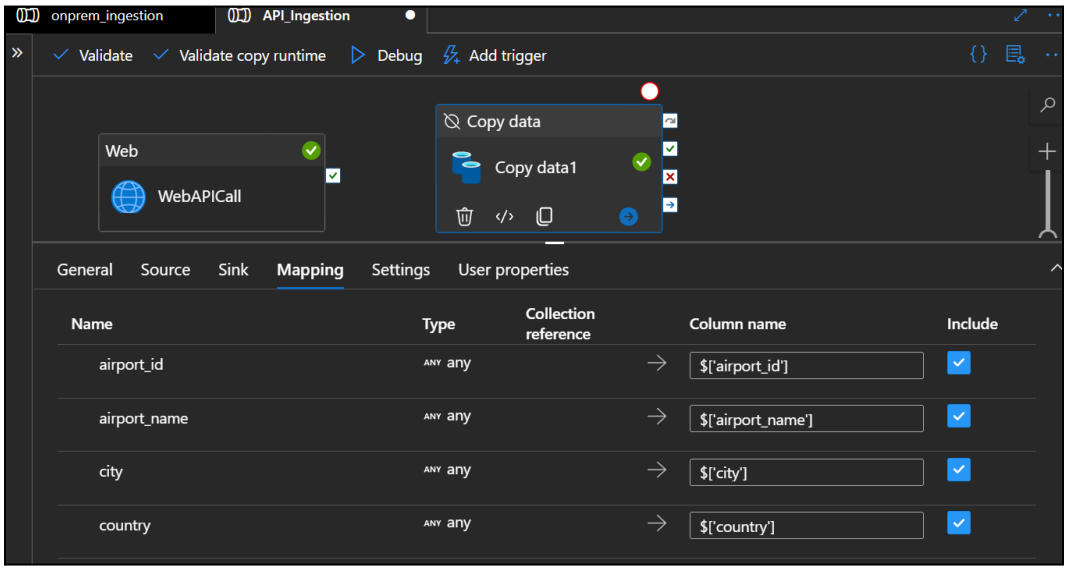

Set properties
Name
ds_api_source
Linked service *
ls_github
Relative URL
anshlambagit/AnshLambaYoutube/refs/heads/main/ADF_Project/DimAirport.json
Import schema
From connection/store
From sample file
None
Advanced
OK
Back
Cancel

For sink :

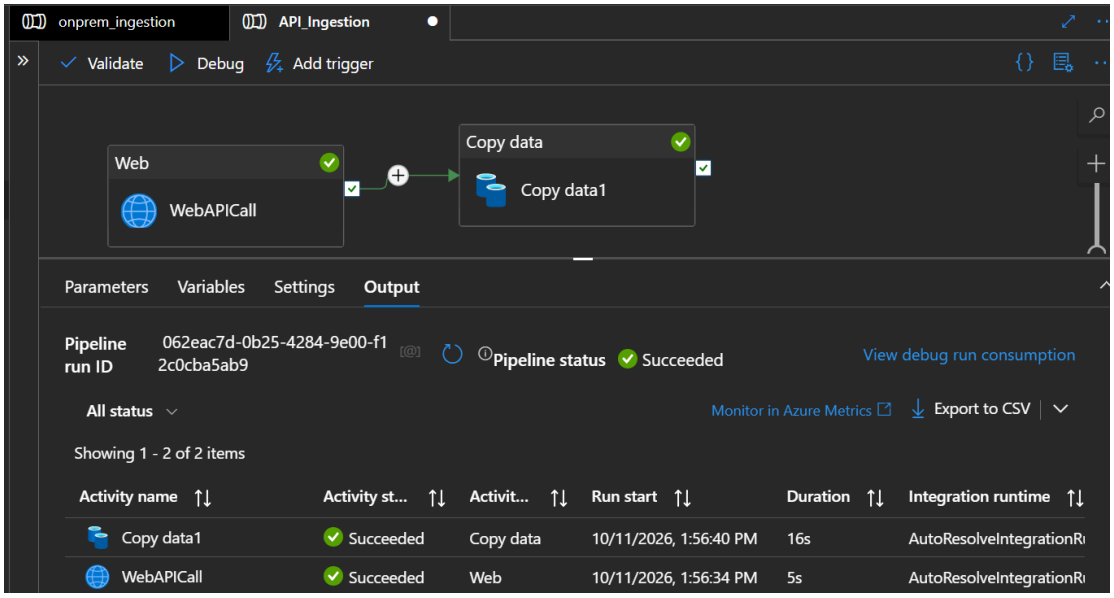

Microsoft Azure
Data Factory
ketakidatafactory
Search
kolad.river.rafting2025@gmail.com
DEFAULT DIRECTORY
Factory Resources
Pipelines
onprem_ingestion
API_Ingestion
Pipeline1
Change Data Capture (preview)
Datasets
Data flows
Power Query
Validate
Validate copy runtime
Web
WebAPICall
General
Source
Sink
Mapping
Sink dataset *
New dataset
In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. Learn more
Select a data store
Search
All
Azure
Database
File
Generic protocol
MongoDB
Azure Cosmos DB for NoSQL
Azure Data Explorer (Kusto)
Azure Data Lake Storage Gen2
Continue
Cancel


Select format
Choose the format type of your data
Avro
Binary
DelimitedText
Iceberg
JSON
ORC
Continue
Back
Cancel

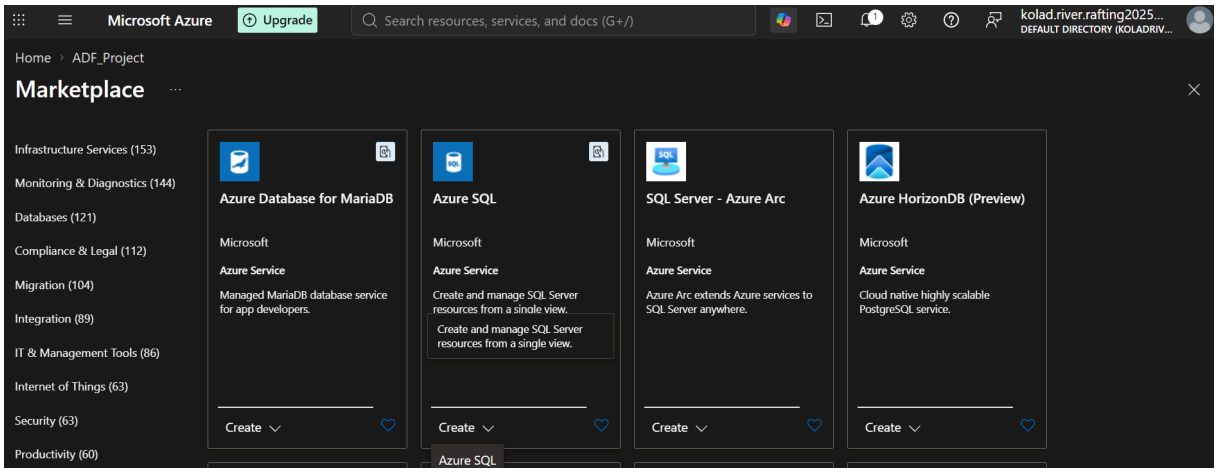

Set properties
Name
ds_api_sink
Linked service *
ls_datalake
File path
bronze
github
DimAirport.json
Import schema
From connection/store
From sample file
None
Advanced
OK
Back
Cancel


onprem_ingestion
API_Ingestion
Validate
Validate copy runtime
Debug
Add trigger
Web
WebAPICall
Copy data
Copy data1
General
Source
Sink
Mapping
Settings
User properties
Name
Type
Collection reference
Column name
Include
airport_id
any
$['airport_id']
airport_name
any
$['airport_name']
city
any
$['city']
country
any
$['country']

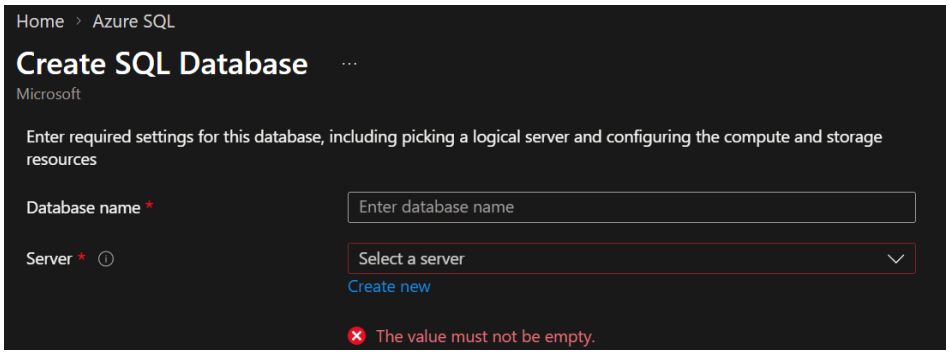
Validate and debug :

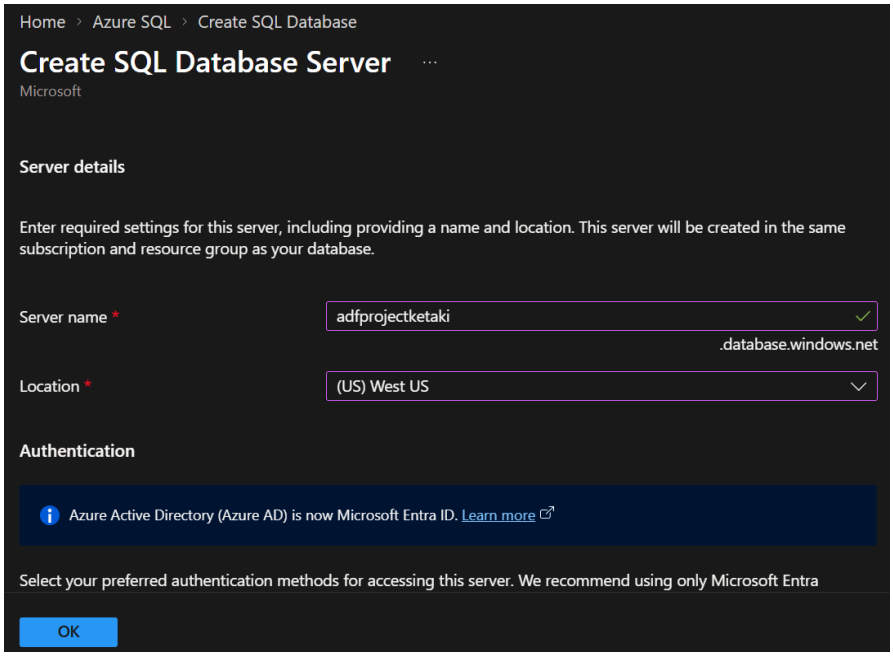


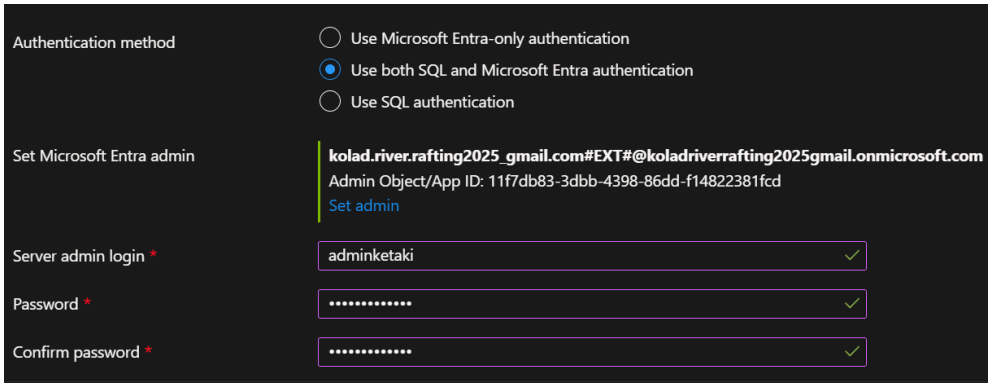
2:12:21 Set up Azure SQL DB



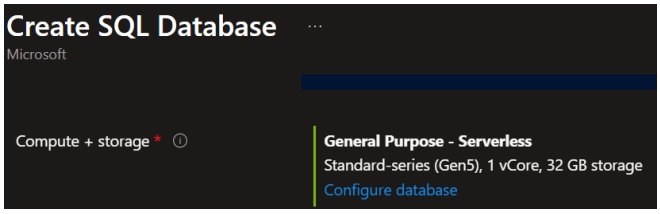
Create new server :

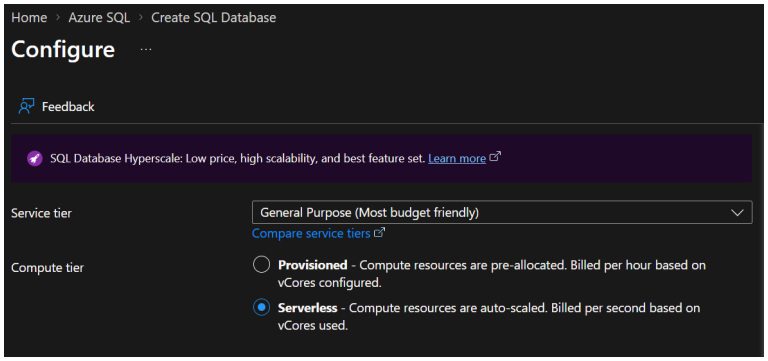




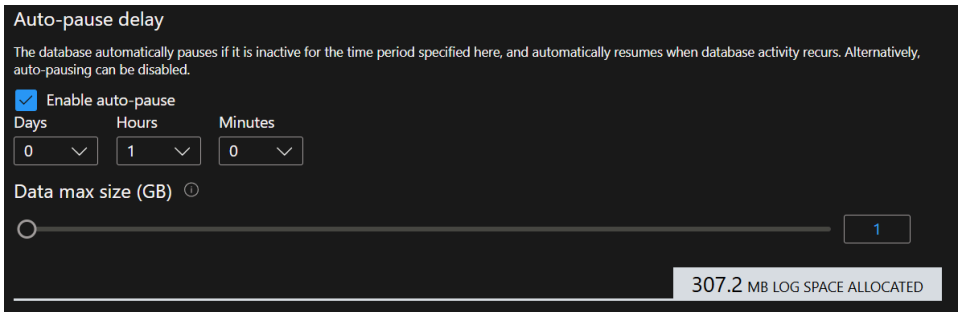


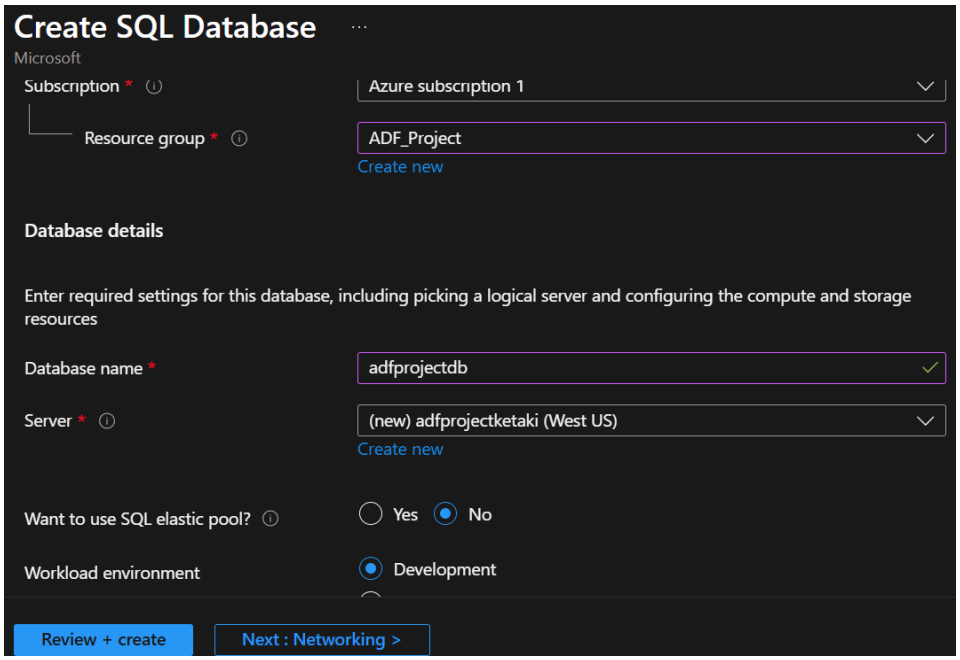
Configure db :

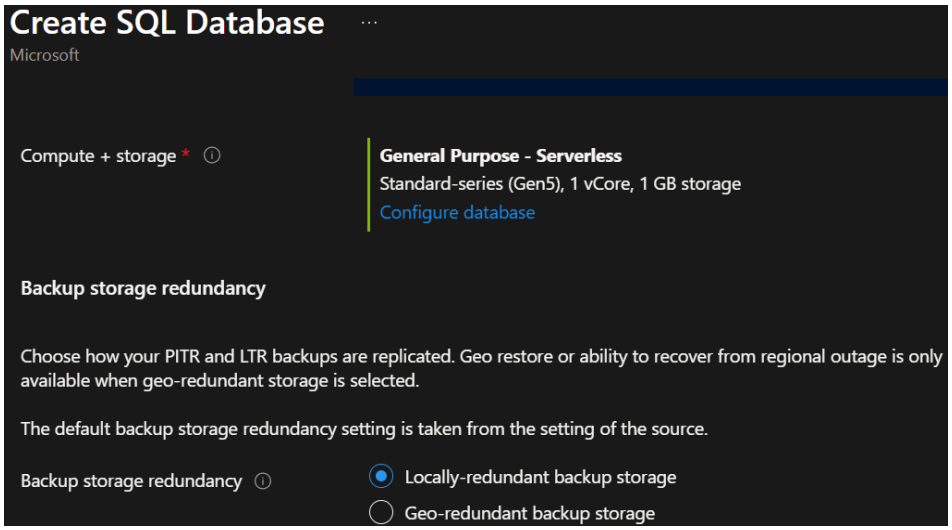


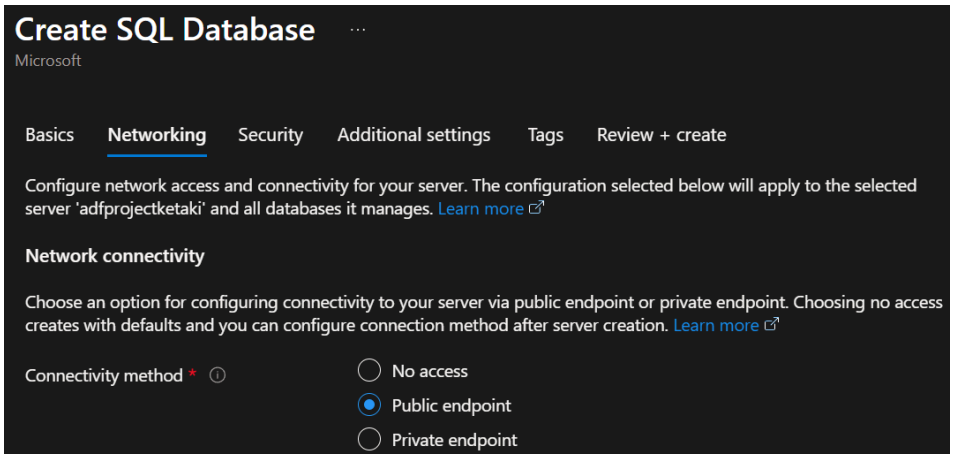


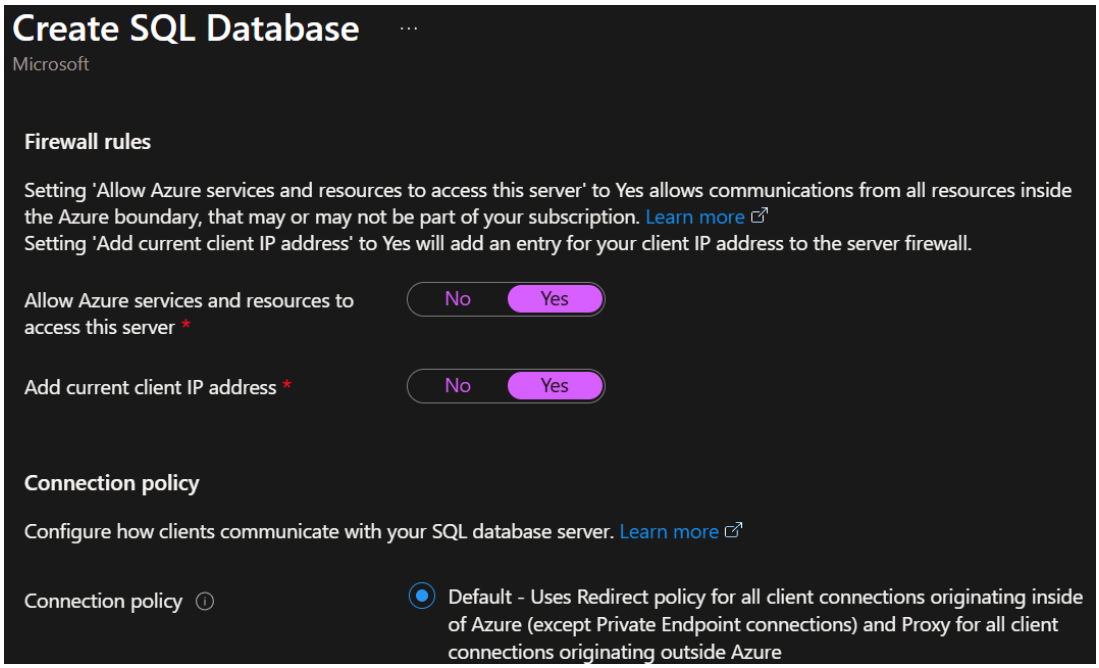
Home › Azure SQL › Create SQL Database
Configure
Feedback
Max vCores
1
Min vCores
0.5 vCores

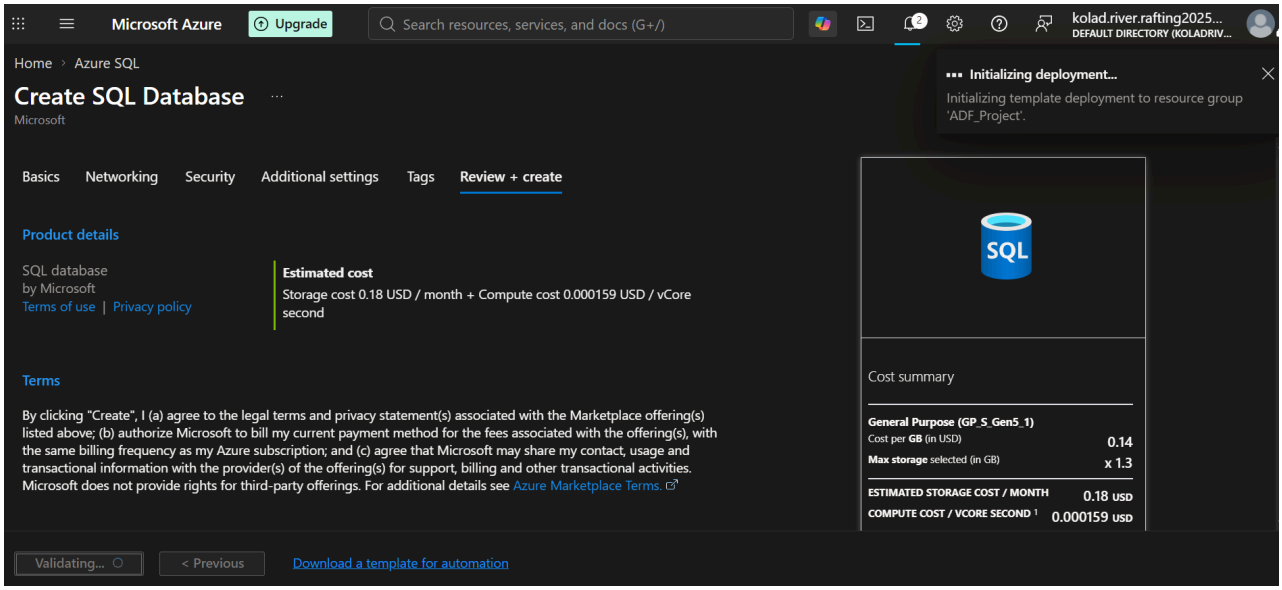

Auto-pause delay
The database automatically pauses if it is inactive for the time period specified here, and automatically resumes when database activity recurs. Alternatively, auto-pausing can be disabled.
Enable auto-pause
Days
Hours
Minutes
0
1
0
Data max size (GB)
1
307.2 MB LOG SPACE ALLOCATED

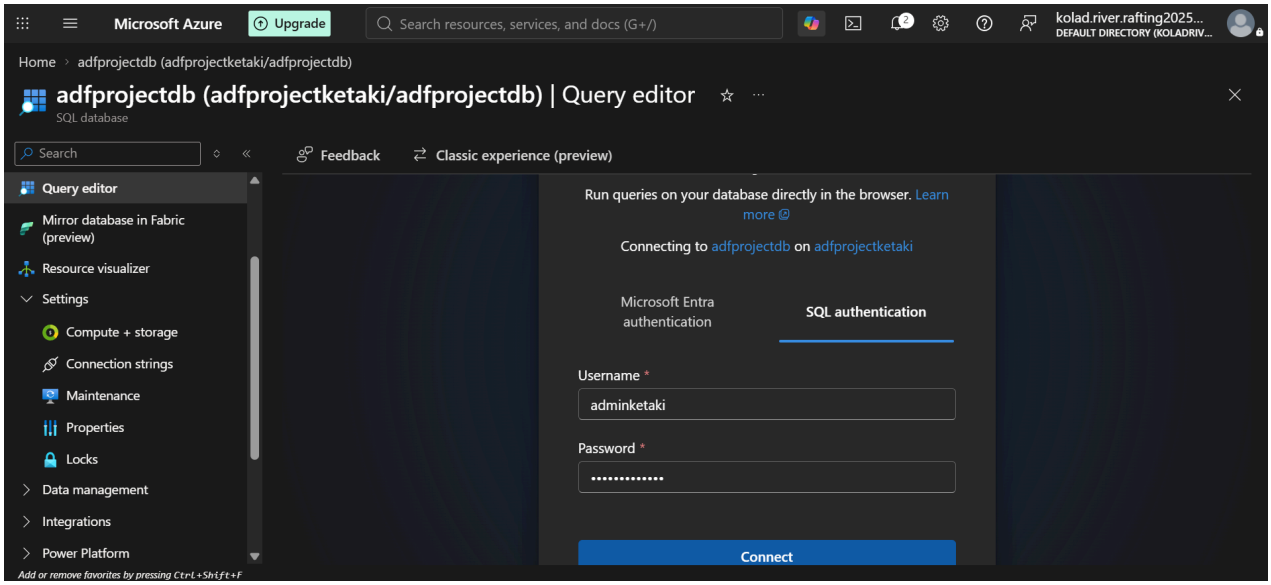

Create SQL Database
Microsoft
Subscription *
Azure subscription 1
Resource group *
ADF_Project
Create new
Database details
Enter required settings for this database, including picking a logical server and configuring the compute and storage resources
Database name *
adfprojectdb
Server *
(new) adfprojectketaki (West US)
Create new
Want to use SQL elastic pool?
Yes
No
Workload environment
Development
Review + create
Next : Networking >

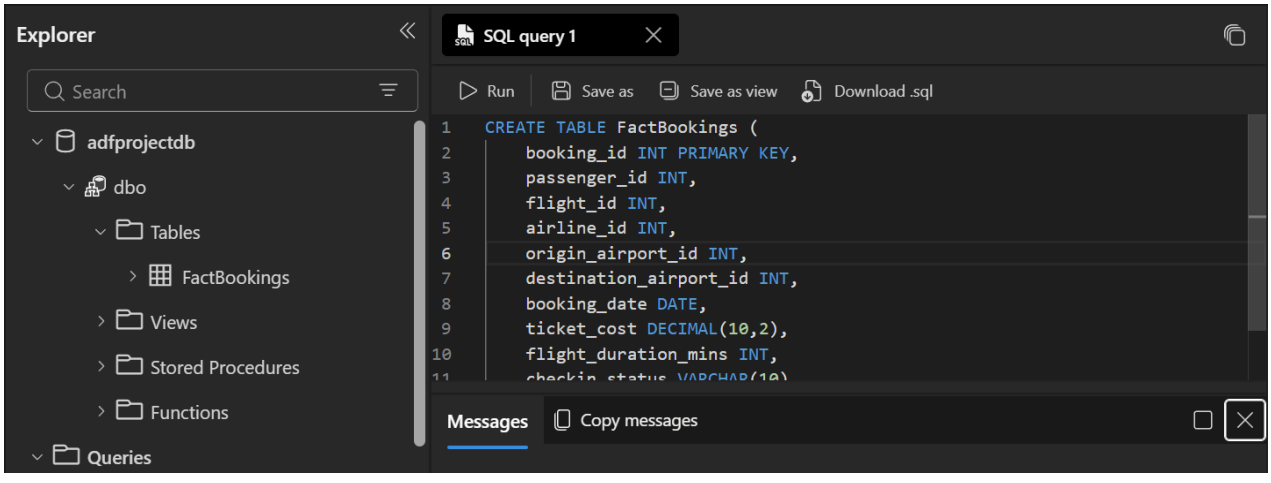

Create SQL Database
Microsoft
Compute + storage *
General Purpose - Serverless
Standard-series (Gen5), 1 vCore, 1 GB storage
Configure database
Backup storage redundancy
Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.
The default backup storage redundancy setting is taken from the setting of the source.
Backup storage redundancy
Locally-redundant backup storage
Geo-redundant backup storage

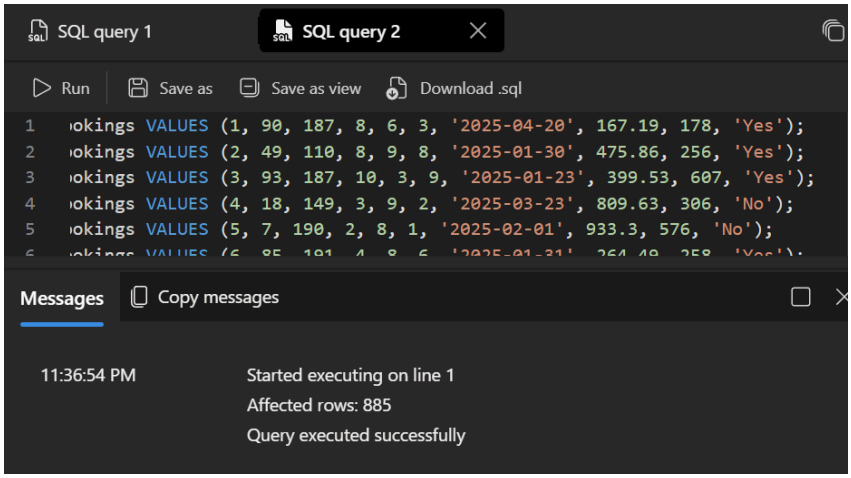

Create SQL Database
Microsoft
Basics
Networking
Security
Additional settings
Tags
Review + create
Configure network access and connectivity for your server. The configuration selected below will apply to the selected server 'adfprojectketaki' and all databases it manages. Learn more
Network connectivity
Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. Learn more
Connectivity method *
No access
Public endpoint
Private endpoint

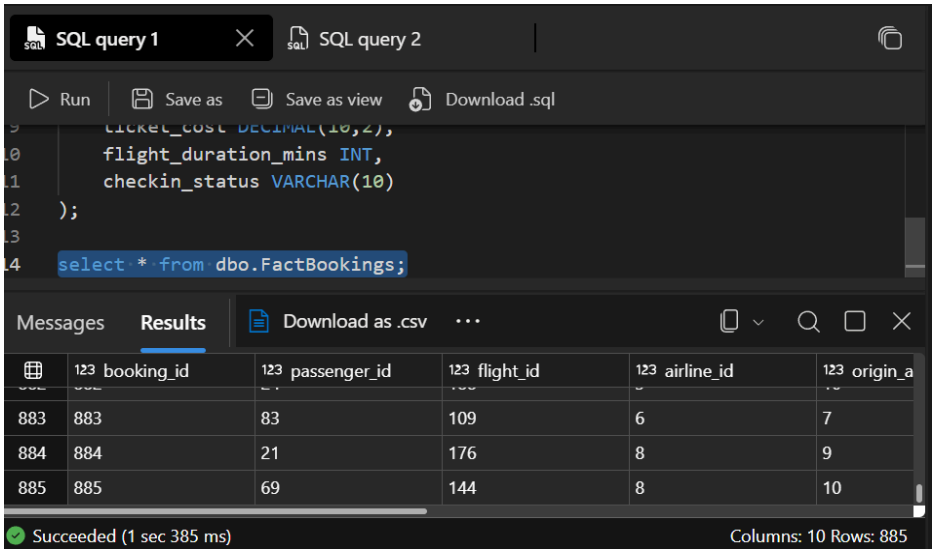

Create SQL Database
Microsoft
Firewall rules
Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. Learn more
Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.
Allow Azure services and resources to access this server *
No
Yes
Add current client IP address *
No
Yes
Connection policy
Configure how clients communicate with your SQL database server. Learn more
Connection policy
Default - Uses Redirect policy for all client connections originating inside of Azure (except Private Endpoint connections) and Proxy for all client connections originating outside Azure


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
kolad.river.rafting2025...
DEFAULT DIRECTORY (KOLADRIV...
Home
Azure SQL
Create SQL Database
Microsoft
Initializing deployment...
Initializing template deployment to resource group 'ADF_Project'.
Basics
Networking
Security
Additional settings
Tags
Review + create
Product details
SQL database
by Microsoft
Terms of use | Privacy policy
Estimated cost
Storage cost 0.18 USD / month + Compute cost 0.000159 USD / vCore second
Terms
By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see Azure Marketplace Terms.
SQL
Cost summary
General Purpose (GP_S_Gen5_1)
Cost per GB (in USD)
0.14
Max storage selected (in GB)
x 1.3
ESTIMATED STORAGE COST / MONTH
0.18 USD
COMPUTE COST / VCORE SECOND
0.000159 USD
Validating...
< Previous
Download a template for automation


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
kolad.river.rafting2025...
DEFAULT DIRECTORY (KOLADRIV...
Home
adfprojectdb (adfprojectketaki/adfprojectdb)
adfprojectdb (adfprojectketaki/adfprojectdb) | Query editor
SQL database
Search
Feedback
Classic experience (preview)
Query editor
Mirror database in Fabric (preview)
Resource visualizer
Settings
Compute + storage
Connection strings
Maintenance
Properties
Locks
Data management
Integrations
Power Platform
Add or remove favorites by pressing Ctrl+Shift+F
Run queries on your database directly in the browser. Learn more
Connecting to adfprojectdb on adfprojectketaki
Microsoft Entra authentication
SQL authentication
Username *
adminketaki
Password *
Connect

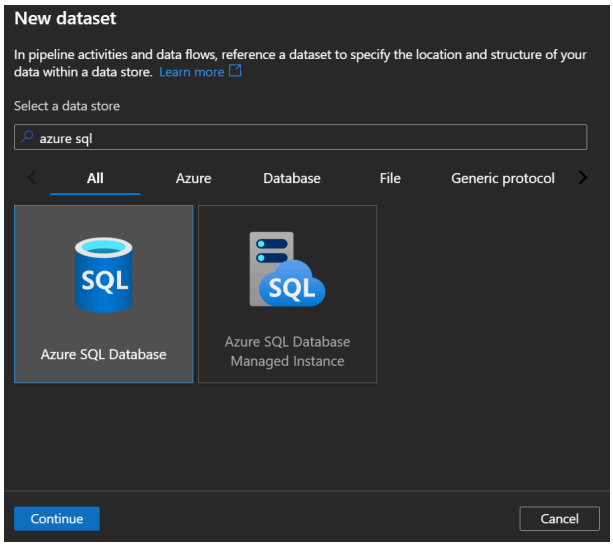

Explorer
Search
adfprojectdb
dbo
Tables
FactBookings
Views
Stored Procedures
Functions
Queries
SQL query 1
Run
Save as
Save as view
Download .sql
CREATE TABLE FactBookings (
booking_id INT PRIMARY KEY,
passenger_id INT,
flight_id INT,
airline_id INT,
origin_airport_id INT,
destination_airport_id INT,
booking_date DATE,
ticket_cost DECIMAL(10,2),
flight_duration_mins INT,
Messages
Copy messages



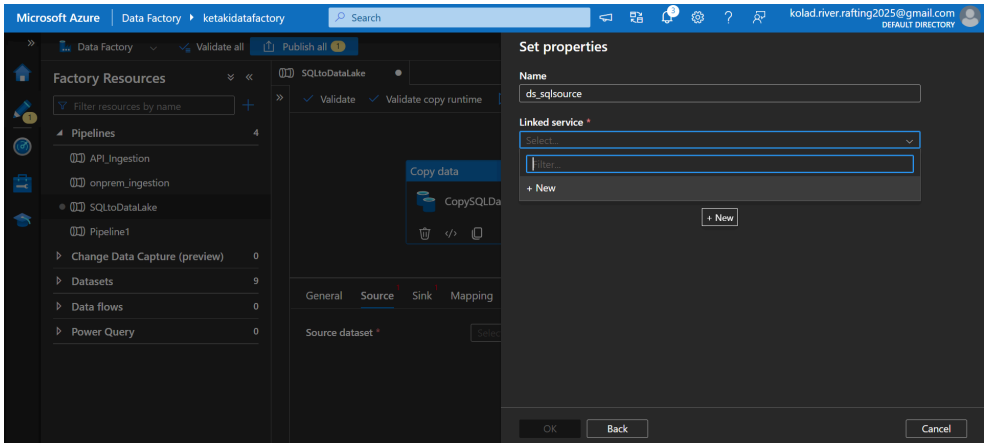


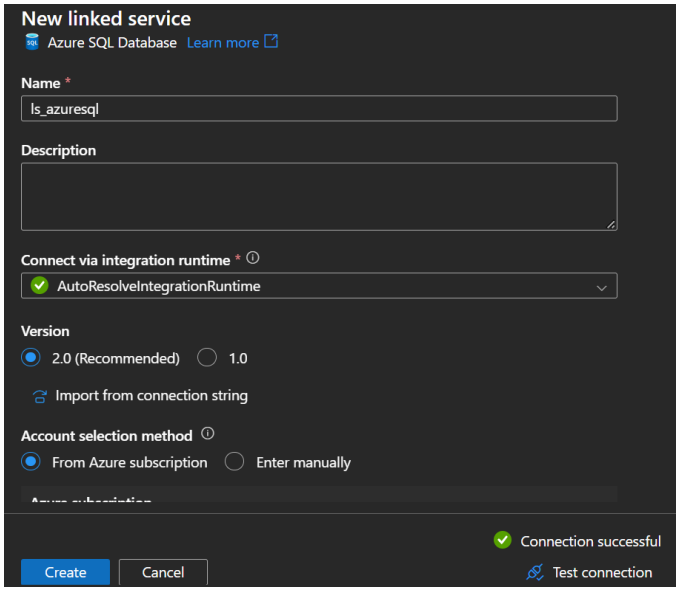
2:21:02 Incremental Data Loading In Azure Data Factory

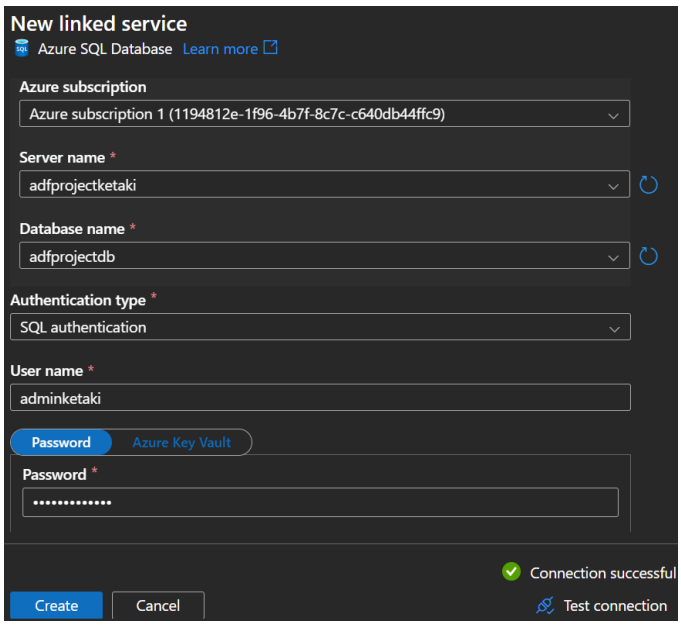
Create new pipeline : , Source :

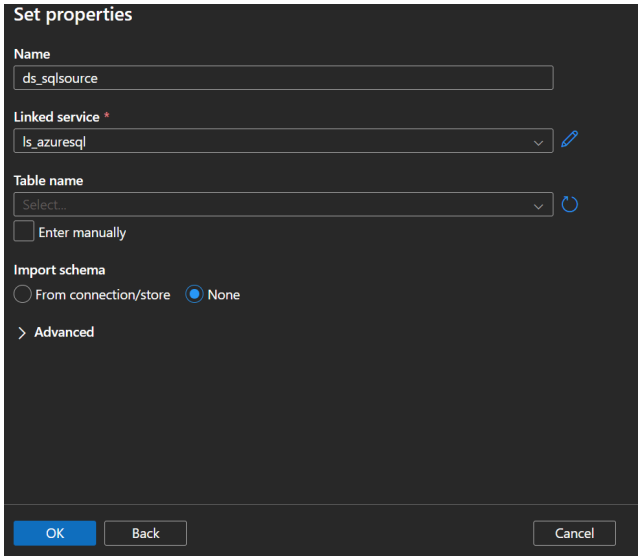


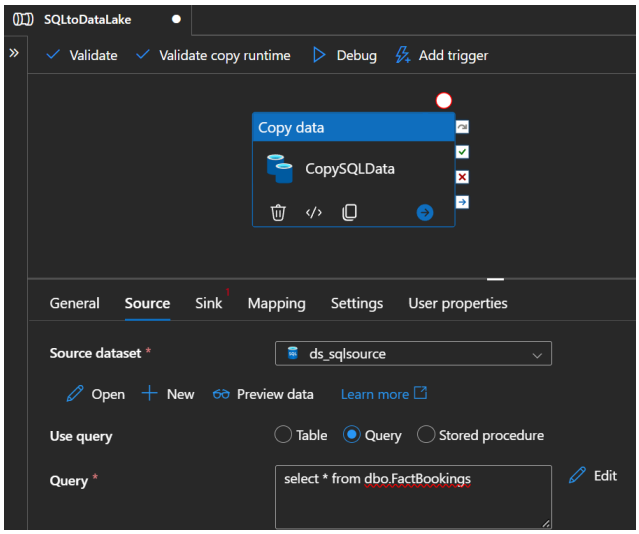
Creating new linked service :



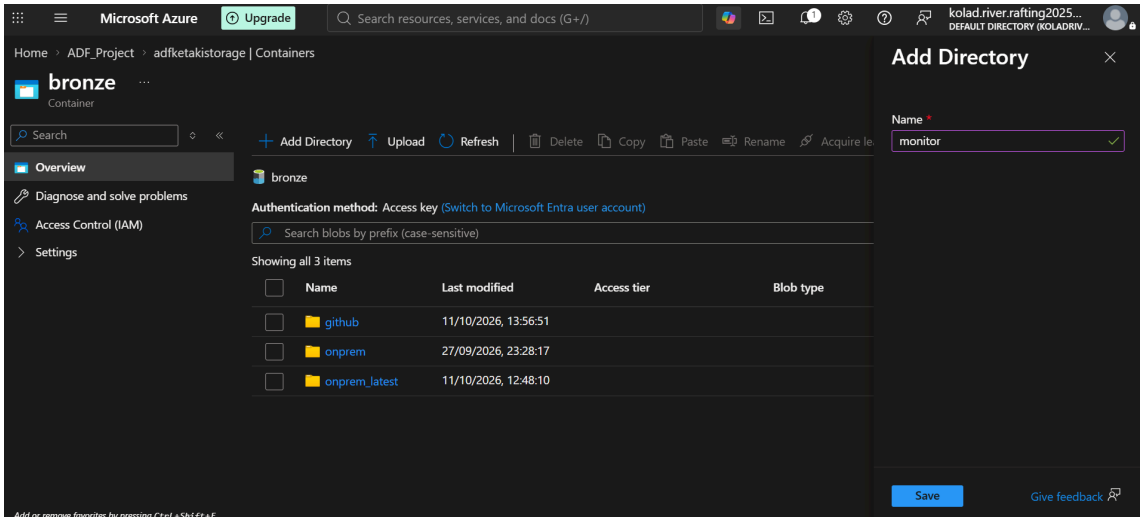




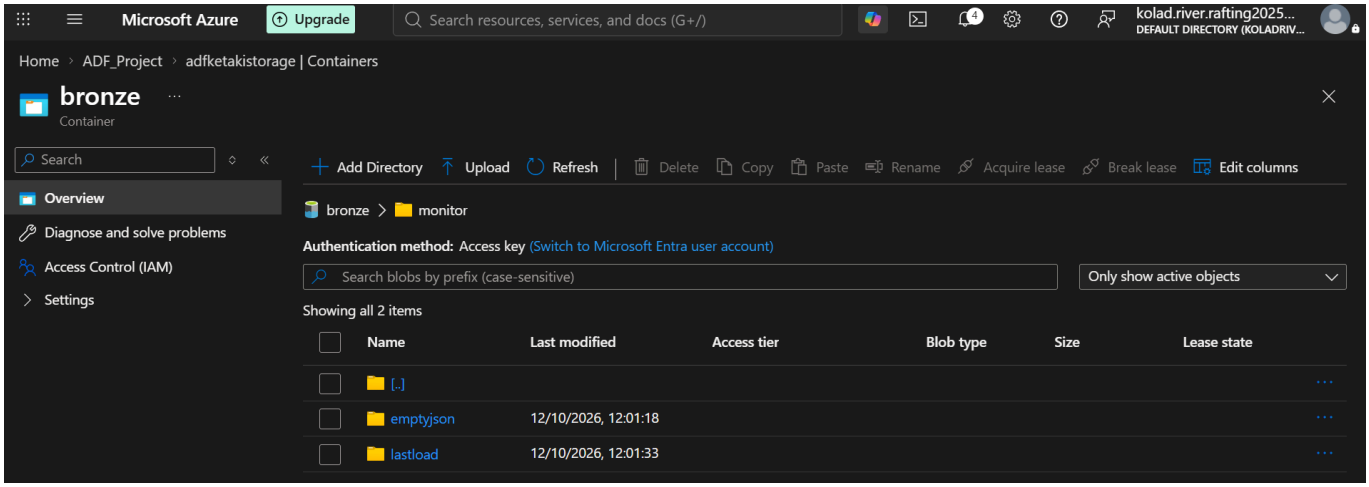




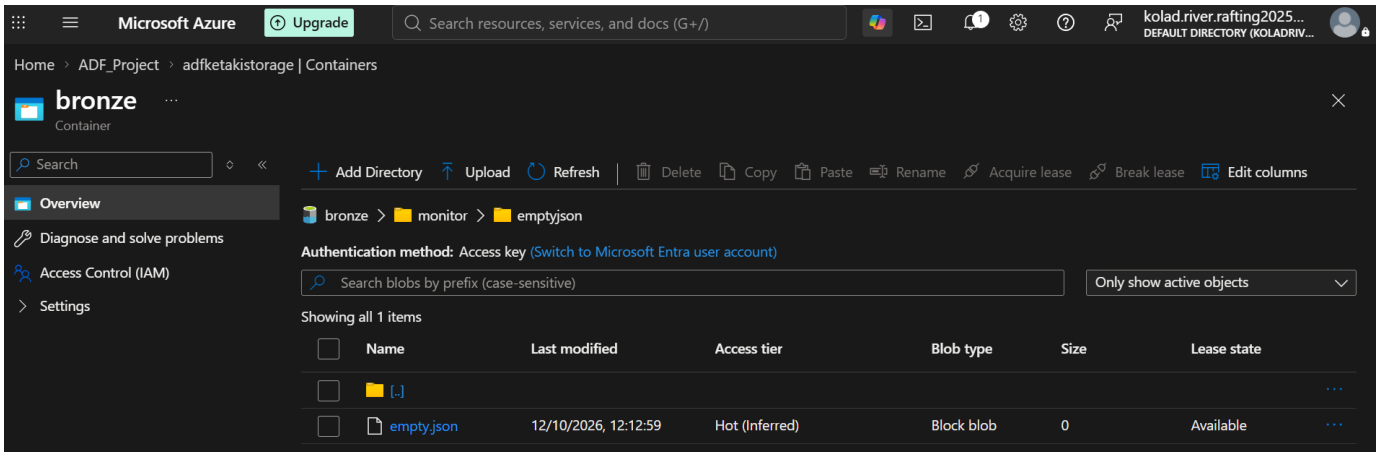
Creating new directory in adf



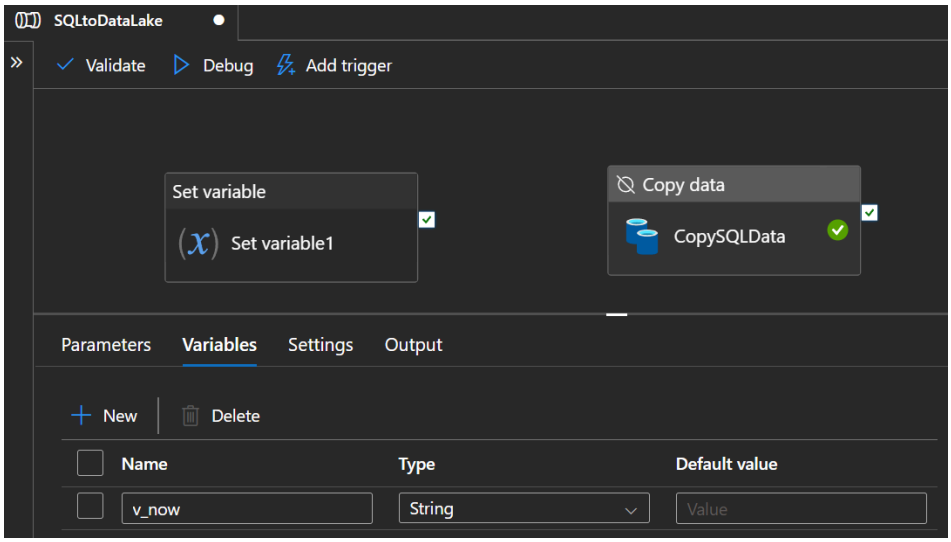
Creating 2 new directories inside monitor directory :



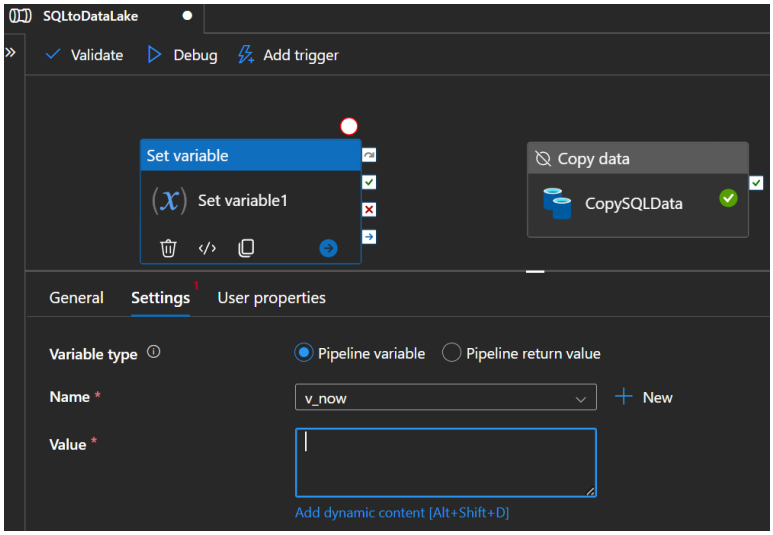
Upload json file inside emptyjson folder :

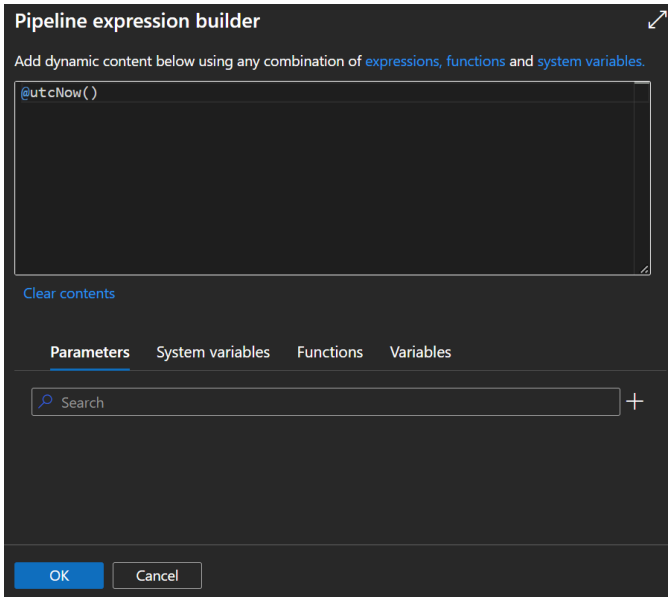


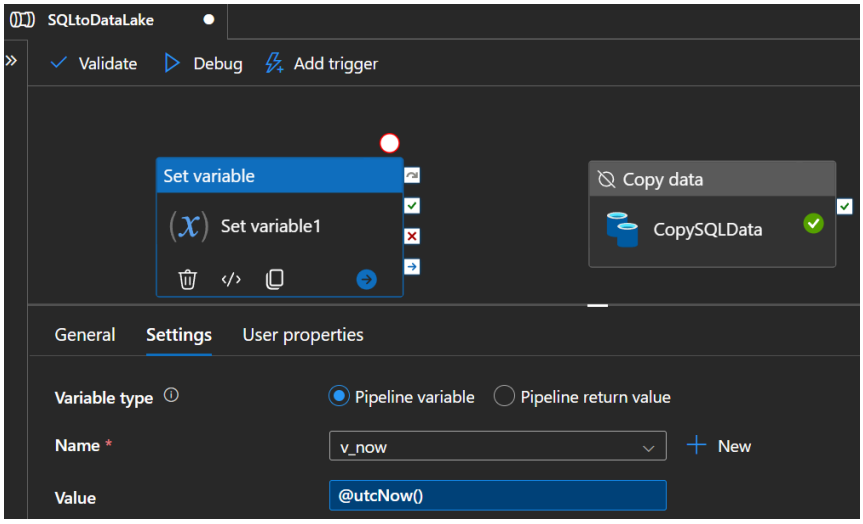
In adf :

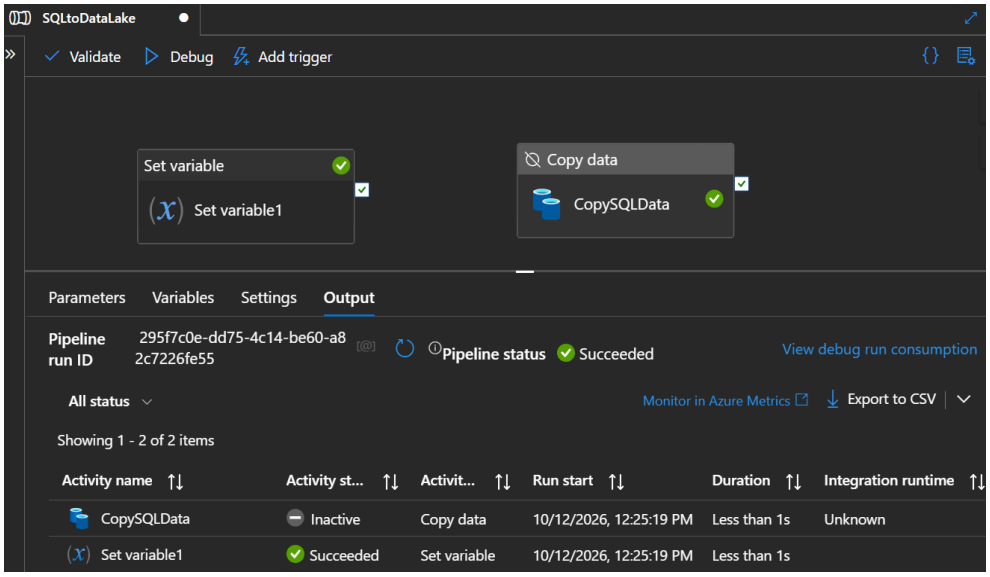


Click on add dynamic content :

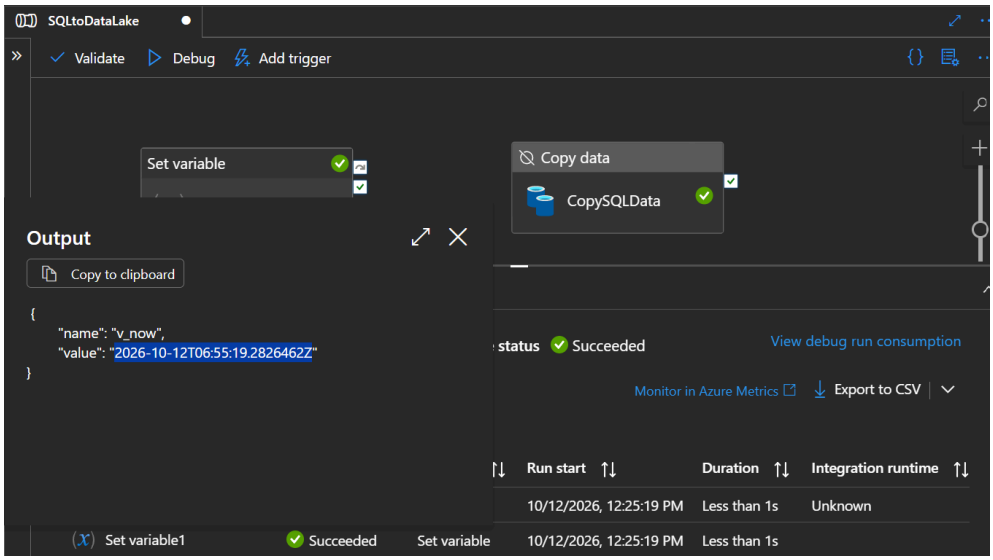






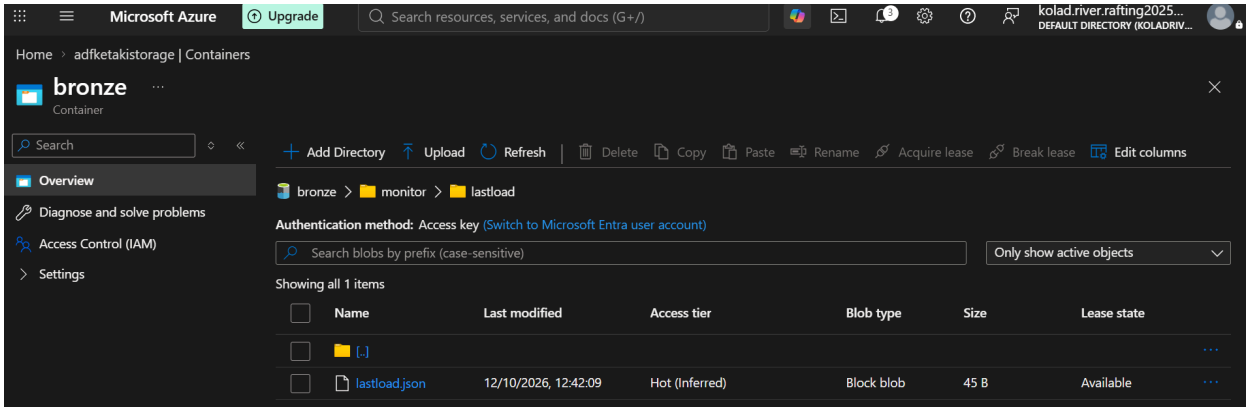


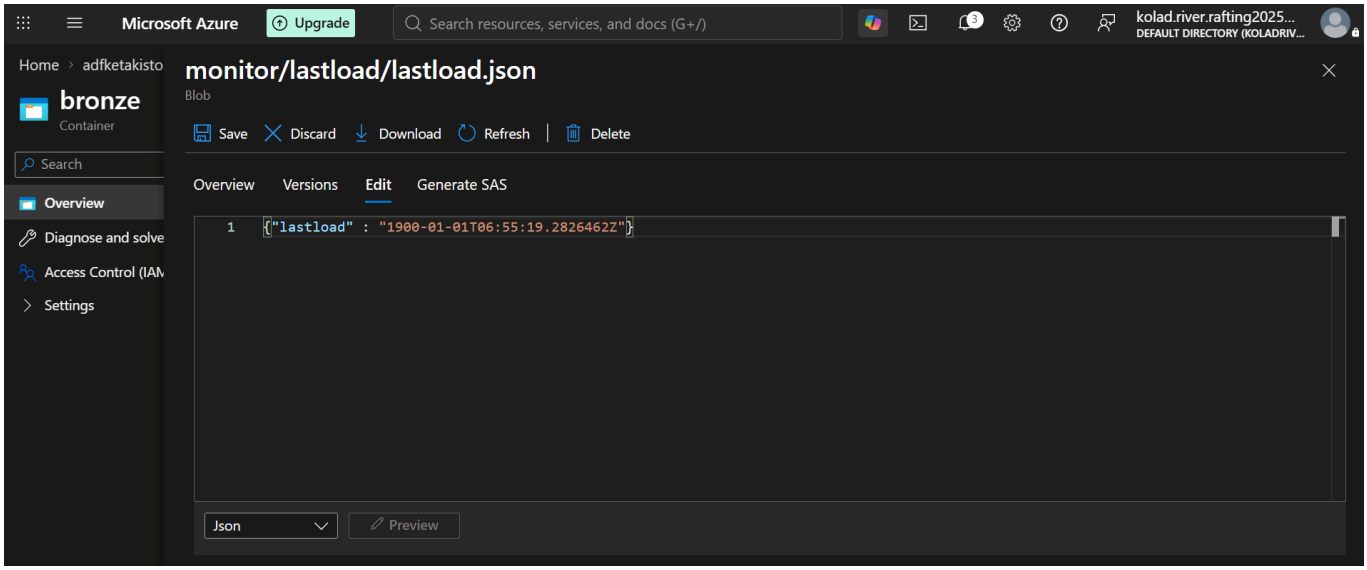
Copy this value :



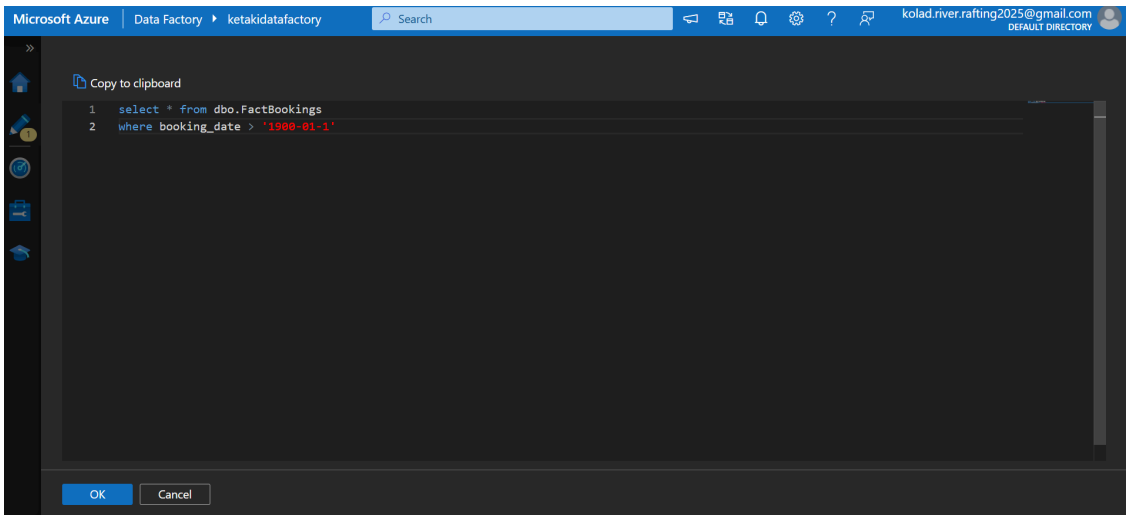
2026-10-12T06:55:19.2826462Z

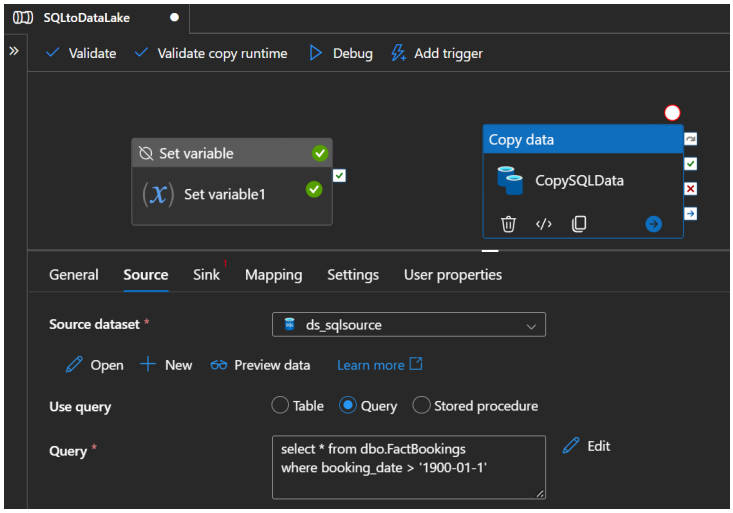
In last load create empty json file :



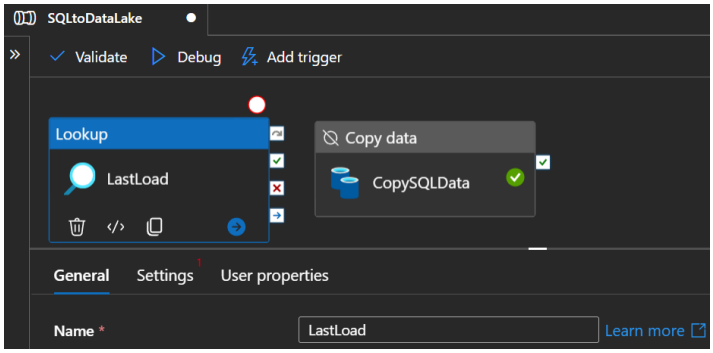


Hard code in copydata

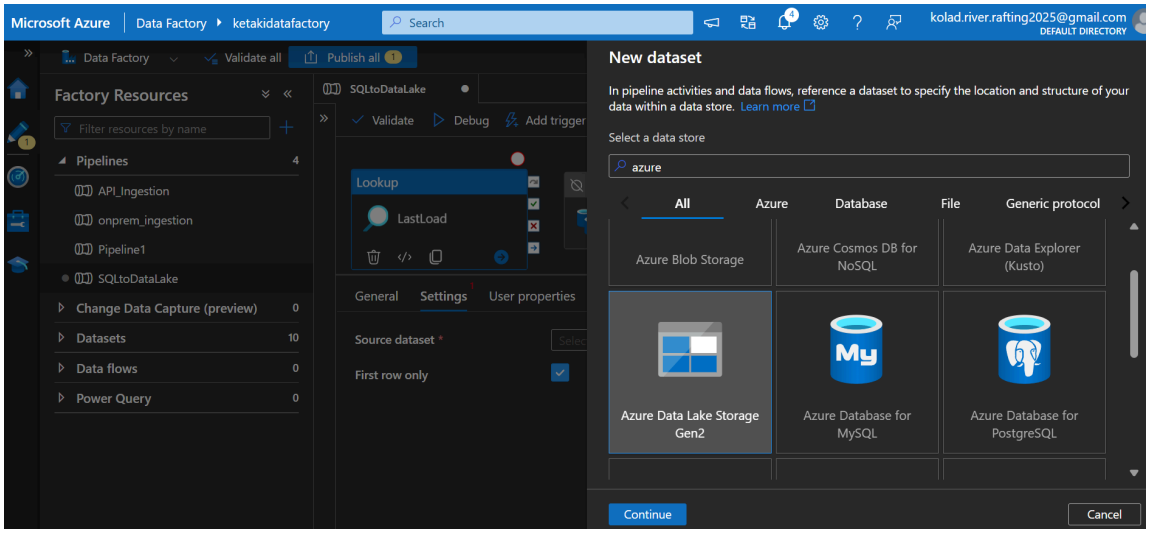


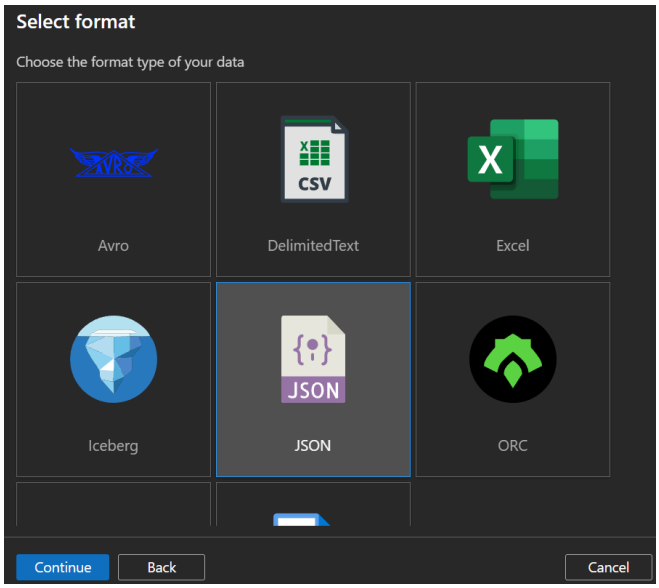


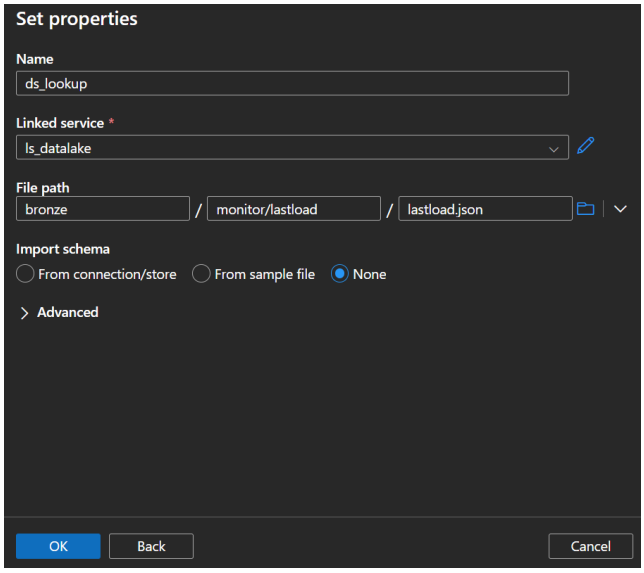
create lookup :

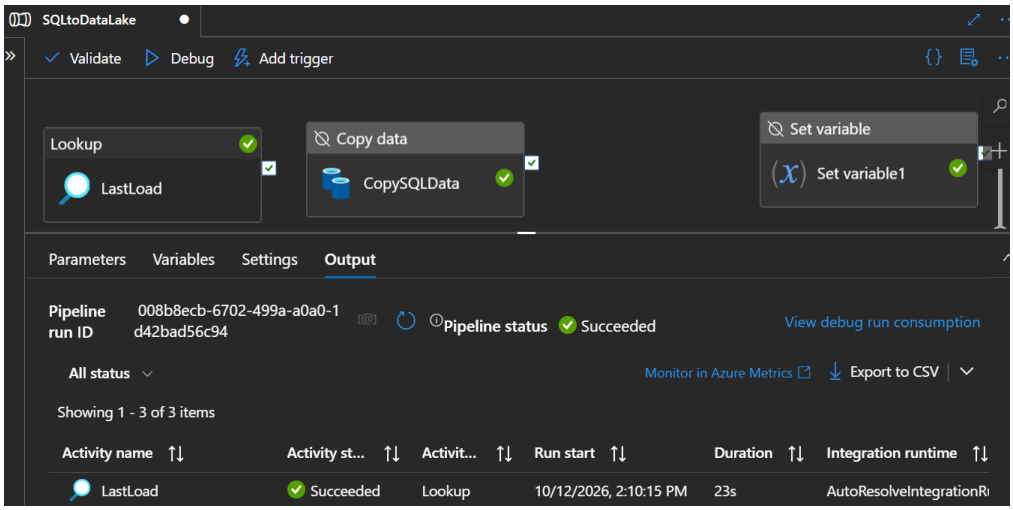


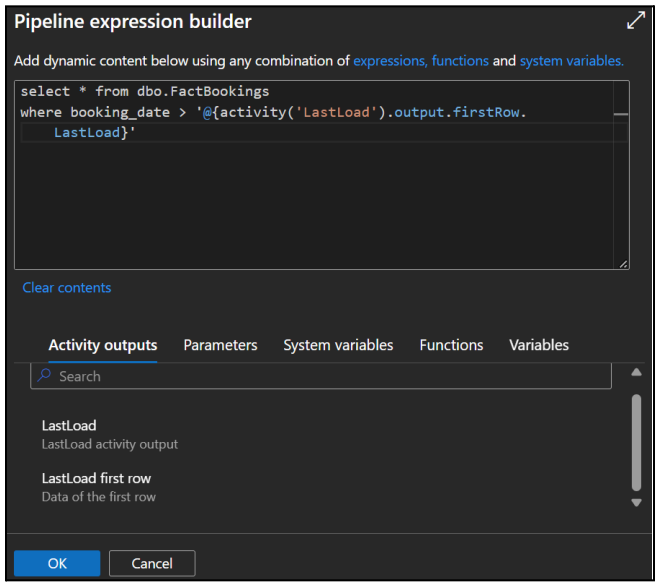
Settings :



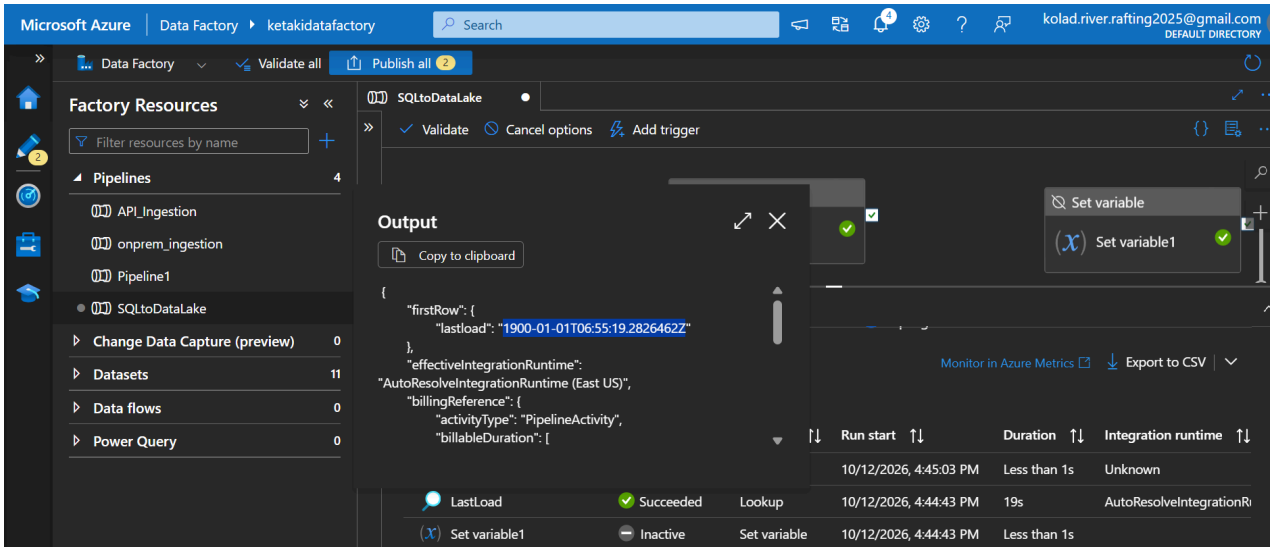


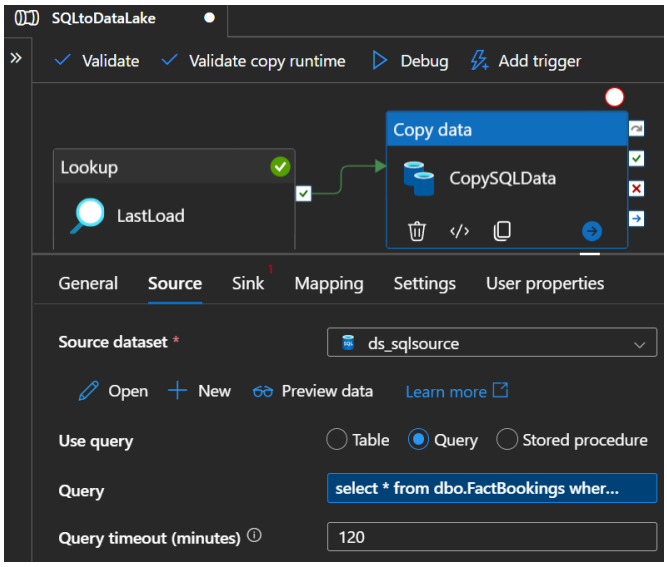

Set properties
Name
ds_lookup
Linked service *
ls_datalake
File path
bronze
monitor/lastload
lastload.json
Import schema
From connection/store
From sample file
None
Advanced
OK
Back
Cancel


SQLtoDataLake
Validate
Debug
Add trigger
Lookup
LastLoad
Copy data
CopySQLData
Set variable
Set variable1
Parameters
Variables
Settings
Output
Pipeline run ID
008b8ecb-6702-499a-a0a0-1d42bad56c94
Pipeline status
Succeeded
View debug run consumption
All status
Monitor in Azure Metrics
Export to CSV
Showing 1 - 3 of 3 items
Activity name
Activity st...
Activit...
Run start
Duration
Integration runtime
LastLoad
Succeeded
Lookup
10/12/2026, 2:10:15 PM
23s
AutoResolveIntegrationR

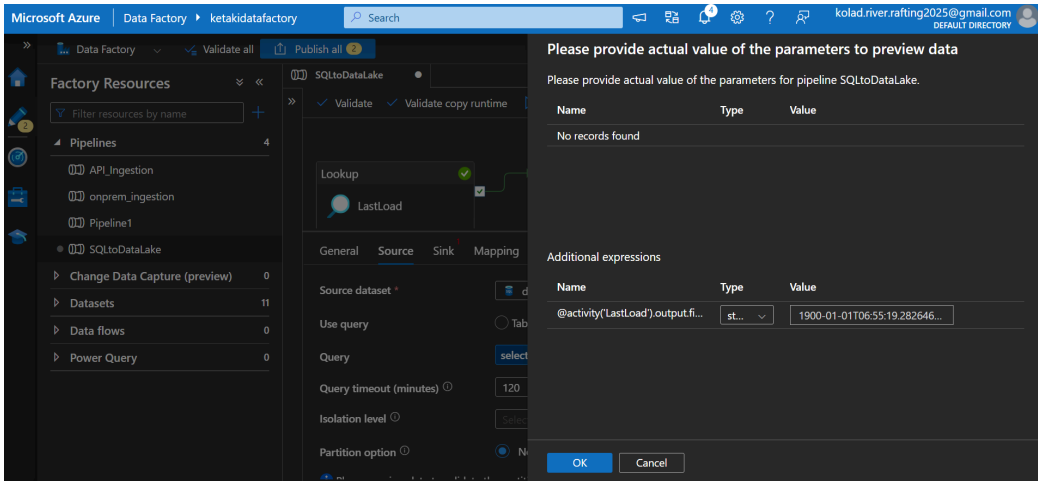

Pipeline expression builder
Add dynamic content below using any combination of expressions, functions and system variables.
select * from dbo.FactBookings
where booking_date > '@{activity('LastLoad').output.firstRow.
LastLoad}'
Clear contents
Activity outputs
Parameters
System variables
Functions
Variables
Search
LastLoad
LastLoad activity output
LastLoad first row
Data of the first row
OK
Cancel

Copy the value from lastload :

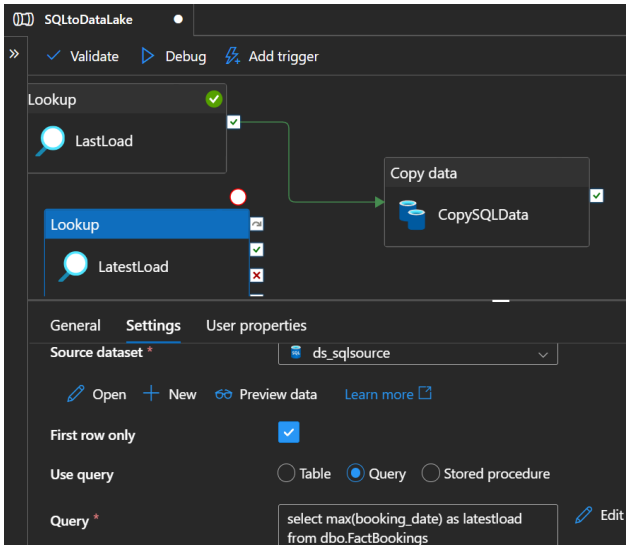


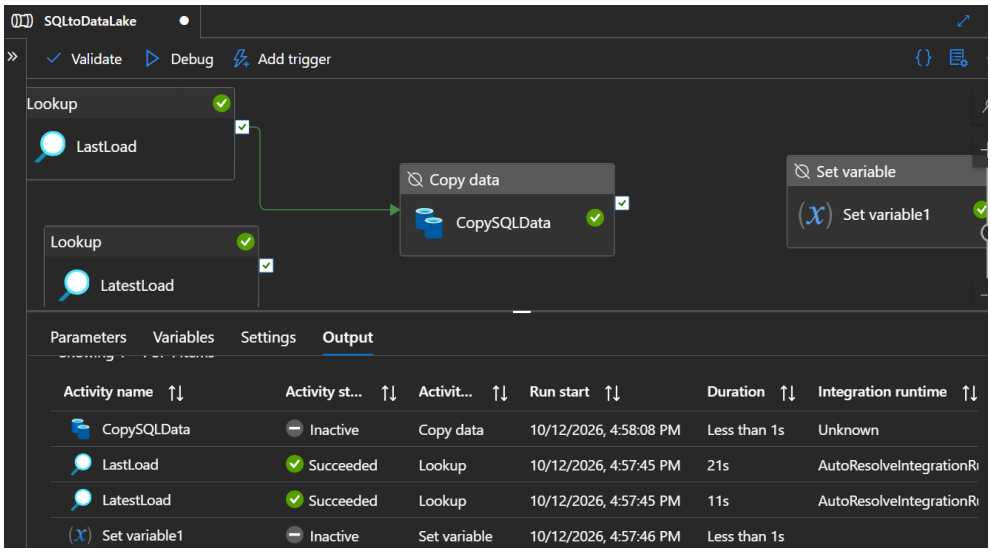


While previewing data in copydata :

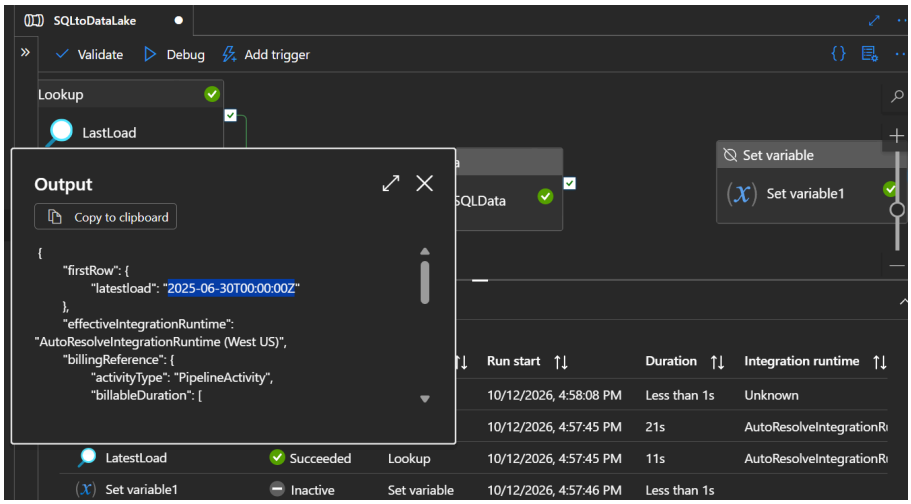


LatestLoad :

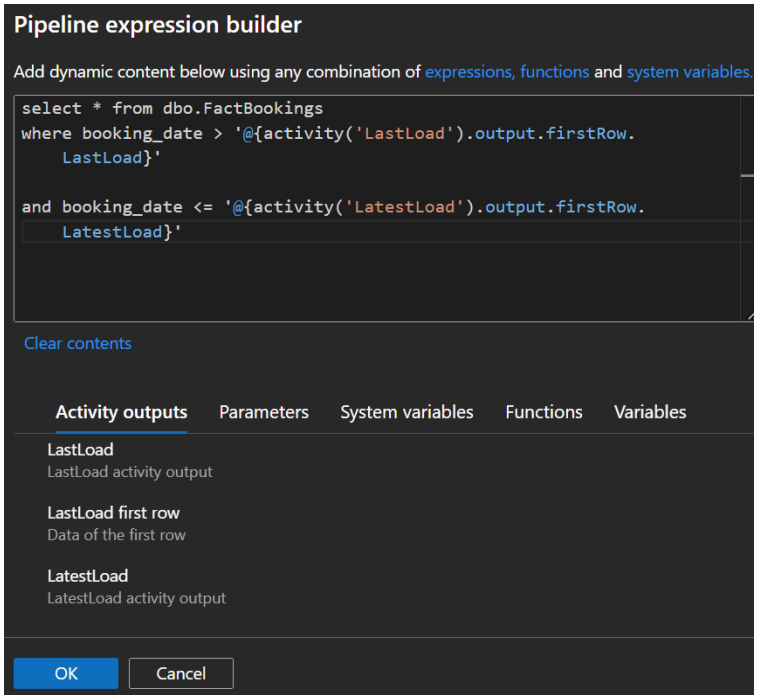




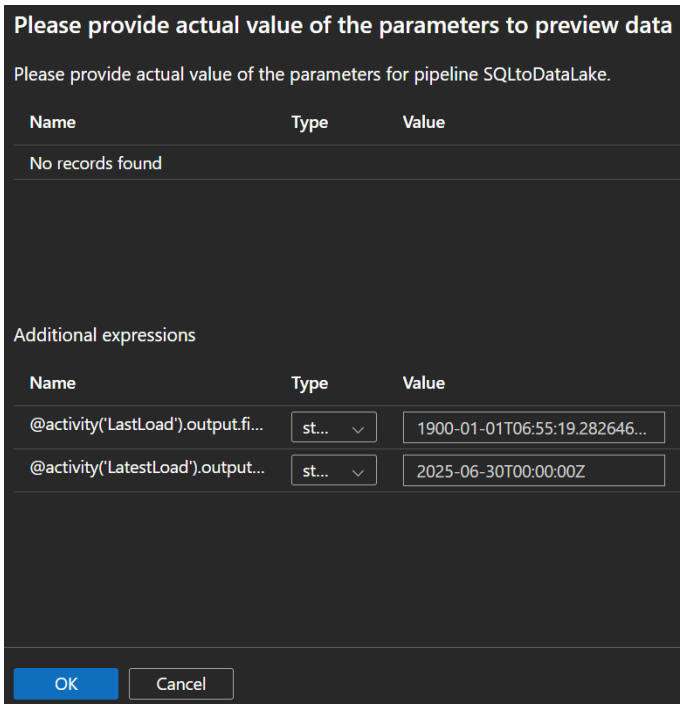
Copy this value of latestload for previewing :



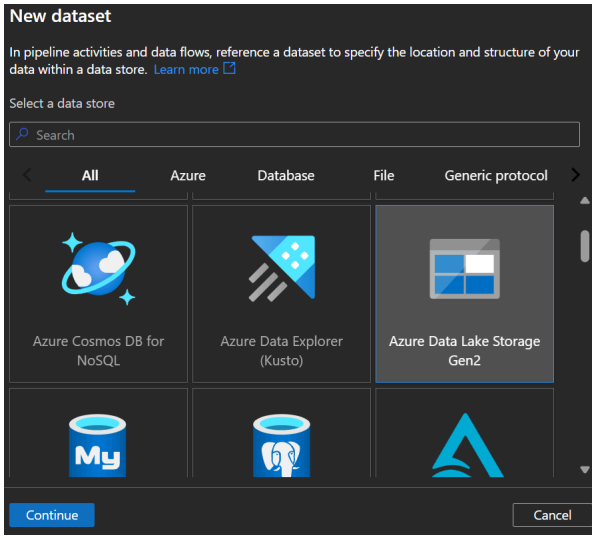
In copydata :

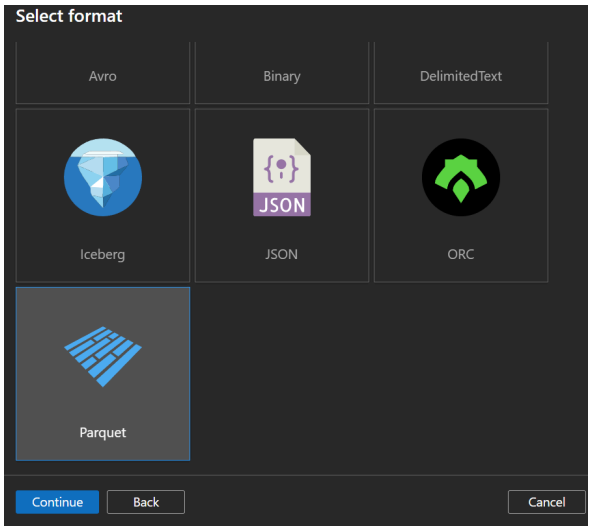


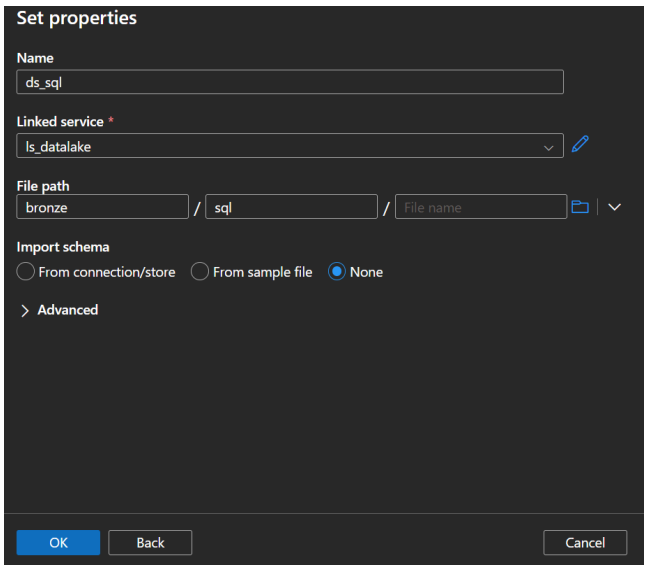
In preview paste that LatestLoad value : 1900-01-01T06:55:19Z , 2025-06-30T00:00:00Z



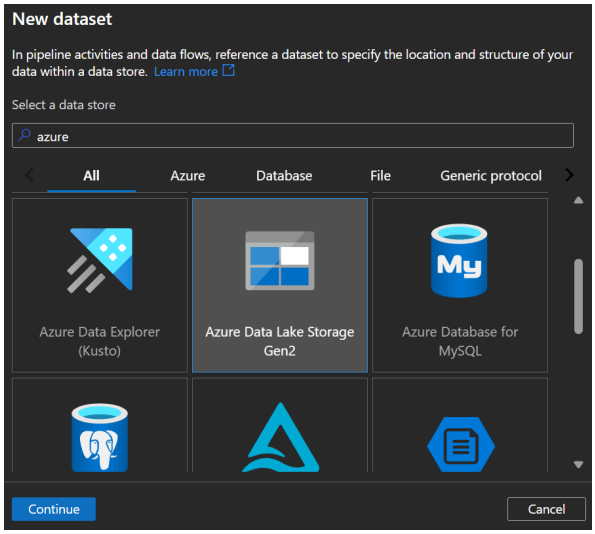
Copydata sink :

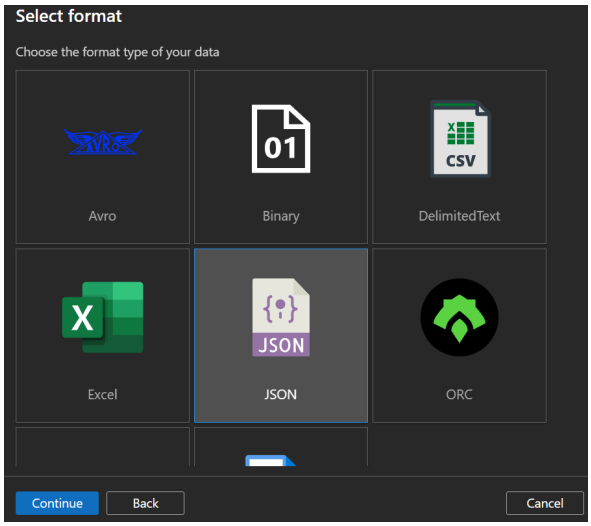


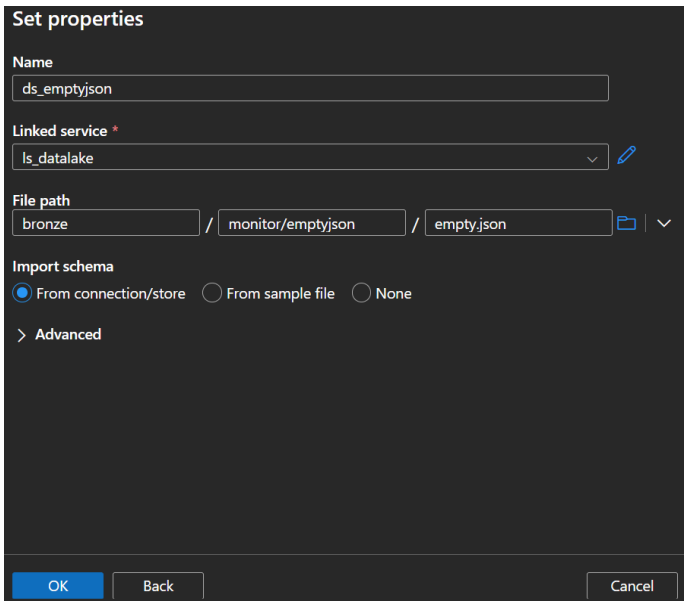




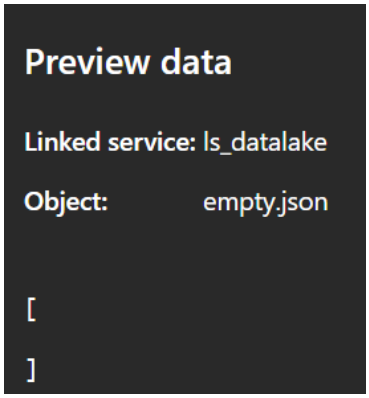
Watermark :

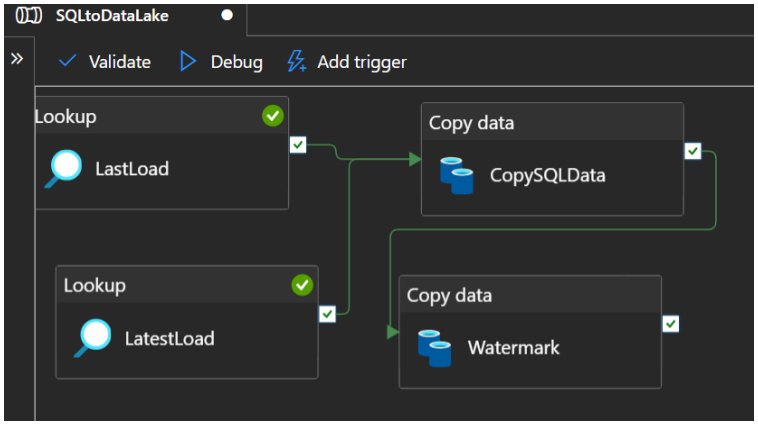




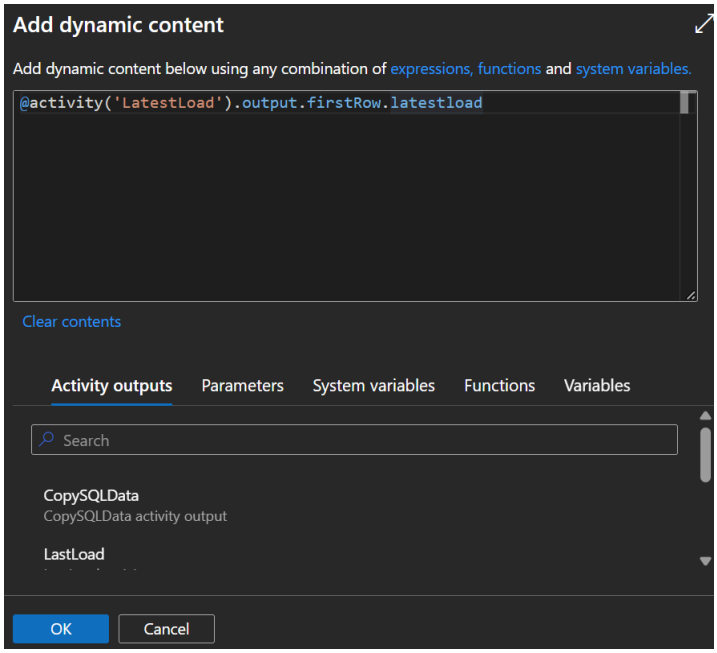


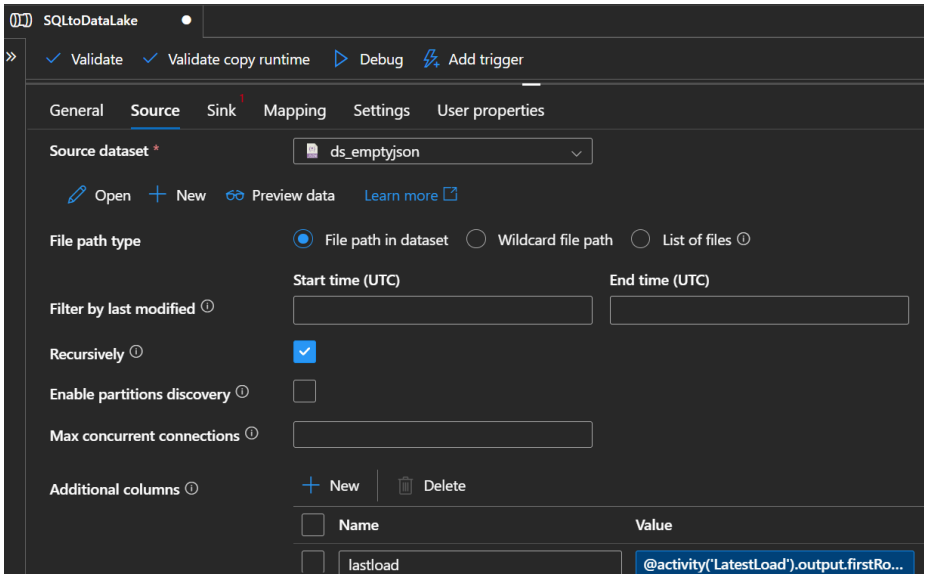
Preview data in watermark :



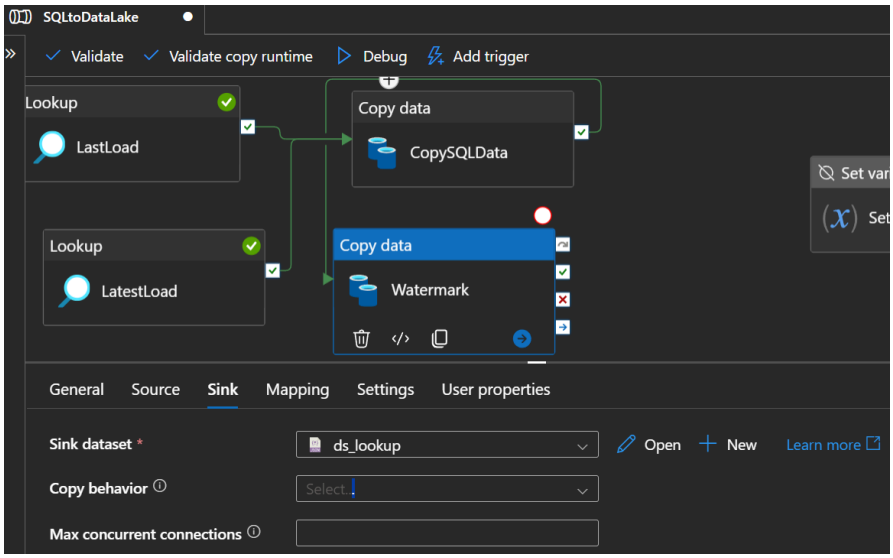


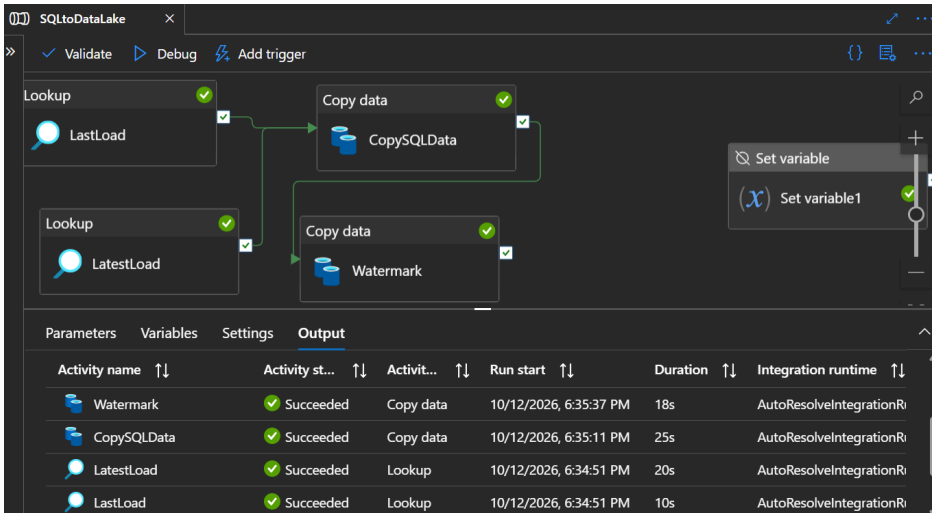
Watermark source :



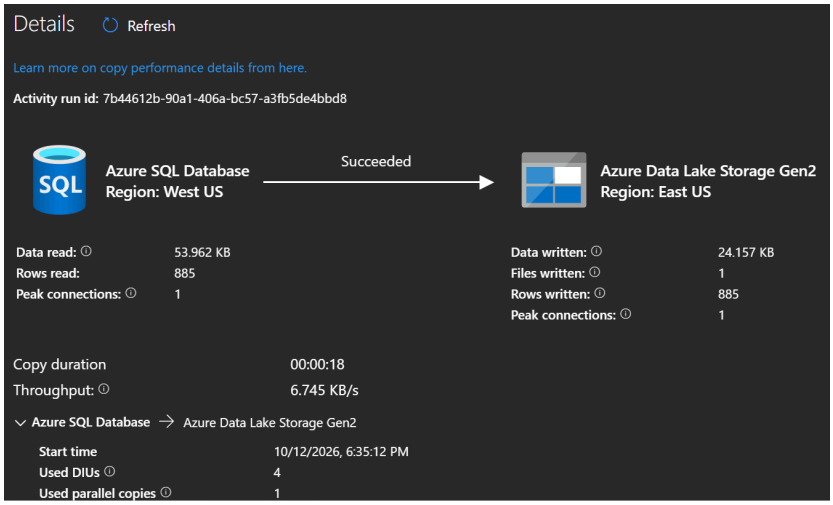


Watermark sink :





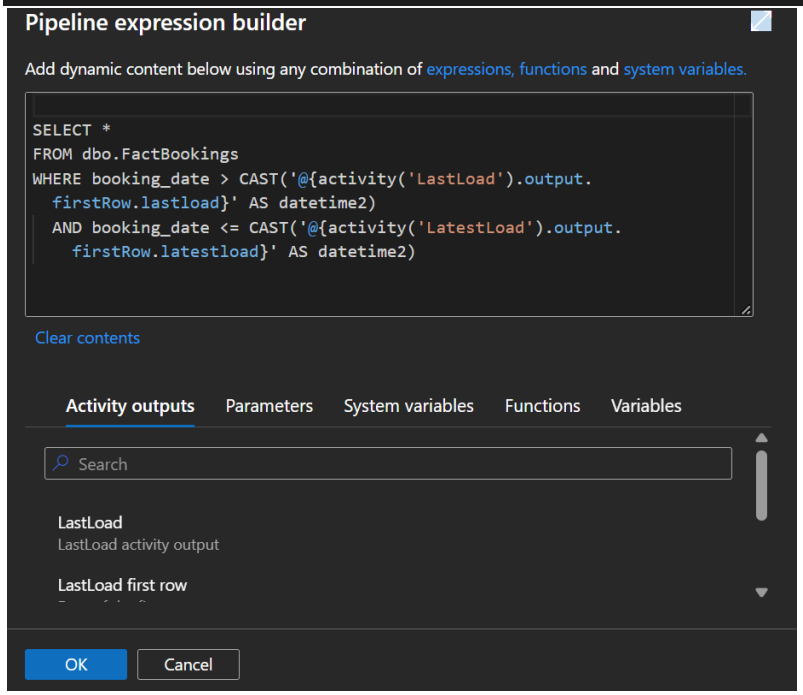
CopySQLData Details :



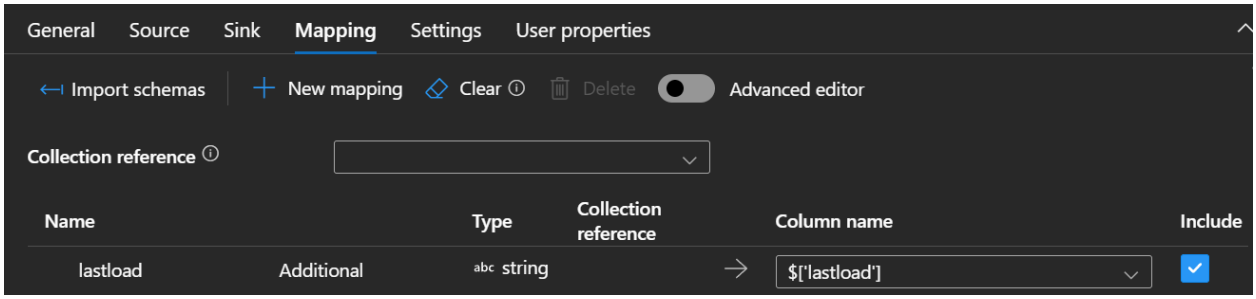
Solving issues : I changed empty.json to contain [{}], configured the lastload Additional column in the Watermark Copy activity, corrected the mapping to avoid the unwanted nested JSON structure, and updated the Lookup expression to read firstRow.lastload directly. This fixed the JSON structure and property-access issue.

Copysqldata query :

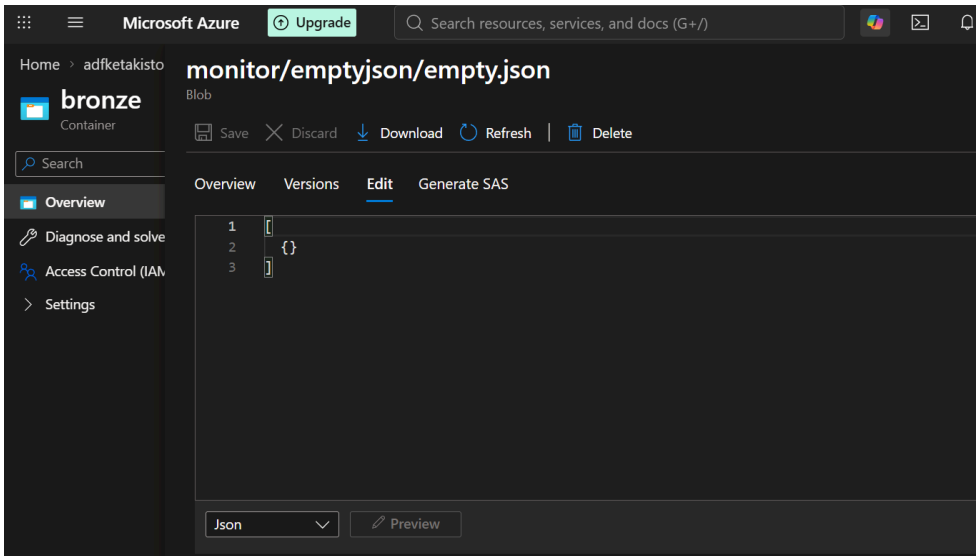
```
SELECT *
FROM dbo.FactBookings
WHERE booking_date > CAST('@{activity('LastLoad').output.firstRow.lastload}' AS datetime2)
  AND booking_date <= CAST('@{activity('LatestLoad').output.firstRow.latestload}' AS
datetime2)
```



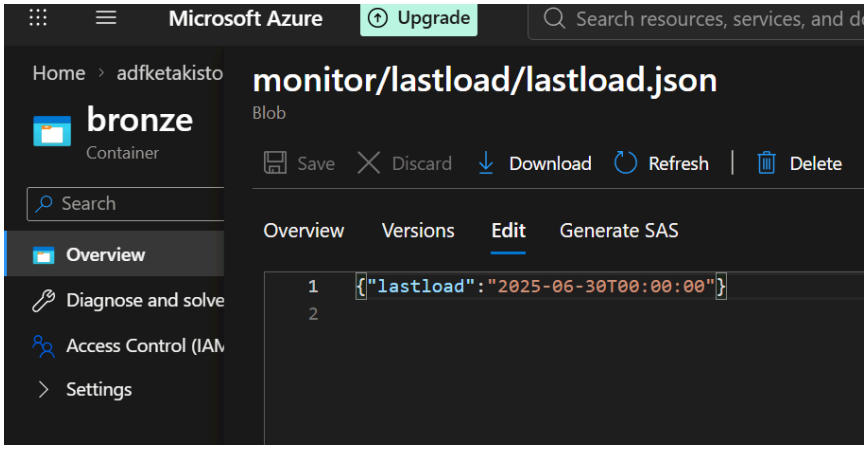
Watermark mapping :



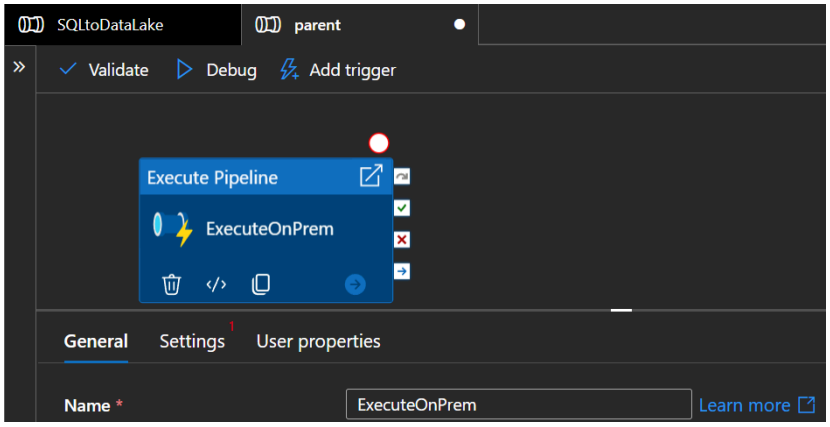
Empty.json in storage :



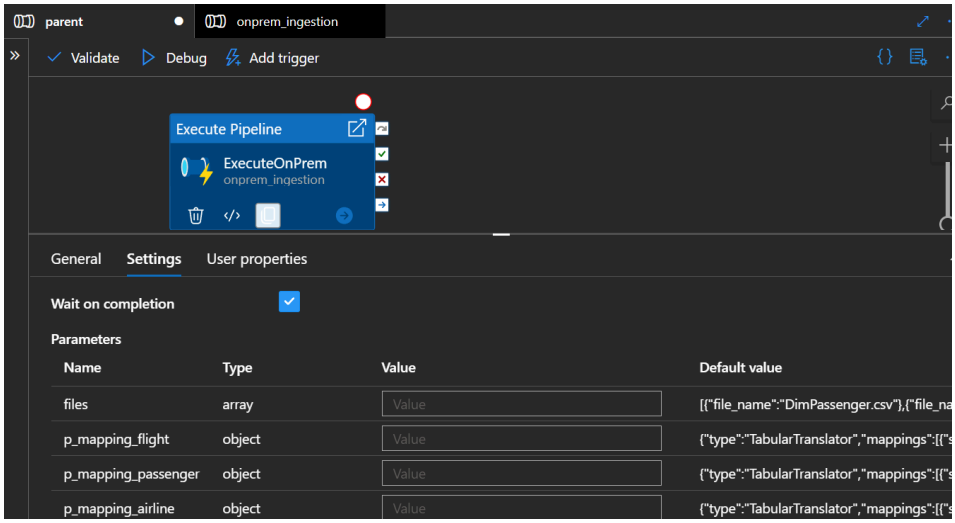
Problem solved as it was updated :



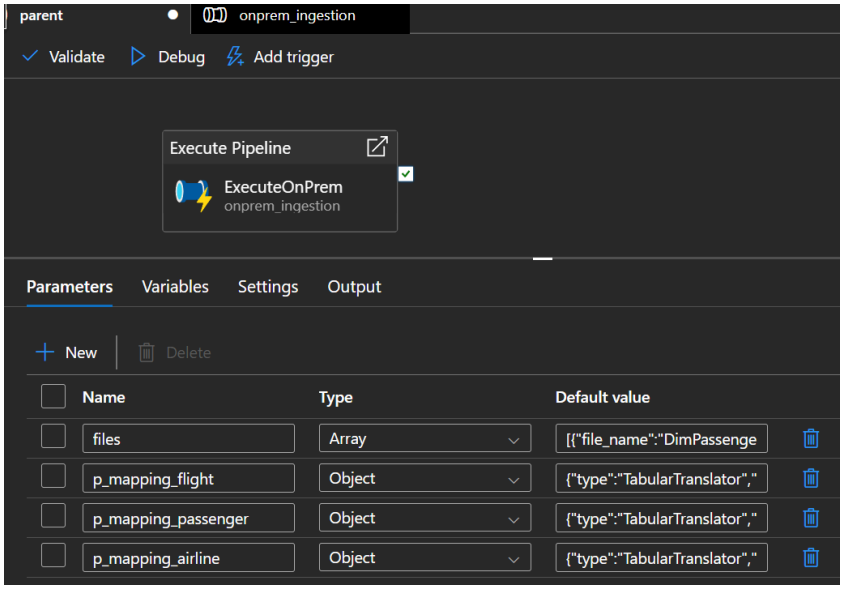
2:59:05 Data Orchestration in Azure Data Factory
Create new parent pipeline :



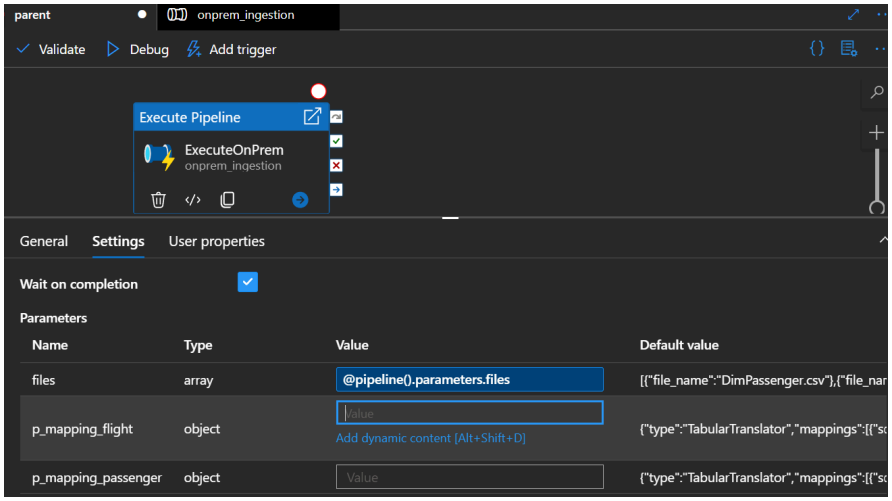
Best practices : create parameters in parent and pass them to child
If you pass the values here it will convert array and object into string so use parameters

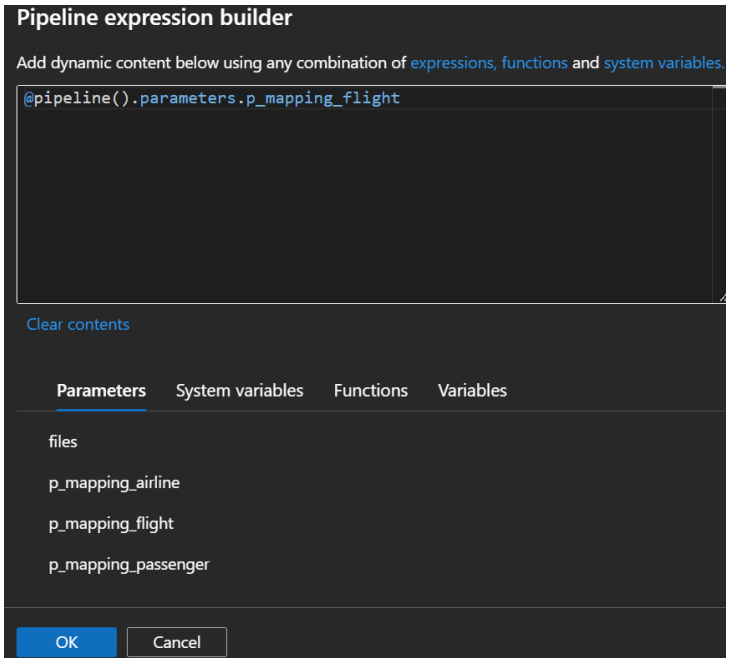


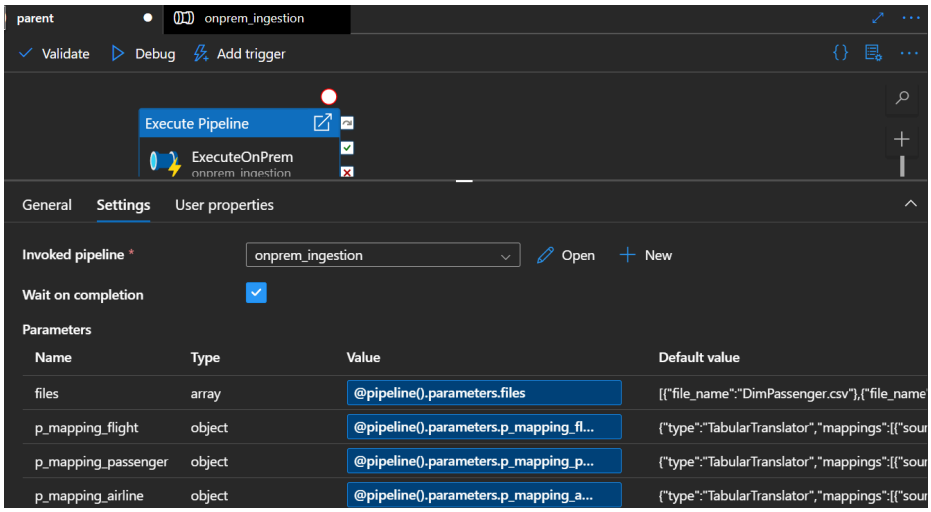
Copy the parameters from onprem pipeline :

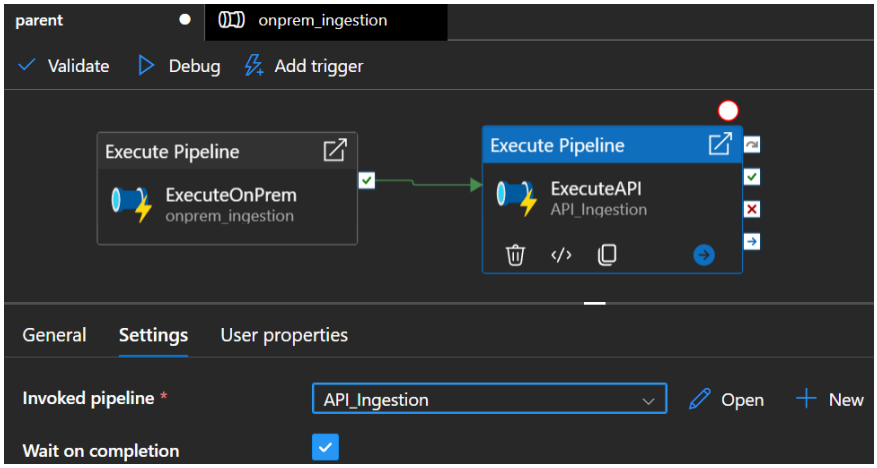


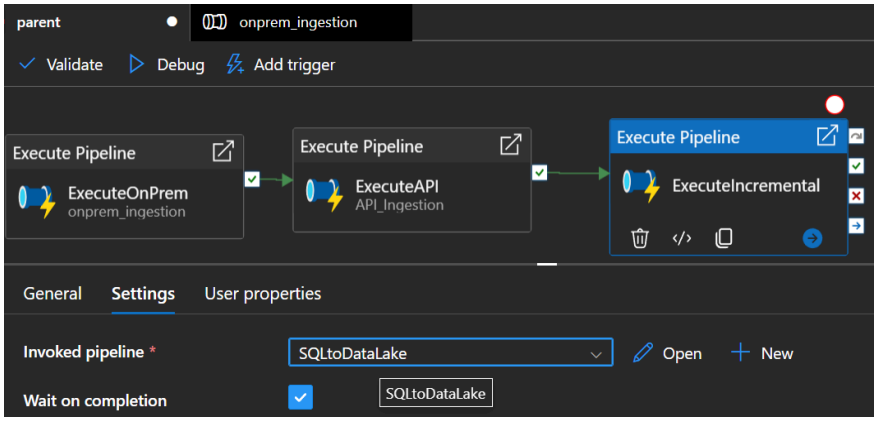
Click on add dynamic content :



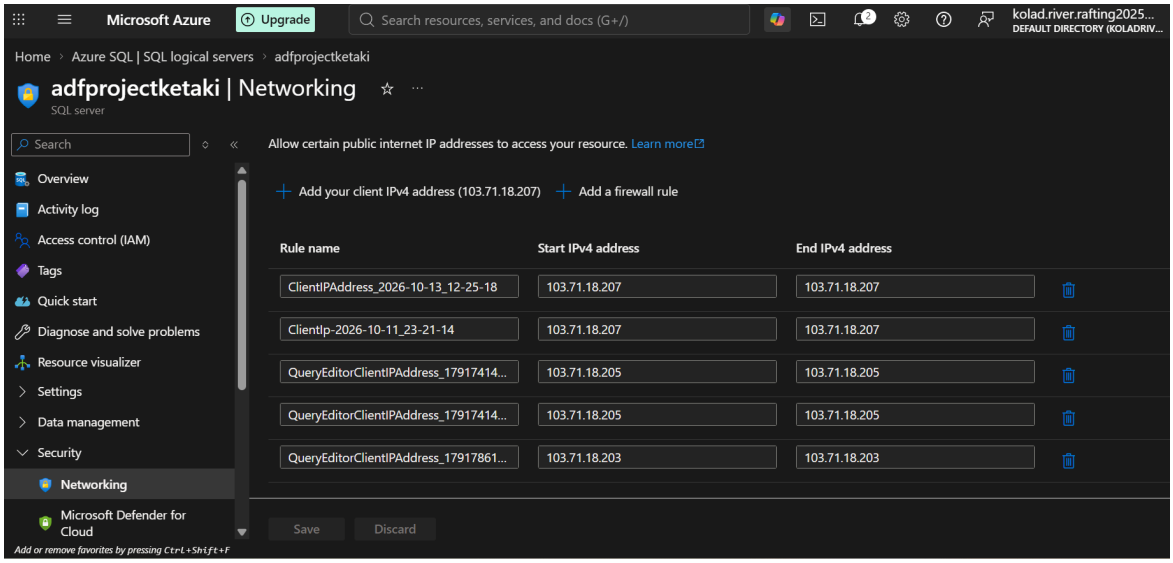


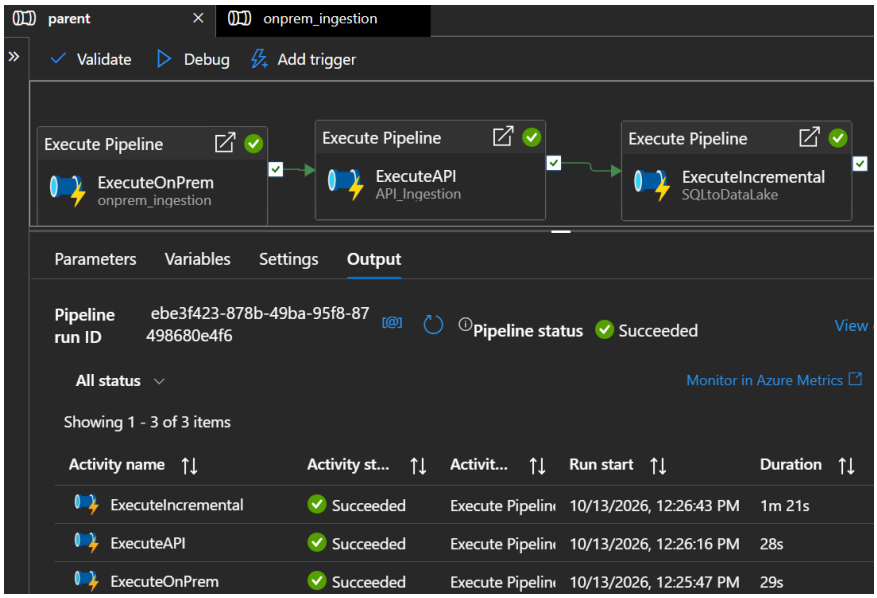






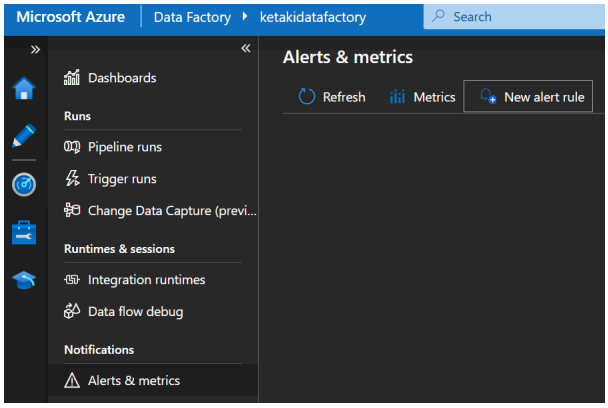
There was issue with 3rd execute pipeline so : added client ipv4 address





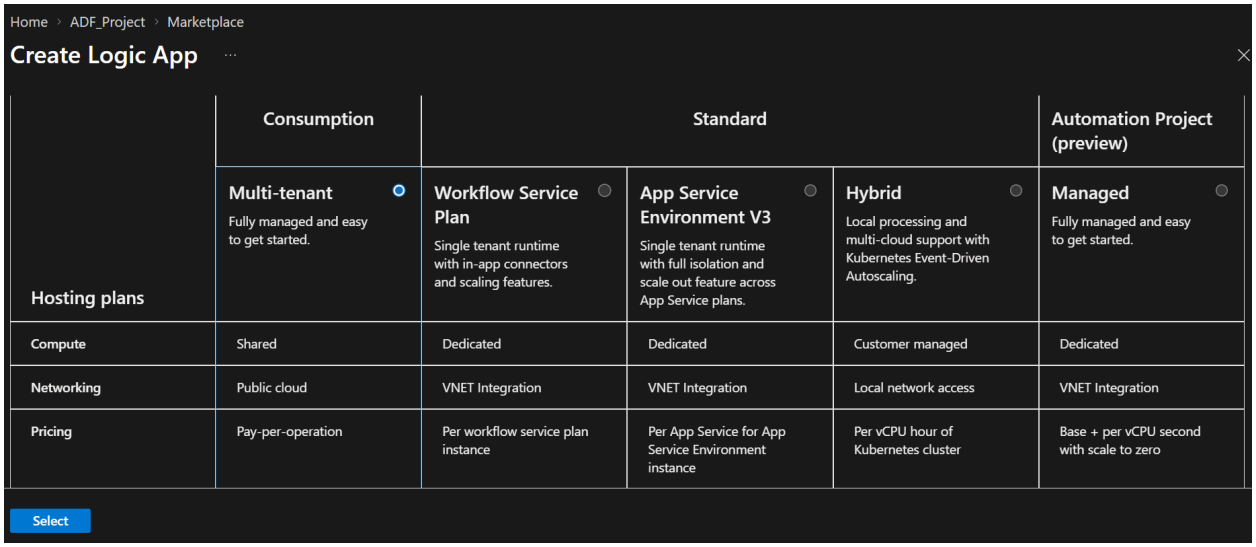
3:11:58 Logic Apps Using Azure Data Factory

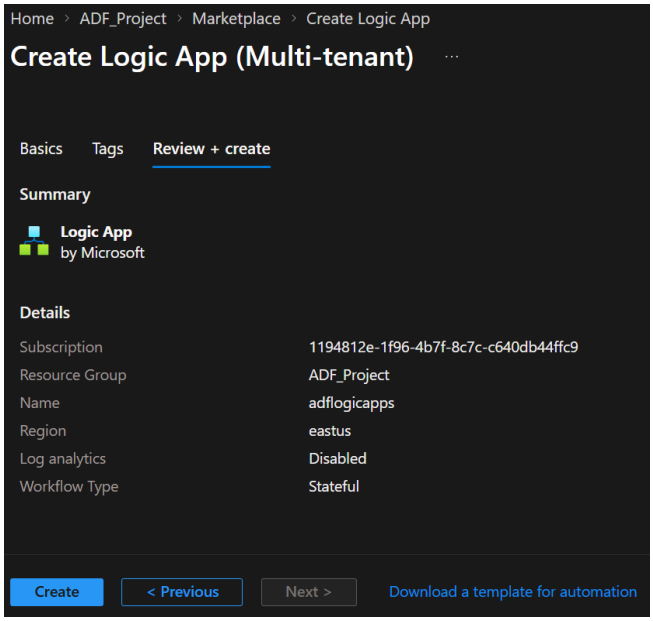
Not easy to leverage for bigger pipelines :

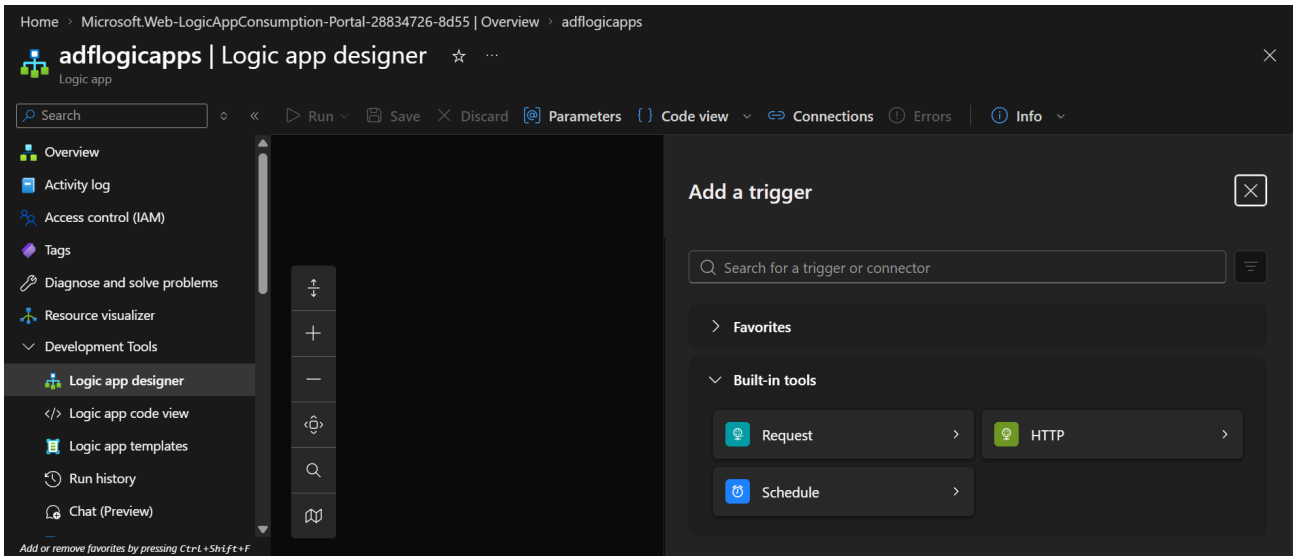


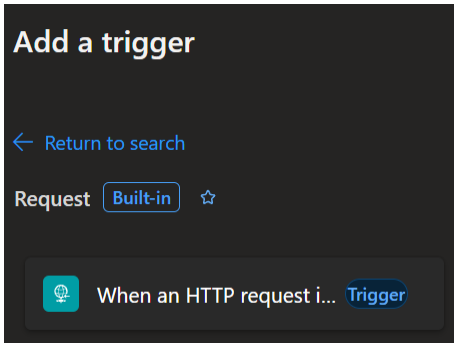
Logic app receives the rest api request.

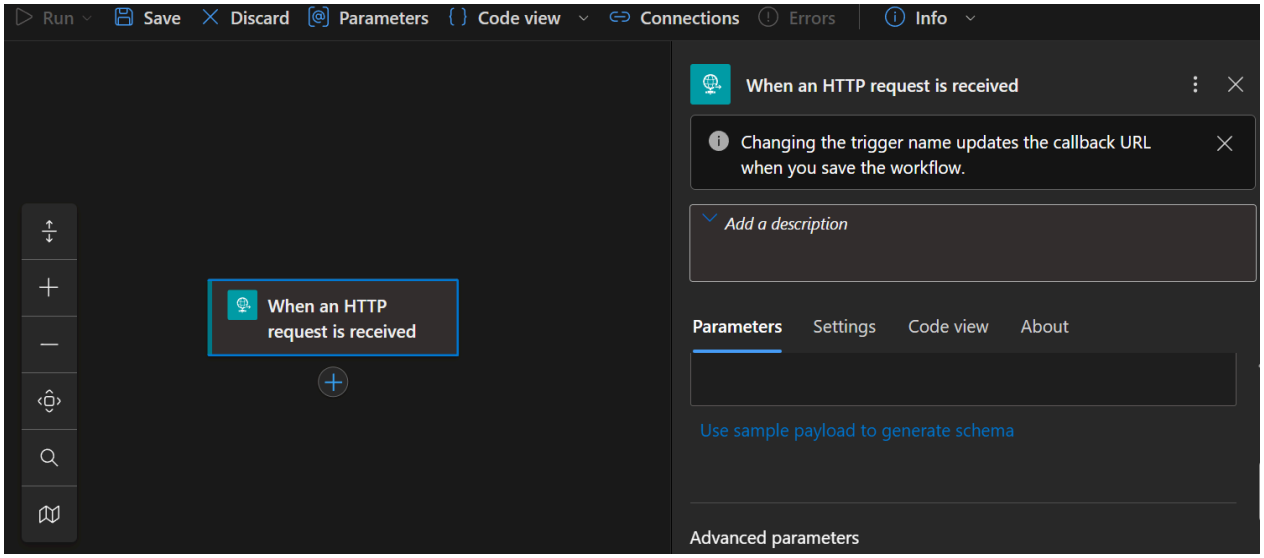
Go to marketplace and search for logic apps : Consumption

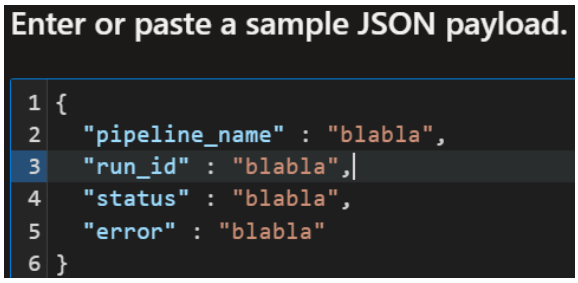


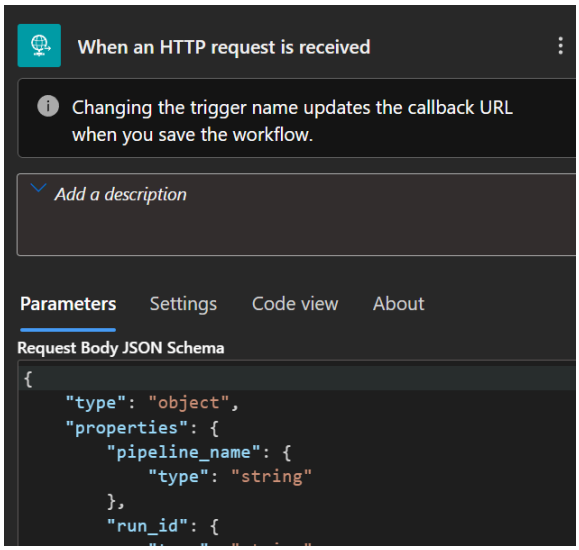

Home › ADF_Project › Marketplace › Create Logic App
Create Logic App (Multi-tenant)
Basics
Tags
Review + create
Summary
Logic App
by Microsoft
Details
Subscription
1194812e-1f96-4b7f-8c7c-c640db44ffc9
Resource Group
ADF_Project
Name
adflogicapps
Region
eastus
Log analytics
Disabled
Workflow Type
Stateful
Create
< Previous
Next >
Download a template for automation


Home › Microsoft.Web-LogicAppConsumption-Portal-28834726-8d55 | Overview › adflogicapps
adflogicapps | Logic app designer
Logic app
Search
Run
Save
Discard
Parameters
Code view
Connections
Errors
Info
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Development Tools
Logic app designer
Logic app code view
Logic app templates
Run history
Chat (Preview)
Add or remove favorites by pressing Ctrl+Shift+F
Add a trigger
Search for a trigger or connector
Favorites
Built-in tools
Request
HTTP
Schedule


Add a trigger
Return to search
Request
Built-in
When an HTTP request i...
Trigger



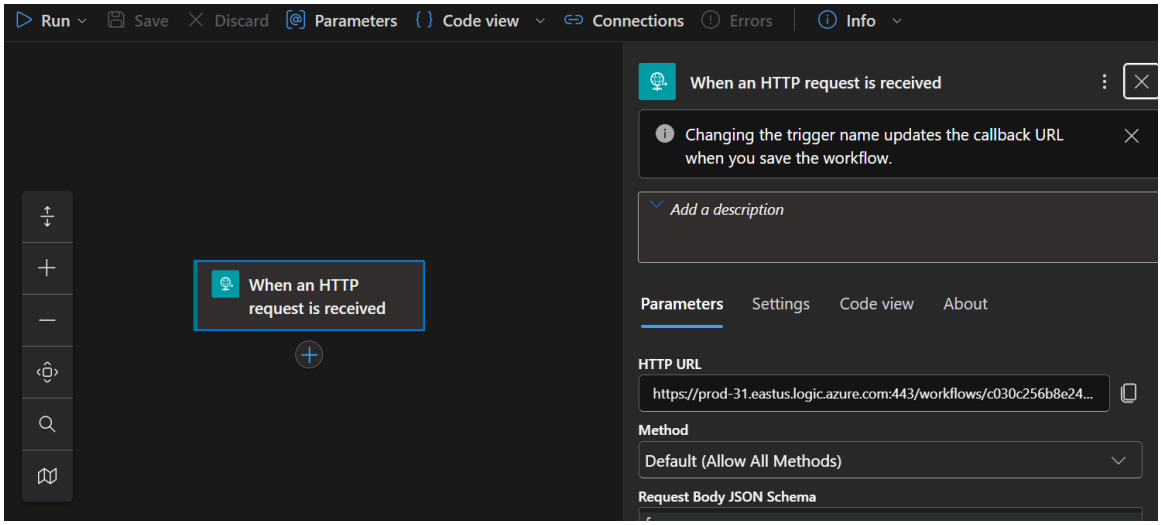




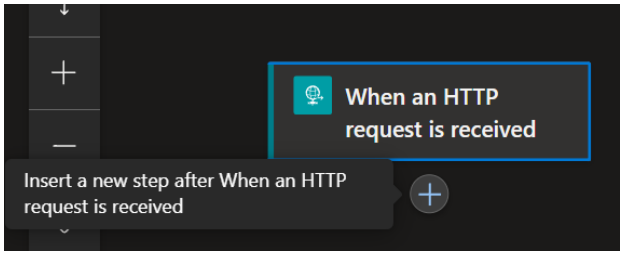

```
{
  "type": "object",
  "properties": {
    "pipeline_name": {
      "type": "string"
    },
    "run_id": {
      "type": "string"
```

```
        },
        "status": {
            "type": "string"
        },
        "error": {
            "type": "string"
        }
    }
}
```

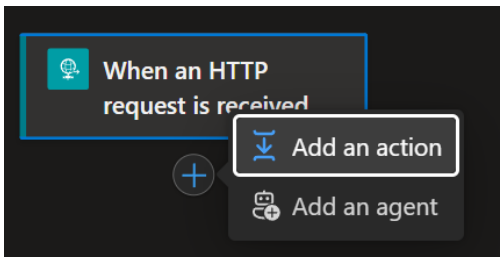
After saving the workflow , http url is created.

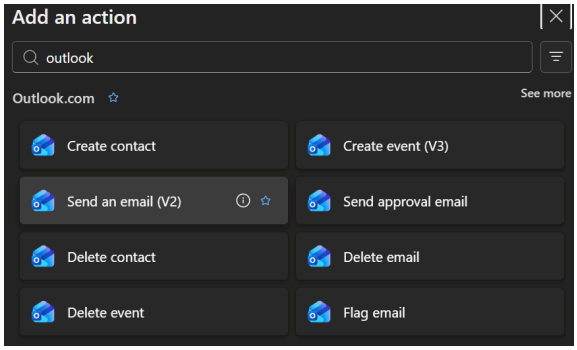


Click on + :

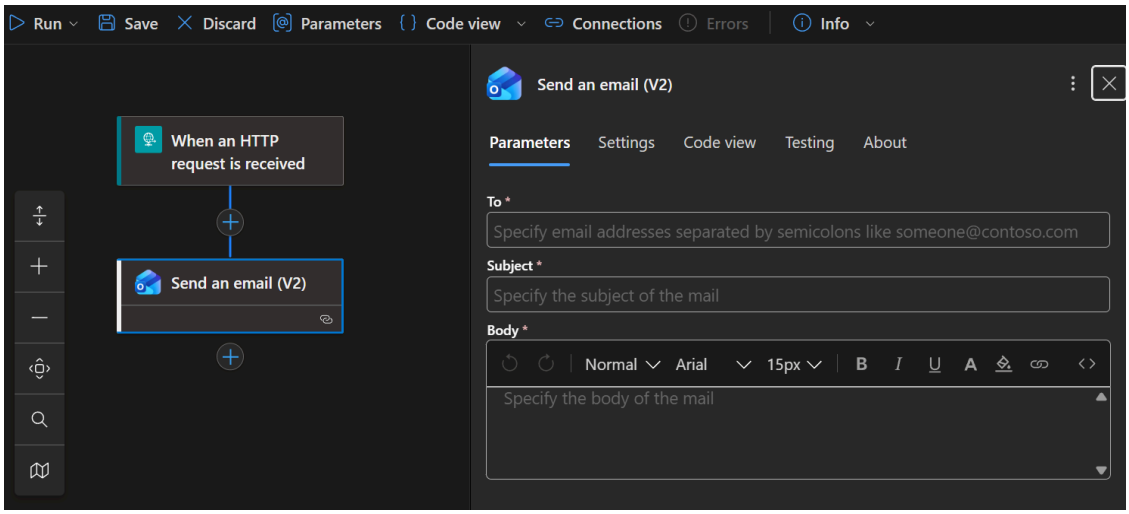


Click add action :

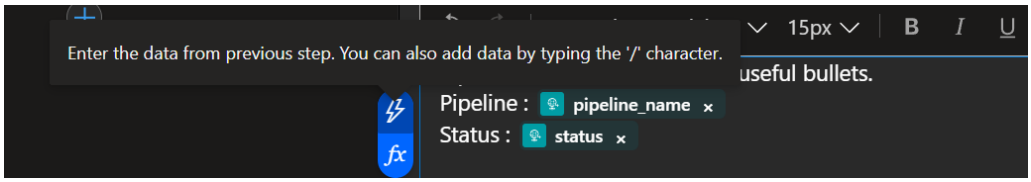


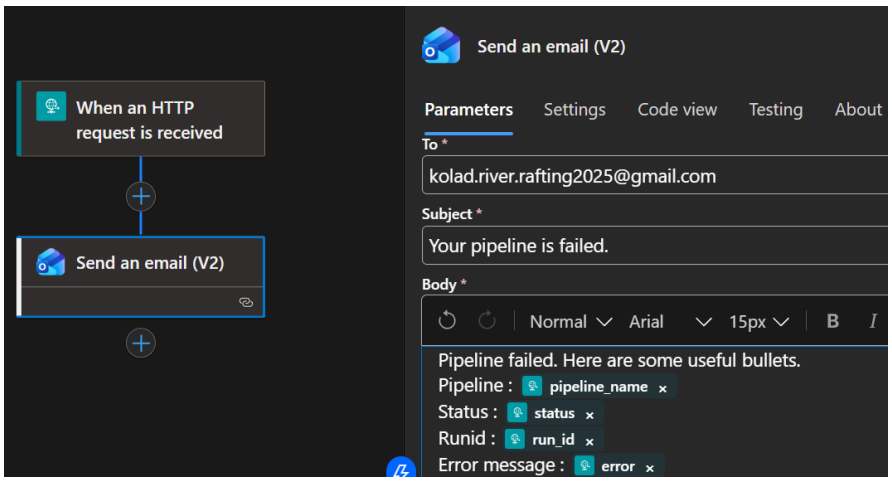


After signing in :

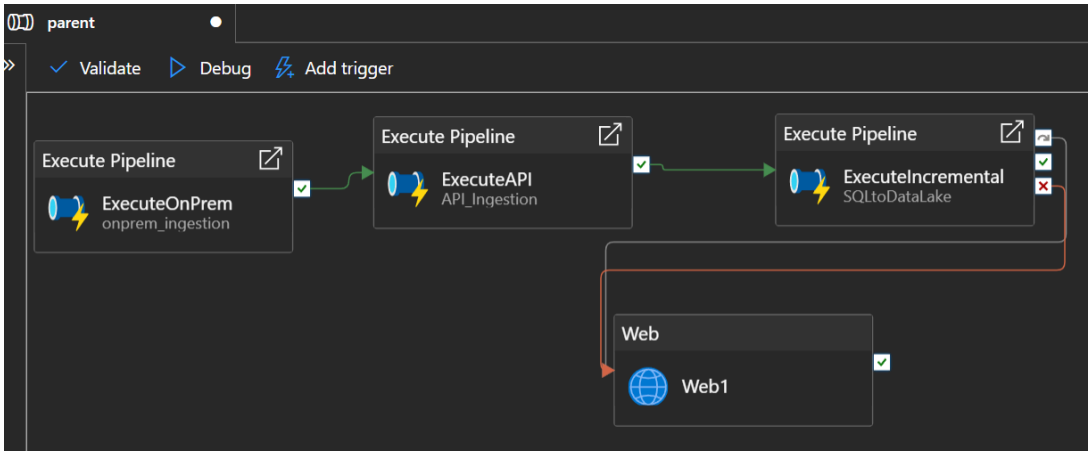


Click on this thundering sign :



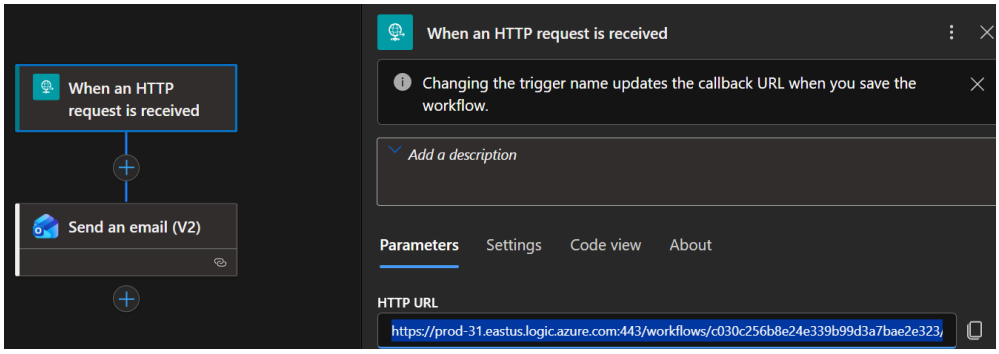


On skip and failure :

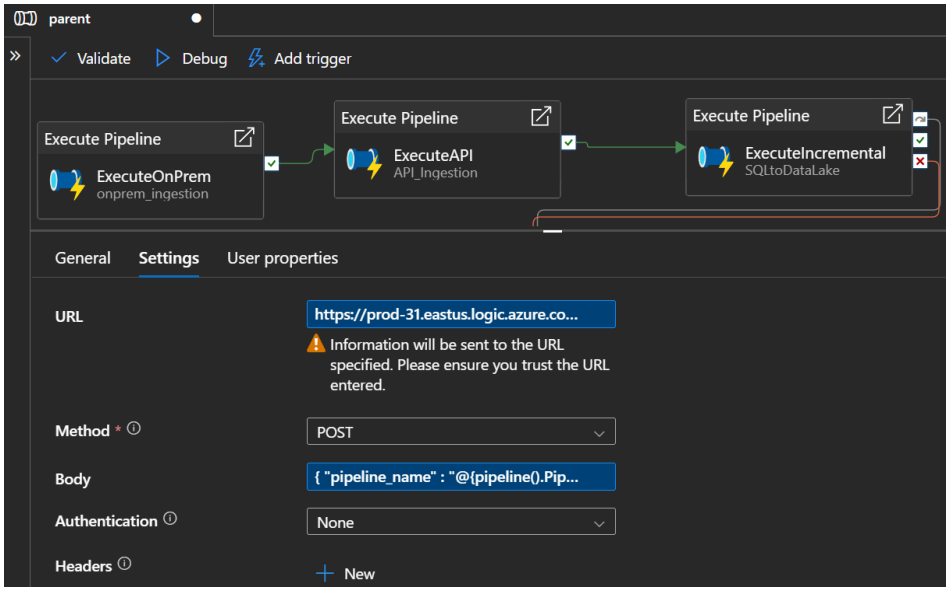


Copy the http url :

https://prod-31.eastus.logic.azure.com:443/workflows/c030c256b8e24e339b99d3a7bae2e323/triggers/When_an_HTTP_request_is_received/paths/invoke?api-version=2016-10-01&sp=%2Ftriggers%2FWhen_an_HTTP_request_is_received%2Frun&sv=1.0&sig=d3cPGjOc13e24fLnrX1X3_q8jli_hdfcUxkWfwTVoYo



And paste in the settings of the web :

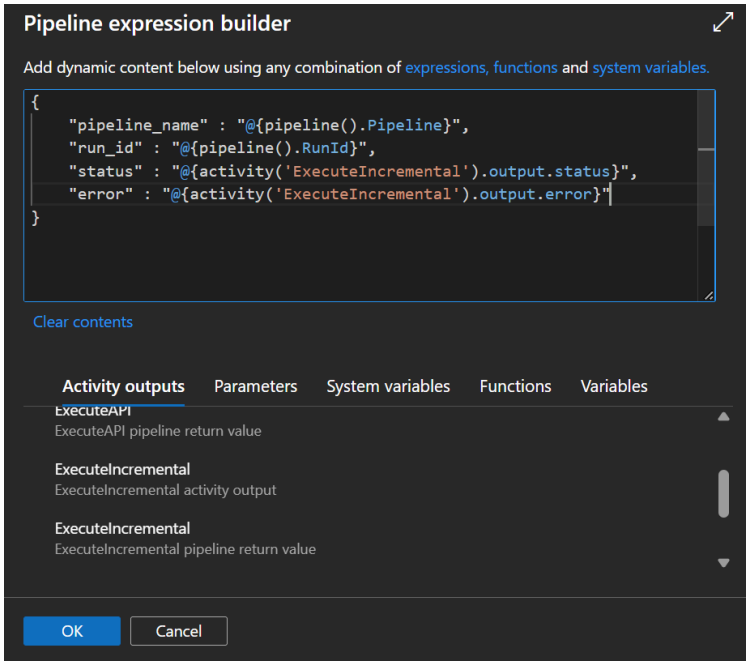


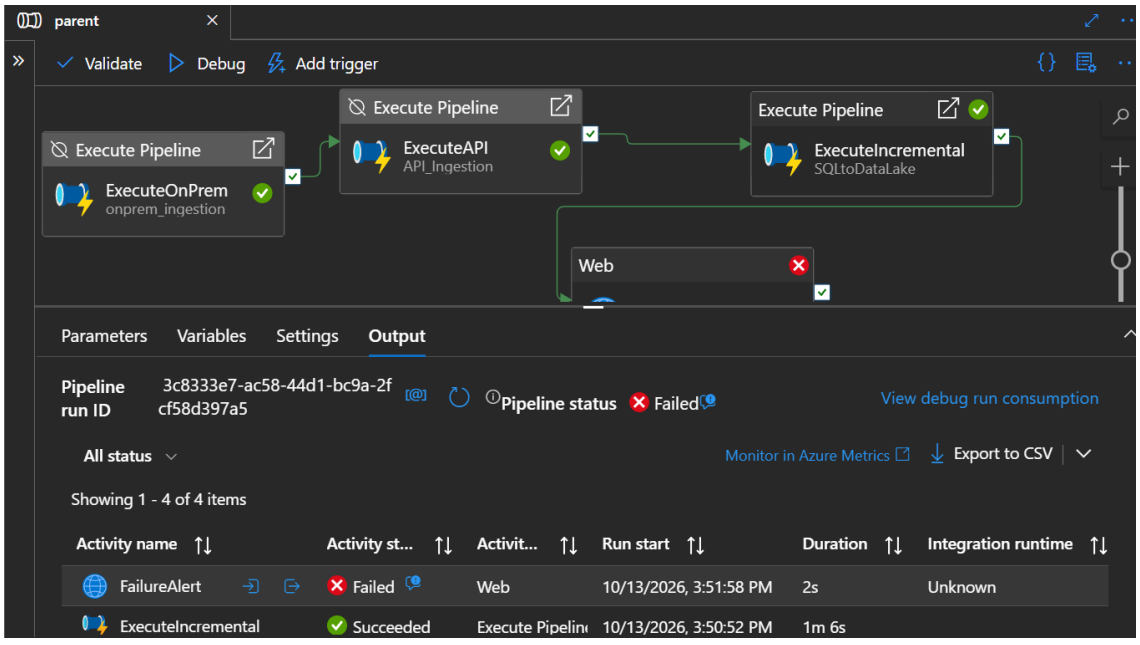
POST body :

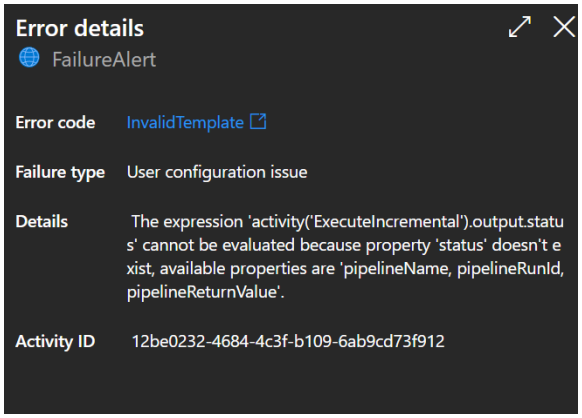
```
{
    "pipeline_name" : "@{pipeline().Pipeline}",
    "run_id" : "@{pipeline().RunId}",
    "status" : "@{activity('ExecuteIncremental').output.status}",
    "error" : "@{activity('ExecuteIncremental').output.error}"
}
```

1,2 : system variables

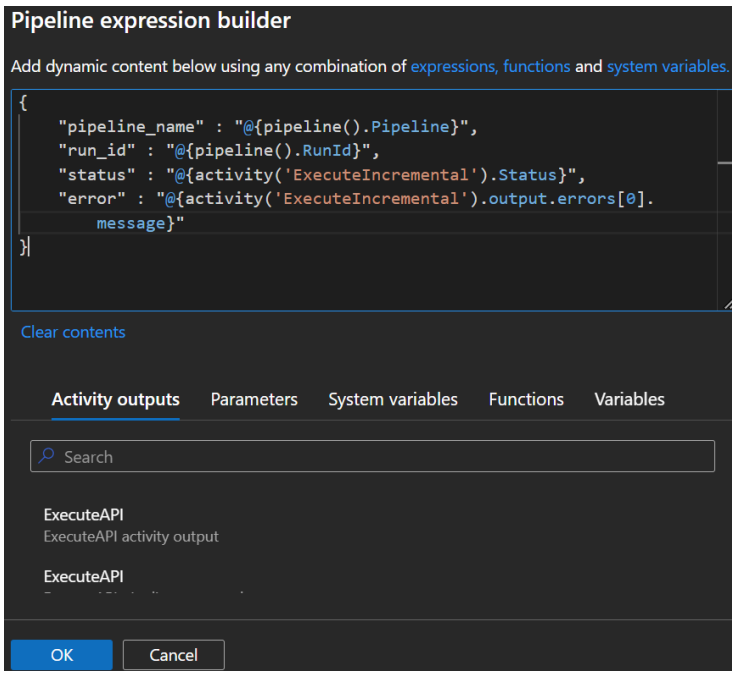
3,4 : Activity outputs







Changes :

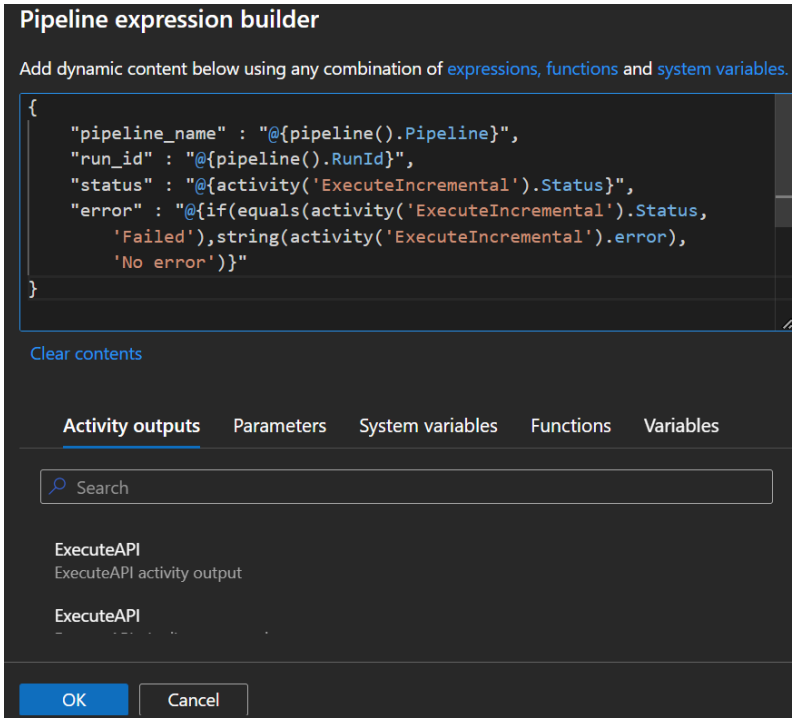



```
"status" : "@{activity('ExecuteIncremental').Status}",
    "error" : "@{activity('ExecuteIncremental').output.errors[0].message}"
```

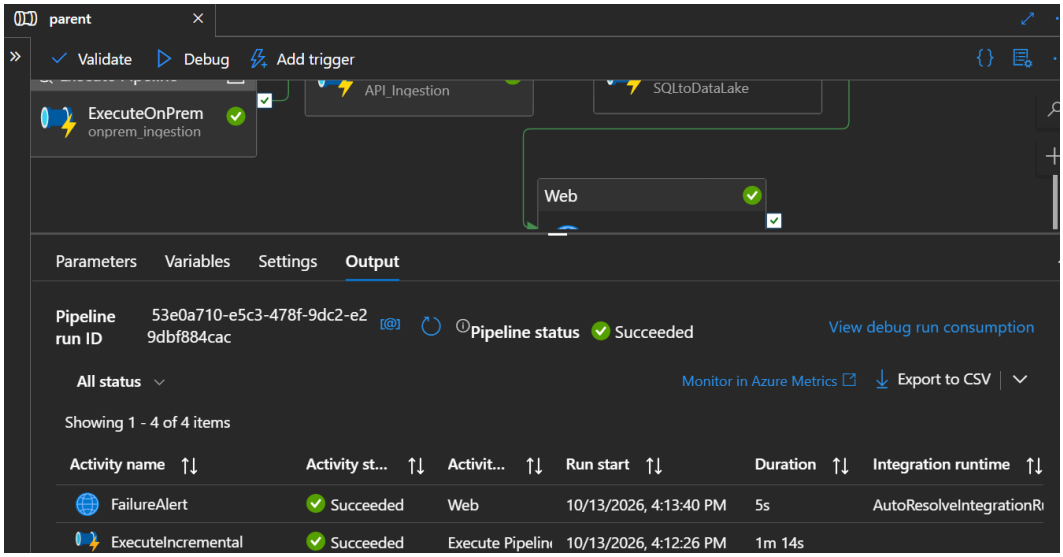
Yet it failed so `"error" : "@{activity('ExecuteIncremental').output.error`

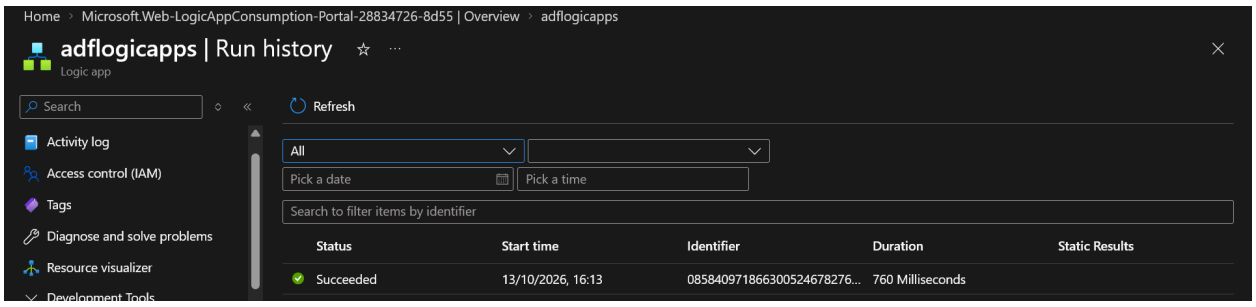
Final correction :

```
{
    "pipeline_name" : "@{pipeline().Pipeline}",
    "run_id" : "@{pipeline().RunId}",
    "status" : "@{activity('ExecuteIncremental').Status}",
    "error" :
"@{if(equals(activity('ExecuteIncremental').Status,'Failed'),string(activity('ExecuteIncremental').error),'No error')}"
}
```



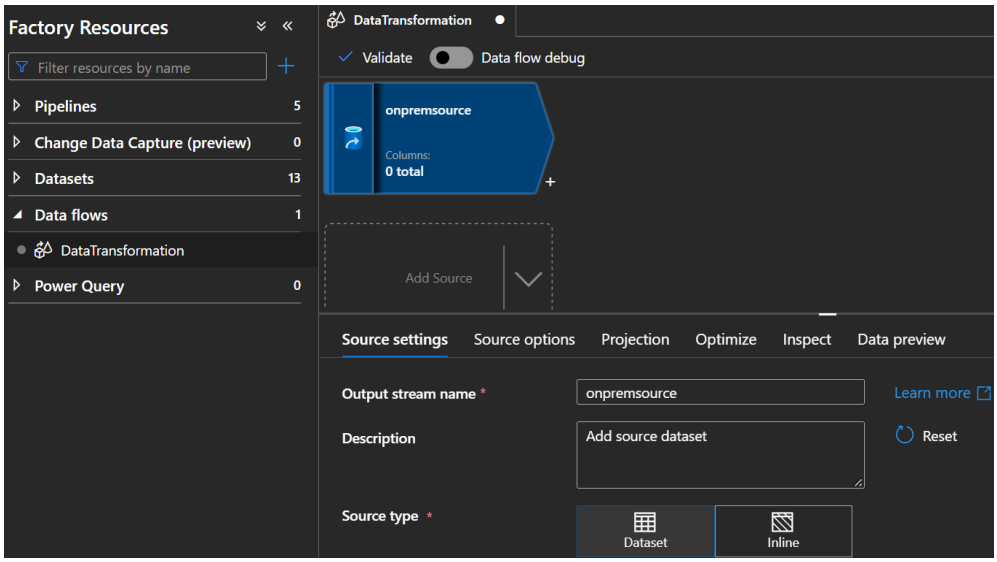
Use try catch for success :



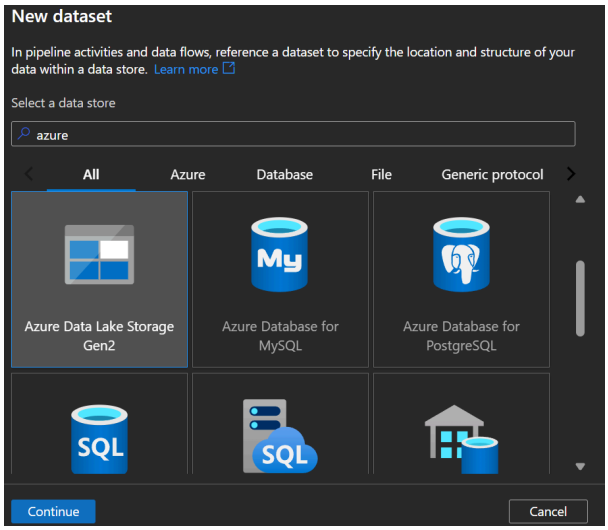


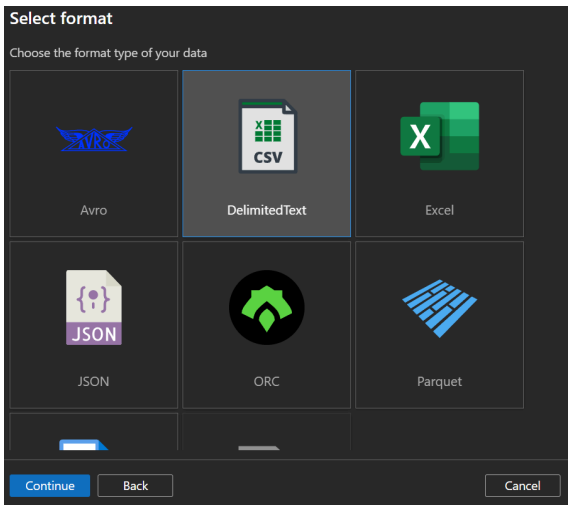
3:45:00 PySpark Transformation Using Azure Data Factory

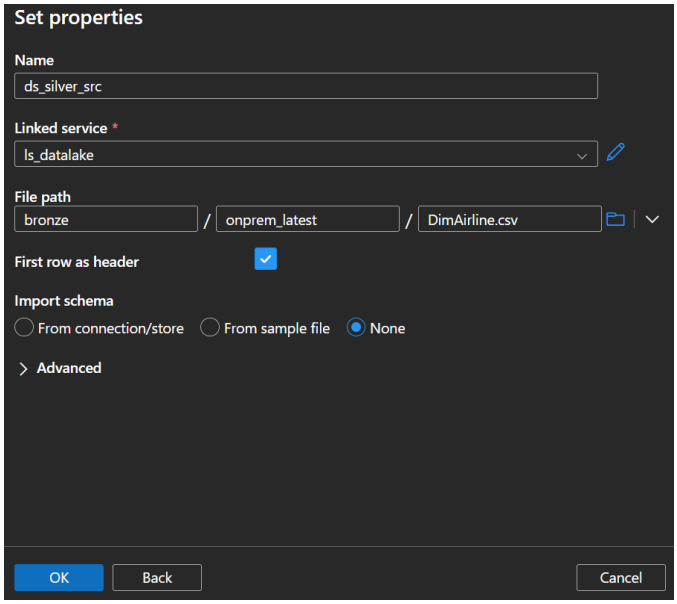
Create dataflow and click on add source : DimAirline



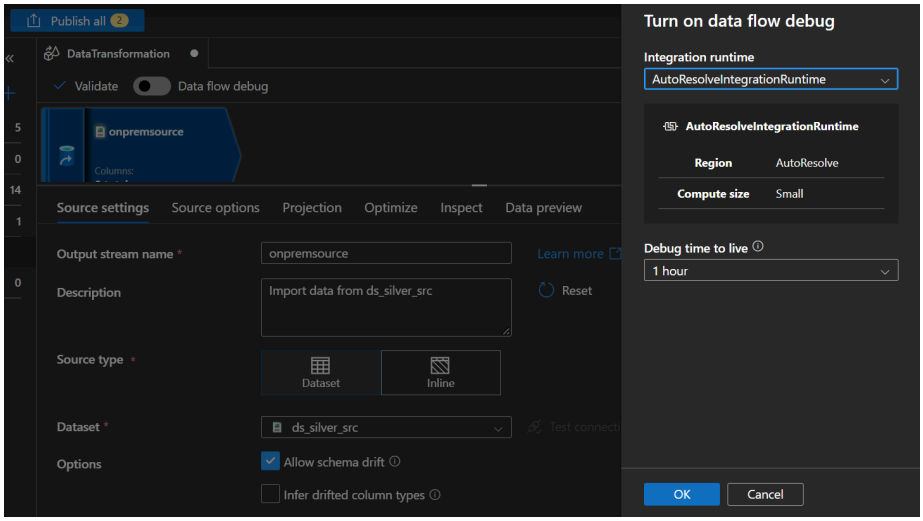
Creating new dataset :



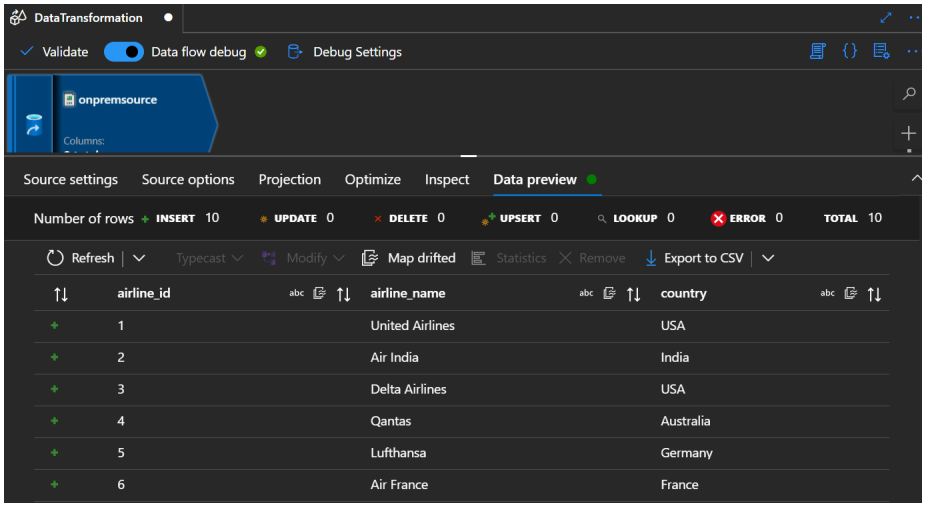




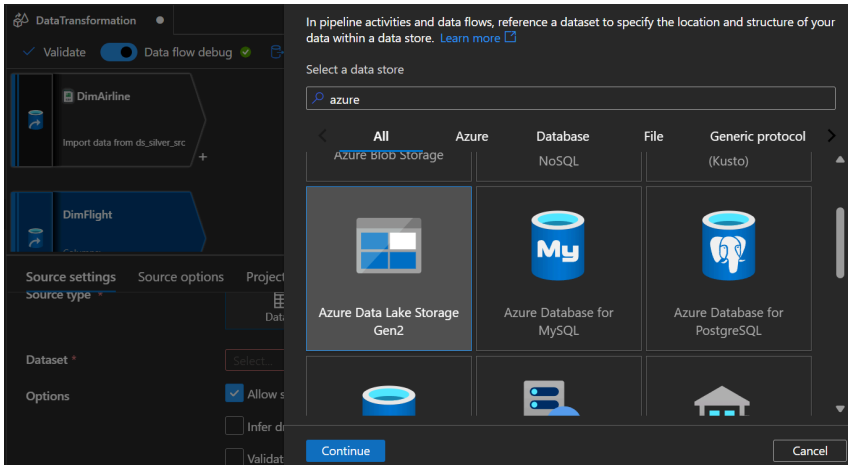
Click on data flow debug :

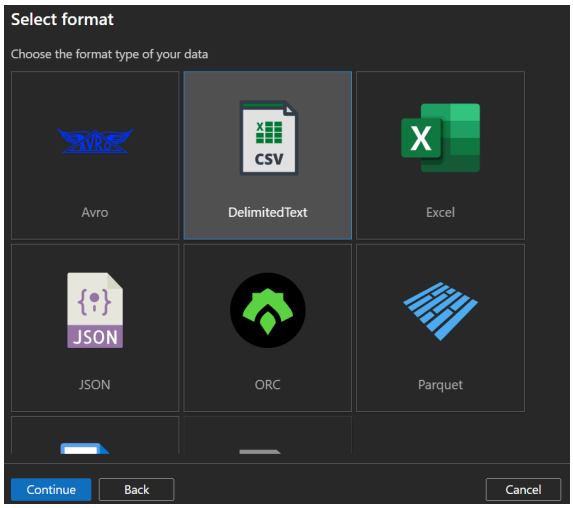


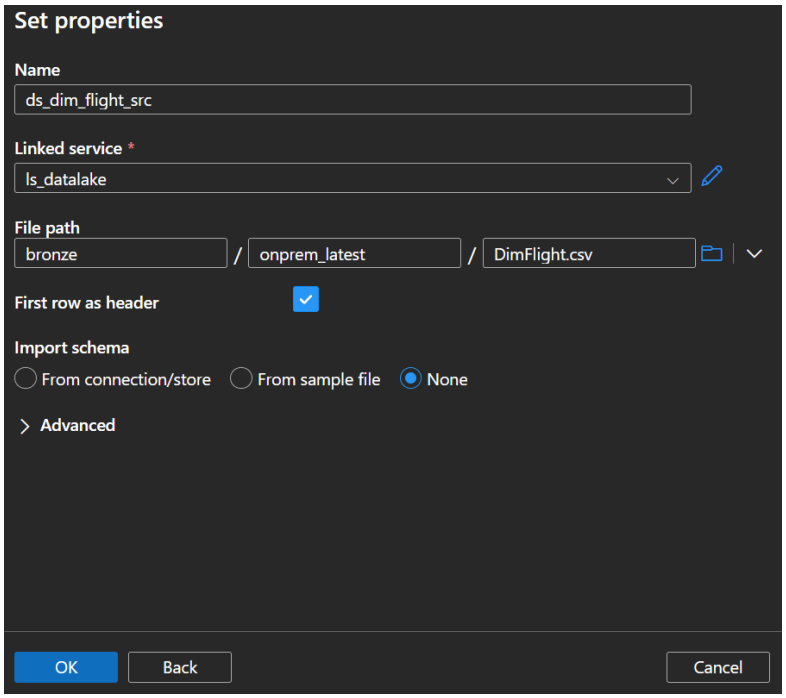
Data flow debug is for previewing :



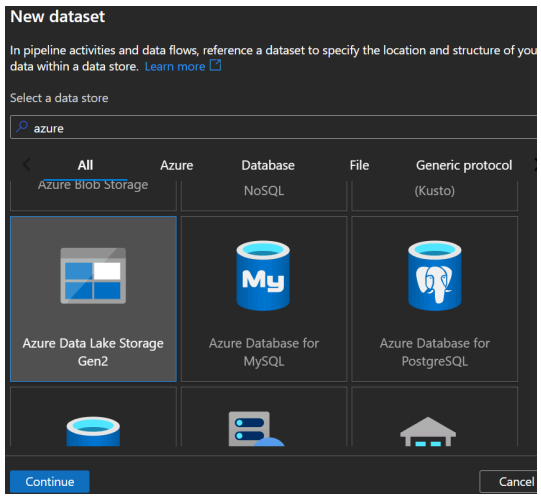
For dimflight source :

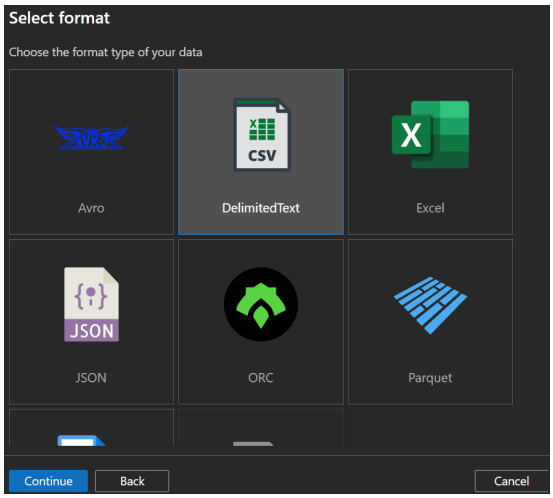


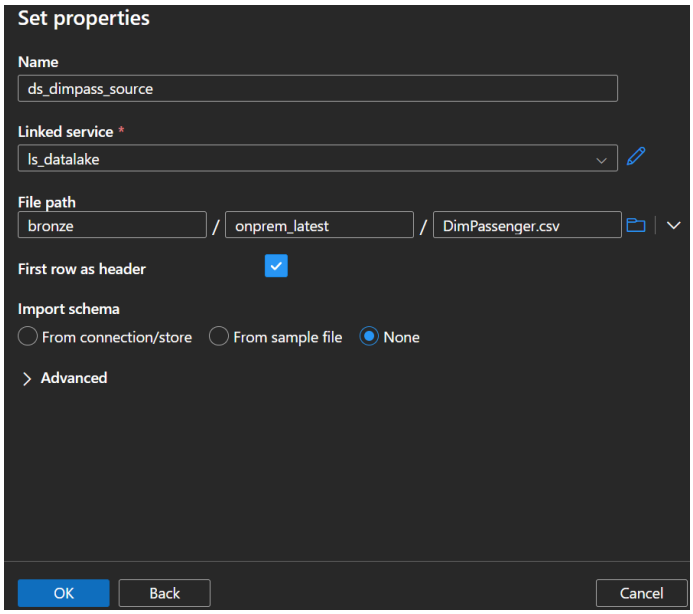




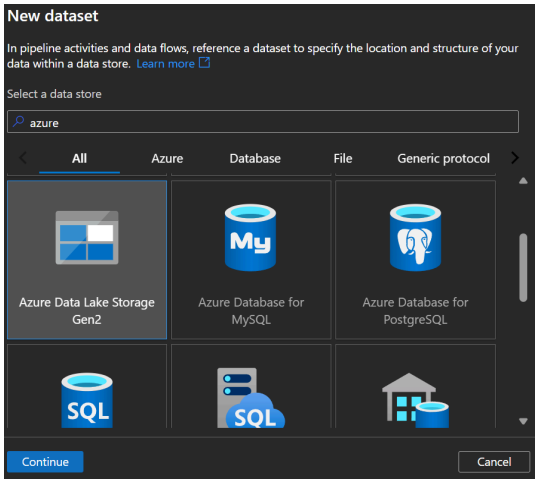
For dimpassenger source :

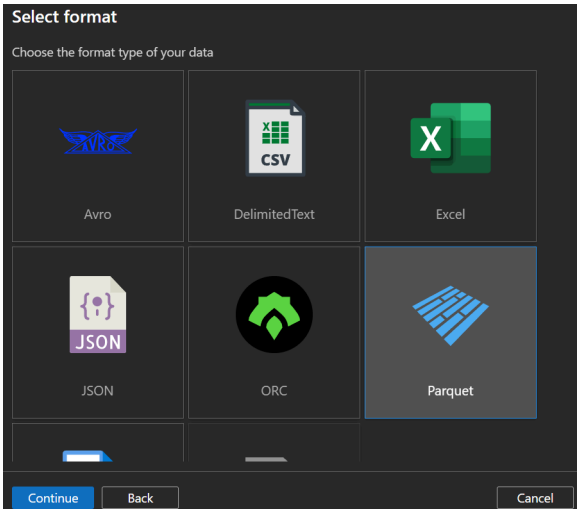


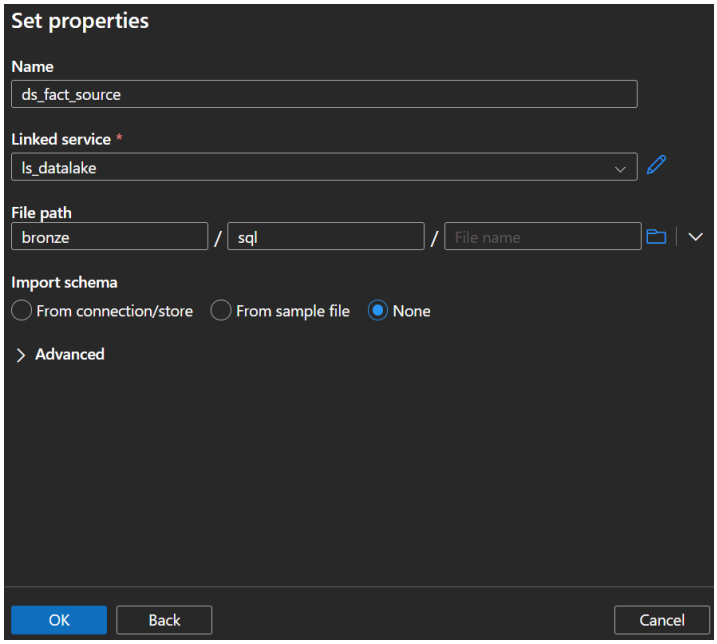




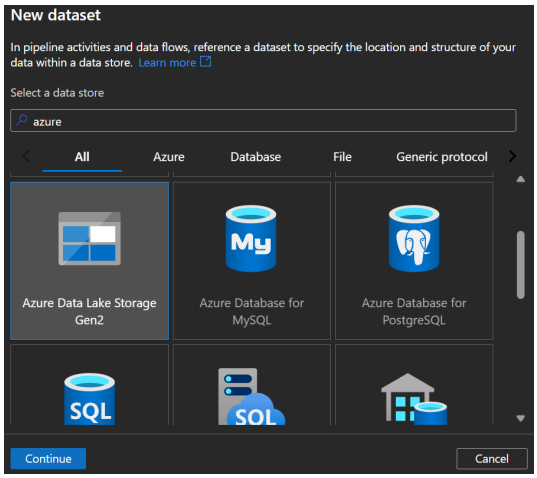
For factbookings source :

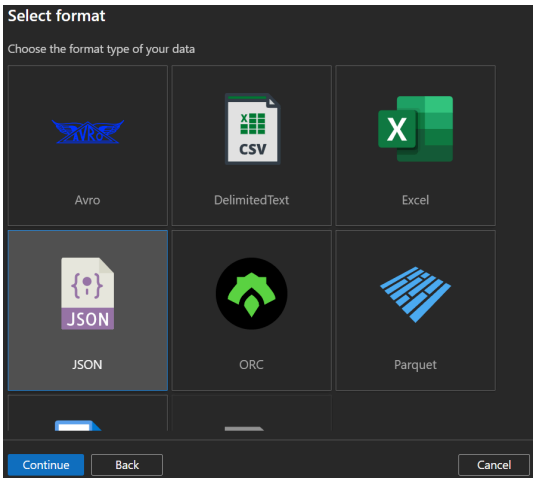


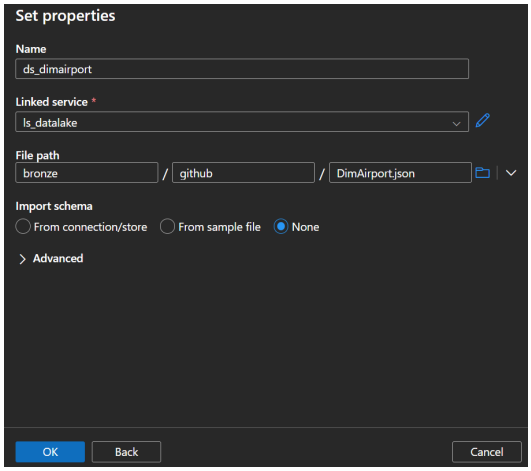




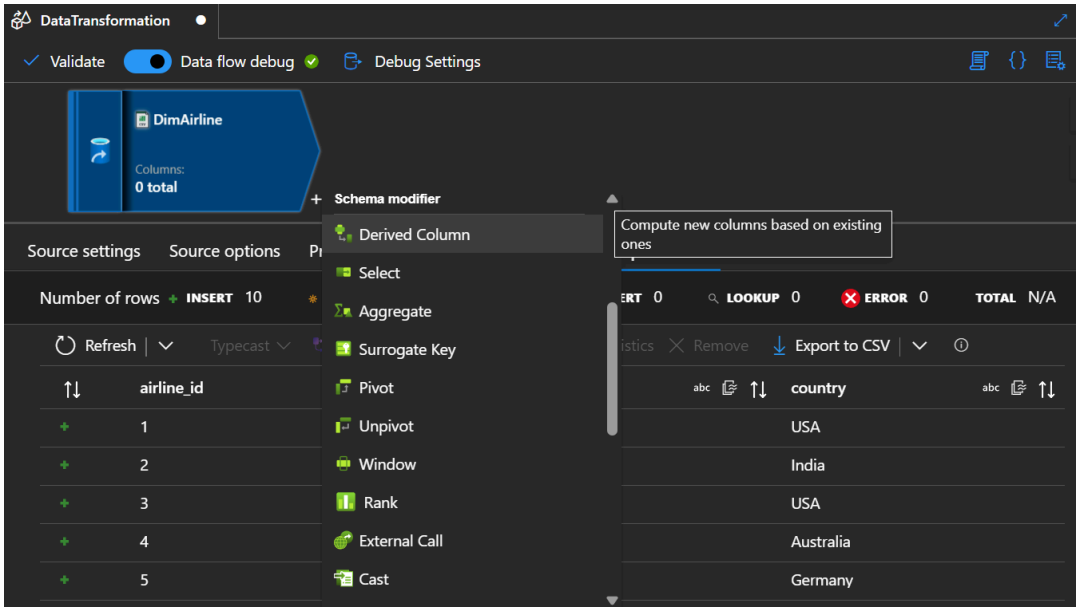
For dimairport source :



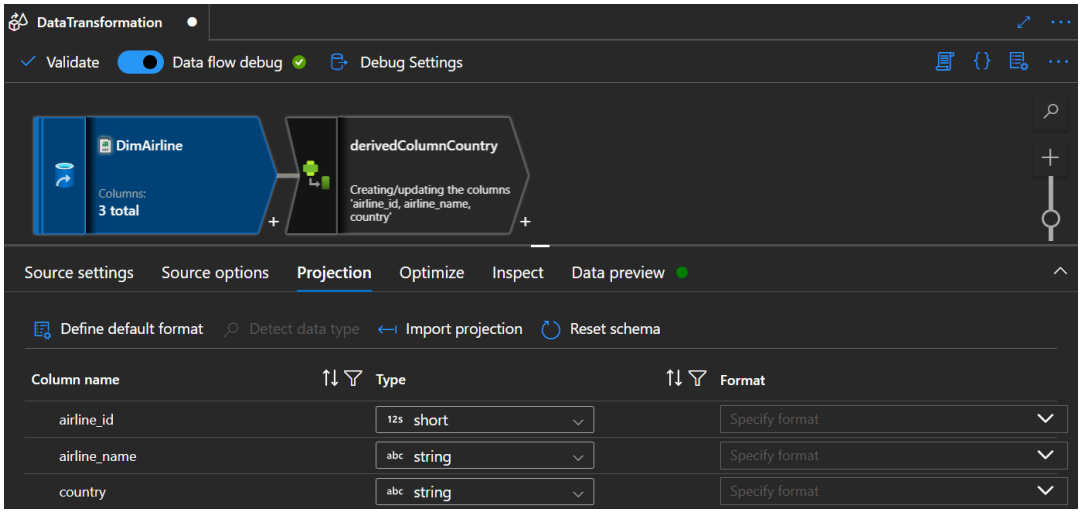




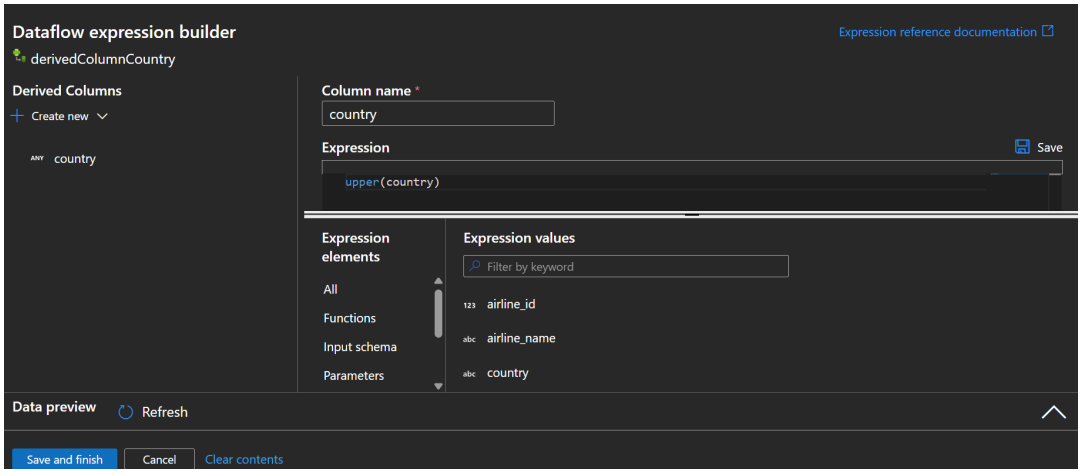
Click on the + and select derived column :



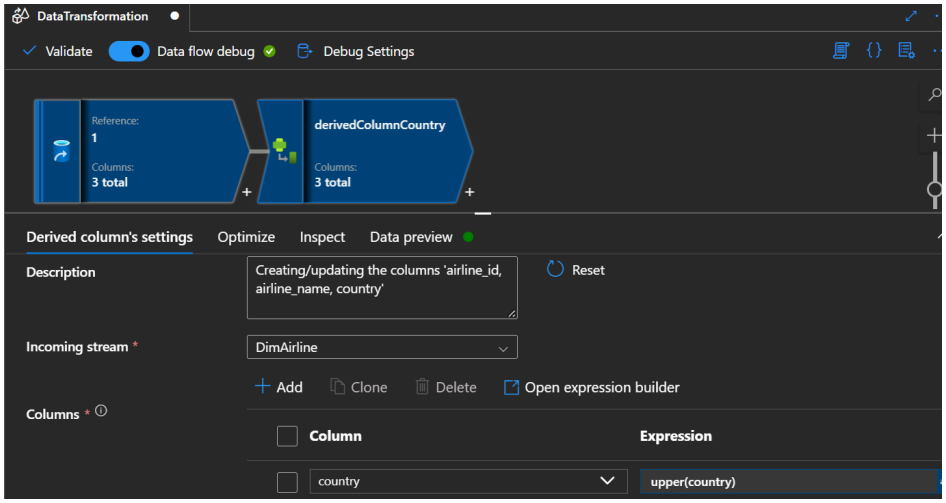
Importing projections for dimairline :



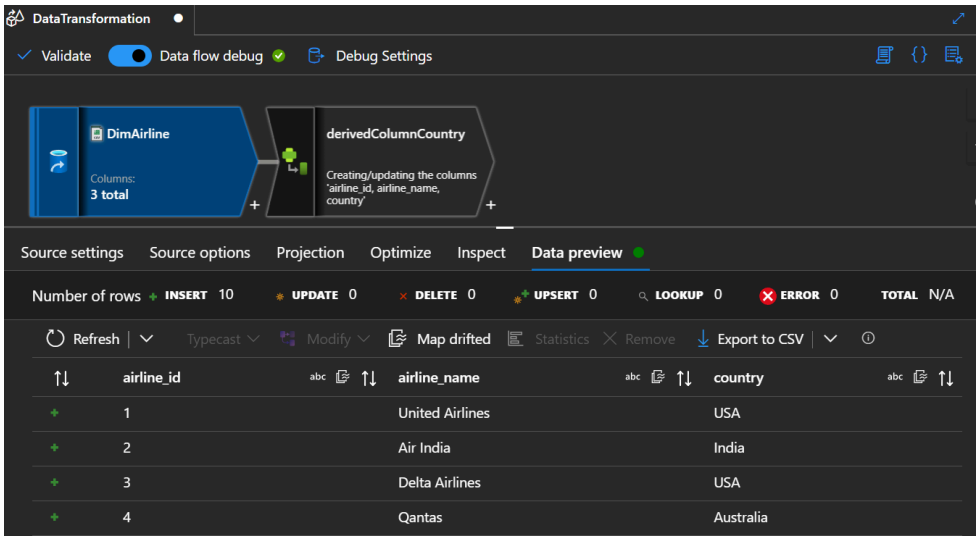
Click on functions -> upper , input schema -> country

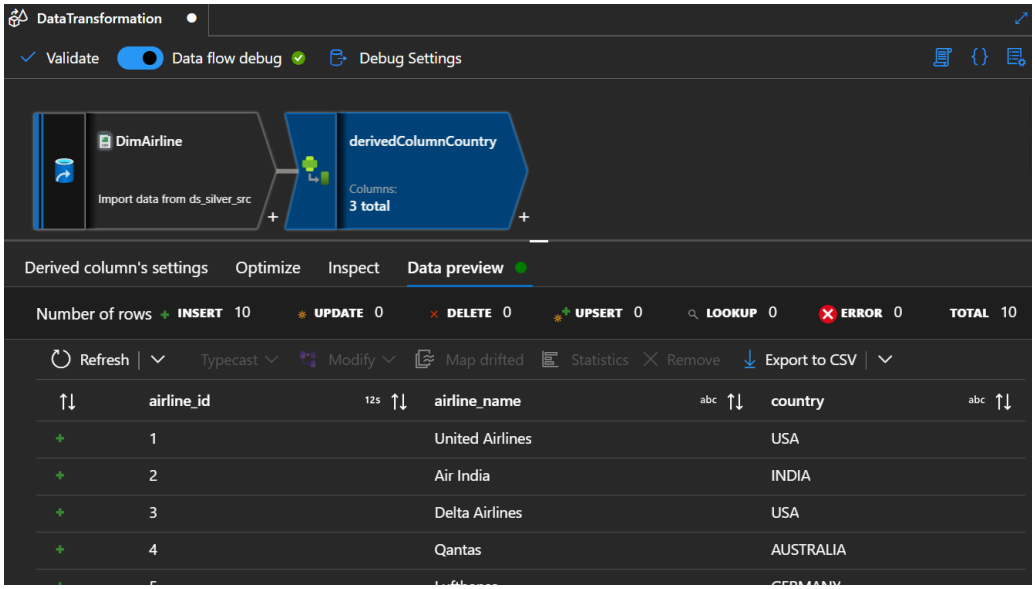


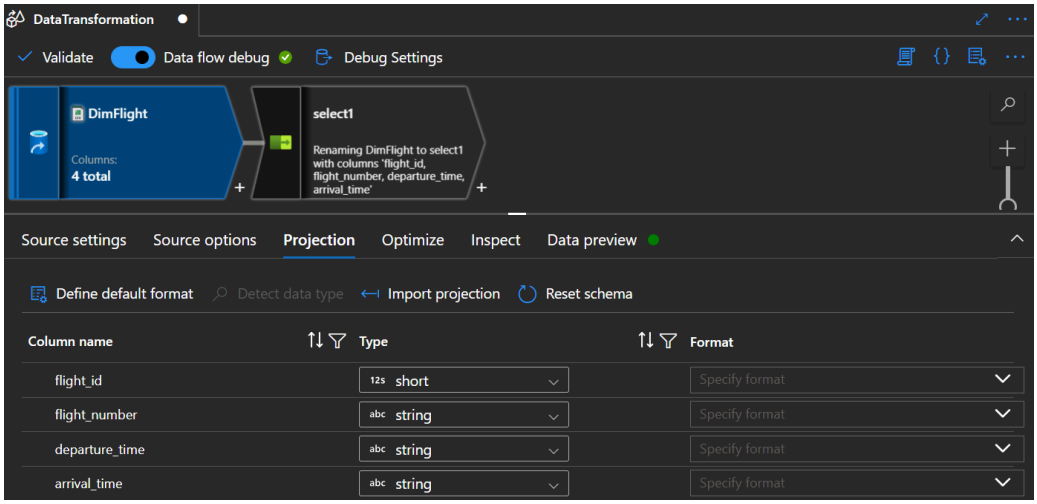
See column and expression :



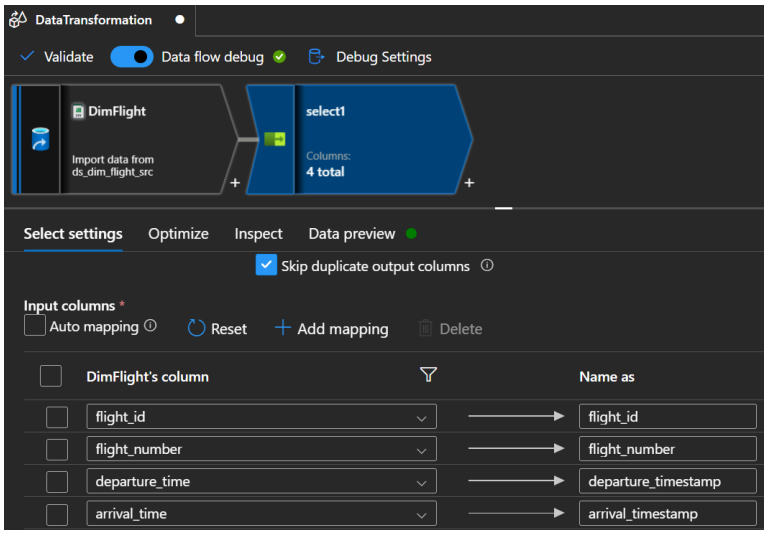
See the changes :



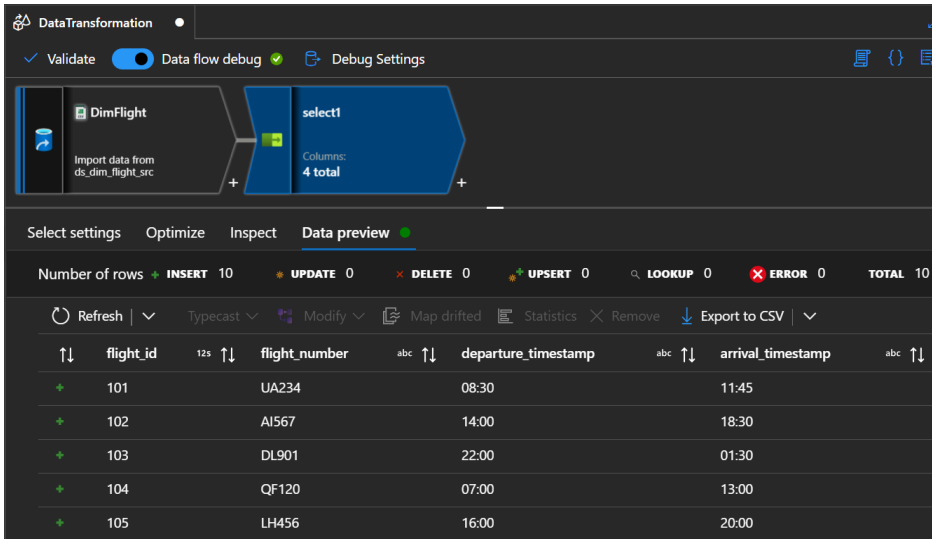




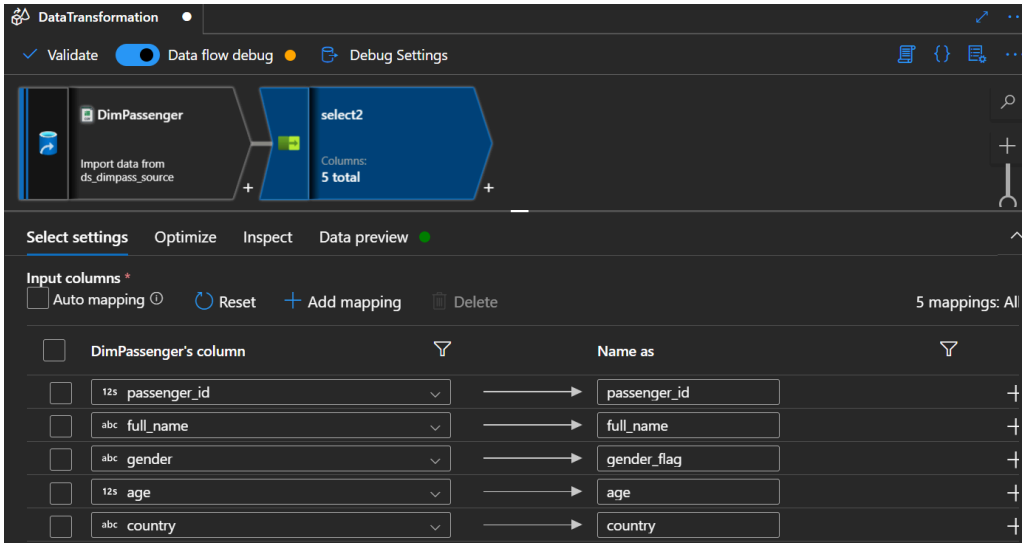
Uncheck automapping and rename the column name :

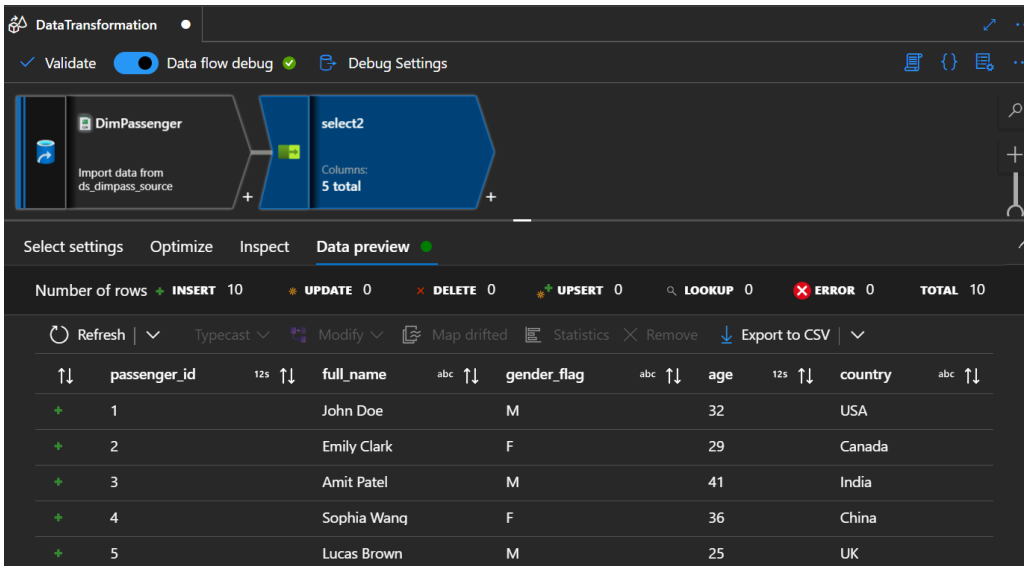


Renaming successful :

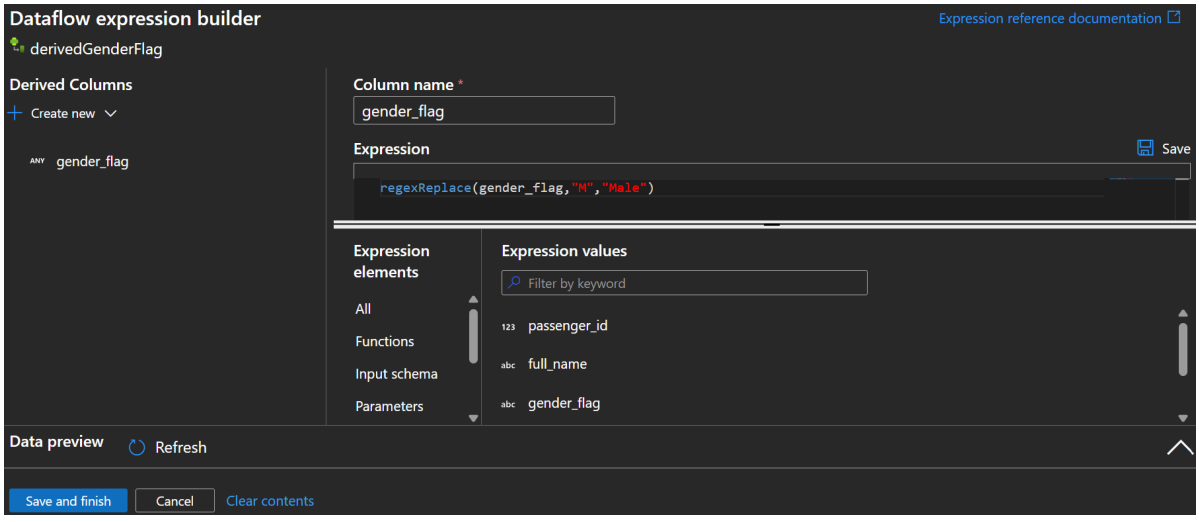


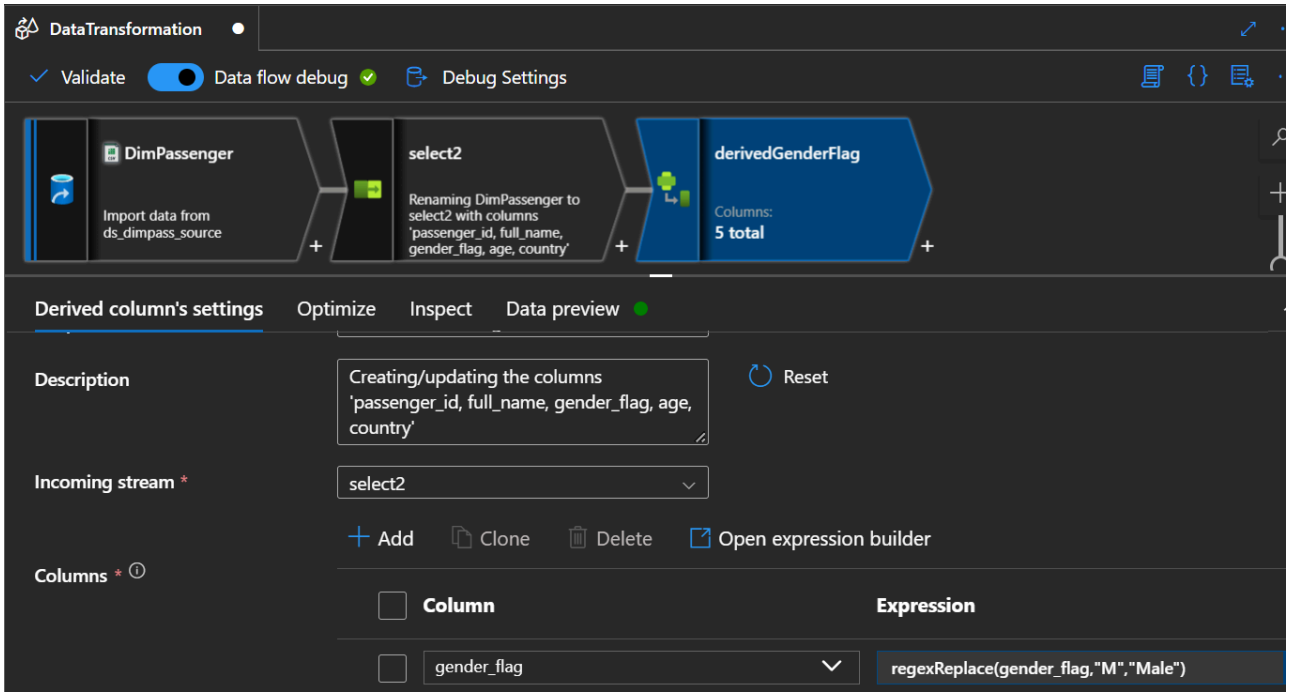
Rename gender col :



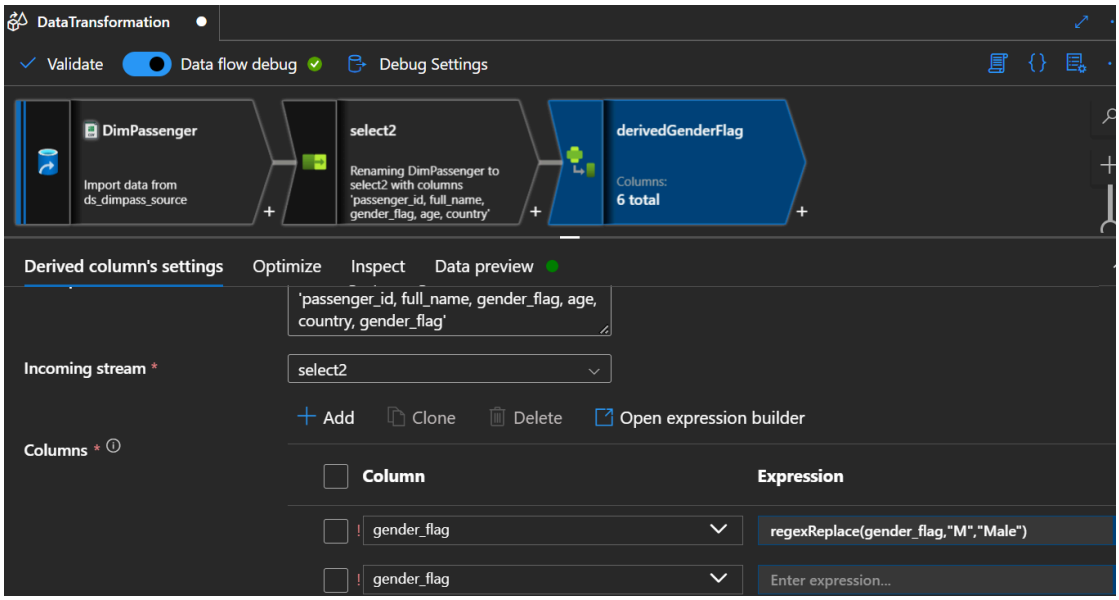


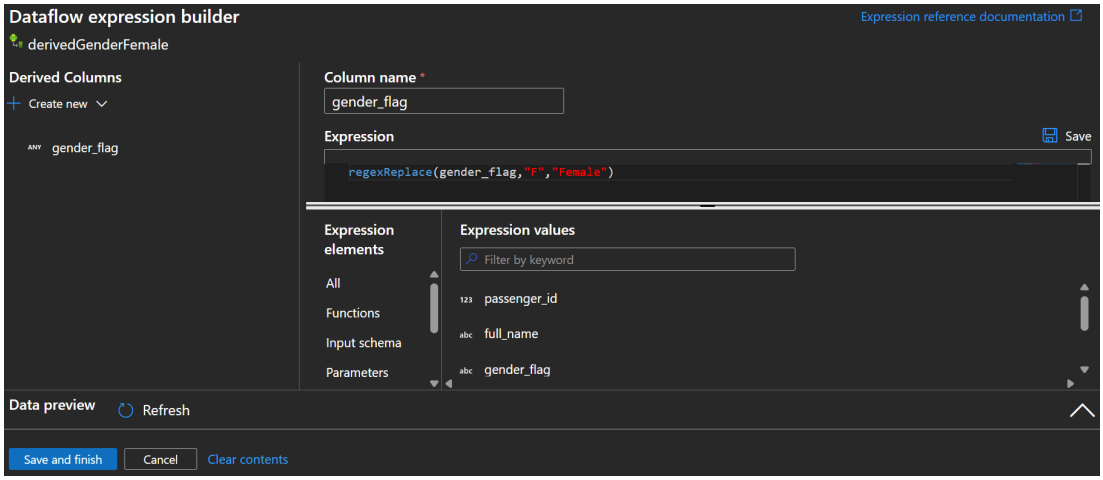
Create a derived column and here is the expression for gender_flag :

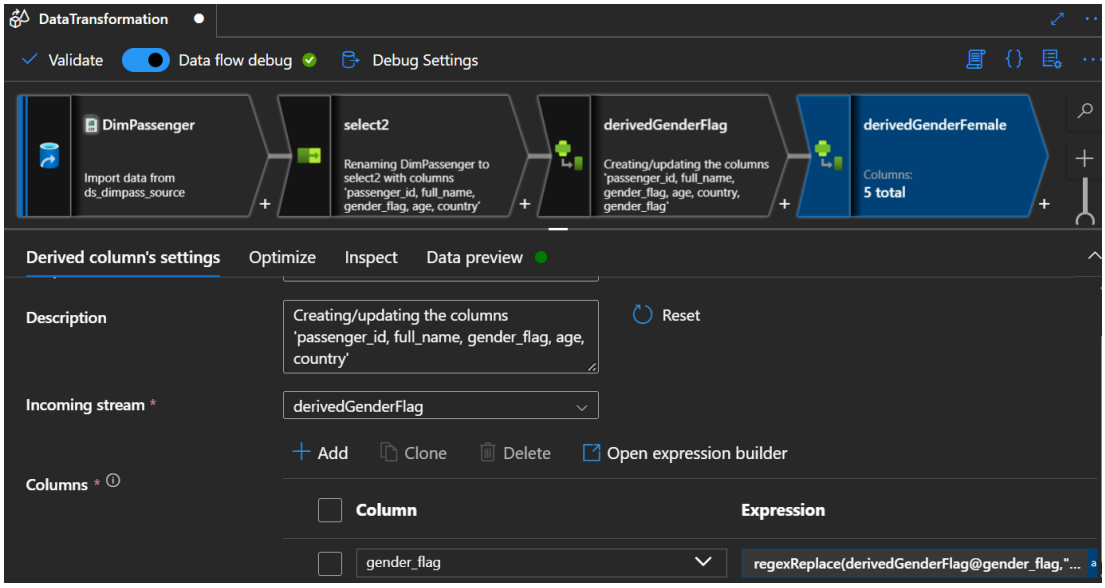


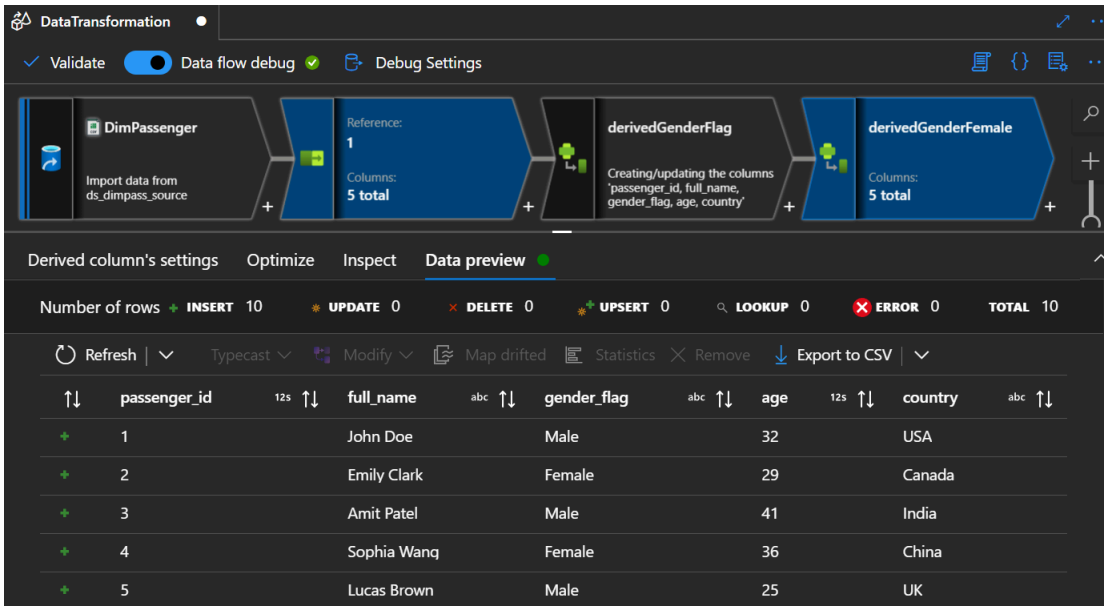


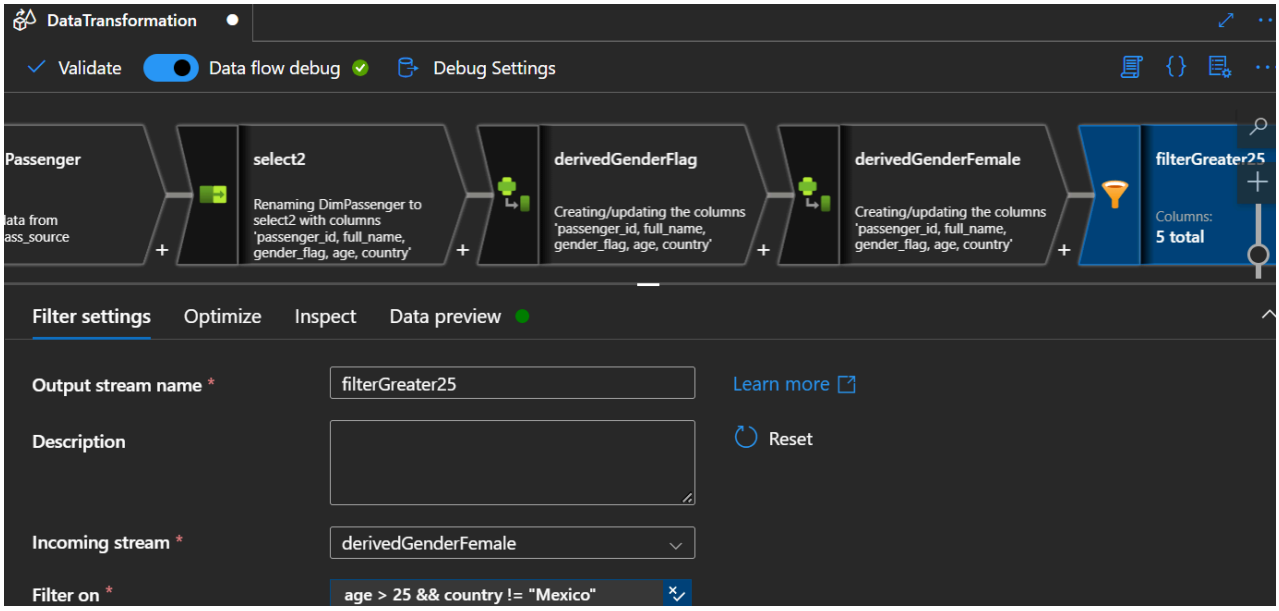
Cannot have same column in one transformation :



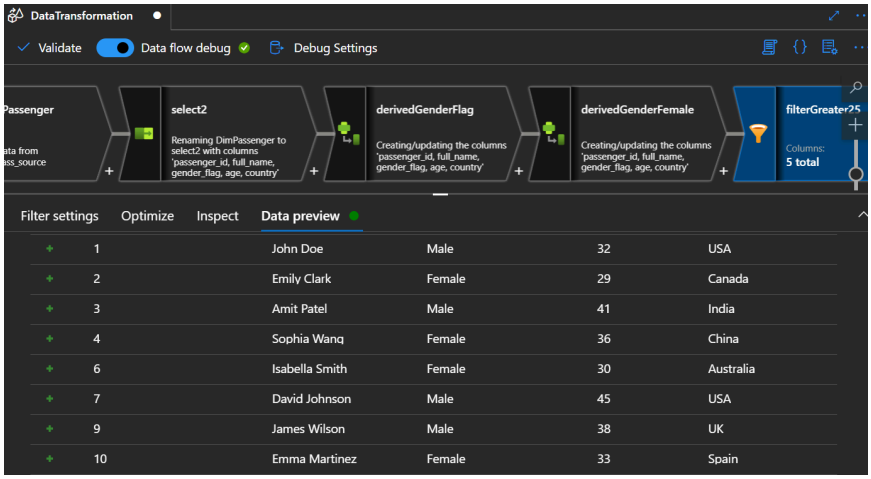

Dataflow expression builder
Expression reference documentation
derivedGenderFemale
Derived Columns
Create new
gender_flag
Column name *
gender_flag
Expression
Save
regexReplace(gender_flag,"F","Female")
Expression elements
All
Functions
Input schema
Parameters
Expression values
Filter by keyword
passenger_id
full_name
gender_flag
Data preview
Refresh
Save and finish
Cancel
Clear contents


DataTransformation
Validate
Data flow debug
Debug Settings
DimPassenger
Import data from ds_dimpass_source
select2
Renaming DimPassenger to select2 with columns 'passenger_id, full_name, gender_flag, age, country'
derivedGenderFlag
Creating/updating the columns 'passenger_id, full_name, gender_flag, age, country, gender_flag'
derivedGenderFemale
Columns:
5 total
Derived column's settings
Optimize
Inspect
Data preview
Description
Creating/updating the columns 'passenger_id, full_name, gender_flag, age, country'
Reset
Incoming stream *
derivedGenderFlag
Add
Clone
Delete
Open expression builder
Columns *
Column
Expression
gender_flag
regexReplace(derivedGenderFlag@gender_flag,"...


DataTransformation
Validate
Data flow debug
Debug Settings
DimPassenger
Import data from ds_dimpass_source
Reference:
1
Columns:
5 total
derivedGenderFlag
Creating/updating the columns 'passenger_id, full_name, gender_flag, age, country'
derivedGenderFemale
Columns:
5 total
Derived column's settings
Optimize
Inspect
Data preview
Number of rows
INSERT 10
UPDATE 0
DELETE 0
UPSERT 0
LOOKUP 0
ERROR 0
TOTAL 10
Refresh
Typecast
Modify
Map drifted
Statistics
Remove
Export to CSV
passenger_id
full_name
gender_flag
age
country
1
John Doe
Male
32
USA
2
Emily Clark
Female
29
Canada
3
Amit Patel
Male
41
India
4
Sophia Wang
Female
36
China
5
Lucas Brown
Male
25
UK

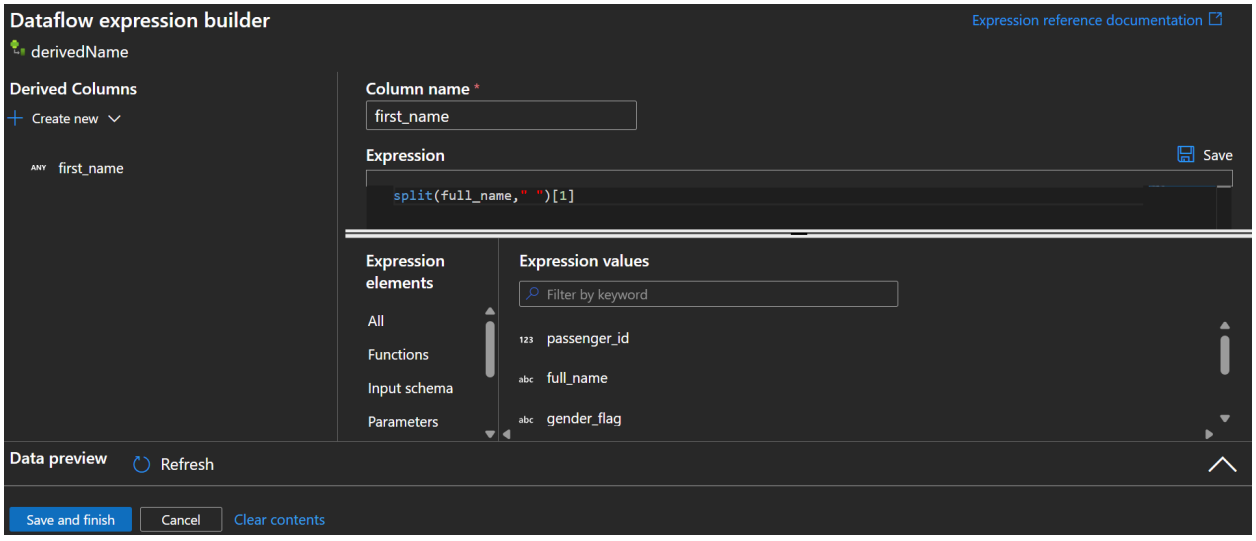


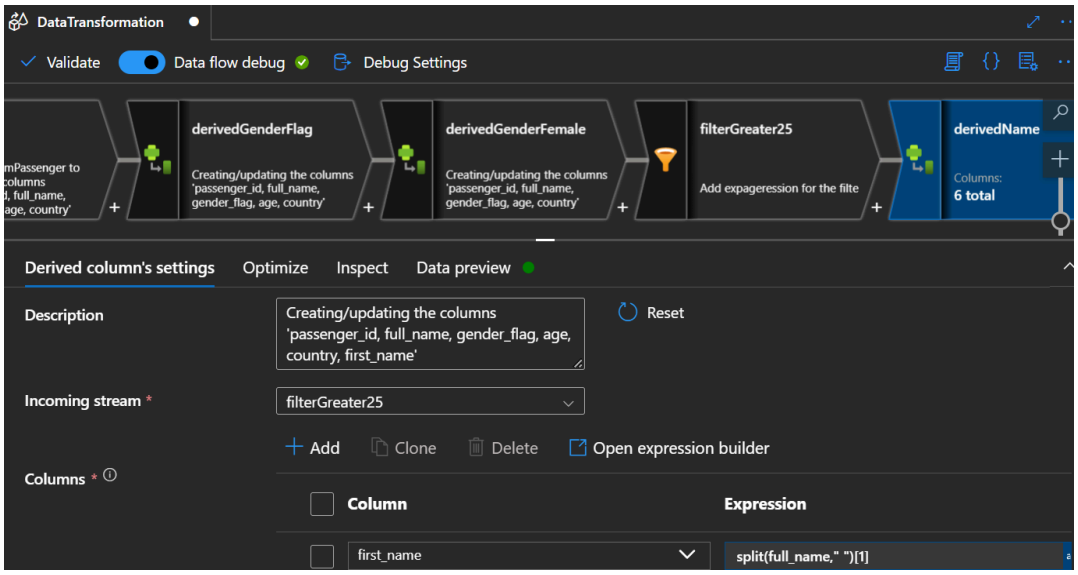
Filtration successful :

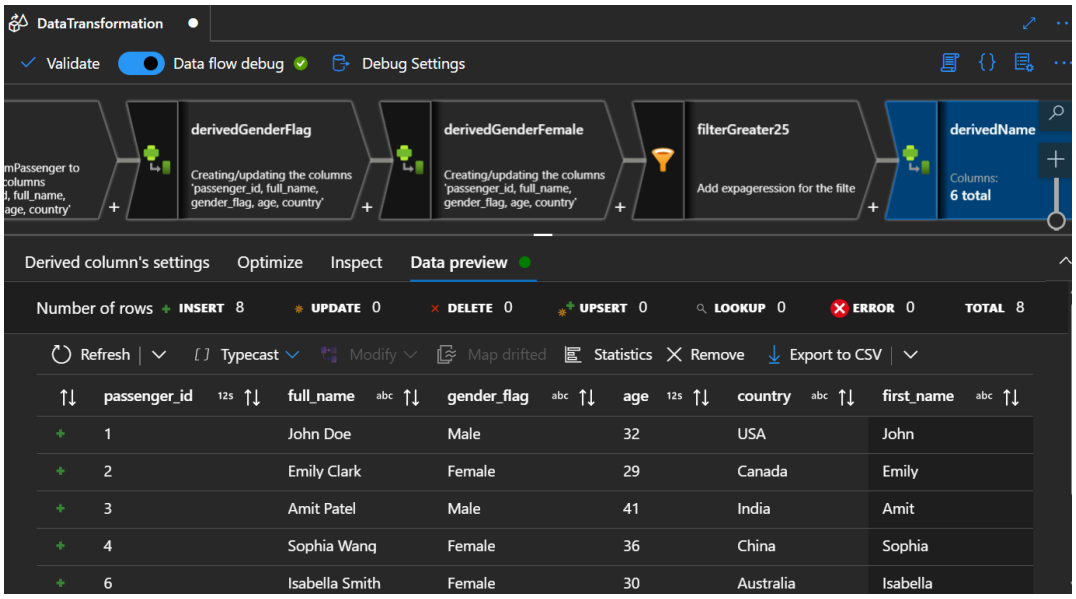


+	1	John Doe	Male	32	USA
+	2	Emily Clark	Female	29	Canada
+	3	Amit Patel	Male	41	India
+	4	Sophia Wang	Female	36	China
+	6	Isabella Smith	Female	30	Australia
+	7	David Johnson	Male	45	USA
+	9	James Wilson	Male	38	UK
+	10	Emma Martinez	Female	33	Spain

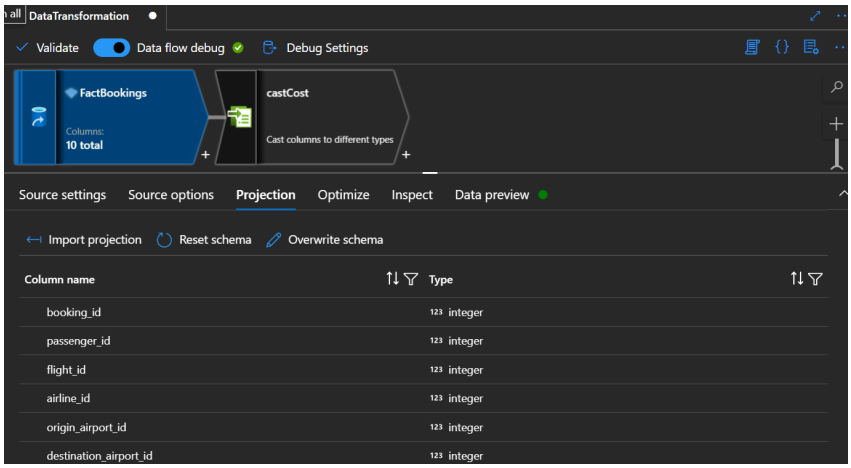
Creating new transformation :

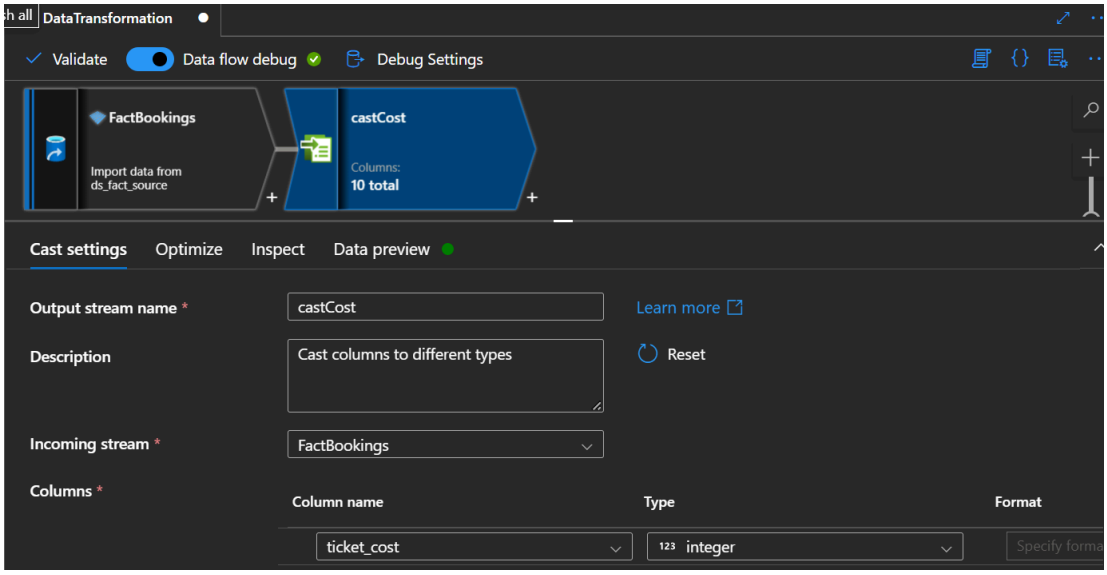




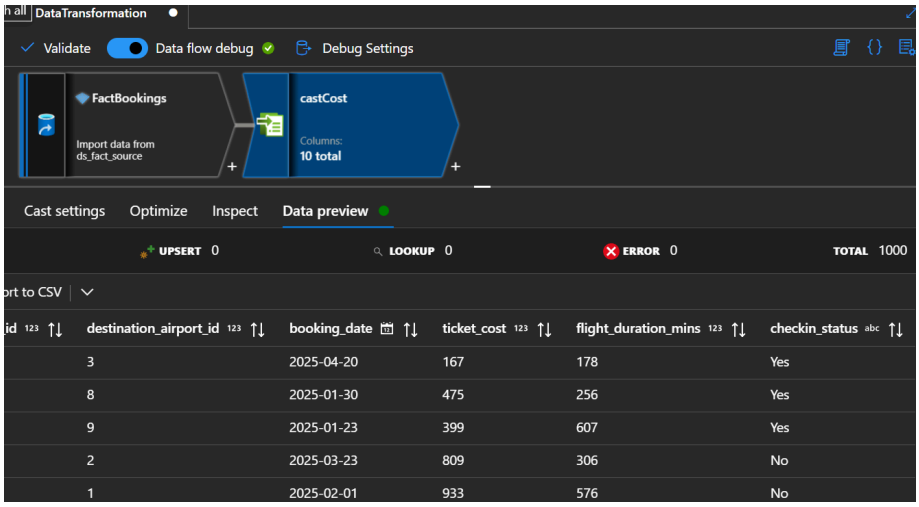


FactBookings : projection



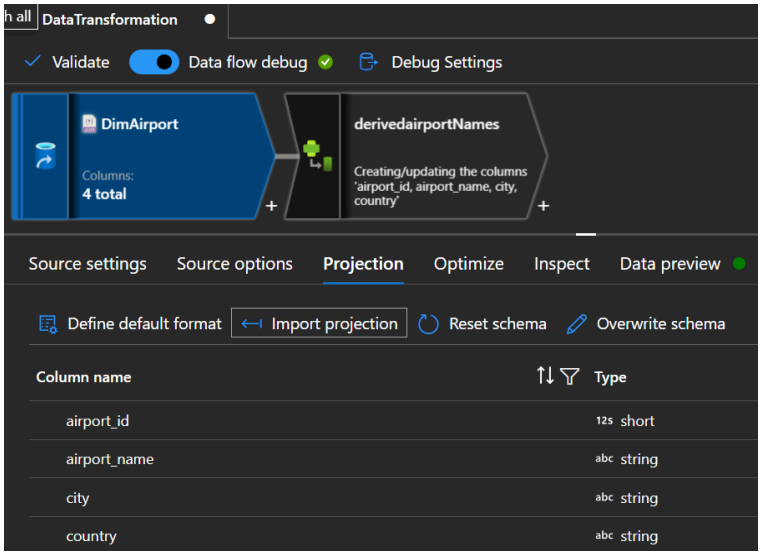


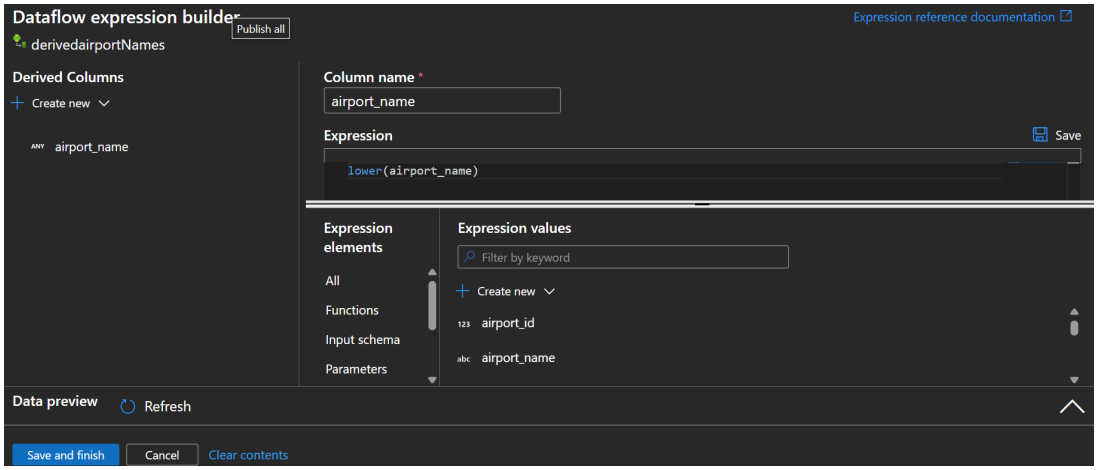
Conversion success :

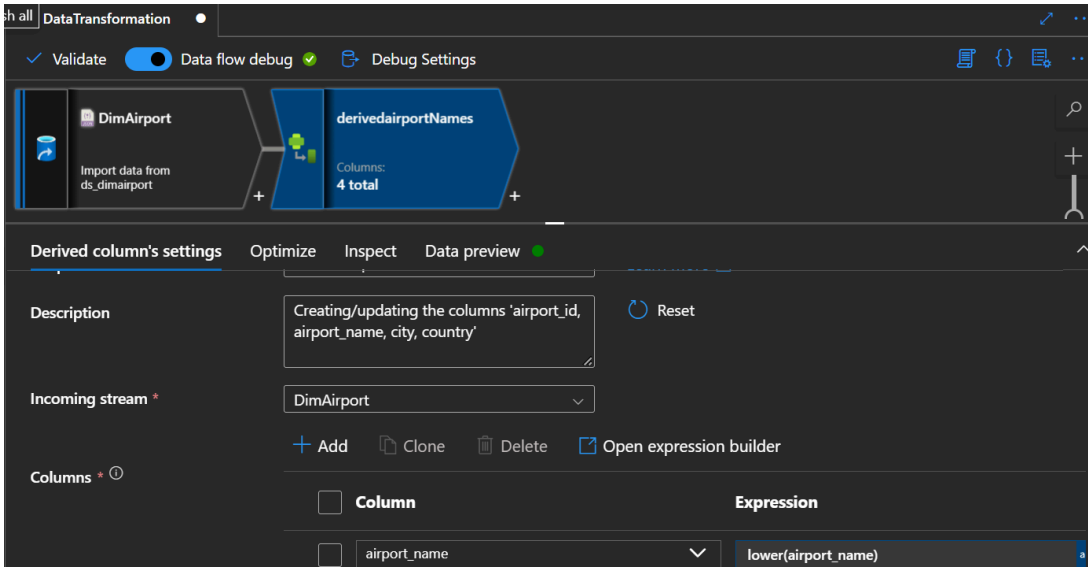


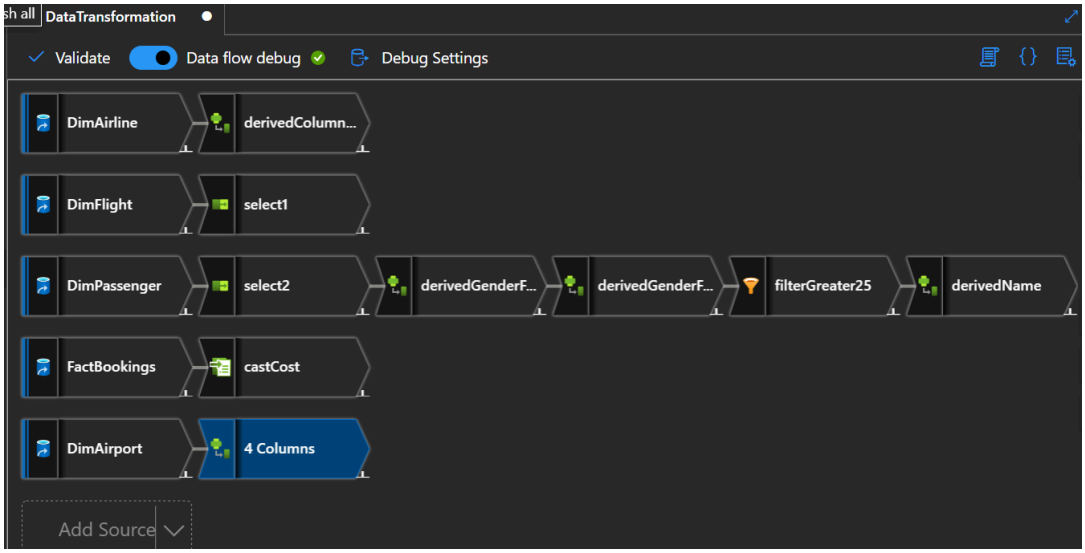
id	destination_airport_id	booking_date	ticket_cost	flight_duration_mins	checkin_status
	3	2025-04-20	167	178	Yes
	8	2025-01-30	475	256	Yes
	9	2025-01-23	399	607	Yes
	2	2025-03-23	809	306	No
	1	2025-02-01	933	576	No

For dimairport source :

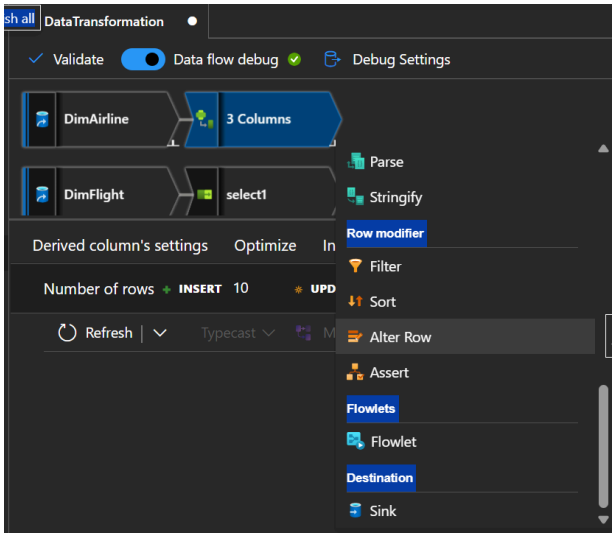


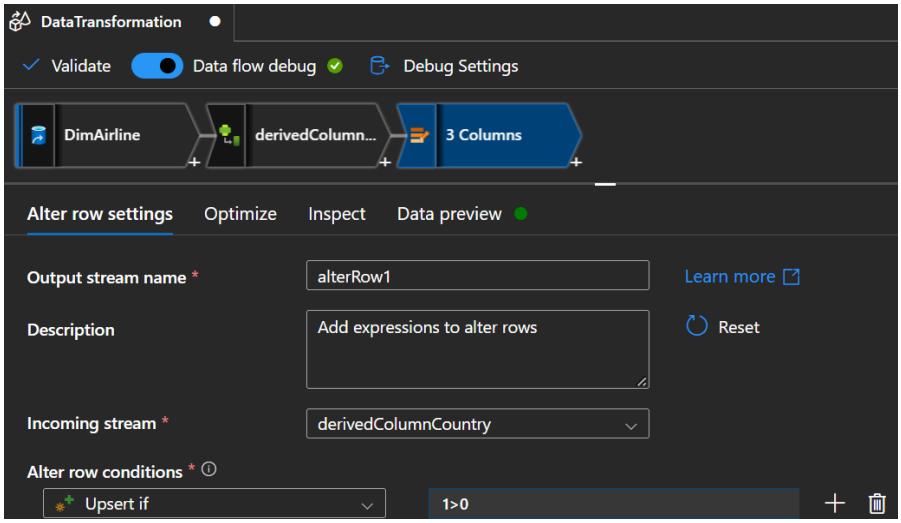

Dataflow expression builder
Publish all
Expression reference documentation
derivedairportNames
Derived Columns
Create new
airport_name
Column name *
airport_name
Expression
Save
lower(airport_name)
Expression elements
All
Functions
Input schema
Parameters
Expression values
Filter by keyword
Create new
airport_id
airport_name
Data preview
Refresh
Save and finish
Cancel
Clear contents

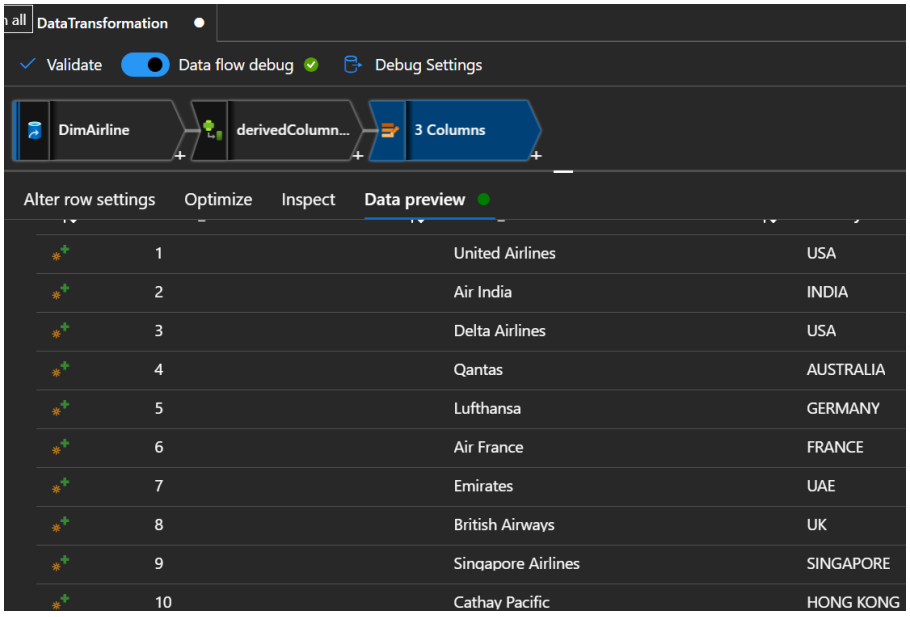

DataTransformation
Validate
Data flow debug
Debug Settings
DimAirport
Import data from ds_dimairport
derivedairportNames
Columns:
4 total
Derived column's settings
Optimize
Inspect
Data preview
Description
Creating/updating the columns 'airport_id, airport_name, city, country'
Reset
Incoming stream *
DimAirport
Add
Clone
Delete
Open expression builder
Columns *
Column
Expression
airport_name
lower(airport_name)


DataTransformation
Validate
Data flow debug
Debug Settings
DimAirline
derivedColumn...
DimFlight
select1
DimPassenger
select2
derivedGenderF...
derivedGenderF...
filterGreater25
derivedName
FactBookings
castCost
DimAirport
4 Columns
Add Source

Click on alter row :

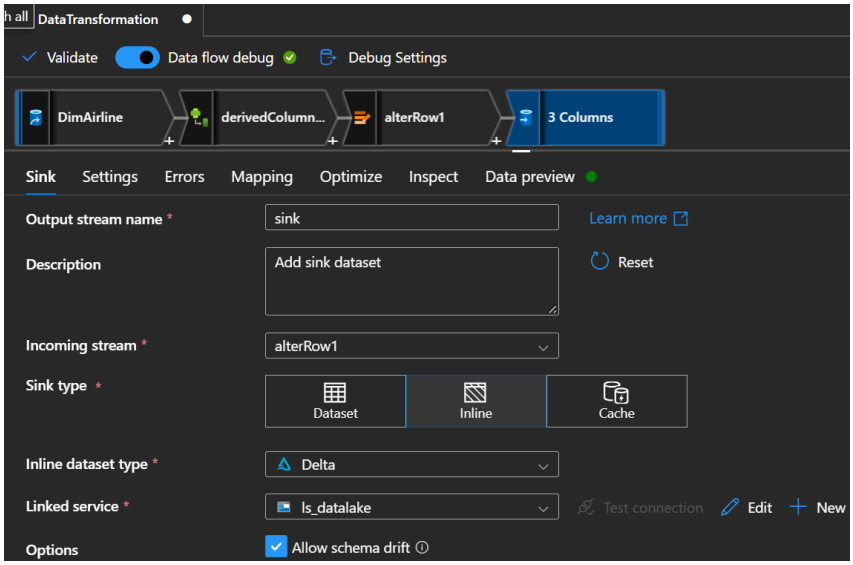


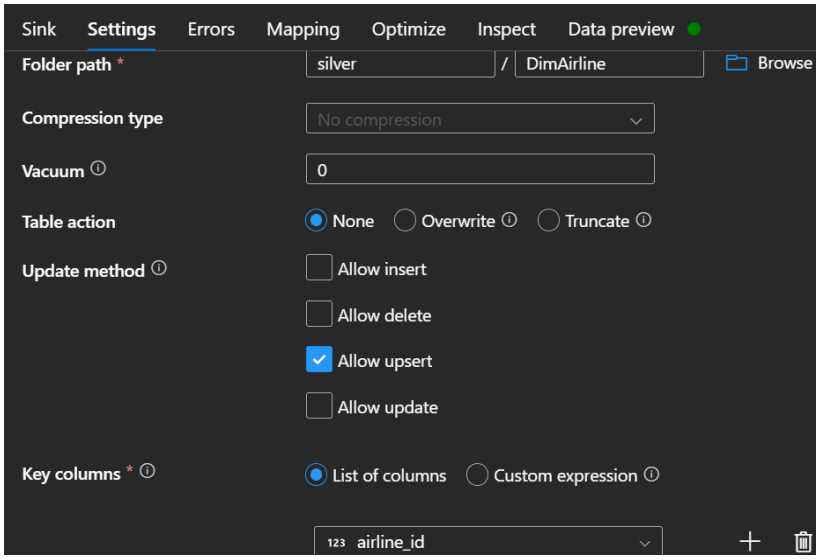


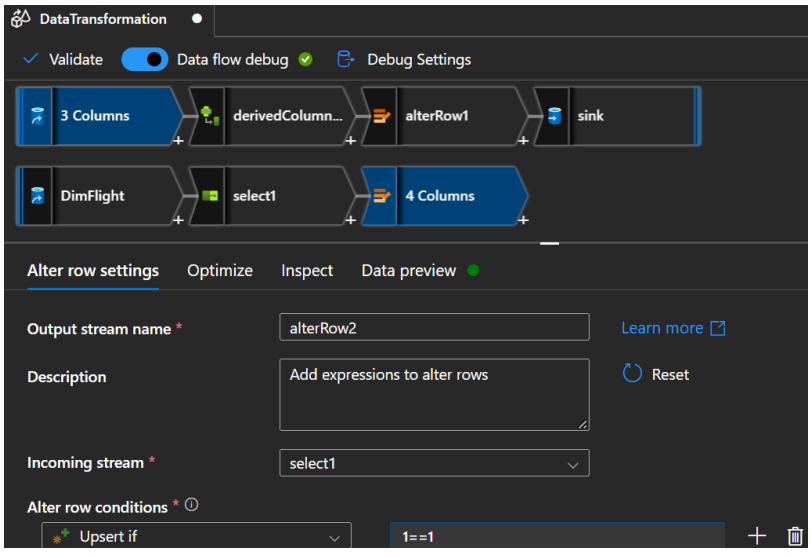


	1	United Airlines	USA
	2	Air India	INDIA
	3	Delta Airlines	USA
	4	Qantas	AUSTRALIA
	5	Lufthansa	GERMANY
	6	Air France	FRANCE
	7	Emirates	UAE
	8	British Airways	UK
	9	Singapore Airlines	SINGAPORE
	10	Cathay Pacific	HONG KONG

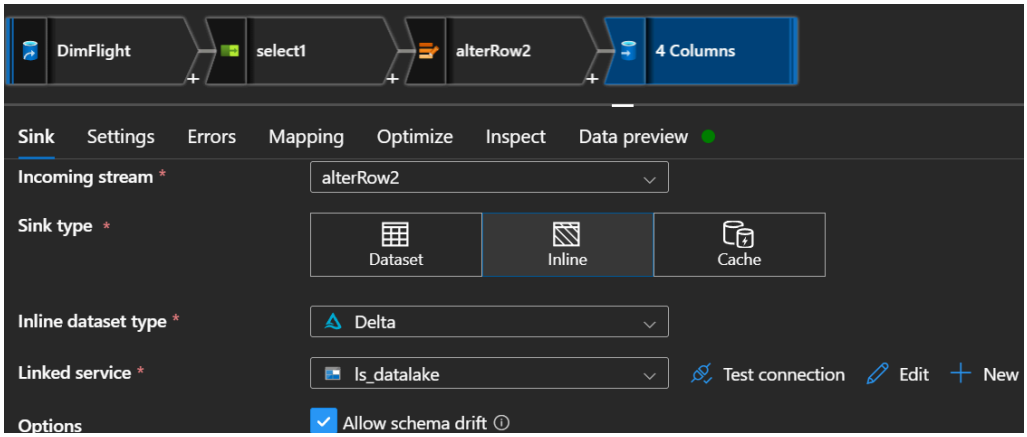
Creating sink :

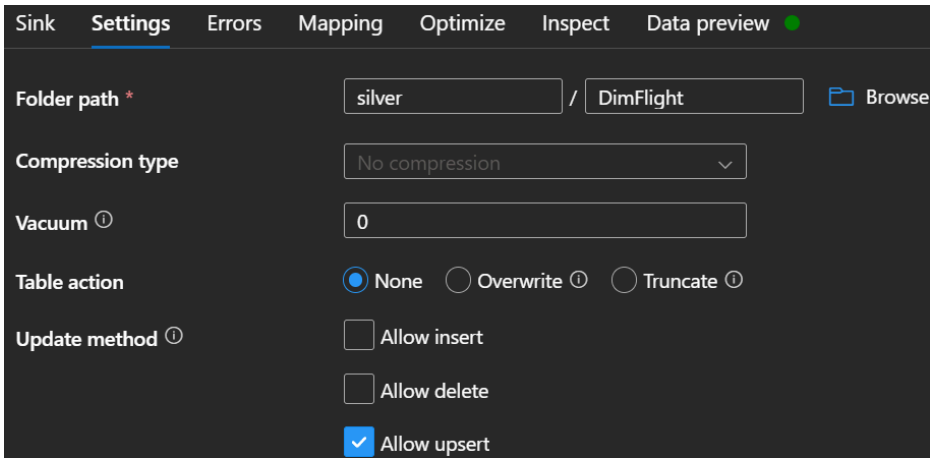






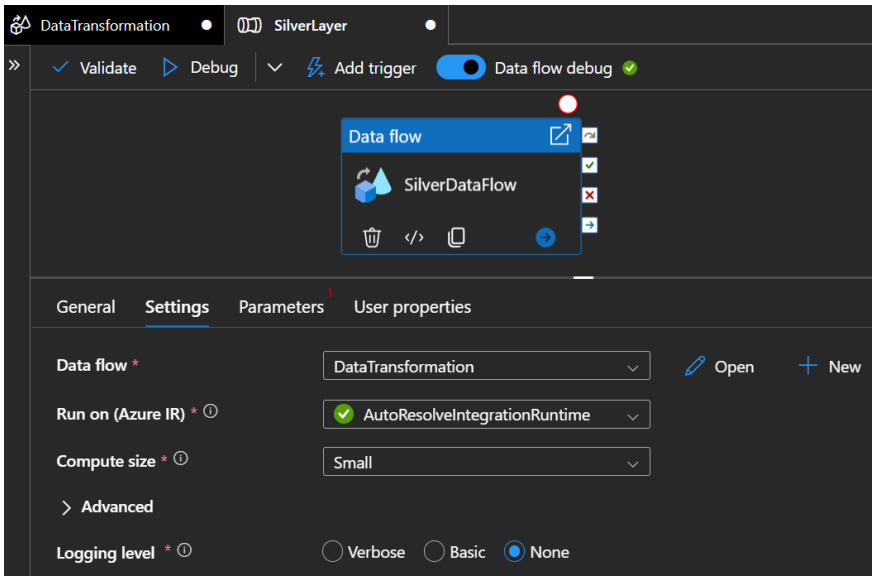
Creating sink for DimFlight :

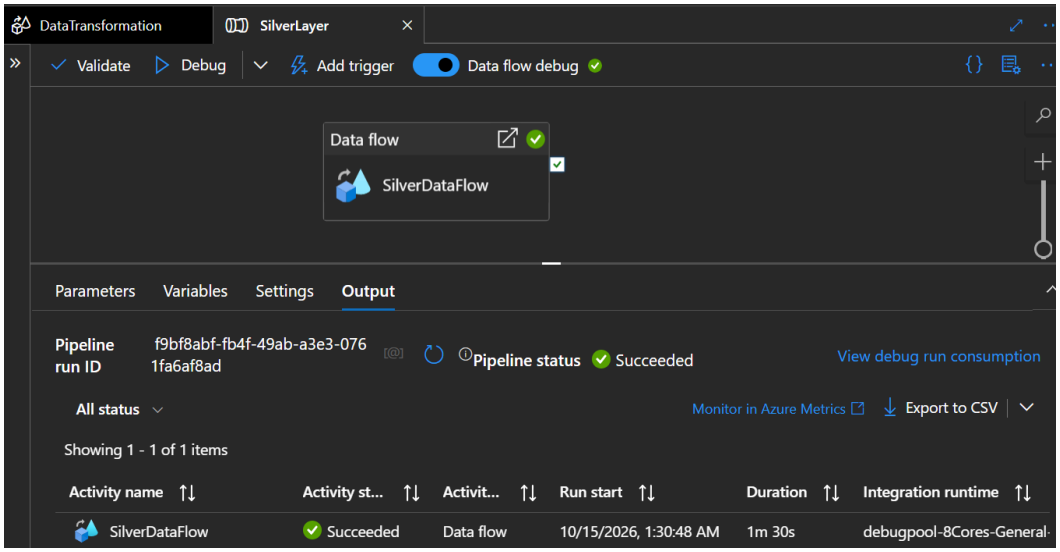


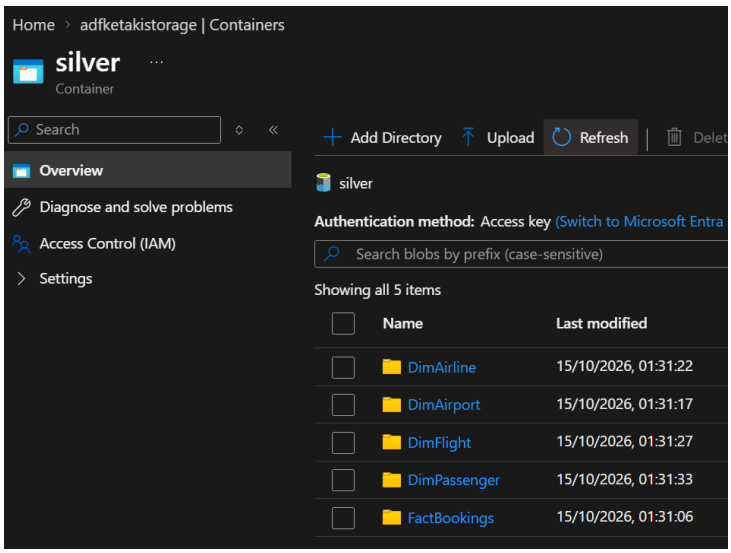


And similarly for every transformation create an alter row and sink.
You cannot add sink activity without an alter row.

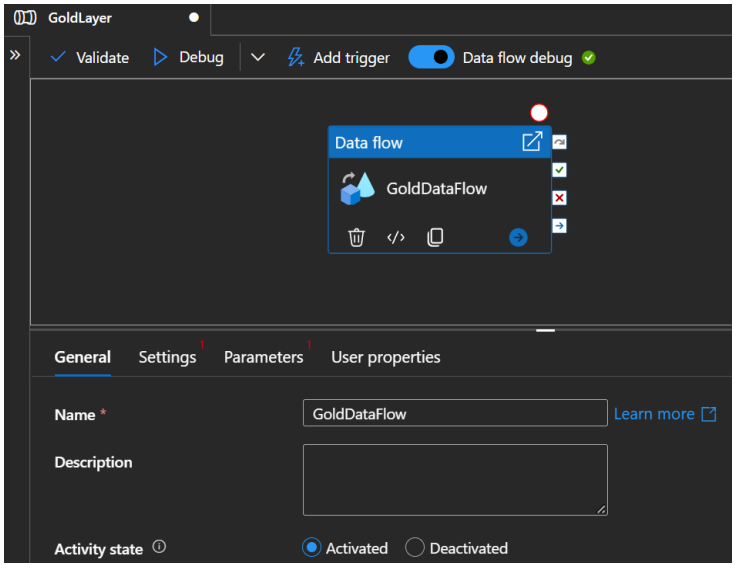
In production use : use activity runtime
Create a new pipeline :



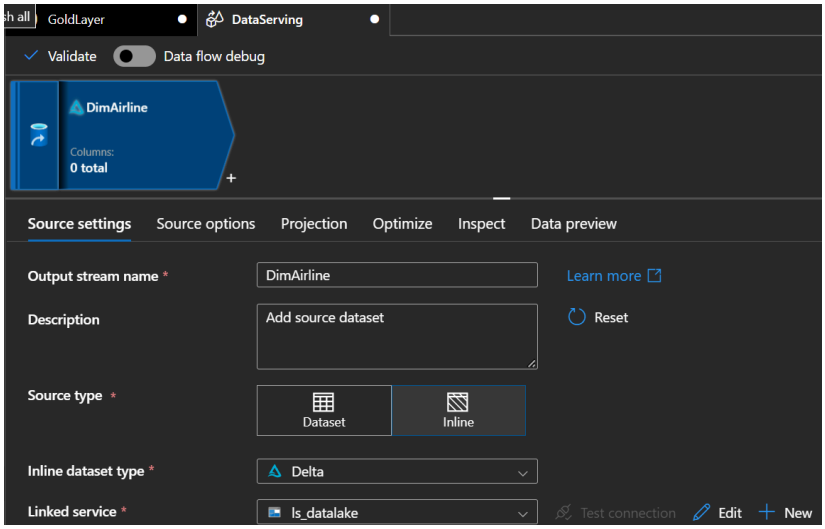


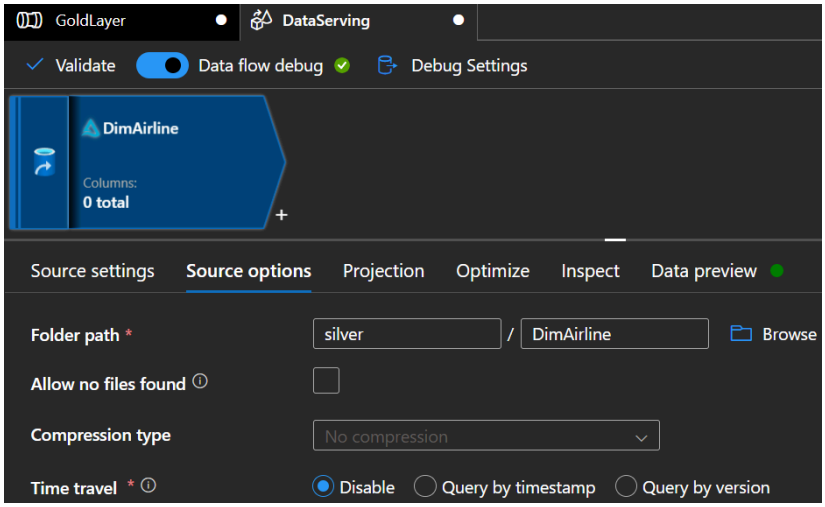


4:40:38 Gold Layer For Business Views , creating new pipeline called gold layer

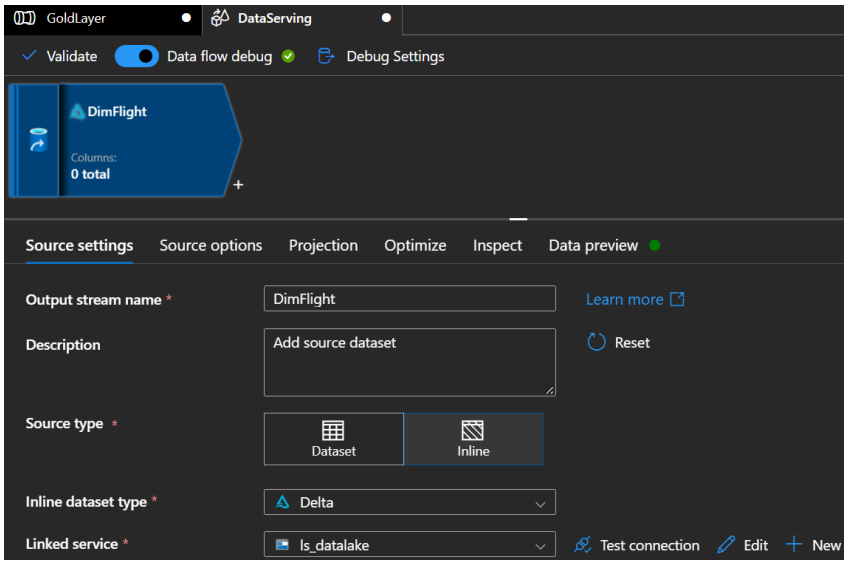


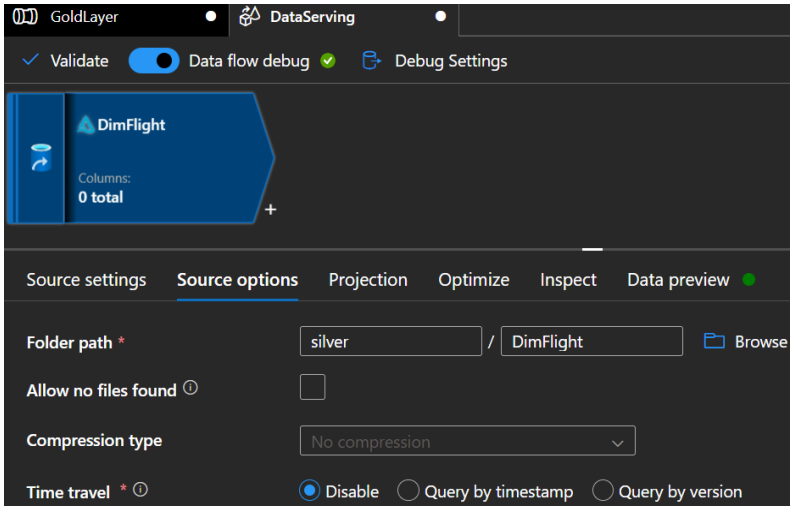
DimAirline :



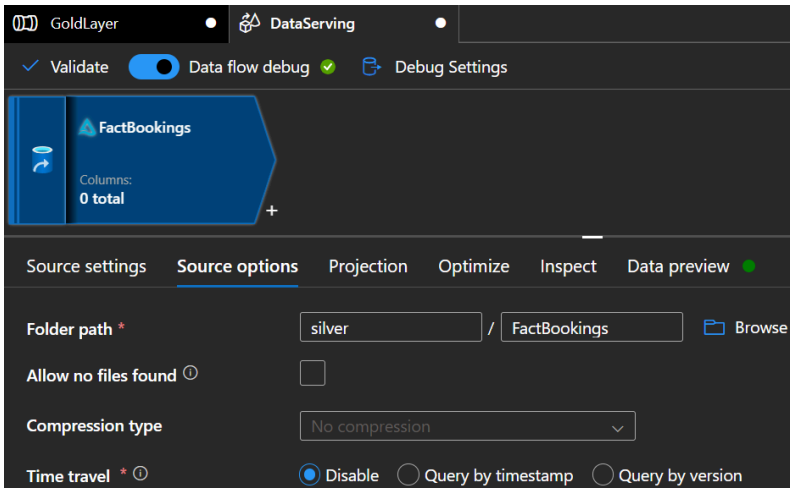


DimFlight :

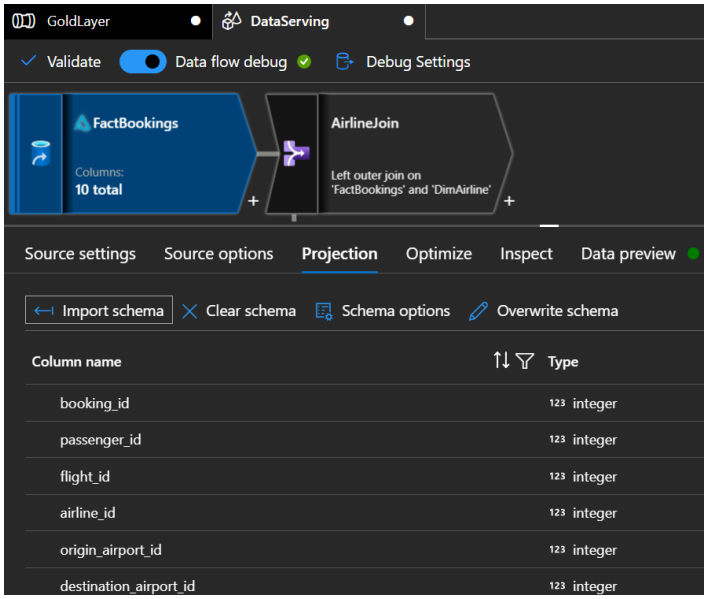




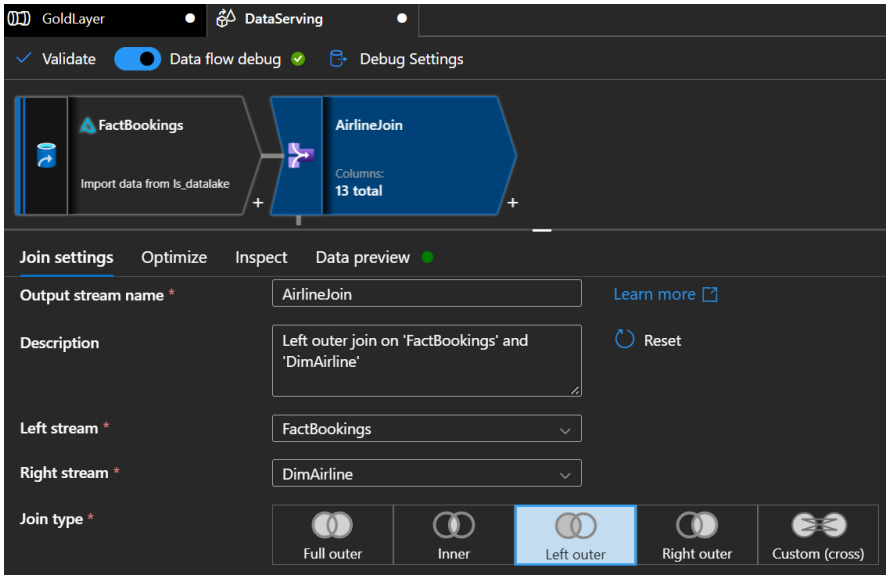
FactBookings :

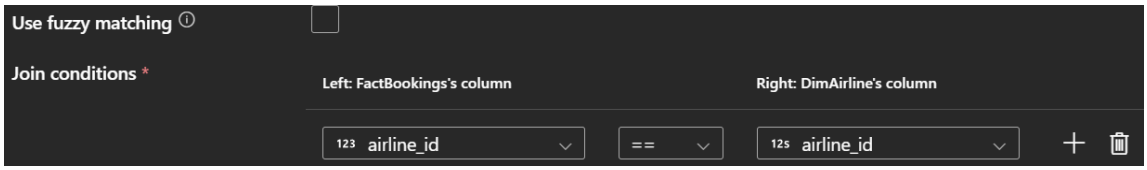


Importing projection :

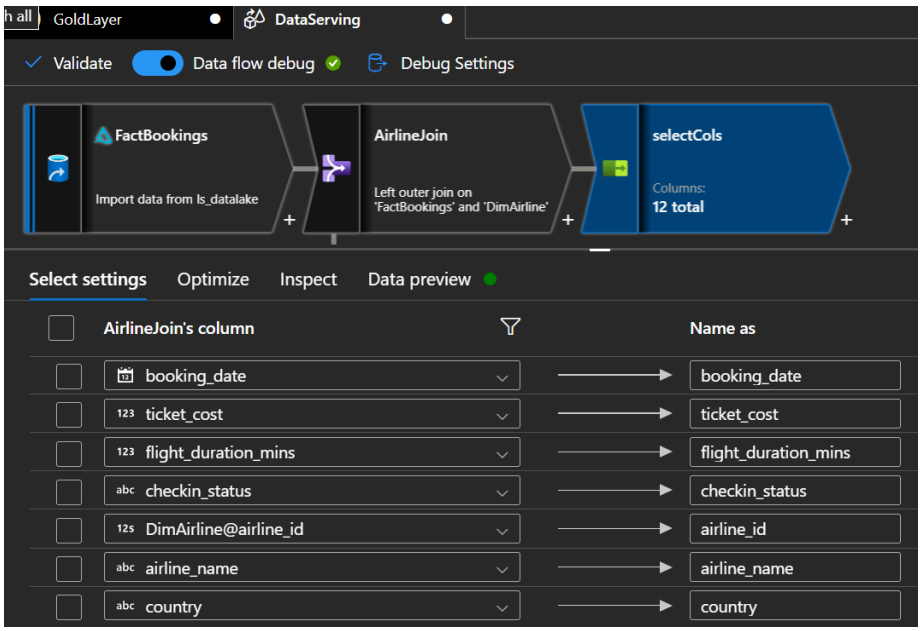


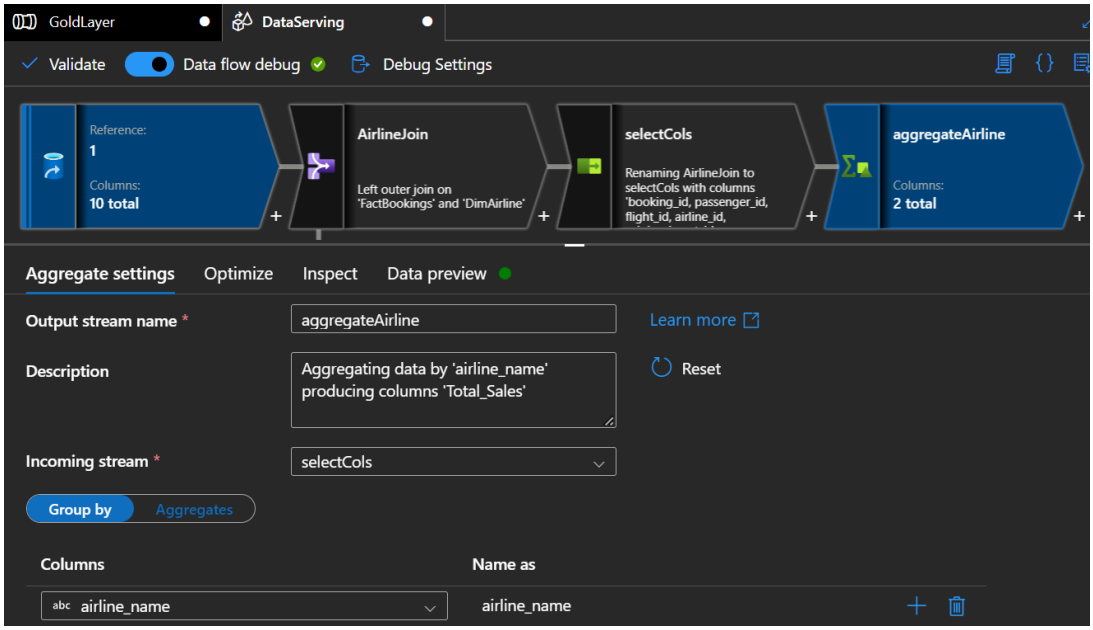
Import projections for DimAirline and DimFlight

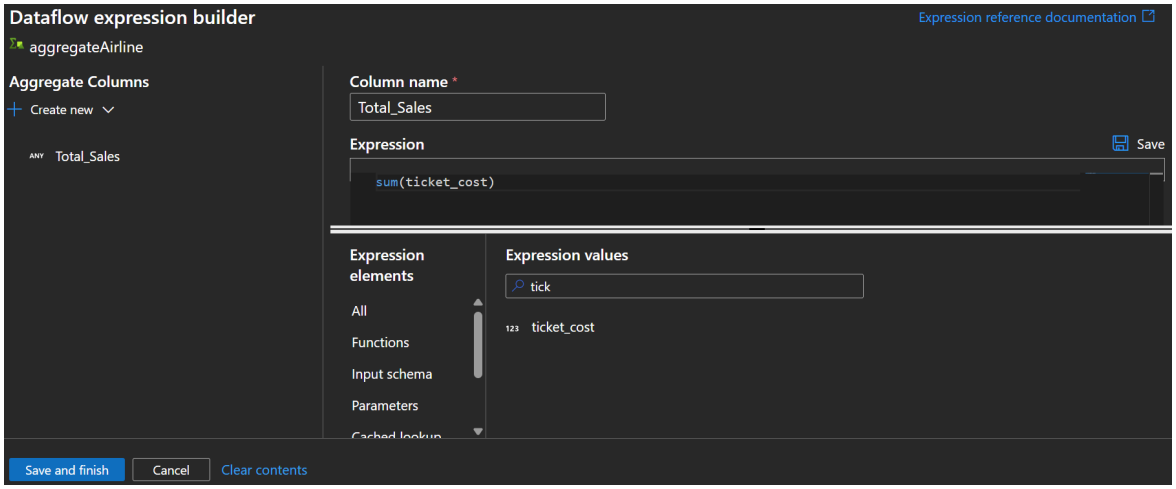


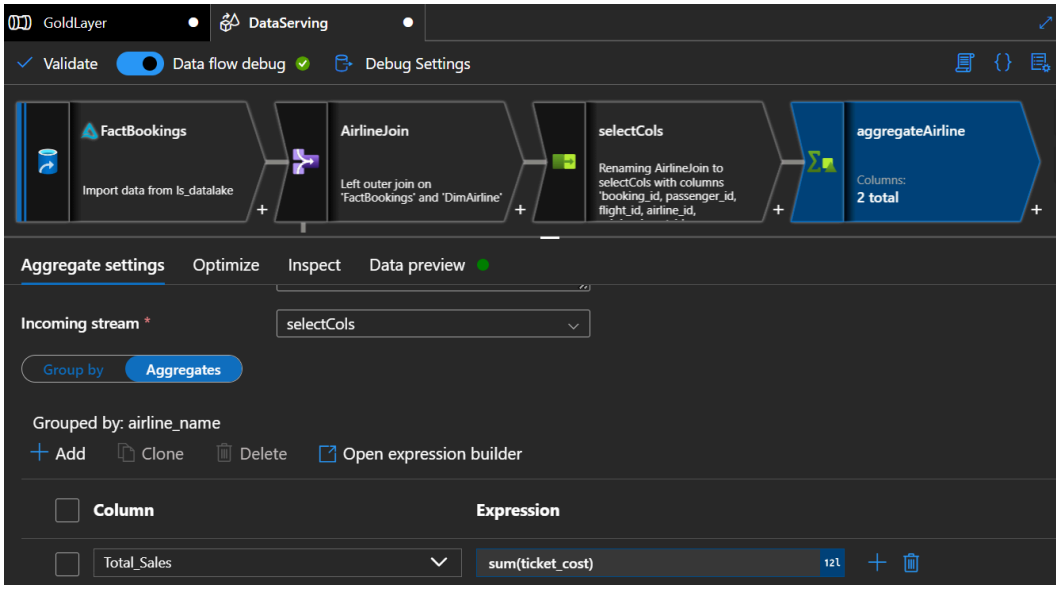


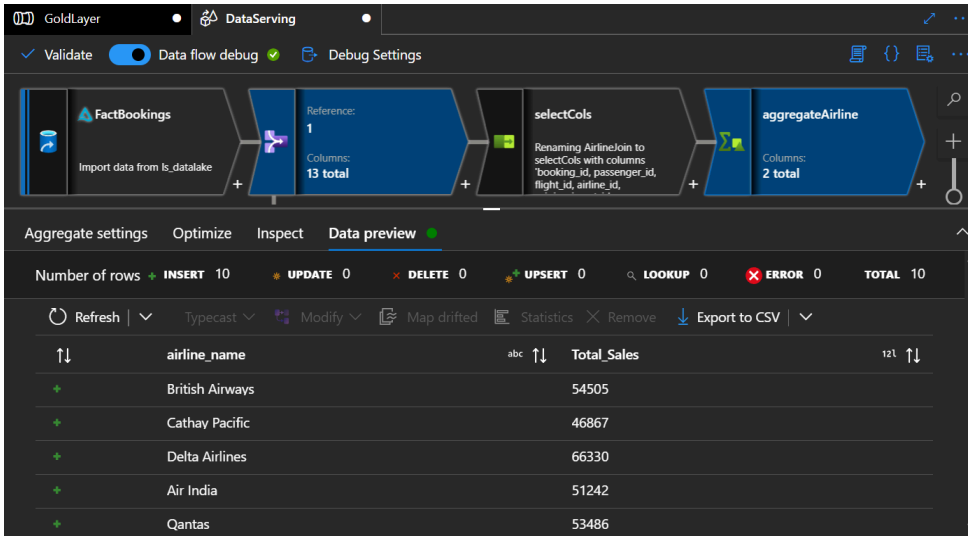
Deleting airlineid :



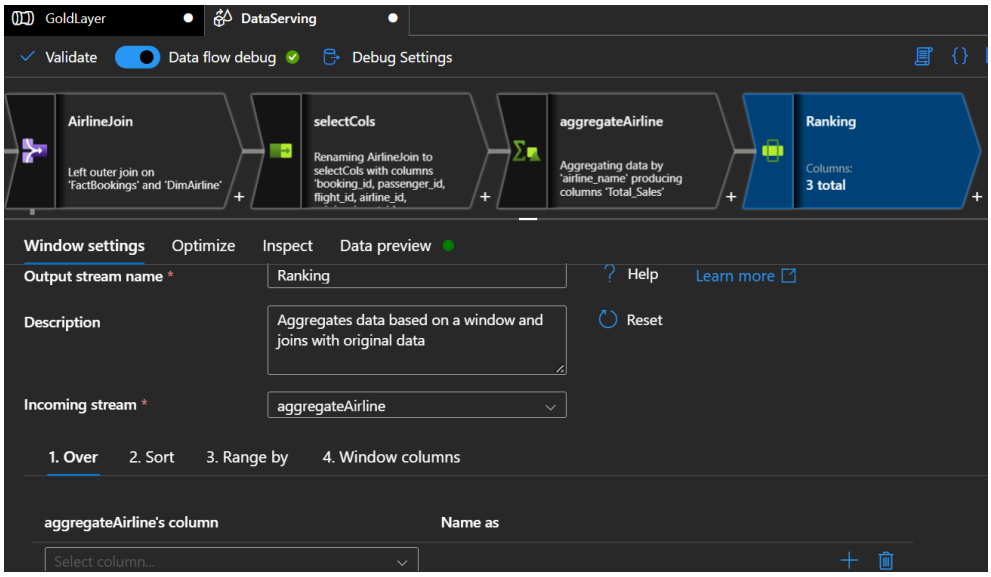

GoldLayer
DataServing
Validate
Data flow debug
Debug Settings
Reference:
1
Columns:
10 total
AirlineJoin
Left outer join on 'FactBookings' and 'DimAirline'
selectCols
Renaming AirlineJoin to selectCols with columns 'booking_id, passenger_id, flight_id, airline_id,
aggregateAirline
Columns:
2 total
Aggregate settings
Optimize
Inspect
Data preview
Output stream name *
aggregateAirline
Learn more
Description
Aggregating data by 'airline_name' producing columns 'Total_Sales'
Reset
Incoming stream *
selectCols
Group by
Aggregates
Columns
Name as
airline_name
airline_name

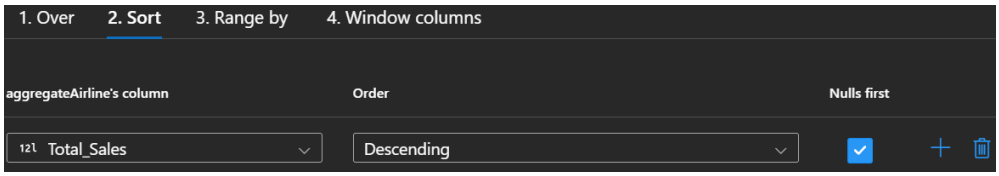

Dataflow expression builder
Expression reference documentation
aggregateAirline
Aggregate Columns
Create new
Total_Sales
Column name *
Total_Sales
Expression
Save
sum(ticket_cost)
Expression elements
All
Functions
Input schema
Parameters
Expression values
tick
ticket_cost
Save and finish
Cancel
Clear contents

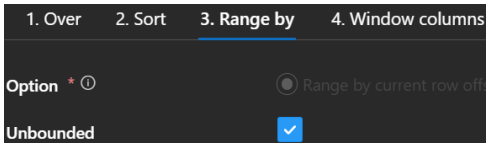

GoldLayer
DataServing
Validate
Data flow debug
Debug Settings
FactBookings
Import data from ls_datalake
AirlineJoin
Left outer join on 'FactBookings' and 'DimAirline'
selectCols
Renaming AirlineJoin to selectCols with columns 'booking_id, passenger_id, flight_id, airline_id,
aggregateAirline
Columns:
2 total
Aggregate settings
Optimize
Inspect
Data preview
Incoming stream *
selectCols
Group by
Aggregates
Grouped by: airline_name
Add
Clone
Delete
Open expression builder
Column
Expression
Total_Sales
sum(ticket_cost)

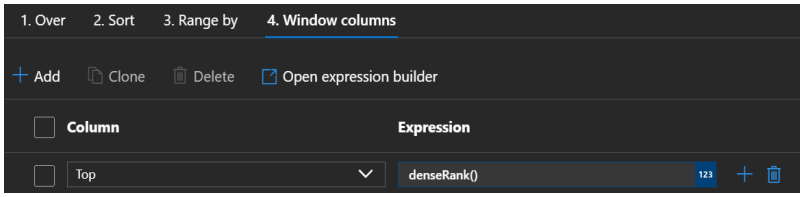


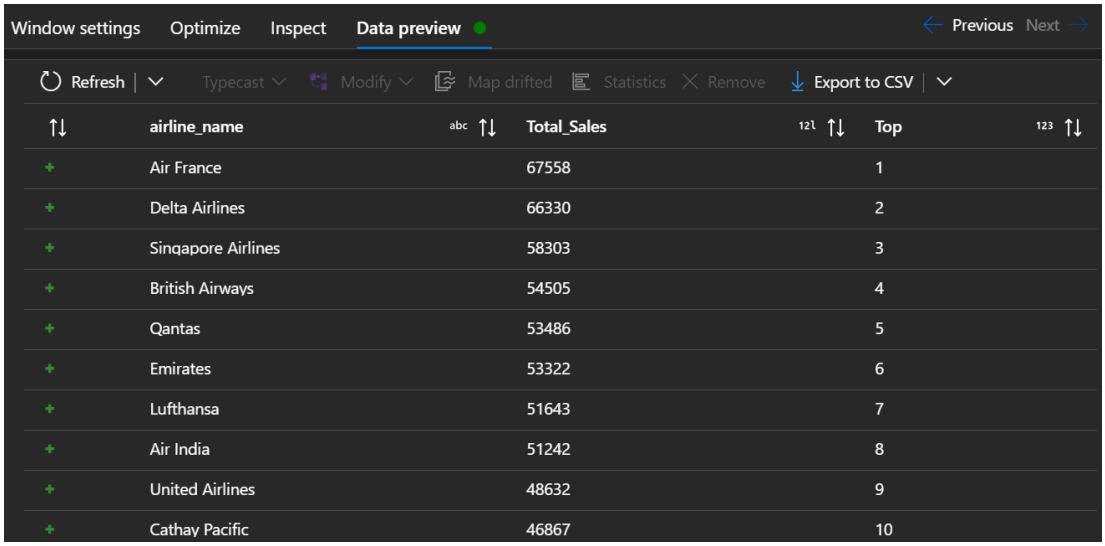
Ranking Transformation :

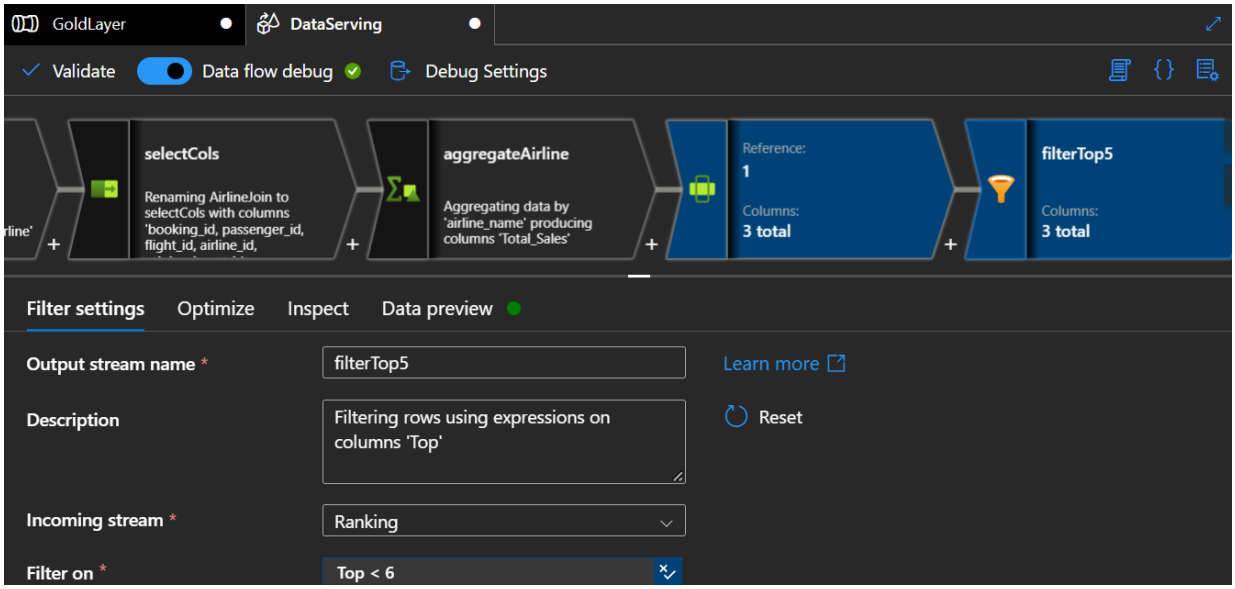


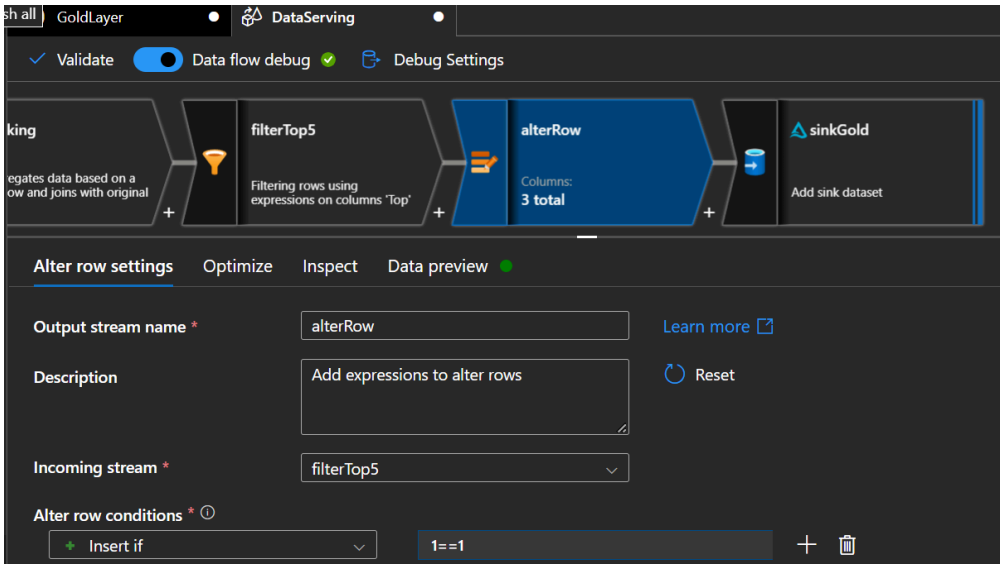


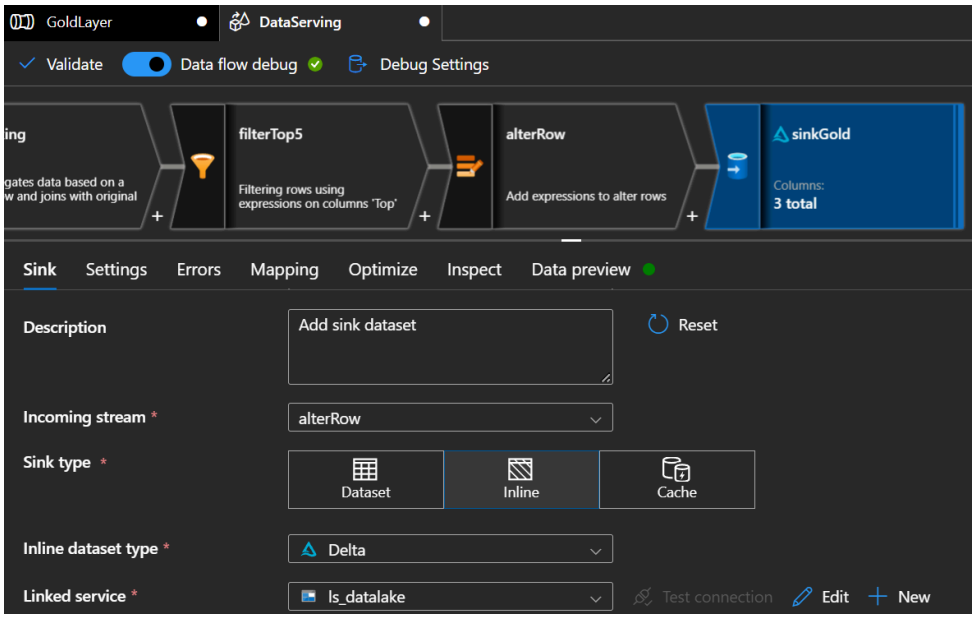


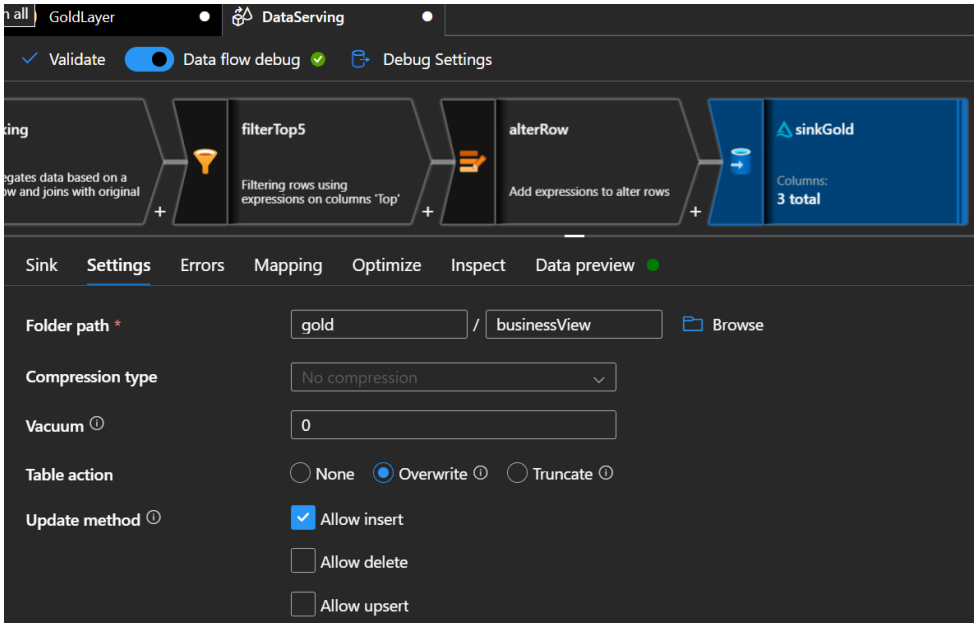


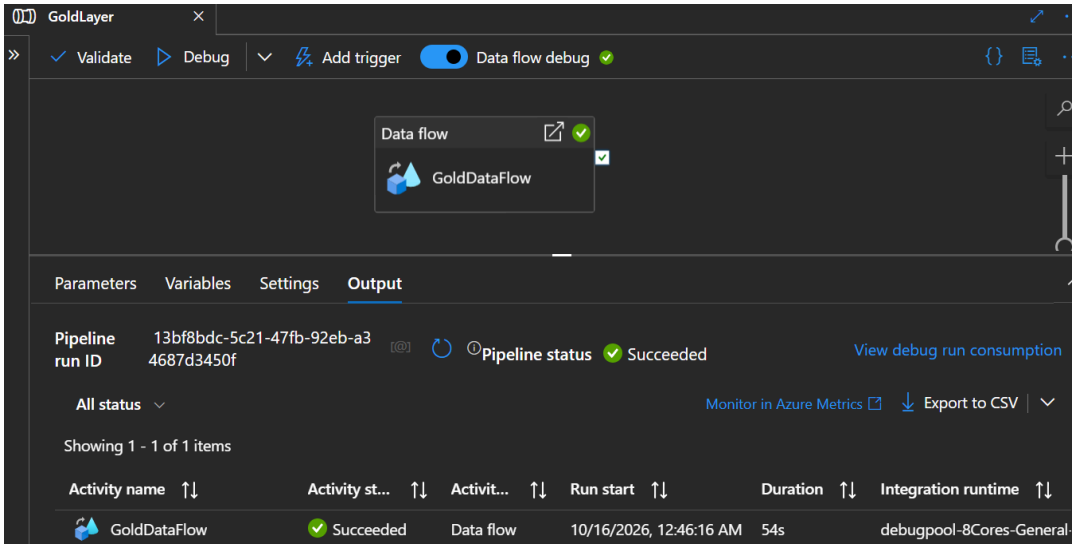

Window settings
Optimize
Inspect
Data preview
Previous
Next
Refresh
Typecast
Modify
Map drifted
Statistics
Remove
Export to CSV
airline_name
Total_Sales
Top
Air France 67558 1
Delta Airlines 66330 2
Singapore Airlines 58303 3
British Airways 54505 4
Qantas 53486 5
Emirates 53322 6
Lufthansa 51643 7
Air India 51242 8
United Airlines 48632 9
Cathay Pacific 46867 10


GoldLayer
DataServing
Validate
Data flow debug
Debug Settings
selectCols
Renaming AirlineJoin to selectCols with columns 'booking_id, passenger_id, flight_id, airline_id,
aggregateAirline
Aggregating data by 'airline_name' producing columns 'Total_Sales'
Reference:
1
Columns:
3 total
filterTop5
Columns:
3 total
Filter settings
Optimize
Inspect
Data preview
Output stream name *
filterTop5
Learn more
Description
Filtering rows using expressions on columns 'Top'
Reset
Incoming stream *
Ranking
Filter on *
Top < 6

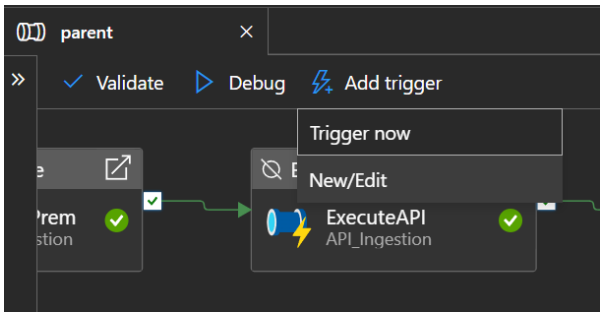

GoldLayer
DataServing
Validate
Data flow debug
Debug Settings
filterTop5
Filtering rows using expressions on columns 'Top'
alterRow
Columns:
3 total
sinkGold
Add sink dataset
Alter row settings
Optimize
Inspect
Data preview
Output stream name *
alterRow
Learn more
Description
Add expressions to alter rows
Reset
Incoming stream *
filterTop5
Alter row conditions *
Insert if
1==1

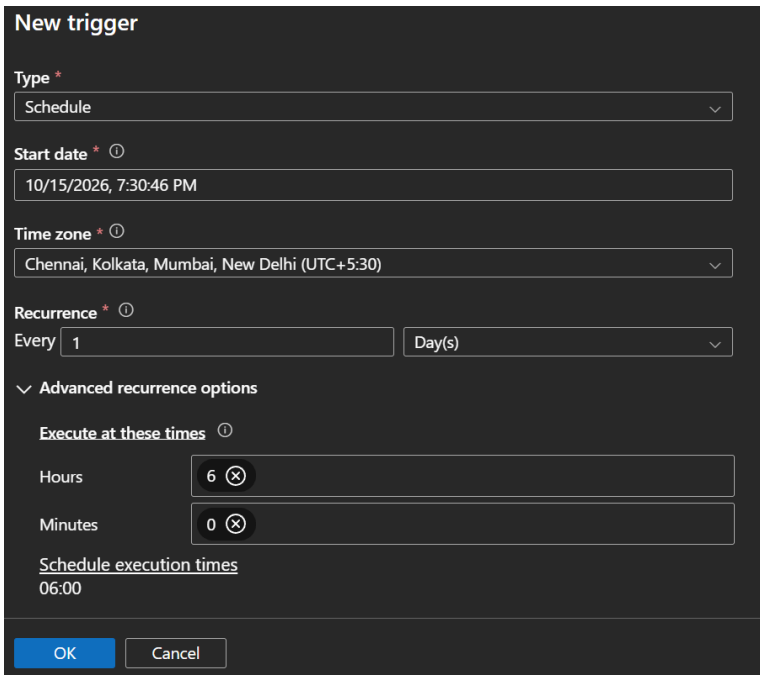

GoldLayer
DataServing
Validate
Data flow debug
Debug Settings
filterTop5
Filtering rows using expressions on columns 'Top'
alterRow
Add expressions to alter rows
sinkGold
Columns:
3 total
Sink
Settings
Errors
Mapping
Optimize
Inspect
Data preview
Description
Add sink dataset
Reset
Incoming stream *
alterRow
Sink type *
Dataset
Inline
Cache
Inline dataset type *
Delta
Linked service *
ls_datalake
Test connection
Edit
New


GoldLayer
DataServing
Validate
Data flow debug
Debug Settings
filterTop5
Filtering rows using expressions on columns 'Top'
alterRow
Add expressions to alter rows
sinkGold
Columns:
3 total
Sink
Settings
Errors
Mapping
Optimize
Inspect
Data preview
Folder path *
gold
businessView
Browse
Compression type
No compression
Vacuum
0
Table action
None
Overwrite
Truncate
Update method
Allow insert
Allow delete
Allow upsert

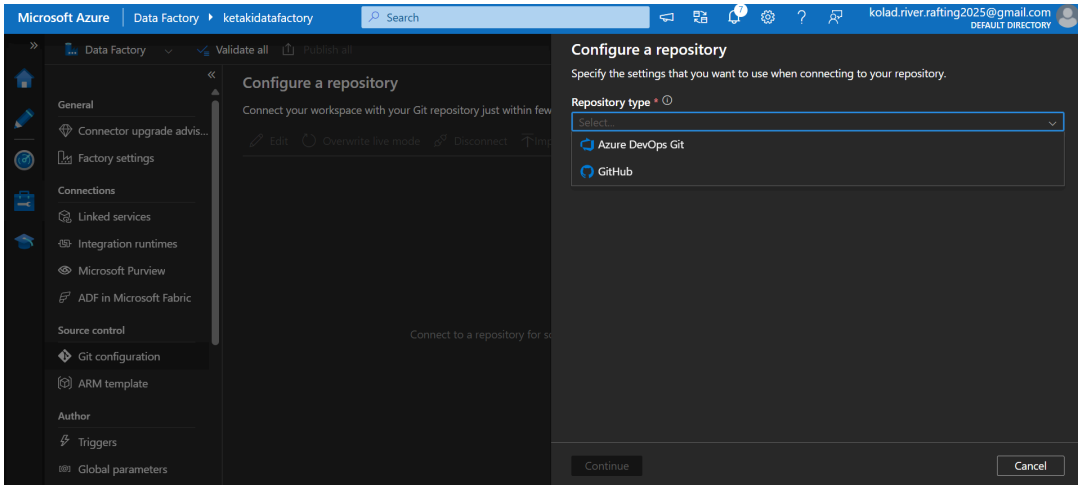


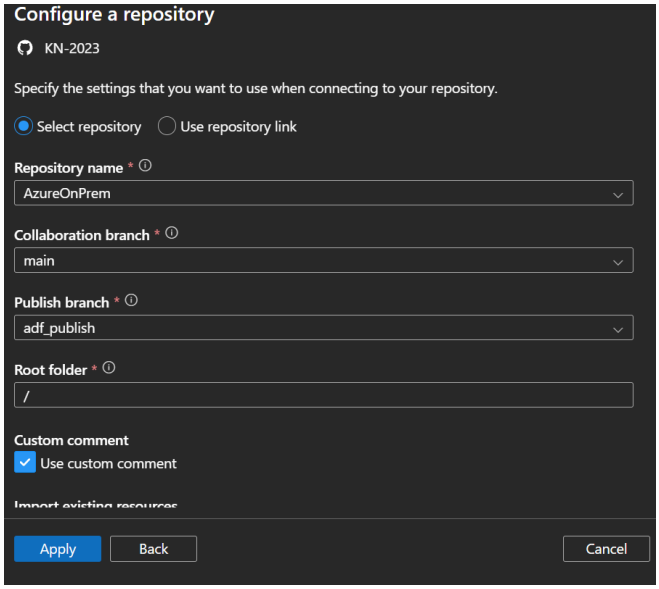
5:10:20 Triggers in Azure Data Factory

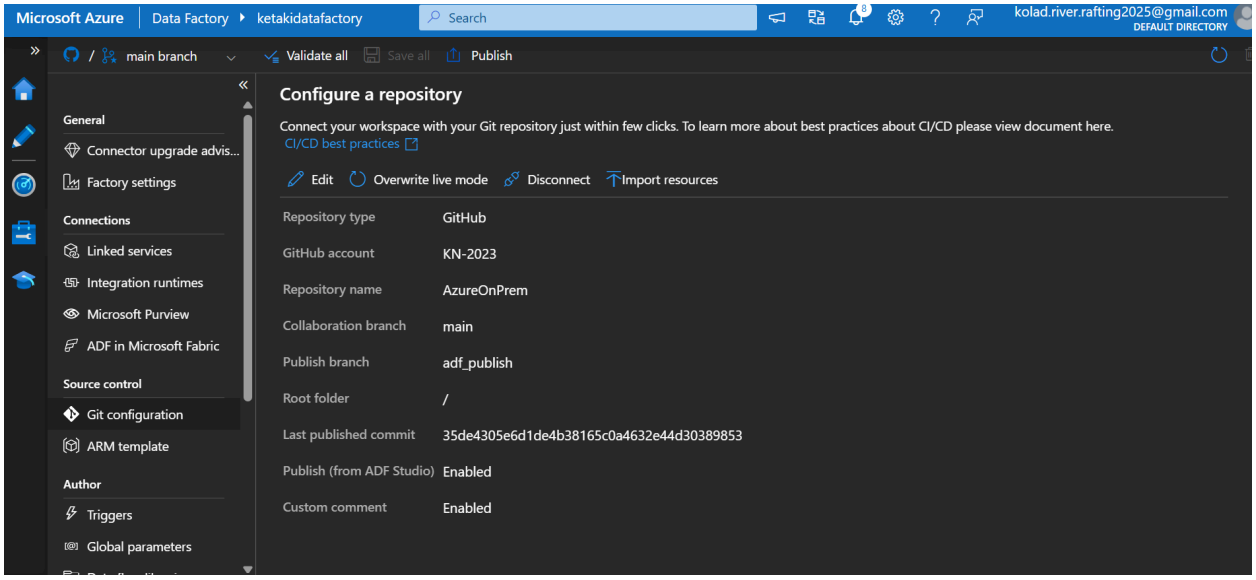




5:13:51 Azure Git Integration with GitHub

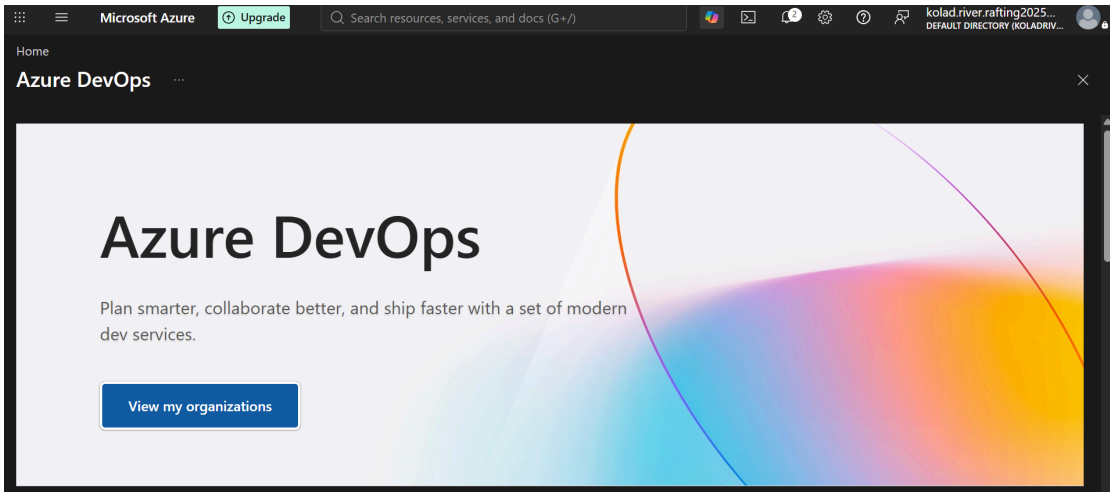




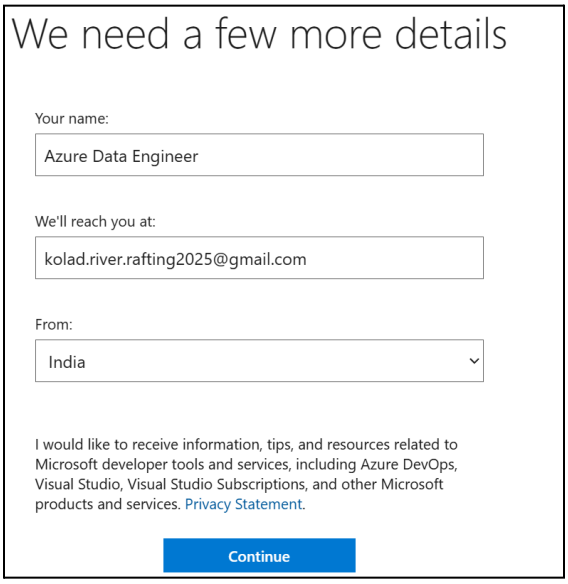


5:18:59 Azure DevOps Free Account

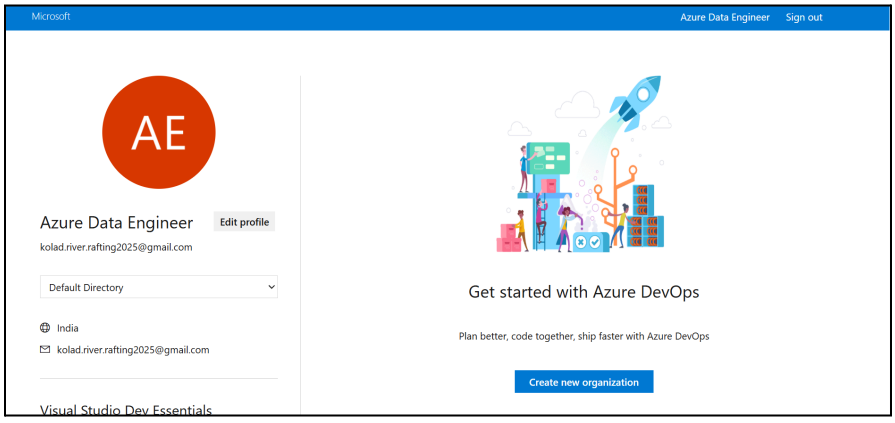
Search azure devops :

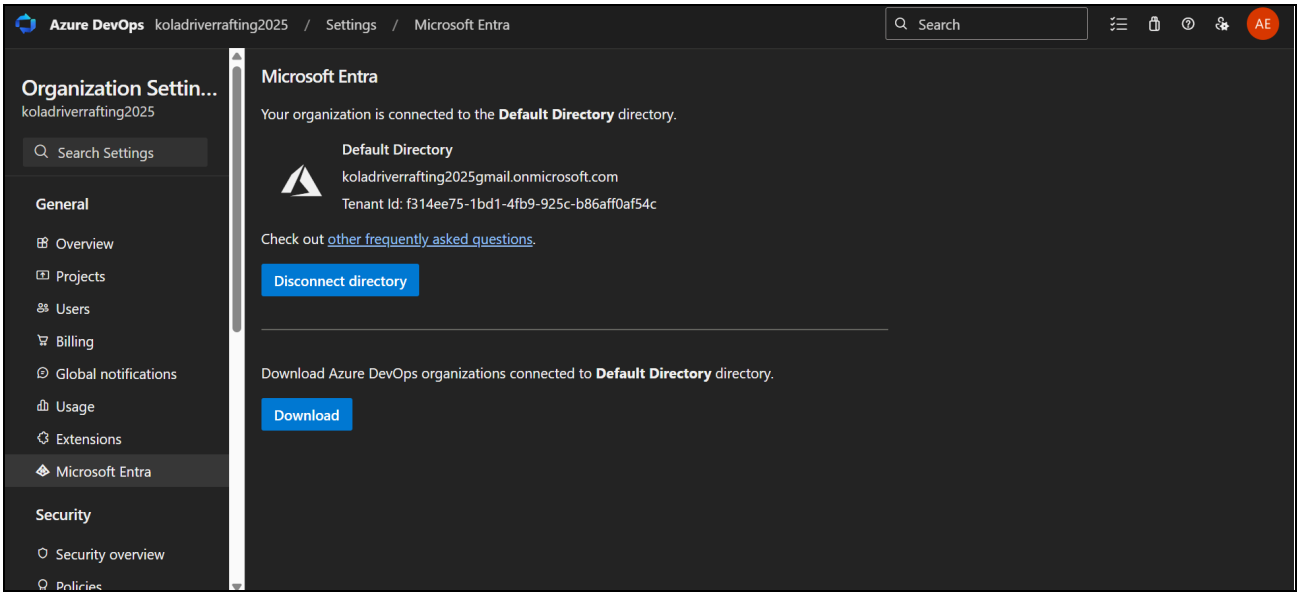


Click on view my organization :

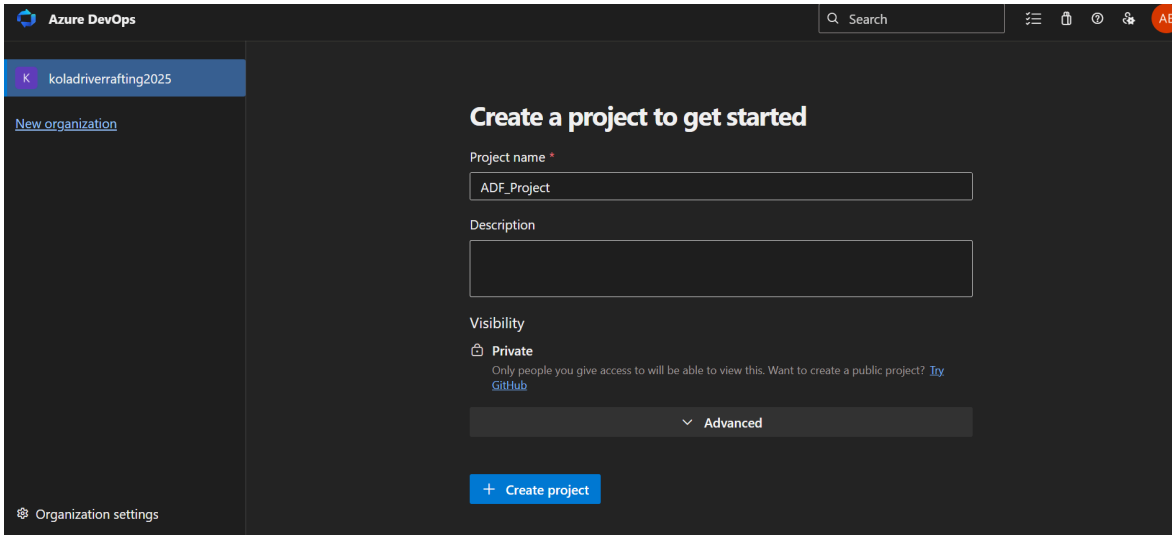


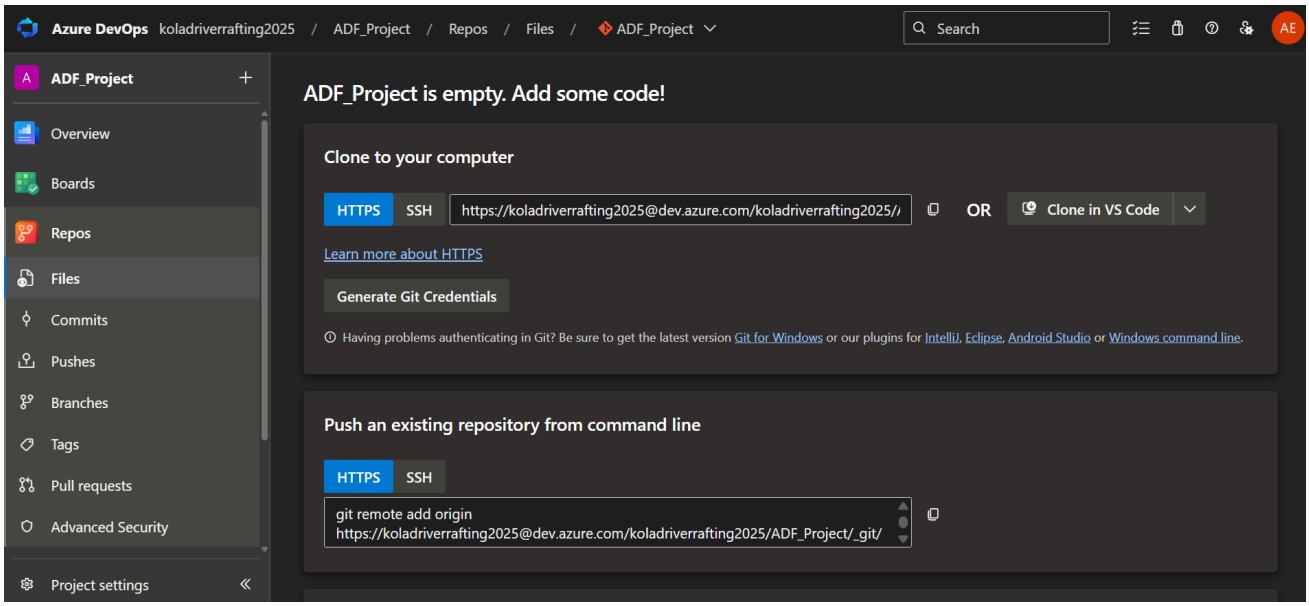
Click on create new organization in default directory :

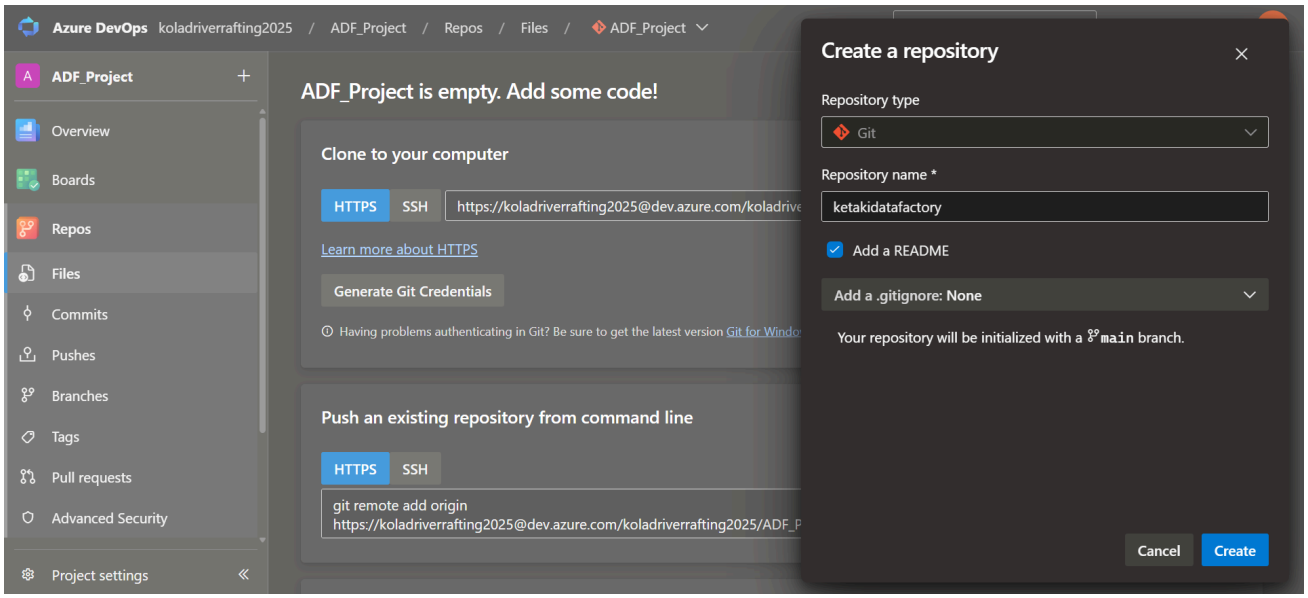


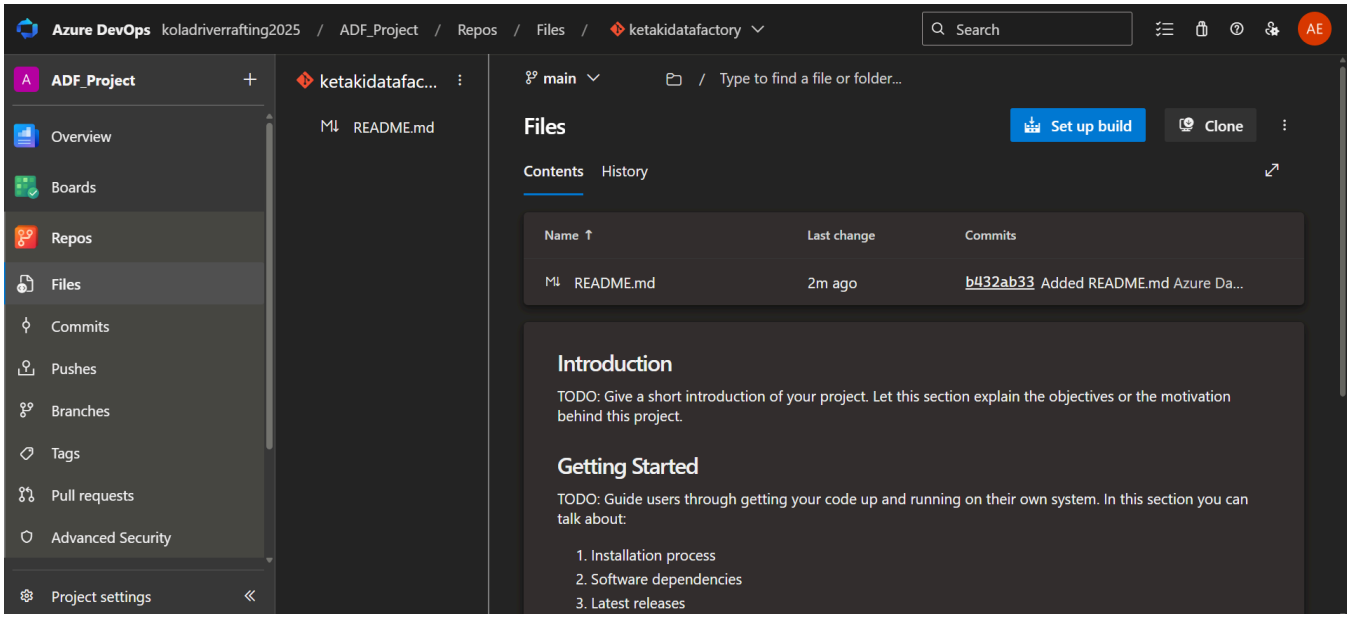


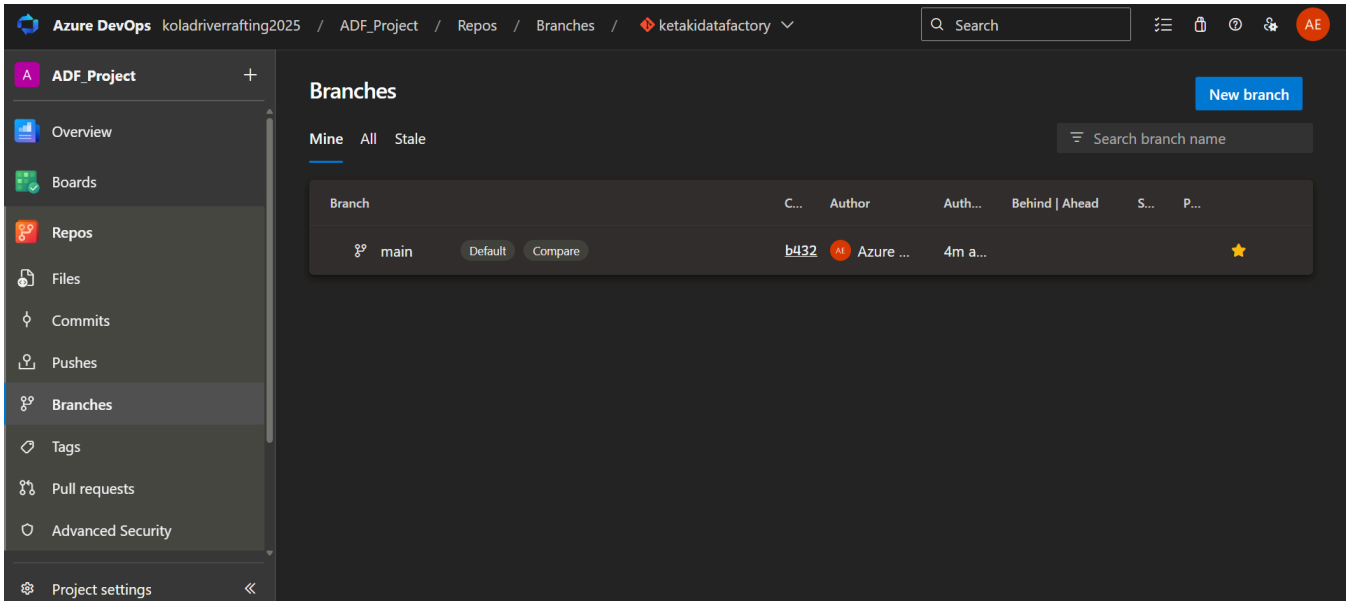
5:21:21 Azure DevOps Tutorial



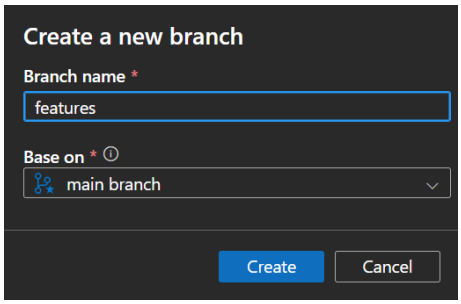

Azure DevOps koladriverrafting2025 / ADF_Project / Repos / Files / ADF_Project
Search
AE
ADF_Project
Overview
Boards
Repos
Files
Commits
Pushes
Branches
Tags
Pull requests
Advanced Security
Project settings
ADF_Project is empty. Add some code!
Clone to your computer
HTTPS
SSH
https://koladriverrafting2025@dev.azure.com/koladriverrafting2025/
OR
Clone in VS Code
Learn more about HTTPS
Generate Git Credentials
Having problems authenticating in Git? Be sure to get the latest version Git for Windows or our plugins for IntelliJ, Eclipse, Android Studio or Windows command line.
Push an existing repository from command line
HTTPS
SSH
git remote add origin
https://koladriverrafting2025@dev.azure.com/koladriverrafting2025/ADF_Project/_git/

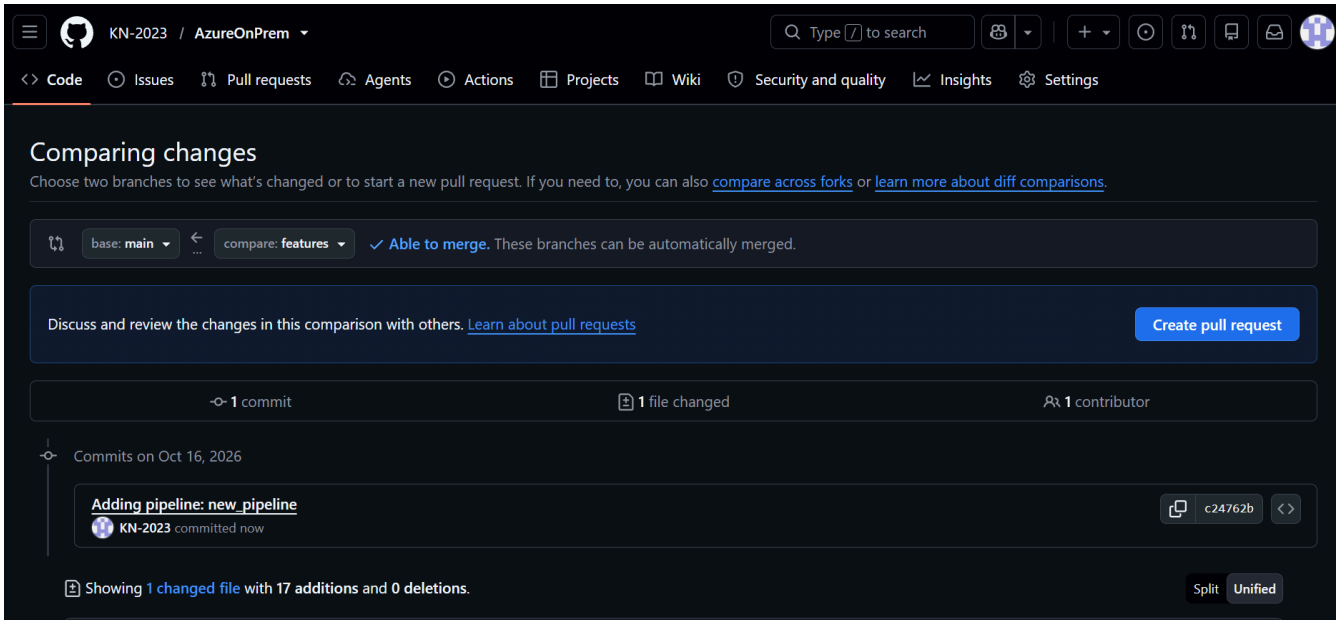

Azure DevOps koladriverrafting2025 / ADF_Project / Repos / Files / ADF_Project
ADF_Project
Overview
Boards
Repos
Files
Commits
Pushes
Branches
Tags
Pull requests
Advanced Security
Project settings
ADF_Project is empty. Add some code!
Clone to your computer
HTTPS
SSH
Learn more about HTTPS
Generate Git Credentials
Push an existing repository from command line
HTTPS
SSH
git remote add origin
Create a repository
Repository type
Git
Repository name *
ketakidatafactory
Add a README
Add a .gitignore: None
Your repository will be initialized with a main branch.
Cancel
Create

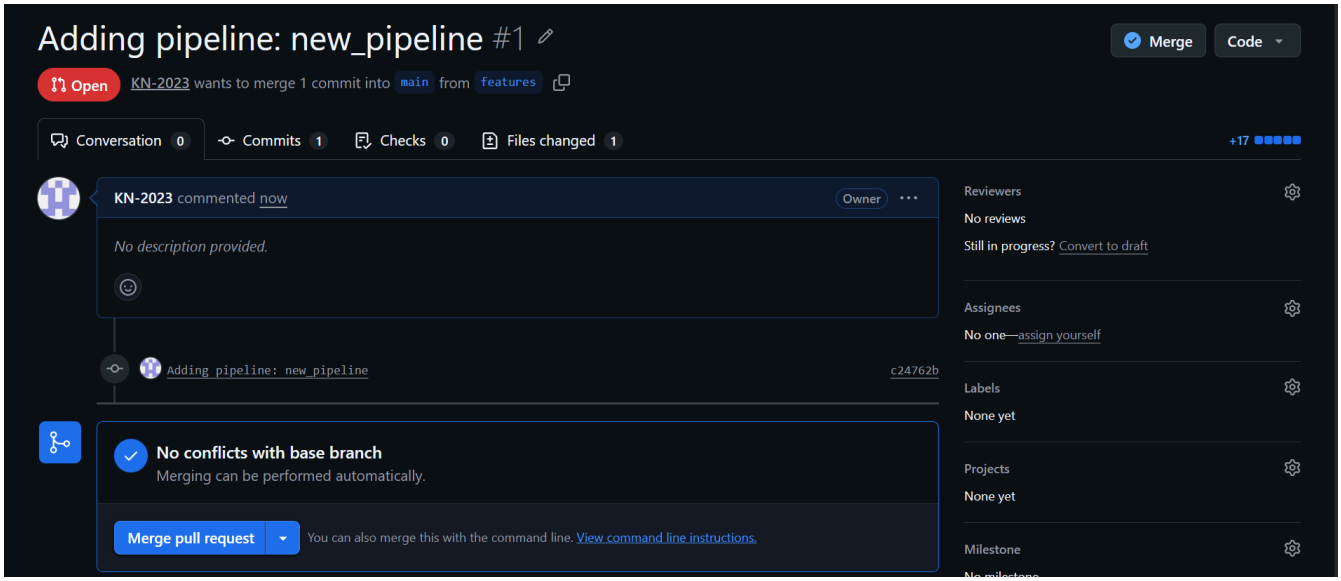


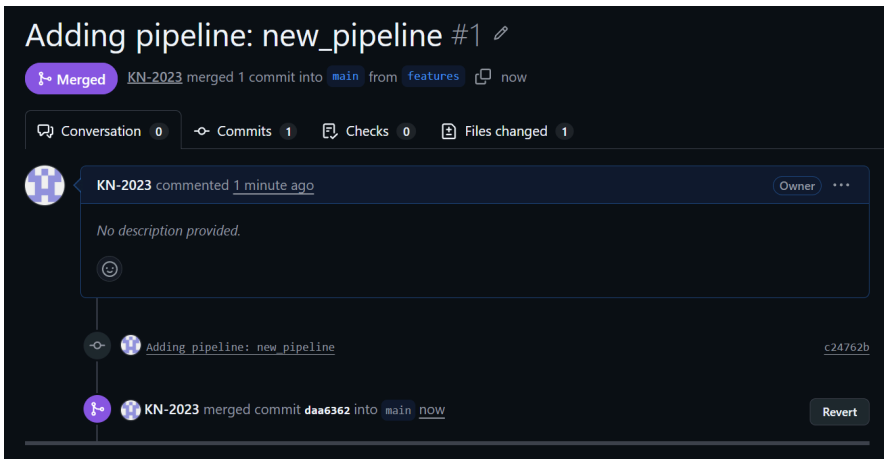


5:23:31 Feature Branch & Pull Request In ADF

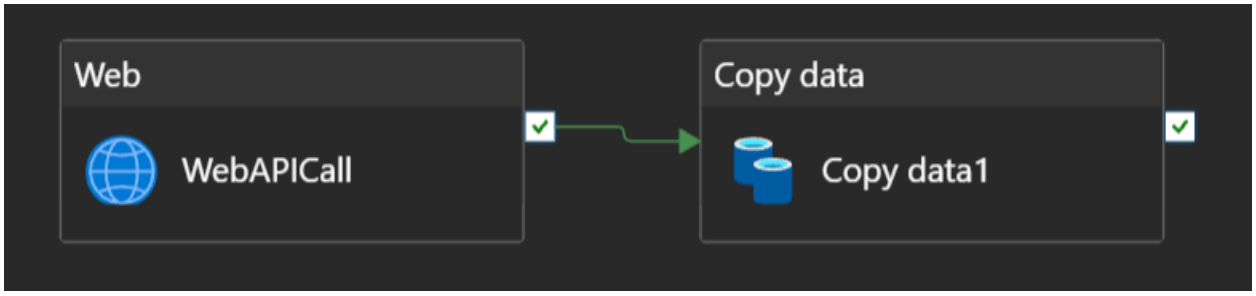



KN-2023 / AzureOnPrem
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Comparing changes
Choose two branches to see what's changed or to start a new pull request. If you need to, you can also compare across forks or learn more about diff comparisons.
base: main
compare: features
Able to merge. These branches can be automatically merged.
Discuss and review the changes in this comparison with others. Learn about pull requests
Create pull request
1 commit
1 file changed
1 contributor
Commits on Oct 16, 2026
Adding pipeline: new_pipeline
KN-2023 committed now
c24762b
Showing 1 changed file with 17 additions and 0 deletions.
Split
Unified

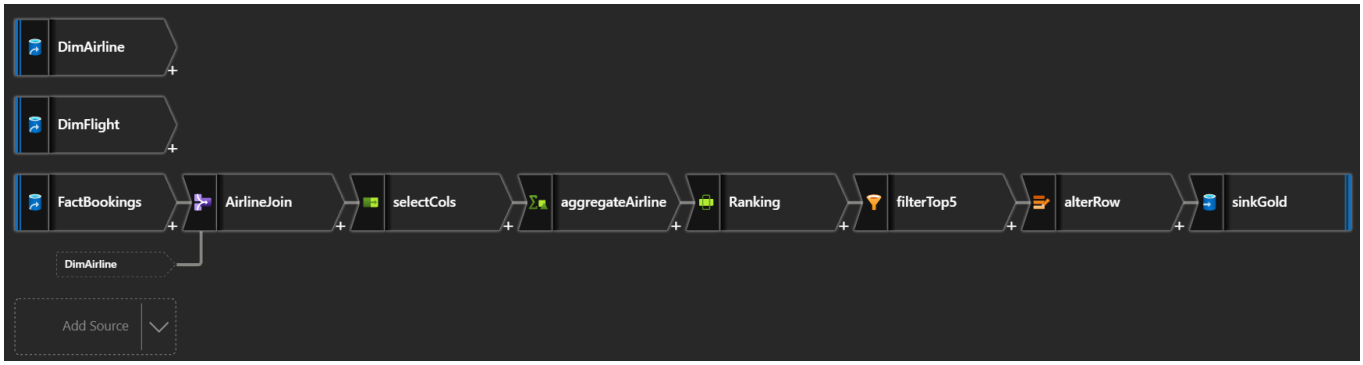

Adding pipeline: new_pipeline #1
Merge
Code
Open
KN-2023 wants to merge 1 commit into main from features
Conversation 0
Commits 1
Checks 0
Files changed 1
+17
KN-2023 commented now
Owner
No description provided.
Adding pipeline: new_pipeline
c24762b
No conflicts with base branch
Merging can be performed automatically.
Merge pull request
You can also merge this with the command line. View command line instructions.
Reviewers
No reviews
Still in progress? Convert to draft
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone


Adding pipeline: new_pipeline #1
Merged
KN-2023 merged 1 commit into main from features now
Conversation 0
Commits 1
Checks 0
Files changed 1
KN-2023 commented 1 minute ago
Owner
No description provided.
Adding pipeline: new_pipeline
c24762b
KN-2023 merged commit daa6362 into main now
Revert

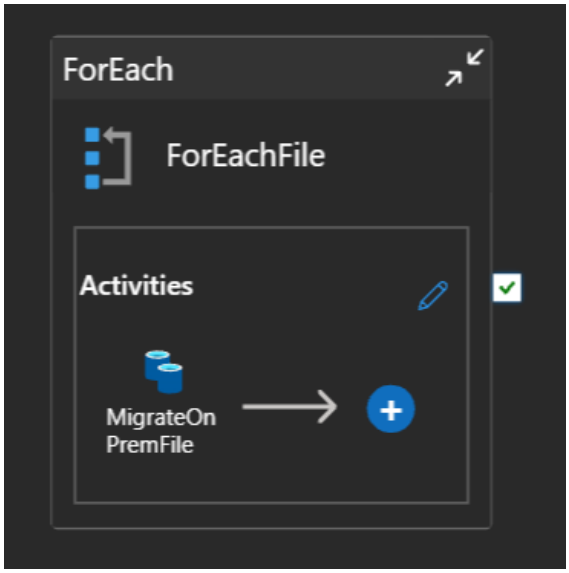
API_ingestion :



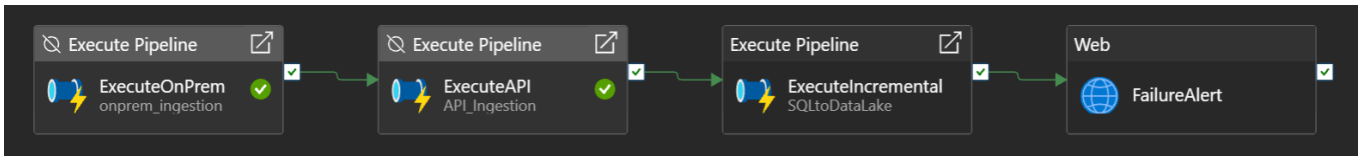
Gold layer :



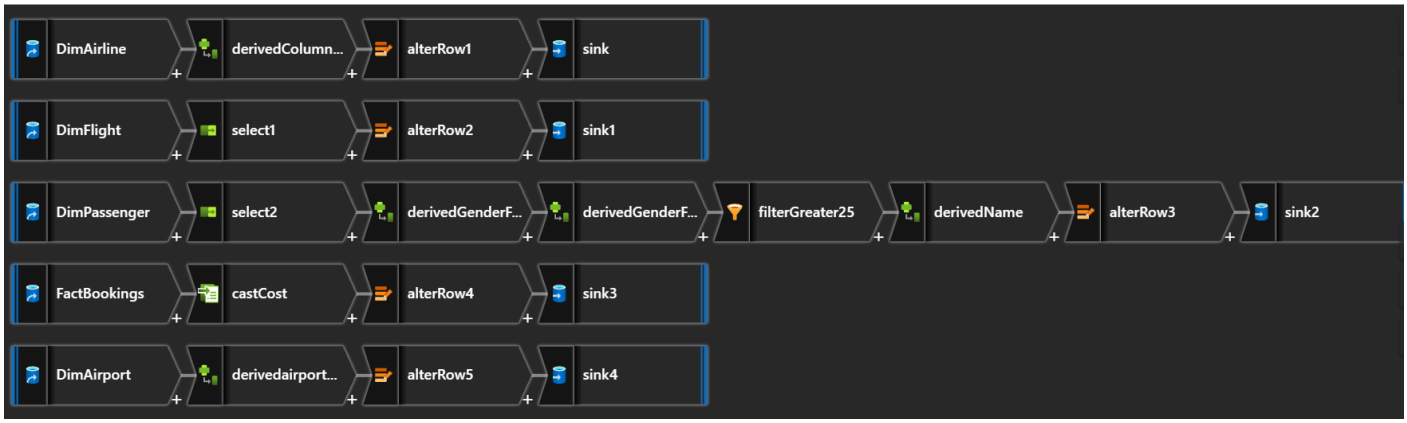
Onprem_ingestion :



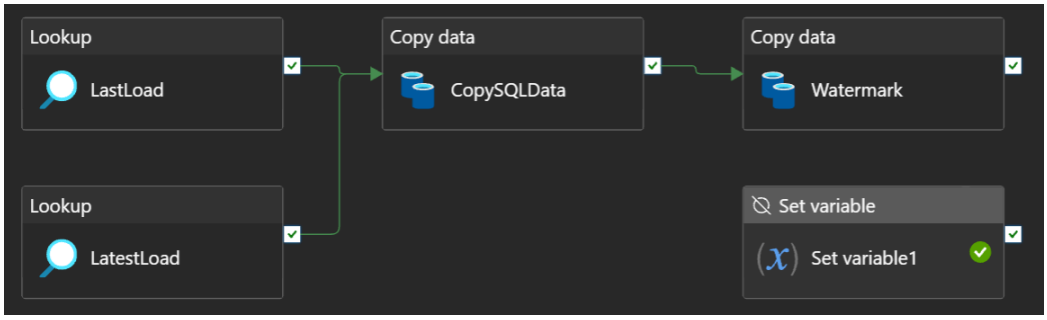
Parent :



Silver layer :

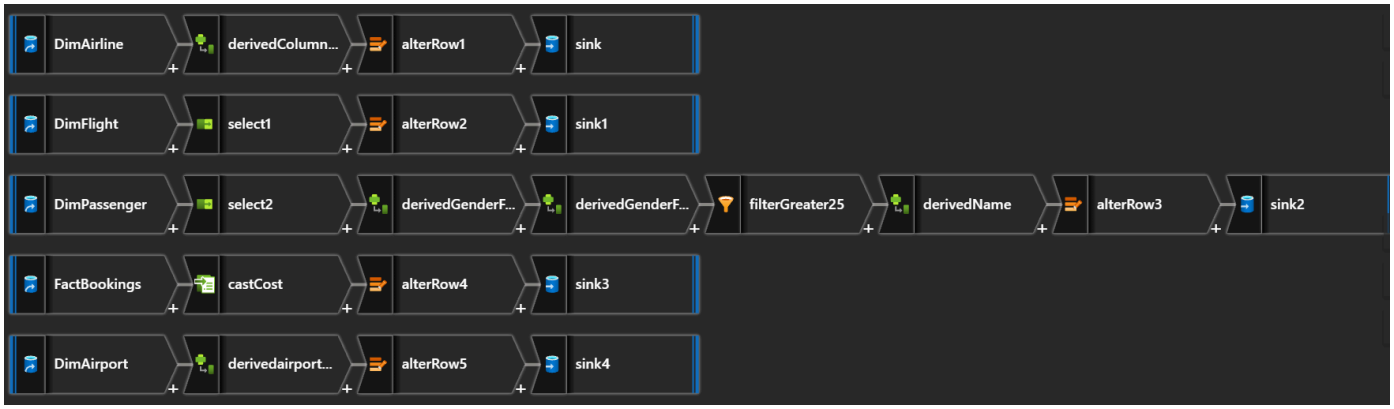


Sqltodatalake :

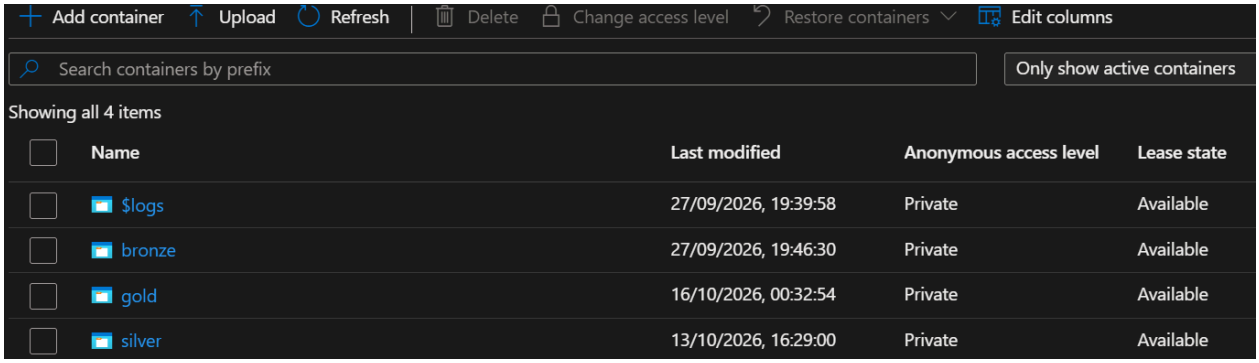


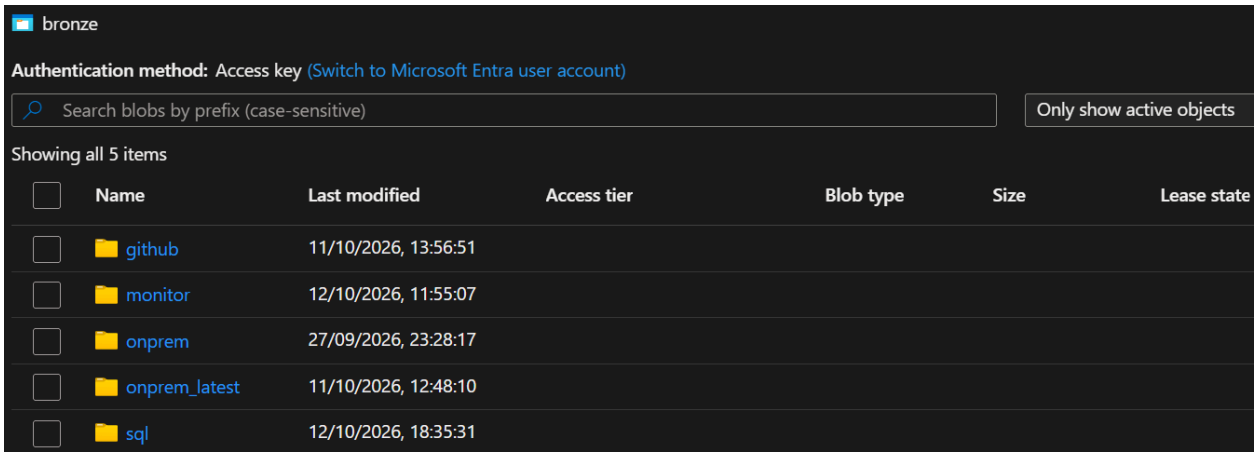
Data serving :

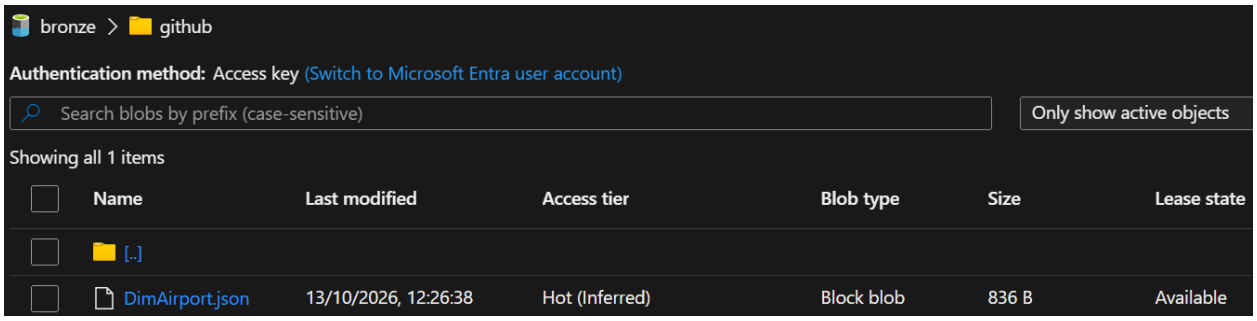
Data Transformation :

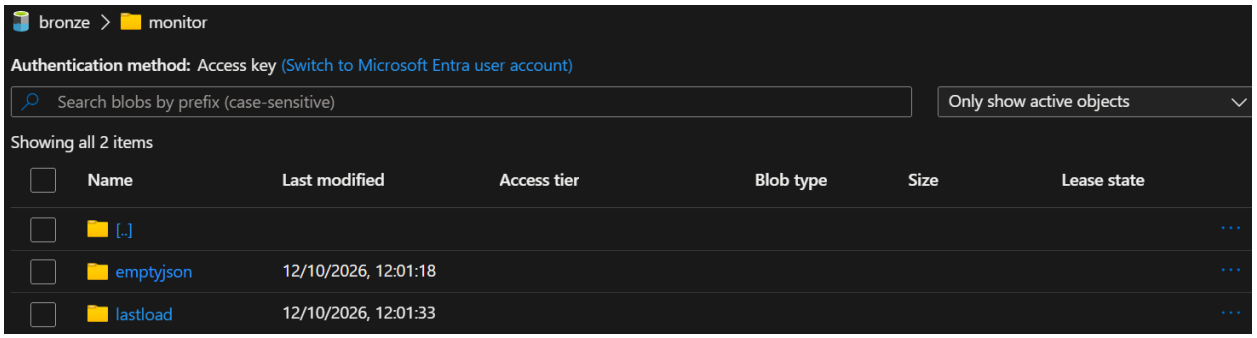


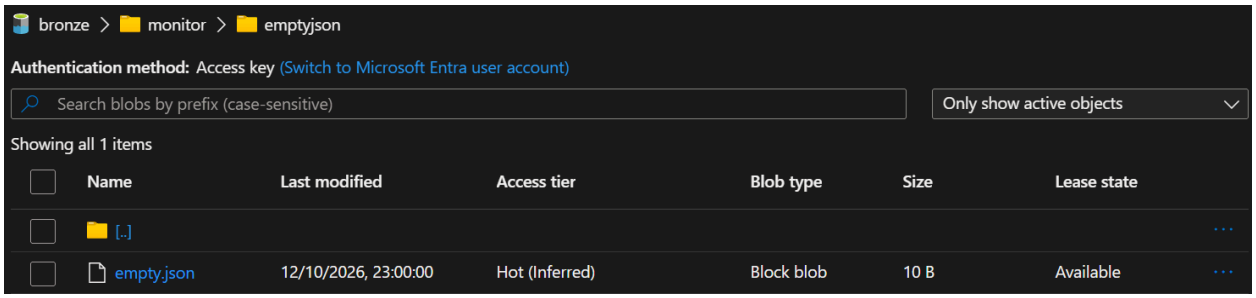
Containers :

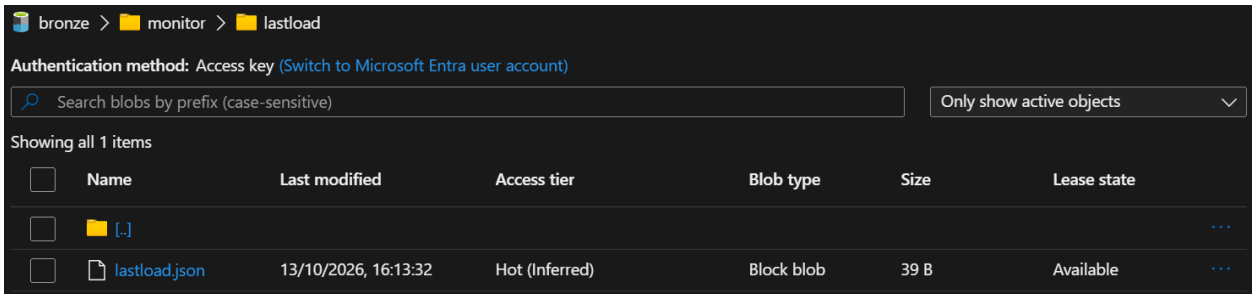

Add container
Upload
Refresh
Delete
Change access level
Restore containers
Edit columns
Search containers by prefix
Only show active containers
Showing all 4 items
Name
Last modified
Anonymous access level
Lease state
$logs
27/09/2026, 19:39:58
Private
Available
bronze
27/09/2026, 19:46:30
Private
Available
gold
16/10/2026, 00:32:54
Private
Available
silver
13/10/2026, 16:29:00
Private
Available

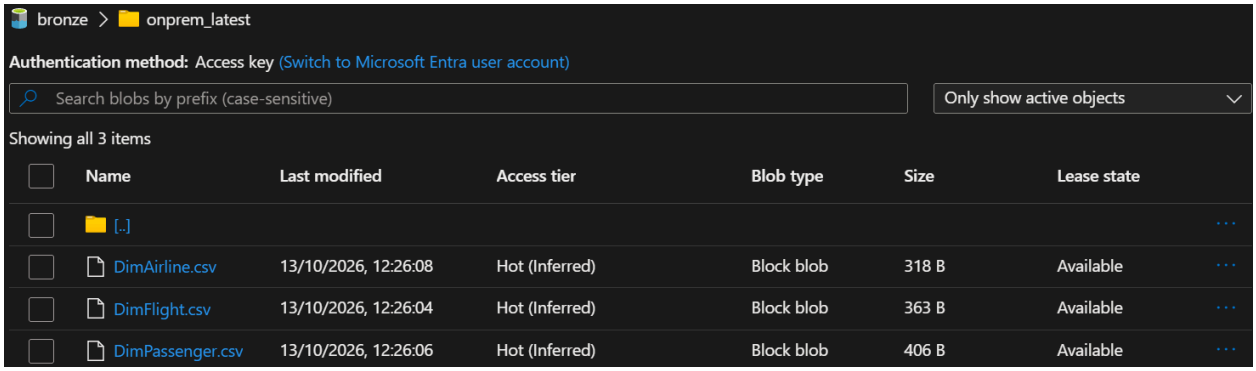

bronze
Authentication method: Access key (Switch to Microsoft Entra user account)
Search blobs by prefix (case-sensitive)
Only show active objects
Showing all 5 items
Name
Last modified
Access tier
Blob type
Size
Lease state
github
11/10/2026, 13:56:51
monitor
12/10/2026, 11:55:07
onprem
27/09/2026, 23:28:17
onprem_latest
11/10/2026, 12:48:10
sql
12/10/2026, 18:35:31

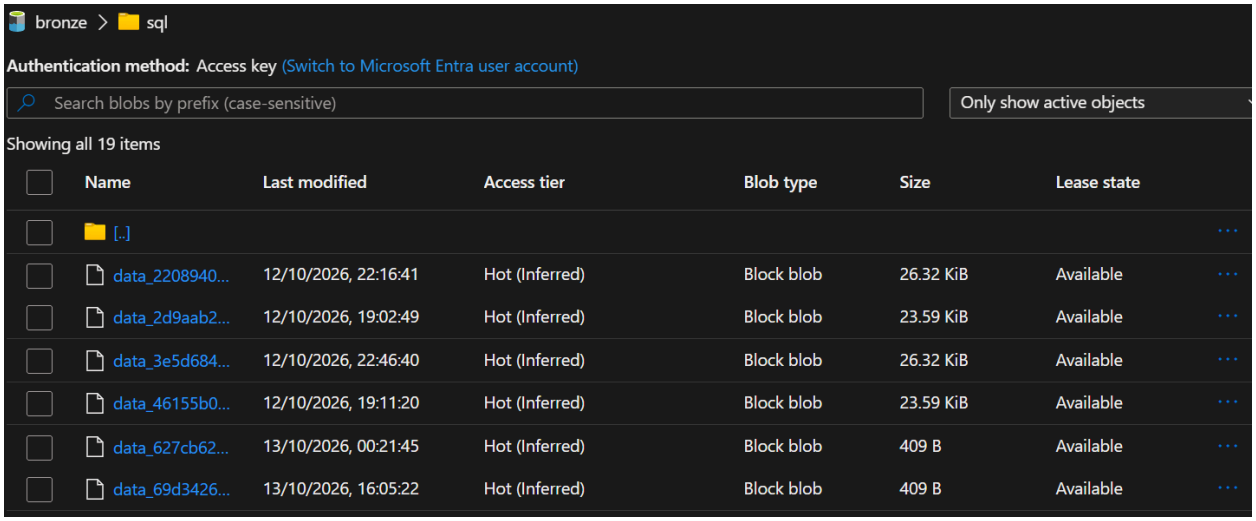

bronze
github
Authentication method: Access key (Switch to Microsoft Entra user account)
Search blobs by prefix (case-sensitive)
Only show active objects
Showing all 1 items
Name
Last modified
Access tier
Blob type
Size
Lease state
[..]
DimAirport.json
13/10/2026, 12:26:38
Hot (Inferred)
Block blob
836 B
Available

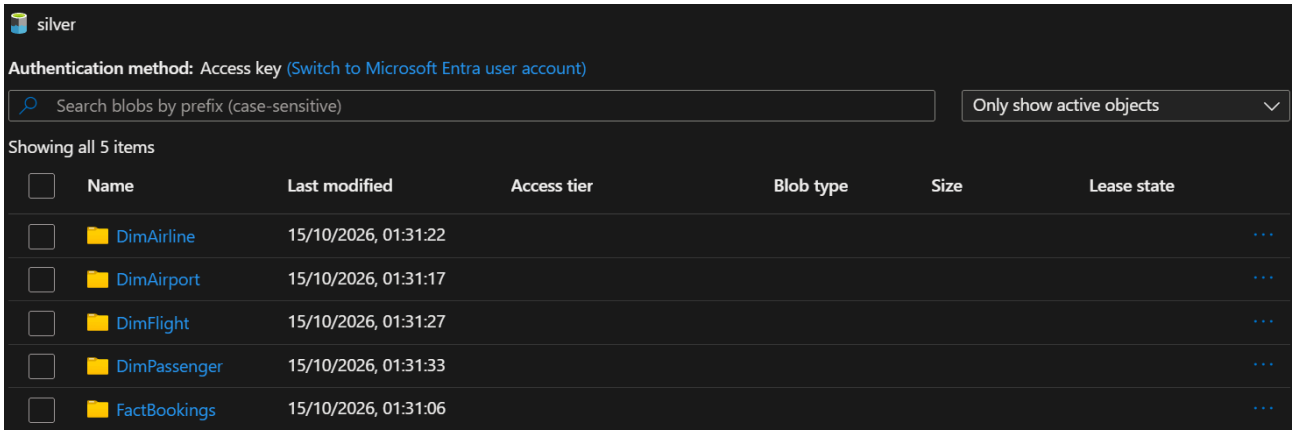

bronze
monitor
Authentication method: Access key (Switch to Microsoft Entra user account)
Search blobs by prefix (case-sensitive)
Only show active objects
Showing all 2 items
Name
Last modified
Access tier
Blob type
Size
Lease state
[..]
emptyjson
12/10/2026, 12:01:18
lastload
12/10/2026, 12:01:33

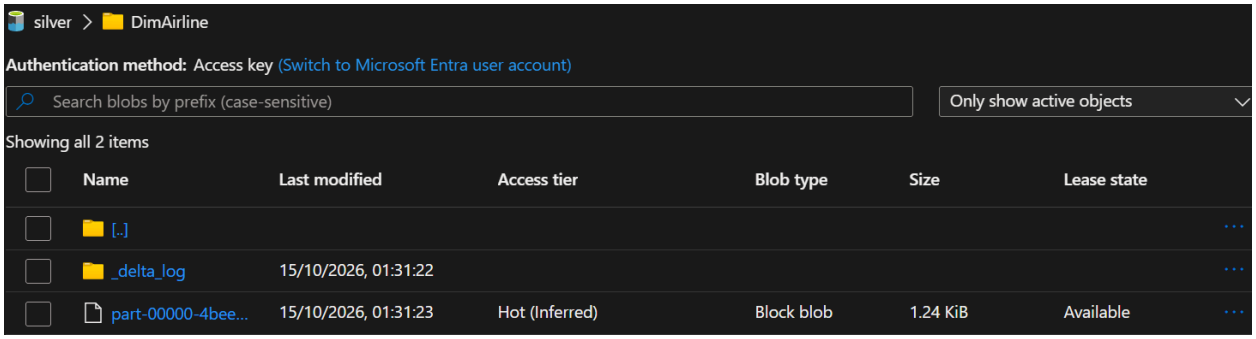

bronze > monitor > emptyjson
Authentication method: Access key (Switch to Microsoft Entra user account)
Search blobs by prefix (case-sensitive)
Only show active objects
Showing all 1 items
Name Last modified Access tier Blob type Size Lease state
[..]
empty.json 12/10/2026, 23:00:00 Hot (Inferred) Block blob 10 B Available

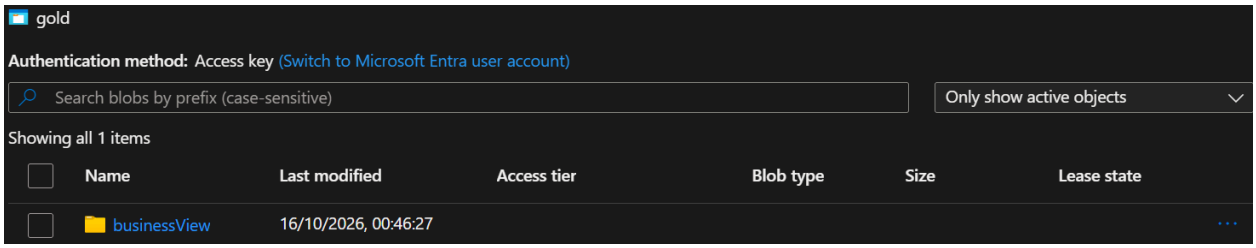

bronze > monitor > lastload
Authentication method: Access key (Switch to Microsoft Entra user account)
Search blobs by prefix (case-sensitive)
Only show active objects
Showing all 1 items
Name Last modified Access tier Blob type Size Lease state
[..]
lastload.json 13/10/2026, 16:13:32 Hot (Inferred) Block blob 39 B Available

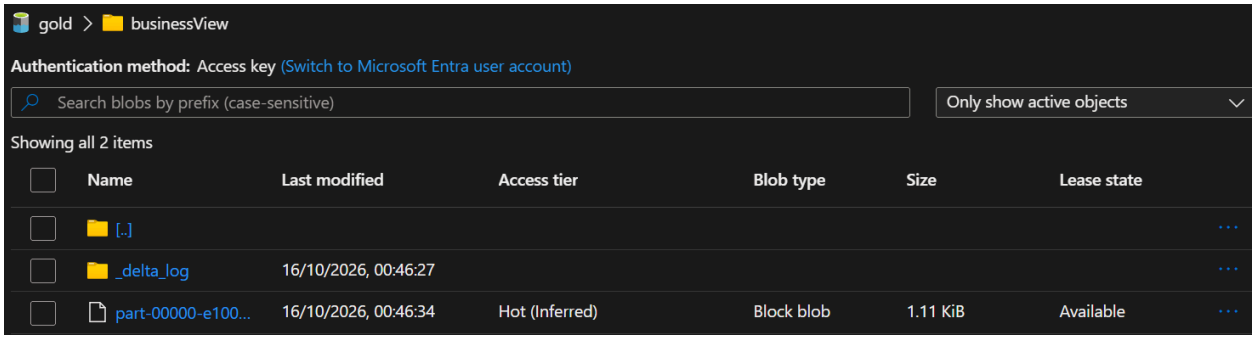

bronze > onprem_latest
Authentication method: Access key (Switch to Microsoft Entra user account)
Search blobs by prefix (case-sensitive)
Only show active objects
Showing all 3 items
Name Last modified Access tier Blob type Size Lease state
[..]
DimAirline.csv 13/10/2026, 12:26:08 Hot (Inferred) Block blob 318 B Available
DimFlight.csv 13/10/2026, 12:26:04 Hot (Inferred) Block blob 363 B Available
DimPassenger.csv 13/10/2026, 12:26:06 Hot (Inferred) Block blob 406 B Available


bronze > sql
Authentication method: Access key (Switch to Microsoft Entra user account)
Search blobs by prefix (case-sensitive)
Only show active objects
Showing all 19 items
Name Last modified Access tier Blob type Size Lease state
[..]
data_2208940... 12/10/2026, 22:16:41 Hot (Inferred) Block blob 26.32 KiB Available
data_2d9aab2... 12/10/2026, 19:02:49 Hot (Inferred) Block blob 23.59 KiB Available
data_3e5d684... 12/10/2026, 22:46:40 Hot (Inferred) Block blob 26.32 KiB Available
data_46155b0... 12/10/2026, 19:11:20 Hot (Inferred) Block blob 23.59 KiB Available
data_627cb62... 13/10/2026, 00:21:45 Hot (Inferred) Block blob 409 B Available
data_69d3426... 13/10/2026, 16:05:22 Hot (Inferred) Block blob 409 B Available


silver
Authentication method: Access key (Switch to Microsoft Entra user account)
Search blobs by prefix (case-sensitive)
Only show active objects
Showing all 5 items
Name
Last modified
Access tier
Blob type
Size
Lease state
DimAirline
15/10/2026, 01:31:22
DimAirport
15/10/2026, 01:31:17
DimFlight
15/10/2026, 01:31:27
DimPassenger
15/10/2026, 01:31:33
FactBookings
15/10/2026, 01:31:06


silver > DimAirline
Authentication method: Access key (Switch to Microsoft Entra user account)
Search blobs by prefix (case-sensitive)
Only show active objects
Showing all 2 items
Name
Last modified
Access tier
Blob type
Size
Lease state
[..]
_delta_log
15/10/2026, 01:31:22
part-00000-4bee...
15/10/2026, 01:31:23
Hot (Inferred)
Block blob
1.24 KiB
Available


gold
Authentication method: Access key (Switch to Microsoft Entra user account)
Search blobs by prefix (case-sensitive)
Only show active objects
Showing all 1 items
Name
Last modified
Access tier
Blob type
Size
Lease state
businessView
16/10/2026, 00:46:27


gold > businessView
Authentication method: Access key (Switch to Microsoft Entra user account)
Search blobs by prefix (case-sensitive)
Only show active objects
Showing all 2 items
Name
Last modified
Access tier
Blob type
Size
Lease state
[..]
_delta_log
16/10/2026, 00:46:27
part-00000-e100...
16/10/2026, 00:46:34
Hot (Inferred)
Block blob
1.11 KiB
Available